Chapter 1

Introduction

Frequency synthesizers are an integral part of various applications, such as wireless [1, 2, 3] and wireline [4, 5, 6] communication systems and radars [7, 8, 9]. In this chapter, we will look at how frequency synthesizers are used in these systems, and how specific performance metrics of the synthesizer effect the overall system performance.

The generation of frequency sources with low phase noise at a cost in terms of power, area, and many other factors has been an on-going challenge over the years. This performance improvement is motivated by the fact that frequency sources are used in various applications, and the performance of the frequency source often limits that of the overall system and application.

1.1 Application of Frequency Synthesizers

Fig. 1.1 shows a block diagram of a conventional serial link, how some frequency sources are used in the system to generate data retiming clocks in the TX and RX, and the effect of the clock jitter on the accuracy of data transmission. A PLL in the TX path generates a clock to decide when the transmitted signal switches to the next bit of data. In a Non-Return-to-Zero (NRZ) format, this data is either 0 or 1. Then, after the transmitted data travels through the channel and experiences frequency dependent loss, the signal is partially recovered using an equalizer circuit that counteracts the loss from the channel, and then finally the signal is retimed by a clock from a Clock-and-Data-Recovery (CDR) circuit in the RX path i.e. a decision is made about whether the received bit is 0 or 1 whenever the RX clock rises. We can see that the accuracy of this decision is highest when the transition of the RX clock happens at around the middle of two consecutive transitions of the TX clock, since this is where the retimed signal is most clearly a 0 or a 1. A deviation from this optimal timing will cause the signal to be resampled when the signal is close to the data threshold, leading to an increase in Bit-Error-Rate (BER). The effect of DC skew between the data transition and the RX clock, as well as that of low frequency noise from the TX PLL and channel can be suppressed since the CDR generates the RX clock based on the received data. However, the high frequency noise from the TX PLL and channel that is outside the CDR bandwidth, as well the noise from the CDR itself will cause fluctuations in the relative timing of the data transitions and RX clock, and can be difficult to suppress when we must satisfy various requirements, such as low BER, high channel loss, and low power and area cost. Generating low jitter clocks at a low cost is crucial to improve the performance of wireline communication systems. Furthermore, wireline communication systems are also often realized as multi-channel systems with individual clocks accompanying each lane. As such, making frequency synthesizers with low silicon area is also important if we wish to integrate these systems onto a single chip to lower the cost.

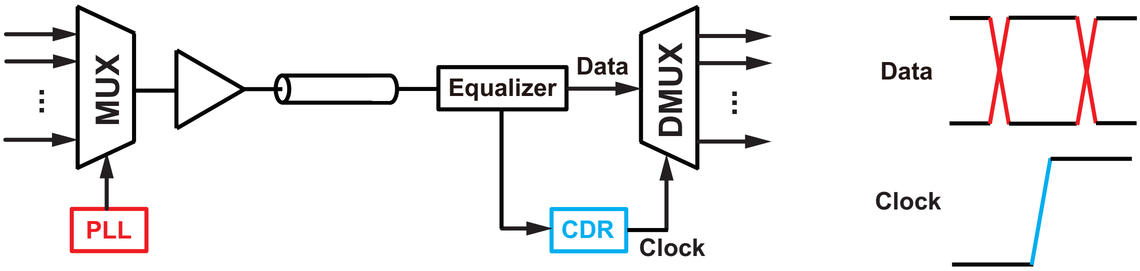


Figure 1.1: A block diagram of a conventional serial link, and the effect of the clock jitter on data transmission

Fig. 1.2(a) shows a simplified view of a wireless receiver where the received signal enters an antenna, and then gets down-converted by mixing the received signal with an LO. The phase noise of the LO leads to two important issues. Firstly, as can be seen in Fig. 1.2(b), a phase error in the LO will cause the constellation of the received signal to rotate, and since the value of the received bits are decided based on the position of the symbol in the I/Q plane, this sort of rotation of the symbol caused by the LO noise will lead to increased BER since the symbol has been moved from the correct spot. The ratio between the deviation caused by noise and the distance between symbols is called the Error-Vector-Magnitude (EVM), and it must be kept low in-order to maintain a satisfactory BER. Furthermore, maintaining a small EVM becomes more difficult as the modulation scheme becomes more complex, since the symbols become more congested in the I/Q plane. Secondly, a phase error in the LO signal can become problematic when combined with a large blocker signal. As shown conceptually in Fig. 1.2(a), the input to the receiver will contain not only the desired signal at around the carrier frequency, but also an unwanted signal component at a certain frequency offset. In an ideal receiver with a perfect single tone LO and linear mixers and amplifiers, the blocker would maintain the same frequency offset from the desired signal and now it can be filtered easily with a Low-Pass-Filter (LPF) because the signal is now at DC. However, in reality the LO signal contains phase noise and as a result, the down-converted blocker will also have a skirt on both sides corresponding to the LO phase noise. This unwanted phenomenon is known as reciprocal mixing, and depending on the frequency offset and magnitude of the blocker along with the magnitude of the LO phase noise, these skirts can overlap with and cover the desired signal, degrading the EVM and in turn the signal integrity. The key here is that this noise component cannot be filtered out anymore since it is at the same location of the frequency spectrum as the desired signal. The same issue of reciprocal mixing can be caused by spurs in the LO spectrum, which are discrete tones at specific frequency offsets from the carrier. Lowering the phase noise and spurs in frequency synthesizers is crucial to realize a wireless communication system with high performance. Another important aspect of frequency synthesizer performance in wireless communication systems is the settling time. Wireless transceivers sometimes employ frequency hopping where the frequency synthesizer has to settle to a new frequency very quickly, and they may also be duty-cycled in IoT systems to save power meaning that they must start-up quickly.

Fig. 1.3 shows a conceptual diagram of a Frequency-Modulated-Continuous-Wave (FMCW) radar. The radar sends a signal whose frequency is modulated, often in a way that the transient behaviour of the frequency resembles that of a saw-tooth waveform. This signal (often called the chirp signal) will hit the target, get reflected, and return to the radar. The received signal is the same chirp signal that we originally sent to the target, but shifted by the time it took for the signal to reach the target and come back. If we can obtain the frequency difference between

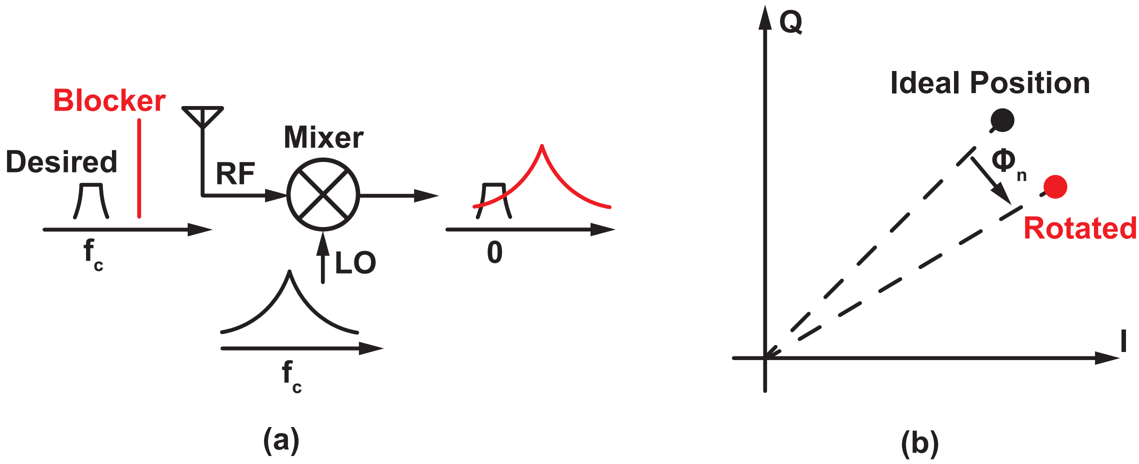


Figure 1.2: A simplified view of a wireless receiver, and the phenomenon of reciprocal mixing

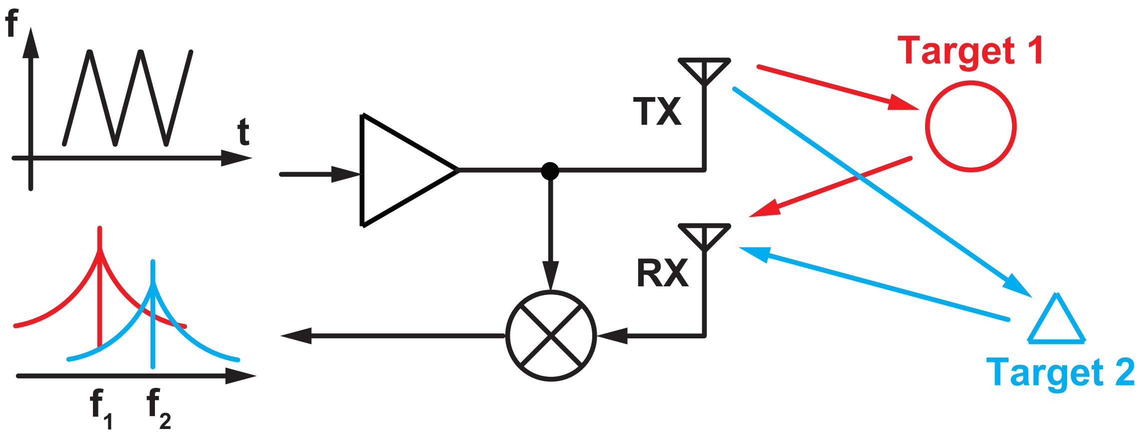


Figure 1.3: A basic FMCW radar system and the effect of phase noise on its performance

the transmitted and received chirp signals, we will be able to calculate the distance to the target since the frequency difference is equal to the slope of the saw-tooth multiplied by the time it takes for the signal to reach the target and come back, and the slope is a well defined value controlled by the radar system while the speed of the signal is equal to the speed of light. As such, the radar system generates a beat signal of the TX and RX signals by mixing the two, and is able to derive the distance to the target as desired. The distance to multiple targets can be measured at the same time based on the same principle, with a corresponding beat signal component for each target. However, an issue occurs when there are two or more signals at similar distances. As shown conceptually in Fig. 1.3, the phase noise contained in the chirp signal can result in noise around the beat signals. When 2 or more beat signals are close to each other, the noise around one of the beat signals can contaminate the other beat signals. This will lead to a lower Signal-to-Noise-Ratio (SNR) and result in a lower resolution in terms of distinguishing between two targets. It is also very important to realize a chirp signal with good linearity, because otherwise the beat signal frequency will not be constant and the spectrum can get smeared, again potentially degrading the SNR.

As can be seen from these examples, frequency synthesizers are an integral part of many different applications. Furthermore, improving their performance in terms of jitter, spurs, power consumption, bandwidth, silicon area, settling time, and chirp linearity among many other metrics is vital to extend the possible applications communication and sensing systems.

1.2 Impact of This Dissertation

Here we will explain the impact of the research introduced in this dissertation, and what type of advantages it provides compared to the prior arts. We have seen that frequency synthesizers are a critical part of various applications. The frequency synthesizers are usually realized using fractional-N PLLs which will look at in detail in Chapter 2 due to their capability of synthesizing signals of different frequencies with fine resolution. Many of the current state-of-the-art fractional-N PLLs employ Digital-to-Time Converters (DTCs) as we will also explain in Chapter 2, and these PLLs can achieve great jitter-power performance. On the other hand, our dissertation deals with fractional-N PLLs employing a technique called Harmonic-Mixer (HM)-based feedback. HM-based PLLs have some distinct advantages over DTC-based ones as we will explain below.

First and foremost, the biggest advantage of the HM-based PLL over DTC-based ones is the fact that background calibration is not required. DTC-based PLLs improve the jitter-power performance of a fractional-N PLL by cancelling the unwanted noise. This requires the DTC itself to have accurate gain and linearity, leading to a need for calibration schemes. This calibration is necesssary not just during start-up but also when switching output frequencies. A major issue with calibration is that it often takes a long time to settle. The transient locking time of a PLL is usually in the order of hundreds of ns to tens of us, depending on the loop bandwidth and frequency change among other things. On the other hand, the settling time for the gain and linearity calibration in a DTC-based PLL will take in the order of tens of us, to even a few ms in the worst case [10, 11, 12]. This has huge implications in communication systems because frequency switching occurs very often and the transmission of data can only begin once the PLL has settled. Another type of application where the reduced locking time improves the overall performance is the duty-cycled radio, often used in Internet-of-Things (IoT) systems to save power. As conceptually shown in Fig. 1.4, power consuming blocks such as the TX and RX are only turned on occassionally for short periods at a time, and the rest of the time, only low power blocks such as those used for biasing and time-keeping are turned on, resulting in low average power consumption [13, 14, 15, 16, 17]. If the PLL takes long to start up however, the on-time of the TX/RX blocks have to be long, meaning that the extent to which the power consumption can be lowered will be limited. Therefore, calibration-free PLLs can be advantageous to improve the performance especially in these situations. Of course, there are many other types of calibration-free architectures for PLLs, including just a very simple single-loop fractional-N PLL with no DTC. However, such architectures tend to suffer from worse jitter-power performance compared to DTC-based architectures. HM-based PLLs on the other-hand, have the potential to achieve close to, or in some cases even better in terms of jitter-power performance than DTC-based PLLs as will be analyzed in detail in chapters 3 and 5. This capability of achieving great jitter-power performance while being calibration-free is a major strength of HM-based PLLs, and is one of the main factors motivating the work done in this dissertation.

Another motivation of our dissertation is the potential to improve multi-channel communication systems. An example of this is Universal Chiplet Interconnect Express (UCIe) for chiplet-based systems. While process scaling has gradually improved the speed and power consumption of circuits overall, the cost of implementation has not gone down as quickly, meaning that the realization of advanced processes is not accessible to everyone, or only affordable in small silicon areas. As such, many systems in recent years are shifting to chiplet-based designs, where different circuit

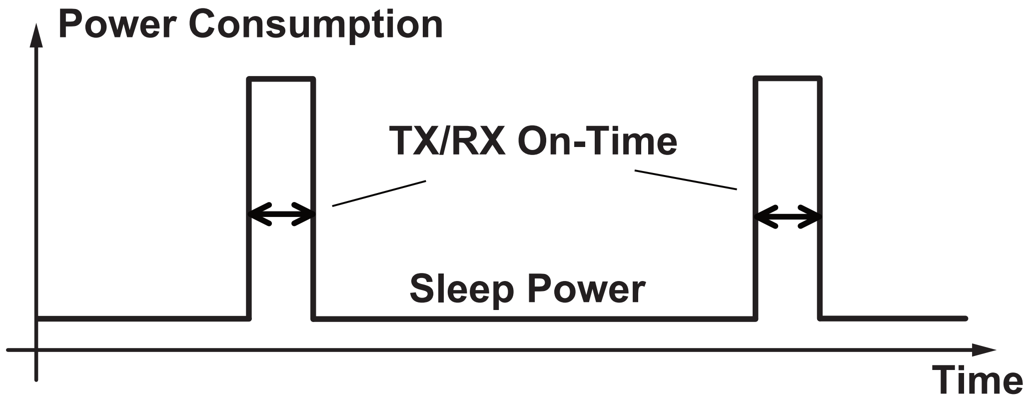


Figure 1.4: The concept of saving power by employing duty-cycled operation for the radio

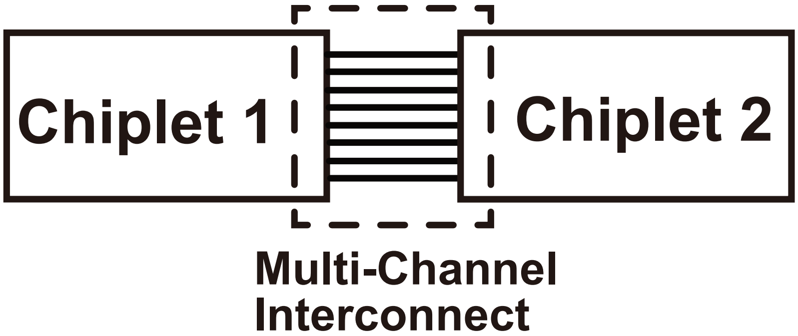


Figure 1.5: The concept of multi-channel interconnects in chiplet-based systems

blocks are realized using different processes from potentially different vendors and are then connected with each other at a higher layer e.g.7nm process for digital circuits such as CPUs, 180nm process for analog blocks, and 65nm process for RF front-ends. The interconnect between different blocks is crucial in such a system, and multi-channel links as shown in Fig. 1.5 are often used. It is preferrable to have individual PLLs for each channel to realize flexible clocking, and as such, PLLs with small area can be highly advantageous [18, 19, 20]. Calibration-intensive PLLs require large Look-Up Tables (LUTs) to store various coefficients and may not be suited for this purpose. HM-based PLLs on the other-hand, do not require such LUTs because they are calibration-free. Furthermore, we have demonstrated in chapter 5 of this dissertation that the HM-based architecture can be used for ring-VCO-based fractional-N PLLs. The use of ring VCOs over LC-based ones can greatly help to reduce the silicon area of PLLs, and its combination with the calibration-free nature of HM-based PLLs allows compact and flexible clocking that is especially suitable for multi-channel communications systems.

Finally, we should note the importance of maintaining a good diversity of solutions for high performance frequency synthesis and circuit design in general. Even in the presence of an existing solution (DTC-based PLLs in this case), it is often worth investigating a new approach since it can lead to even higher performing circuits and applications if there are advances in process technology or innovations in circuit design that are yet to be seen. A good example of this in recent years is the rise of ring amplifiers in Analog-to-Digital-Converter (ADC) designs [21, 22, 23, 24]. It was becoming difficult to implement amplifiers in advanced processes due to the decreasing supply voltages and there were a lot of advances being made in the area of Successive-Approximation-Register (SAR) ADCs which did not rely on amplifiers. As such, some researchers at the time were suggesting that we do away with ADCs such as pipeline or delta-sigma ADCs that rely on amplifiers and focus solely on SAR ADCs. However, by keeping the idea of amplifier-based ADCs alive by introducing the scaling friendly ring-amplifier, we were not only able to maintain the diversity of solutions in ADCs by realizing scaling friendly amplifier-based ADCs, but we were also able to come up with new high performing architectures such as noise shaping SAR ADCs that combined the ideas from SAR ADCs with that from amplifier-based ADCs. By maintaining a good diversity of solutions in our circuit toolbox, we create the possibility of new breakthroughs in the future.

To summarize, the work done in this dissertation has the potential to improve efficiency of duty-cycled radios in IoT and communication systems in general due to its calibration-free nature, provide compact and flexible clock sources without large LUTs for multi-channel applications like UCIe, and also to provide a new direction in PLL research by presenting a different approach from that taken in the current state-of-the-art works.

1.3 Thesis Organization

This dissertation consists of 7 chapters. Chapter 2 reviews the basics of PLL design, with a focus on the trade-offs that occur in terms of phase noise and its relation with the loop transfer functions in integer-N and fractional-N configurations. We will also look at some prior arts that attempt to resolve the issues present in fractional-N PLLs. In chapter 3, we introduce the concept of HM-based feedback and their usage in fractional-N PLLs to reduce phase noise. We also introduce the dual-feedback PLL which is an architecture proposed in our research that makes use of the dual-feedback architecture, and present measurement results and analysis to demonstrate the advantages of our work. In chapter 4, we deal with the phenomenon of offset spurs which is a general issue in HM-based PLLs, and present analysis results that predict the amplitude and location of these spurs as well as provide a guideline on how to design the circuits so that the offset spurs are suppressed. Chapter 5 presents an idea where we look to employ the HM-based PLL to ring VCO based inductorless frequency synthesis by improving on the dual-feedback PLL to allow an even better suppression of noise. Comparisons with other approaches, as well as analysis and measurement results are presented. Chapter 6 presents yet another HM-based architecture where we again look to improve on the dual-feedback architecture, but this time with a more general technique that employs feed-forward noise cancellation. Finally, we conclude the dissertation in chapter 7.

Chapter 2

The Basics of Phase Locked Loops and Prior Arts

The Phase-Locked-Loop (PLL) is a circuit that is used highly often to perform frequency synthesis. In this chapter, we will first show the basic linear model and noise sources in PLLs along with how to deal with these noise sources. Then, we will derive the loop transfer functions and the general effect of noise sources at various points of the loop. We will also discuss the stability requirements and how the poles and zeroes in the loop must be configured for the loop to operate correctly. Finally, we will look specifically at fractional-N PLLs and the additional issues they have over integer-N PLLs, and how prior arts have looked to overcome these issues.

2.1 Basic Linear Model and Noise Sources for Conventional PLLs

Fig. 2.1 shows a basic linear model of a conventional PLL, along with noise sources that contribute noise to the PLL output at various points inside the loop. ϕ_{nREF} is the reference path noise, and is usually composed of Phase Noise (PN) from the reference oscillator itself, as well as that from the input buffer. The PN profile of this noise source can mostly be approximated as white noise at the frequency of interest, although it can be dominated by flicker noise at lower frequencies. i_{nPD} is the noise caused by the Phase-Detector (PD), which can take on various forms depending on the specific configuration chosen. For a Phase-Frequency-Detector/Charge-Pump (PFD/CP)-based configuration, the noise may be caused by the time domain noise from the PFD or the current domain noise from the CP. In a Sampling-Phase-Detector (SPD)-based configuration, the noise can be attributed to the kT/C noise at the sampling capacitor and the flicker noise from the switches. In a Digital-PLL (DPLL), the noise can be attributed to the quantization noise from the Time-to-Digital-Converter (TDC). This noise source can also often be approximated as white noise, but may be dominated by flicker noise at lower frequencies. v_{nLF} is the noise caused by the Loop-Filter (LF). This may come from the kT/C noise in an RC network. Again, the noise profile is usually white with some flicker noise at lower frequencies, but the LF noise can often be designed to be negligibly low and may be ignored in many cases. ϕ_{nVCO} is the PN from the Voltage-Controlled-Oscillator (VCO), and is often one of the primary sources of noise that decides the design of the overall PLL. The PN profile is mainly proportional to $1/f^2$, although at lower frequencies it may be $1/f^3$ due to up-converted flicker noise. ϕ_{nDIV} is the PN from the feedback frequency divider. In a simple integer-N PLL, this may simply be white thermal noise from the divider, or the output

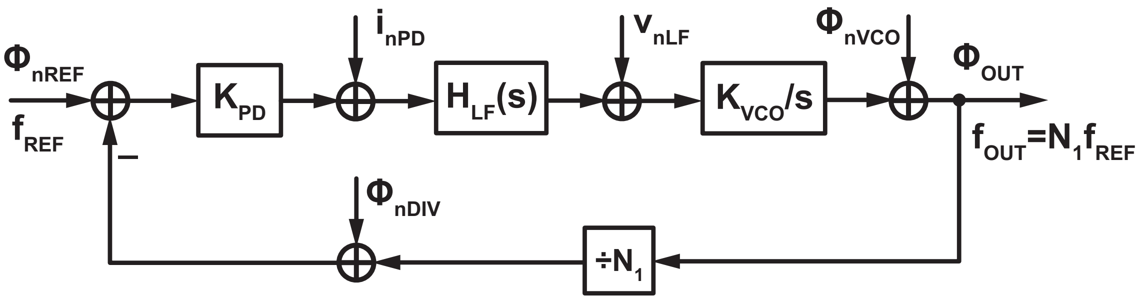


Figure 2.1: A basic linear model of a PLL along with noise sources at different points in the loop

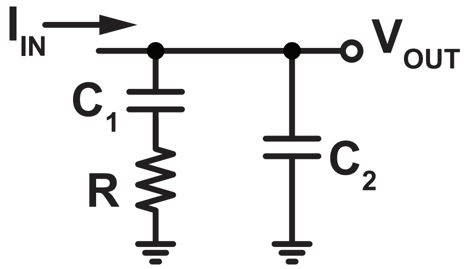


Figure 2.2: A second order LF assuming a current input and a voltage output

re-timer if we have one. In a fractional-N PLL, this may also contain the shaped quantization noise from modulating the division ratio as we will explain later. In a Sub-Sampling PLL (SSPLL), this may come from just the isolation buffer since there is no frequency divider. ϕ_{OUT} is the resulting PN at the output signal due to all the different noise sources. K_{PD}, $H_{\mathrm{LF}}(s)$, K_{VCO}, and N_1 is the phase detector gain, LF transfer function, VCO frequency gain, and feedback division ratio respectively. Finally, f_{REF} and f_{OUT} are the reference and output frequencies respectively, with $f_{\mathrm{OUT}} = N_1 f_{\mathrm{REF}}$.

2.1.1 Loop Transfer Functions in a PLL and the Effect of Each Noise Source

Based on the linear model shown in Fig. 2.1, we can derive important loop transfer functions and then use them to quantify the effect that each noise source has on the overall output signal. It is useful to first define the open-loop transfer function $H_{\mathrm{OL}}(s)$, which is the transfer function of the entire loop when we cut it at some point. It can be calculated as

$$H_{\mathrm{OL}}(s) = \frac{K_{\mathrm{PD}} K_{\mathrm{VCO}} H_{\mathrm{LF}}(s)}{N_1 s} \tag{2.1}$$

and will be used to describe other transfer functions in subsequent calculations. The actual shape of $H_{\mathrm{OL}}(s)$ depends on $H_{\mathrm{LF}}(s)$. As an example, we can consider a second order RC LF assuming a current input and a voltage output as shown in Fig. 2.2. This kind of configuration is often used in a type-II PFD/CP-based PLL. The transfer function of this LF is given by

$$H_{\mathrm{LF}}(s) \equiv \frac{V_{\mathrm{OUT}}}{I_{\mathrm{IN}}} = \frac{1 + sRC_1}{s(C_1 + C_2) + s^2 RC_1 C_2} = \frac{1}{s(C_1 + C_2)} \frac{1 + s/\omega_{\mathrm{z}}}{1 + s/\omega_{\mathrm{p}}}, \tag{2.2}$$

where V_{OUT} and I_{IN} are the output voltage and input current respectively, and $\omega_{\mathrm{z}} \equiv \frac{1}{RC_1}$ and $\omega_{\mathrm{p}} \equiv \frac{1}{RC_1} + \frac{1}{RC_2}$ are the zero and pole in $H_{\mathrm{LF}}(s)$ respectively. Combining this with the integrator from the transfer function of the VCO, we can see that $H_{\mathrm{OL}}(s)$ contains two poles at DC, a zero at ω_{z} and a third pole at ω_{p}. We can see from equation (2.2) that the open loop gain is infinite at DC, then gradually goes down as the frequency increases and eventually tends towards zero. The point at which the absolute value of $H_{\mathrm{OL}}(s)$ becomes unity is called the unity gain frequency, and will be referred to in this dissertation as ω_{UG}.

We can now calculate the transfer function from each noise source to the overall output. The equations are as shown below.

$$H_{\mathrm{REF,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nREF}}} = \frac{N_1 H_{\mathrm{OL}}(s)}{1 + H_{\mathrm{OL}}(s)} \equiv N_1 H_{\mathrm{LPF}}(s) \tag{2.3}$$

$$H_{\mathrm{PD,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{i_{\mathrm{nPD}}} = \frac{N_1}{K_{\mathrm{PD}}} H_{\mathrm{LPF}}(s) \tag{2.4}$$

$$H_{\mathrm{LF,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{v_{\mathrm{nLF}}} = \frac{K_{\mathrm{VCO}}}{s} \frac{1}{1 + H_{\mathrm{OL}}(s)} \equiv H_{\mathrm{BPF}}(s) \tag{2.5}$$

$$H_{\mathrm{VCO,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCO}}} = \frac{1}{1 + H_{\mathrm{OL}}(s)} \equiv H_{\mathrm{HPF}}(s) \tag{2.6}$$

$$H_{\mathrm{DIV,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nDIV}}} = -H_{\mathrm{LPF}}(s) \tag{2.7}$$

$H_{\mathrm{REF,OUT}}(s)$ is the transfer function from the reference path to the PLL output, and is given by equation (2.3). At low frequencies, $H_{\mathrm{OL}}(s)$ has a very large absolute value and $H_{\mathrm{REF,OUT}}(s)$ can be approximated as just N_1 since $1 + H_{\mathrm{OL}}(s) \approx H_{\mathrm{OL}}(s)$. At frequencies above ω_{UG}, $H_{\mathrm{REF,OUT}}(s)$ can be approximated as $N_1 H_{\mathrm{OL}}(s)$ since $1 + H_{\mathrm{OL}}(s) \approx 1$. As a result, $H_{\mathrm{REF,OUT}}(s)$ can be viewed as a low-pass characteristic with a DC value of N_1, a bandwidth of ω_{UG}, and a roll-off of 40dB/dec. In equation (2.3), we extracted the unity gain portion of $H_{\mathrm{REF,OUT}}(s)$ and defined it as $H_{\mathrm{LPF}}(s)$.

$H_{\mathrm{PD,OUT}}(s)$ is the transfer function from the PD output to the PLL output, and is given by equation (2.4). This expression is similar to equation (2.3), except for the fact that it is a factor of K_{PD} smaller because the PD gain is not contained in the feed-forward path. As a result, $H_{\mathrm{PD,OUT}}(s)$ is a low-pass characteristic with a DC value of $\frac{N_1}{K_{\mathrm{PD}}}$ and a bandwidth of ω_{UG}.

$H_{\mathrm{LF,OUT}}(s)$ is the transfer function from the LF output to the PLL output, and is given by equation (2.5). At low frequencies, $|H_{\mathrm{LF}}(s)|$ has a 20dB/dec slope in relation to the frequency because $\frac{1}{1+H_{\mathrm{OL}}(s)} \approx \frac{1}{H_{\mathrm{OL}}(s)} \propto s^2$, and therefore $H_{\mathrm{LF,OUT}}(s) \propto s$. On the other hand, at frequencies higher than ω_{UG}, $|H_{\mathrm{LF}}(s)|$ has a -20dB/dec slope in relation to the frequency because $\frac{1}{1+H_{\mathrm{OL}}(s)} \approx 1$ and therefore $H_{\mathrm{LF,OUT}(s)} \propto \frac{1}{s}$. As such, $H_{\mathrm{LF,OUT}}(s)$ has a band-pass characteristic with a center frequency around ω_{UG} and a roll-off of 20dB/dec. We will rename this characteristic as $H_{\mathrm{BPF}}(s)$.

$H_{\mathrm{VCO,OUT}}(s)$ is the transfer function from the VCO output to the PLL output. At low frequencies, $H_{\mathrm{VCO,OUT}}(s)$ has a 40dB/dec slope in relation to the frequency because $\frac{1}{1+H_{\mathrm{OL}}(s)} \approx \frac{1}{H_{\mathrm{OL}}(s)} \propto s^2$. At frequencies higher than ω_{UG}, $H_{\mathrm{VCO,OUT}}(s) = \frac{1}{1+H_{\mathrm{OL}}(s)} \approx 1$. As such, $H_{\mathrm{VCO,OUT}}(s)$ exhibits a high-pass characteristic with a cut-off frequency of ω_{UG} and a roll-off of 40dB/dec. In equation 2.6, we redefine $H_{\mathrm{VCO,OUT}}(s)$ as $H_{\mathrm{HPF}}(s)$.

Finally, $H_{\mathrm{DIV,OUT}}(s)$ is the transfer function from the divider output to the PLL output. It has basically the same characteristic as $H_{\mathrm{REF,OUT}}(s)$, except that the polarity is reversed.

By using these equations, we can derive the effect of each noise source on the output PN spectrum based on the fact that the Power Spectral Density (PSD) of the noise source can be multiplied by the absolute value of the transfer function squared to obtain the contribution of that noise source to the overall output. Different noise sources can be considered within a PLL depending on the conditions. Fig. 2.3 shows a simplified version of the linear model shown in Fig. 2.1. The simplification is done in two ways. Firstly, the LF noise v_{nLF} has been removed since it is usually negligible and also since we want to make the main trade-off clearer. Secondly, the noise from the reference path, divider, and PD output have been merged into a single reference-referred equivalent noise source ϕ_{nEQUIV}. We can see from equations (2.3), (2.4), and (2.7) that the transfer functions from these noise sources to the PLL output are all similar low-pass characteristics, but just with different DC values. Furthermore, all of these noise sources can be roughly approximated as white noise. As such, these noise sources can be merged into one large white noise source that experiences a low-pass characteristic to the PLL output. Specifically, ϕ_{nEQUIV} can be calculated as

$$\phi_{\mathrm{nEQUIV}} = \phi_{\mathrm{nREF}} - \phi_{\mathrm{nDIV}} + \frac{1}{K_{\mathrm{PD}}} i_{\mathrm{nPD}} \tag{2.8}$$

and the PSD of ϕ_{nEQUIV} which we refer to as S_{nEQUIV} can be calculated as

$$S_{\mathrm{nEQUIV}} = S_{\mathrm{nREF}} + S_{\mathrm{nDIV}} + \frac{1}{K_{\mathrm{PD}}^2} S_{\mathrm{nPD}} \tag{2.9}$$

where S_{nREF}, S_{nDIV}, and S_{nPD} are the PSDs of ϕ_{nREF}, ϕ_{nDIV}, and i_{nPD} respectively. As a result, we end up with two noise sources, ϕ_{nEQUIV} at the PLL input and ϕ_{nVCO} at the VCO output. ϕ_{nEQUIV} is mostly white noise, while ϕ_{nVCO} has a profile that is mostly proportional to $1/f^2$. Furthermore, ϕ_{nEQUIV} gets low-pass filtered by the PLL while ϕ_{nVCO} gets high-pass filtered. Since the cut-off frequencies for the low-pass and high-pass characteristics are the same, this leads to a trade-off between ϕ_{nEQUIV} and ϕ_{nVCO} i.e. reducing ω_{UG} to reduce the noise contribution from ϕ_{nEQUIV} will lead to an increased noise contribution from ϕ_{nVCO} and vice-versa. This is one of the most basic and fundamental trade-offs in PLL design, and it is important to choose the loop bandwidth so that all the noise sources are suppressed in a balanced way and the total noise at the output is minimized. Fig. 2.4 shows an example demonstrating this concept where we calculated the output PN of an integer-N PLL (yellow) and the noise contribution from the reference-referred noise (blue) and VCO (red) for different unity gain bandwidths of the loop. The reference frequency is 100MHz, the output frequency is 2.4GHz, the equivalent reference-referred noise is white noise with a floor of -150dBc/Hz, and the VCO has a -120dBc/Hz phase noise at 1MHz. K_{PD} is equal to $100 \cdot 10^{-6}/(2\pi)$ (we are assuming a PFD/CP with a CP current equal to 100μA), K_{VCO} is equal to $2\pi \cdot 30 \cdot 10^6$, and the LF is designed to achieve the desired bandwidth. Fig. 2.4(a) shows the PN when the bandwidth is around 1.5MHz. The contribution from the two noise sources are well balanced, and neither one is dominating the other. In contrast, Fig. 2.4(b) shows the output PN when the bandwidth is only 300kHz. Even though the equivalent reference path noise is better suppressed, the VCO noise contributes a large amount of noise and the total noise as a result is increased. Similarly, in Fig. 2.4(c), the bandwidth is a very wide 7.5MHz and even though the

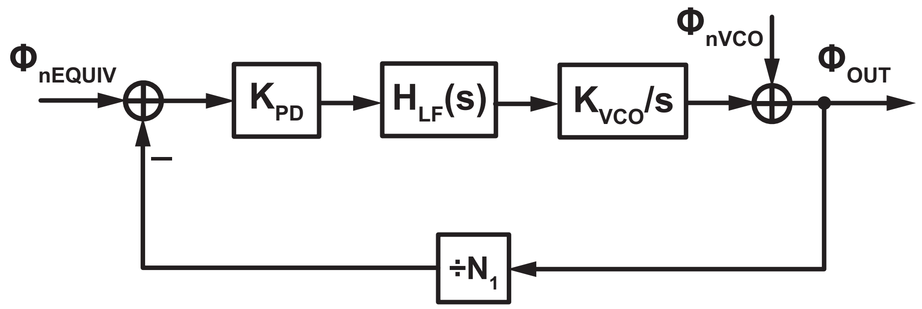


Figure 2.3: A simplified version of the linear noise model

VCO noise is better suppressed, the equivalent reference path noise contributes a great amount of noise and the total noise as a result is increased.

We mentioned the "total noise" of the PLL output multiple times. A common way to quantify the total noise of a frequency signal is to calculate its integrated jitter, which is the standard deviation of the zero crossing timing error of the frequency signal when compared to an ideal one. The integrated jitter can of course be calculated by measuring the timing error of a frequency signal for a period of time and calculating the standard deviation directly. A more common way however, is to integrate the PN spectrum over an appropriate range to obtain the variance of the phase error, and then convert this to the standard deviation of the timing error. Such a method is possible because of the Wiener-Khinchin relation [25]. Fig. 2.5 shows the relationship between the unity gain bandwidth of the PLL and the resulting integrated jitter. The conditions are the same as that used for the calculation in Fig. 2.4, the bandwidth was swept from 100kHz to 10MHz in increments of 100kHz, and the integration range is from 10kHz to 1GHz. We can see that the optimal jitter of around 175fs is achieved at around 1.5MHz as expected from the PN spectrum in Fig. 2.4 where we can see that the PN contribution from the two sources are well balanced. As the bandwidth gets narrower or wider than this optimal value, the integrated jitter degrades.

2.1.2 Stability Requirements

We saw that the loop transfer functions can be used to calculate the PLL output noise, and to adjust the choose an optimal value for the loop bandwidth to minimize the noise. Here we will use the transfer function to make sure that the loop is stable, with a sufficient phase margin. Furthermore, we have so far only looked at the loop characteristic based on a highly simplified framework consisting of just the loop bandwidth. Here, we will take a closer look at the poles and zeroes in the loop transfer function, how they should be positioned in relation to the unity gain bandwidth, and how the LF should be designed to meet these requirements.

Based on equations (2.1) and (2.2), the open-loop transfer function for a PLL using the second order filter shown in Fig. 2.2 can be calculated as

$$H_{\mathrm{OL}}(s) = \frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{N_1(C_1+C_2)}\frac{1+sRC_1}{s^2(1+s\frac{RC_1C_2}{C_1+C_2})} = \frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{N_1(C_1+C_2)}\frac{1+s/\omega_{\mathrm{z}}}{s^2(1+s/\omega_{\mathrm{p}})}. \tag{2.10}$$

As we explained before, $H_{\mathrm{OL}}(s)$ contains two poles at DC, a zero at $\omega_{\mathrm{z}} = \frac{1}{RC_1}$, and then another pole at a higher frequency $\omega_{\mathrm{p}} = \frac{1}{RC_1} + \frac{1}{RC_2}$. Once ω_{z} and ω_{p} are decided, the unity gain bandwidth ω_{UG} can be tuned by adjusting the absolute value of $H_{\mathrm{OL}}(s)$ through the gain $\frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{C_1+C_2}$. K_{PD} and K_{VCO} are usually decided based on the noise requirements and tuning range among other things,

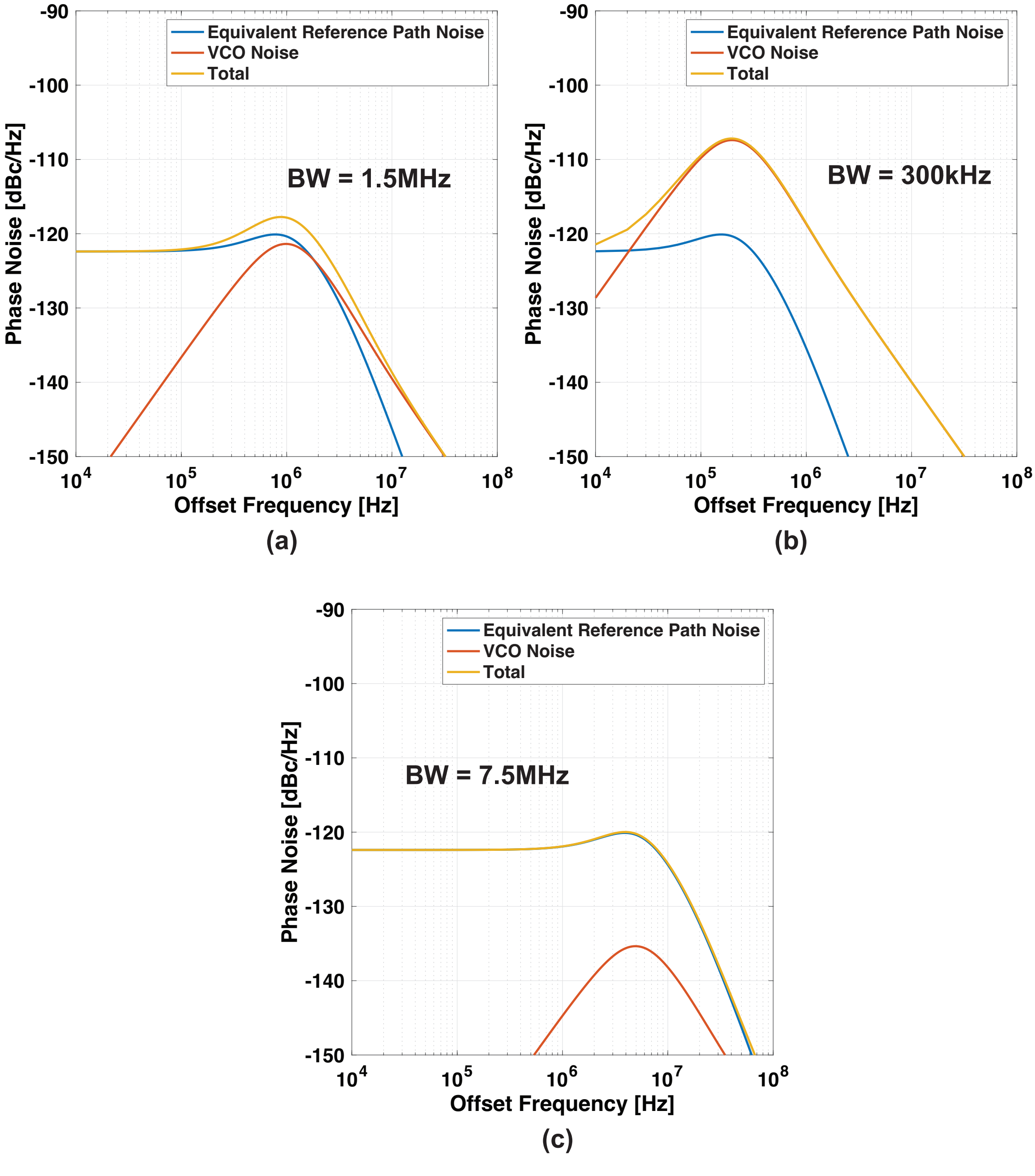


Figure 2.4: The calculated PN of the integer-N PLL (yellow) along with the noise contribution from the equivalent reference referred noise (blue) and the VCO (red) with a loop bandwidth of (a) 1.5MHz (b) 300kHz and (c) 7.5MHz

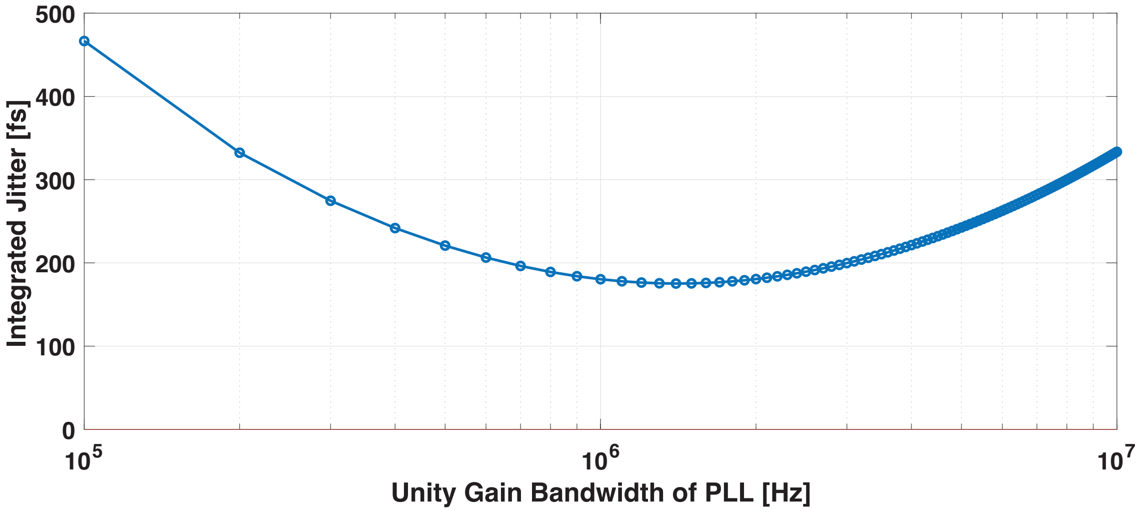


Figure 2.5: The relationship between the unity gain bandwidth of the PLL and the resulting integrated jitter

so the goal is to achieve the desired frequency characteristic by adjusting R, C_1, and C_2. Fig. 2.6 shows an example of a Bode diagram of the open loop transfer function of a PLL. We can see that ω_{UG} is located between ω_{z} and ω_{p}. This is very important to maintain stability since we have to maintain a large phase margin at ω_{UG}. In fact, a common way to design the PLL is to make the ratio between ω_{p} and ω_{UG} the same as the ratio between ω_{UG} and ω_{z}. Denoting this ratio as a, the parameters for the LF can be calculated as

$$R = \frac{a^2}{a^2-1}\frac{N_1\omega_{\mathrm{UG}}}{K_{\mathrm{PD}}K_{\mathrm{VCO}}} \approx \frac{N_1\omega_{\mathrm{UG}}}{K_{\mathrm{PD}}K_{\mathrm{VCO}}} \tag{2.11}$$

$$C_1 = \frac{a^2-1}{a}\frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{N_1\omega_{\mathrm{UG}}^2} \approx \frac{aK_{\mathrm{PD}}K_{\mathrm{VCO}}}{N_1\omega_{\mathrm{UG}}^2} \tag{2.12}$$

$$C_2 = \frac{K_{\mathrm{PD}}K_{\mathrm{VCO}}}{aN_1\omega_{\mathrm{UG}}^2} \tag{2.13}$$

where the approximations are based on the assumption that $a \gg 1$.

It is important to note that the analysis done so far has been based on the assumption that the PD continuously generates an output that is proportional to the input phase difference. In reality however, the PD only generates an output proportional to the input phase difference at the rising (or falling) edge of the input signals, and therefore contains a sampling operation in addition to phase subtraction. As such, phase noise components at large frequency offsets compared to the reference frequency will get aliased by the sampling at the PD and get down-converted in-band. This type of phenomenon can be ignored in many cases since the loop bandwidth is often much narrower than the reference frequency, but in some cases where the loop bandwidth is designed to be very wide, this effect must be taken into consideration [26]. The effect of sampling in the PD is given consideration in a slightly different context in chapter 4.

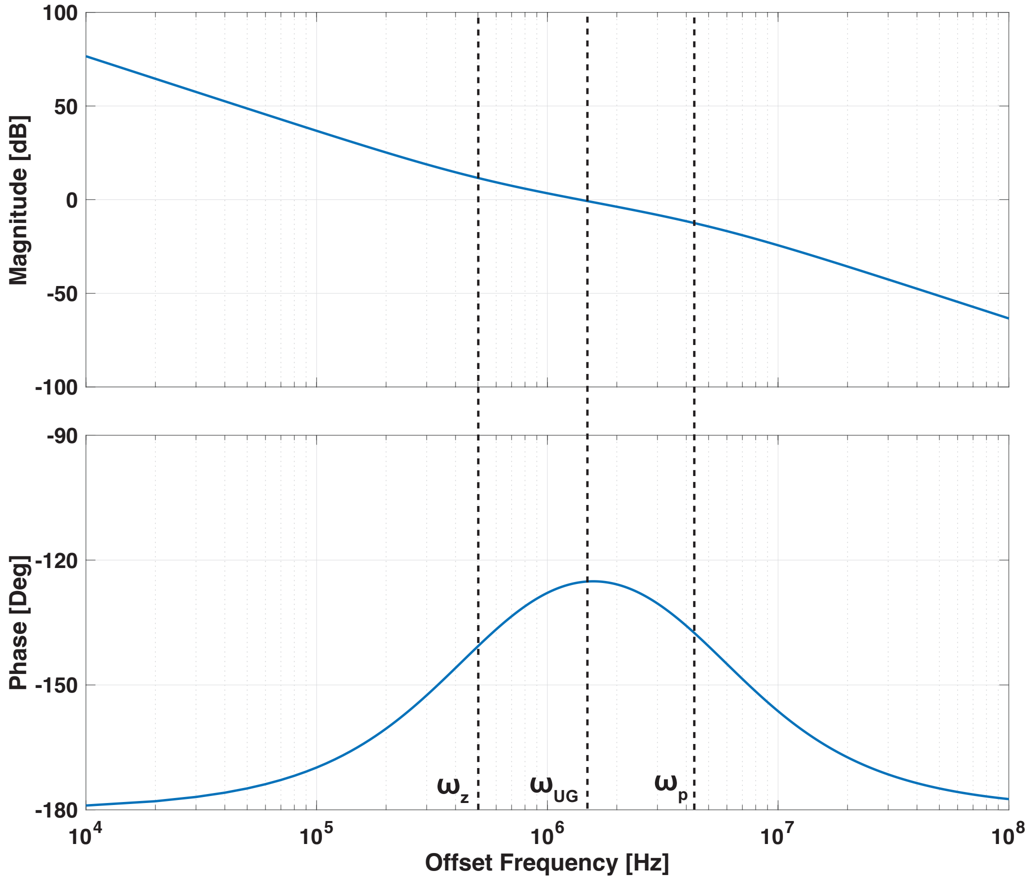


Figure 2.6: A Bode Plot of the open loop transfer function of a PLL

2.2 Quantization Noise and Fractional Spurs in Fractional-N PLLs

The frequency gain of the PLL is decided by the division ratio of the feedback frequency divider N_1. The frequency divider in turn, can be realized based on CMOS logic, Current-Mode-Logic (CML) for higher speeds [27, 28], and Injection-Locked-Frequency-Dividers (ILFD) that employ injection locked oscillators for very high speeds [29, 30]. In all of these cases, only an integer value can be realized for the division ratio because in the dividers are driven by the rising and/or falling edges of the input clock in CMOS logic and CML based dividers, and the ILFD can only lock in a state where the input frequency coincides with a harmonic of the output frequency. This becomes an issue since in many cases we wish to have a PLL with very fine frequency tuning resolution. A PLL whose output frequency can only be integer multiples of the reference frequency are called integer-N PLLs. Lowering the reference frequency is a solution with very limited scope because it will greatly limit the loop bandwidth, although there have been some works that attempt to overcome this issue by employing reference over-sampling techniques [31, 32]. In this section, we will look at the fractional-N PLL, which is a classical technique to realize PLLs with non-integer frequency multiplication ratios by modulating the frequency division ratio over time. We will also discuss the additional issues that come with fractional-N operation, such as quantization noise and non-linearity induced fractional spurs. Finally, we will also look at various techniques that have been proposed over the years to mitigate the aforementioned difficulties of fractional-N operation.

2.2.1 Delta-Sigma Modulation for Fractional-N Operation

To realize a PLL with a non-integer frequency multiplication ratio, we must realize a feedback divider with a non-integer frequency division ratio. However, frequency dividers can only be realized with integer division ratios. To overcome this issue, fractional-N PLLs employ a technique where the feedback division ratio of a Multi-Modulus-Divider (MMD) is varied over time, so that the desired non-integer division ratio can be achieved after averaging. Fig. 2.7 shows an example where we try to achieve a division ratio of 2.25. By switching the instantaneous division ratio between 2 and 3. Assuming that the division ratio continues to cycle through the sequence 2, 3, 2, 2, the phase error $\Delta\phi$ between the feedback and reference signals (ignoring the PLL output PN) will cycle through the sequence $-0.125T_{\mathrm{PLL}}$, $-0.375T_{\mathrm{PLL}}$, $0.375T_{\mathrm{PLL}}$, $0.125T_{\mathrm{PLL}}$, whose average is zero. Hence, we have achieved a non-integer, effective division ratio. The actual implementation of a fractional divider is slightly more complex. Even though the average of the phase error is zero, it is constantly varying unlike an ideal fractional divider whose output will only contain the PLL output PN that we wish to see. This can be seen as the fractional divider adding noise to the feedback signal, which is a form of quantization noise since the edge of the divider output has to be aligned with that of the divider input, and will result in a degradation of the overall output phase noise. The characteristic of this noise depends on how the division ratio is varied. In the example shown in Fig. 2.7, the division ratio is varied in a periodic manner, in this case a period of $9T_{\mathrm{PLL}}$ or a frequency of $0.25f_{\mathrm{REF}}$. This will result in a huge spurious tone at the PLL output which will degrade its performance to the point where it becomes non-usable. Therefore, a better method of varying the division ratio is needed. The division ratio can be randomized with white noise while maintaining an average equal to the desired non-integer division ratio. This will result in white quantization noise being added to the divider output. An interesting thing here is that the high frequency components will be filtered out by the low-pass characteristic of the PLL which we derived in equation (2.7). We can take further advantage of this fact by shaping the noise so that it gets pushed to higher frequencies: then a larger portion of the noise will be filtered out by the low-pass characteristic of the PLL, resulting in less phase noise overall. This sort of sequence with a shaped quantization noise can be realized by using a Delta-Sigma-Modulator (DSM) which takes a continuous valued input corresponding to the fractional part of the non-integer division ratio, and converts it to a sequence of discrete valued outputs, whose average is equal to the input and contains shaped quantization noise. Technically, the input in this case is also discretely valued since it is a digital signal with finite resolution representing the fractional part of the division ratio (This type of DSM is sometimes referred to as a Digital-DSM or DDSM). Nonetheless, the concept is the same since we are converting a digital value with fine resolution into one with a coarse resolution, and so we will get quantization noise that is shaped by the DSM. Fig. 2.8 shows a block diagram of the MASH 1-1-1 DSM which is a type of DSM often used in fractional-N PLLs. It is a combination of 3 first order DSMs, provides third order noise shaping to the quantization noise, and is robust in terms of stability. The input X represents the fractional part of the non-integer division ratio, and Y represents the sequence of integers that represent X when averaged out. The quantization noise from the first and second stages are cancelled out, and the quantization noise from the third stage Q_{n3} is multiplied by $(1-z^{-1})^3$ which corresponds to a third order discrete time high-pass filter. Q_{n3} itself can usually be assumed to be white noise, although dithering can be added sometimes to ensure this [33, 34, 35]. As a result of noise shaping, the quantization noise at the DSM output is suppressed at low frequencies and in exchange, increases at higher frequencies. The increased high frequency noise then gets filtered by the low-pass characteristic of the PLL, and the PN degradation from the MMD is lowered.

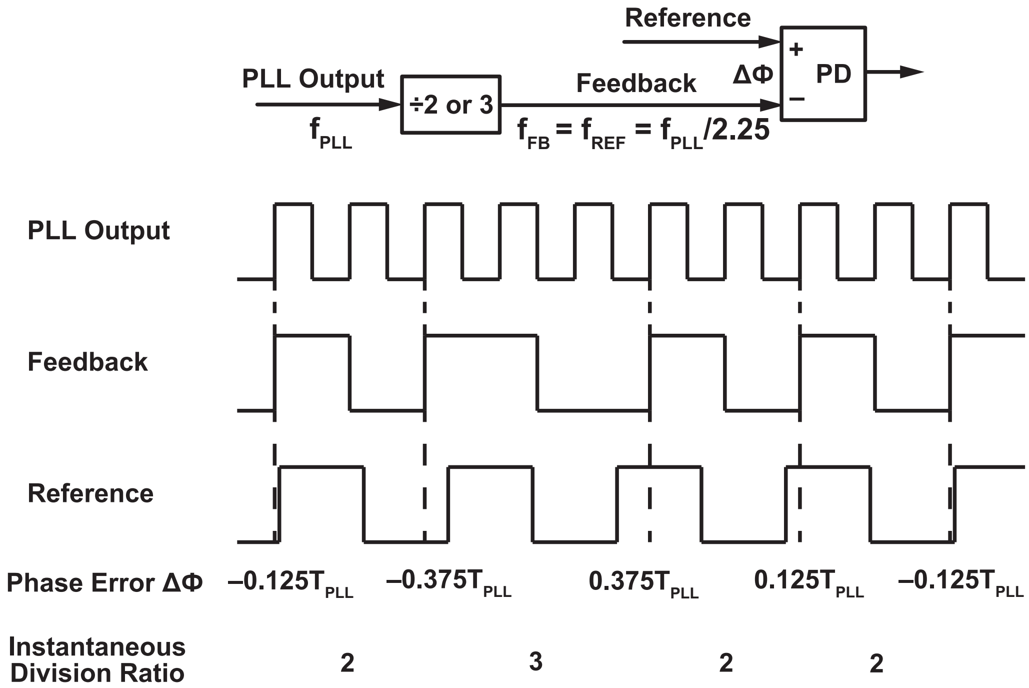


Figure 2.7: The concept of fractional frequency division with an example where the division ratio is 2.25

The realization of a fractional divider using an MMD modulated by a DSM is a technique that has been used for many years. While it helps to make the fractional-N PLL usable however, the issue of quantization noise from the MMD is far from solved and a large degradation in the overall PN spectrum can be seen. Fig. 2.9 shows the calculated output phase noise of a fractional-N PLL with the same conditions as the calculations done in Fig. 2.4, except that the frequency division ratio is slightly different to realize a non-integer value (24.01 instead of 24). In Fig. 2.9(a), we can see the output PN spectrum when the bandwidth is 1.5MHz, which was around the optimal setting for the integer-N PLL. We can see that the noise contribution from the VCO and the equivalent reference path noise are well balanced, as was in the integer-N PLL case. However, the quantization noise from the DSM contributes a huge amount of noise and the overall PN is severely degraded. To reach the optimal condition here, the bandwidth must be narrowed further to suppress the DSM noise. Fig. 2.9(b) shows the output PN spectrum when the loop bandwidth is 300kHz. The noise from the VCO and DSM are relatively well balanced. However, because suppressing the DSM noise required a much narrower bandwidth than it did to suppress the equivalent reference path noise, the total PN is now much worse than it was in an optimally designed integer-N PLL. Fig. 2.10 shows the relationship between the unity gain bandwidth of the fractional-N PLL and the resulting integrated jitter. Here, the optimal jitter of around 290fs is achieved when the loop bandwidth is around 300kHz. This shows that the DSM noise greatly degrades the achievable jitter in a fractional-N PLL when compared to an integer-N PLL, where the optimal jitter was around 175fs. In summary, the quantization noise from the MMD is an issue that has plagued fractional-N PLLs for years, and even with the use of noise shaping with a DSM, the performance degradation is still significant.

2.2.2 Generation of Fractional Spurs

In addition to the quantization noise that can degrade the output PN, a fractional-N PLL can also suffer from fractional spurs, which are periodic tones resulting from the fractional operation. We

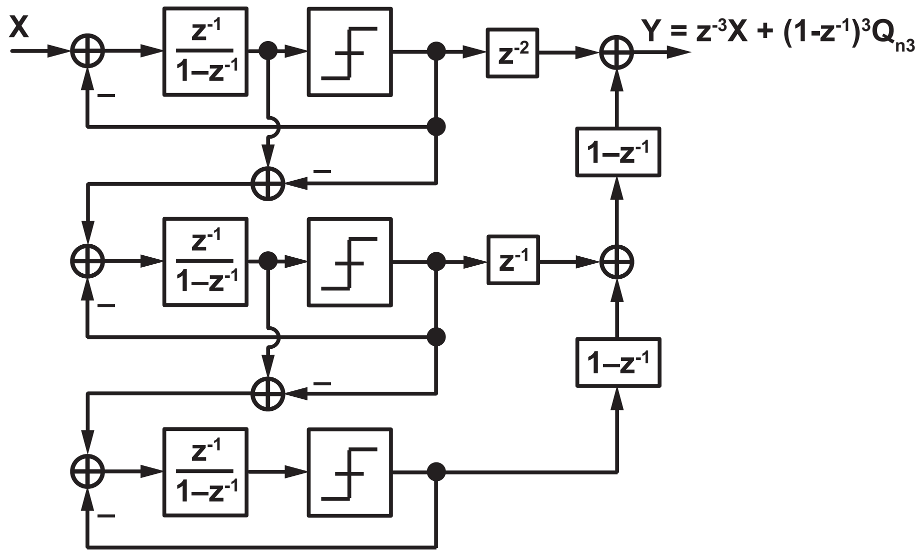


Figure 2.8: A DSM based on the MASH 1-1-1 architecture

mentioned before that by randomizing the sequence controlling the instantaneous division ratio of the frequency divider, the huge spurs resulting from the periodicity of the sequence can be avoided. However, periodic spurs can still be generated as a result of fractional operation. Firstly, the randomization by the DSM is not perfect, and its output can contain some periodicity albeit much smaller than if we had just used a simple periodic sequence. This lack of randomization can be reduced in a few ways such as adding dither to the DSM, although this may degrade the PN somewhat. Secondly, when the quantization noise enters a non-linearity inside the PLL, it can experience noise folding. Depending on how large the non-linearity is, this can result in fractional spurs [36]. The non-linearity in the PD such as the current mismatch in a CP or slope non-linearity in a Sampling-Phase-Detector (SPD) is especially troublesome, because the quantization noise has not yet been suppressed by the LF at this point. This effect of non-linearity can be combated by trying to linearize the blocks through digital calibration or analog linearization techniques.

An issue with fractional spurs that is different from quantization noise is that we cannot rely on low-pass filtering from the PLL to suppress it. If the DSM is clocked at a frequency of f_{DSM} (often equal to the reference frequency) and the fractional part of the frequency division ratio is α, the fractional spur usually appears at a frequency offset of αf_{DSM} and its harmonics. Since fractional-N PLLs often require very fine frequency tuning resolutions, α can take on very small values, which in turn means that the fractional spur can occur at very small offset frequencies. These are often called near-integer fractional spurs, and suppressing them using the low-pass characteristic of the PLL will be impractical since lowering the loop bandwidth excessively will cause the noise contribution from the VCO to become large.

2.2.3 Prior Art for Quantization Noise and Fractional Spur Suppression

We discussed how fractional-N PLLs allow us to realize frequency synthesizers with fine frequency resolution, but suffer from issues such as quantization noise and fractional spurs. Here we will discuss some recently proposed prior arts that look to improve the performance of the fractional-N PLL by suppressing the quantization noise and/or fractional spurs.

The most popular approach for the suppression of quantization noise (especially in recent years) has been to cancel it by adding its inverse at some point in the loop. A key point of the

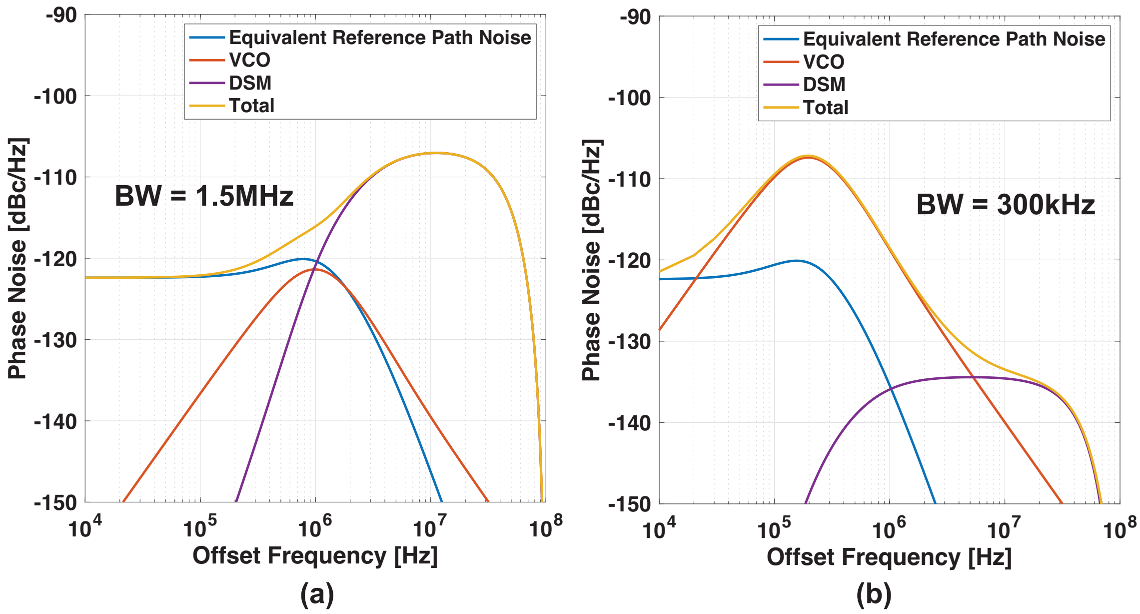


Figure 2.9: The calculated PN of the fractional-N PLL (yellow) along with the noise contribution from the equivalent reference referred noise (blue), the VCO (red), and the DSM (purple) with a loop bandwidth of (a) 1.5MHz and (b) 300kHz

quantization noise at the MMD output is that while it behaves in some sense like random noise, it is easily predictable since we have the sequence of division ratios used to control the MMD. By taking the difference between the instantaneous division ratio and the desired non-integer division ratio and then integrating it, we can anticipate the expected phase error due to quantization noise and cancel it. Fig. 2.11 shows an architecture where a Current Digital-to-Analog Converter (IDAC) is used to cancel the quantization noise at the CP output by subtracting a current that corresponds to that expected from the quantization noise [37, 38, 39]. A similar, but more popular technique in recent years is to cancel the quantization noise in the time-domain using a DTC either in the reference path as shown in Fig. 2.12 or in the feedback path after the MMD [40, 41, 42, 10, 43, 44, 11, 12, 45]. A major advantage of this approach over the IDAC based one is the fact that the quantization noise is cancelled before entering the PD and the PD only has to be linear and/or accurate over a narrow range. Unlike PFD/CPs and XOR based PDs that can operate with the desired linearity over a relatively large input range, PDs used in many modern high-performance PLLs such as the Sampling Phase Detector (SPD) or Time-to-Digital Converter (TDC) in scaling friendly Digital PLLs (DPLL) require a narrow range input to perform well. Specifically, the SPD will suffer greatly from non-linearity if the input is large because signal being sampled has large non-linearity, and TDCs will consume large power if it has to cover a wide range of inputs and achieve fine resolution simultaneously.

An issue with this cancellation-based approach is that the signal generated by the cancellation path must have very good matching with the actual signal corresponding to the quantization noise which we wish to cancel. Whether we use IDACs or DTCs, the blocks used to generate the cancellation signal will inevitably suffer from gain error and/or non-linearity, both as an inherent characteristic and due to Process, Voltage, and Temperature (PVT) variations. Without addressing these issues, the noise-cancellation-based approach can suffer from imperfect cancellation of the quantization noise, or the generation of fractional spurs. The gain error and non-linearity is usually compensated for using digital calibration. Specifically, the gain error is usually calibrated based on the Least-Mean-Square (LMS) algorithm by making use of the PD output to optimize the gain that

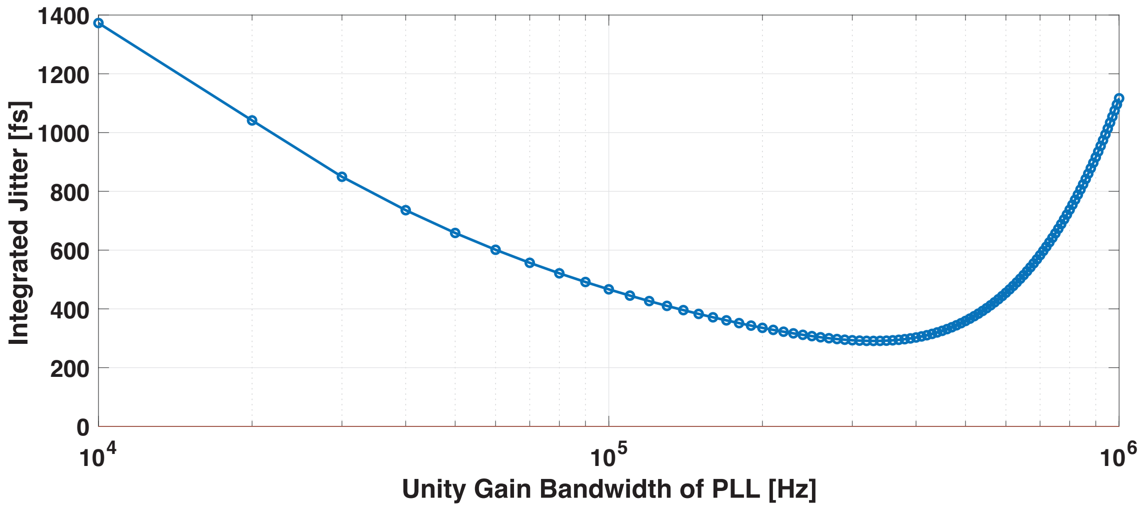


Figure 2.10: The relationship between the unity gain bandwidth of the PLL and the resulting integrated jitter in a fractional-N PLL

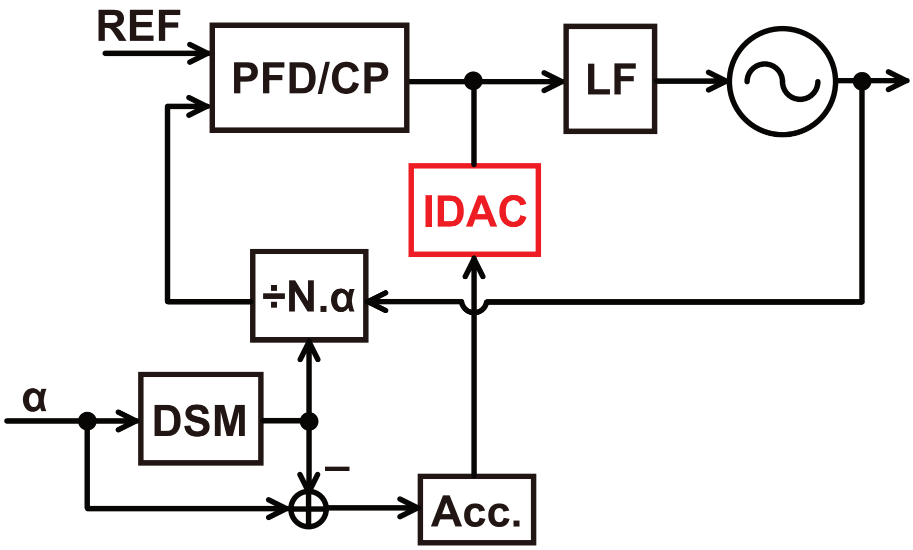


Figure 2.11: The cancellation of the quantization noise at the CP output using an IDAC [37, 38, 39]

is multiplied with the control code for the DTC [45], while the non-linearity is often compensated for using piece-wise pre-distortion using values from a Look-Up-Table (LUT), where the LUT values are updated based on the PD output [40]. It is worth noting that linear DTC architectures such as constant slope DTCs [46] can be used to relax or even remove the linearity calibration, but linear DTCs are usually not as efficient in terms of jitter/power performance as non-linear DTCs. With the use of calibration schemes, fractional-N PLLs based on DTCs have been able to achieve great performance in terms of jitter/power performance and spurs in recent years. An important issue with this approach is that the calibration can take a long time to converge, leading to a longer locking time for DTC-based PLLs compared to calibration-free PLLs whose locking process only consists of the loop transient. To give a rough idea, the loop transient of a PLL will last in the order of hundreds of ns to tens of μs depending on the architecture and loop bandwidth, while the calibration convergence time of DTC-based PLLs reported in prior works can take in the order of tens of μs to even ms in the worst case [10, 11, 12]. This can be a major issue in applications requiring frequent frequency switching and/or system duty cycling where the PLL has to re-lock

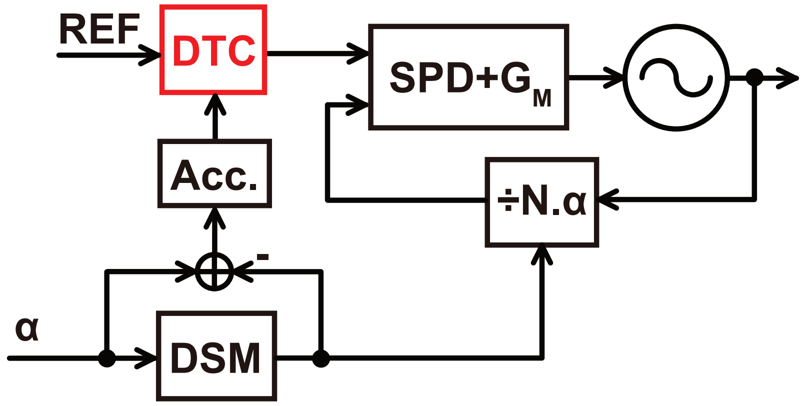


Figure 2.12: The cancellation of the quantization noise in the time-domain using a DTC [40, 41, 42, 10, 43, 44, 11, 12]

after start-up. Another issue is that the residual quantization noise and fractional spurs resulting from noise cancellation based approaches are hard to model and predict, because they result mainly from non-idealities in the circuit and second order effects. In contrast, the noise contribution from the quantization noise in a completely linear, calibration-free PLL can be predicted based on an ideal linear model of the PLL with general loop characteristics, and non-idealities in the circuit and second order effects just cause minor deviations from this prediction. A brute force confirmation using simulation of the entire circuit is difficult for calibration intensive PLLs, since the time scales of the primary loop operation and the calibration are vastly different. Finally, due to the nature of calibration (especially linearity calibration) where we have to correct the entire characteristic of a circuit block rather than deal with just a single input and output, the hardware complexity can be greatly increased as can be seen by the requirement of LUTs and multiple accumulators.

Another approach to suppress the quantization noise is to improve on the low-pass filtering from the PLL characteristic. This approach can be roughly categorized into two parts: increasing the noise shaping frequency and adding another means of filtering the quantization noise other the PLL itself.

Increasing the noise shaping frequency by operating the DSM with a higher frequency clock will push the quantization noise out to higher frequencies, and allow it to be filtered out adequately even if the bandwidth is relatively narrow. In a basic fractional-N PLL, the DSM frequency is equal to the reference frequency, and so increasing the noise shaping frequency will require us to employ a higher reference frequency. Fig. 2.13(a) shows the calculated output PN of a fractional-N PLL with the same conditions as in Fig. 2.9(a) where the bandwidth is 1.5MHz and the reference frequency is 100MHz. As we saw before, the quantization noise heavily degrades the overall PN spectrum and a narrower bandwidth of around 300kHz is required to sufficiently suppress it. On the other hand, Fig. 2.13(b) shows the PN spectrum with similar conditions except the reference frequency is now 600MHz. The equivalent reference path noise has been scaled up by 20log6 = 15.5dB for a fair comparison. We can see that now, the PN degradation from the DSM quantization noise is much smaller, even though the loop bandwidth is the same at 1.5MHz. The integrated jitter is around 189fs, which is not too worse compared with the integrated jitter of 175fs that was achieved with the integer-N PLL. From this, we can see that increasing the noise shaping frequency by increasing the reference frequency is an effective way to suppress the quantization noise from the DSM. That being said, it is not a simple task to obtain a high frequency reference. Crystal-Oscillators (XO) based on quartz crystals are often used as reference signals for PLLs, but

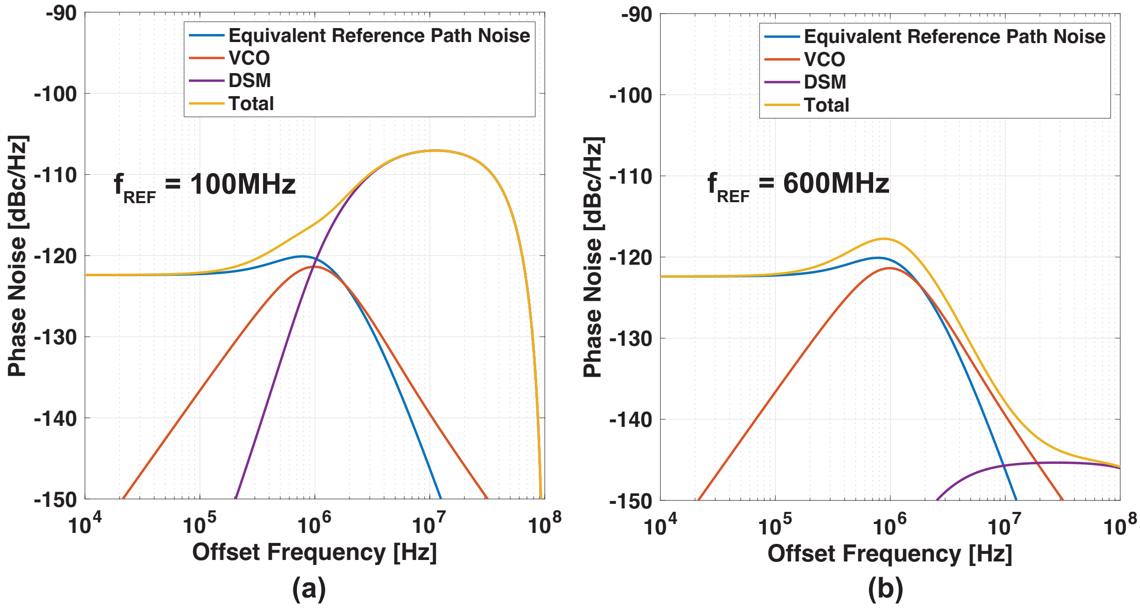


Figure 2.13: The calculated PN of the fractional-N PLL (yellow) along with the noise contribution from the equivalent reference referred noise (blue), the VCO (red), and the DSM (purple) with a reference frequency of (a) 100MHz and (b) 600MHz

XOs tend to become costly as their frequency increases. In fact, XOs tend to be manufactured with frequencies in the order of tens of MHz and XOs operating at hundreds of MHz are often not available. Still, prior works have been able to increase the noise shaping frequency in a few ways. A straightforward way is to employ a cascaded PLL architecture [47] or a frequency multiplier [48, 49, 50] at the input of the fractional-N PLL. Fig. 2.14 shows a block diagram of a fractional-N PLL with reference multiplication at its input. The reference multiplication can be realized either by using an integer-N PLL, or a reference multiplier, which is used to refer to circuits for frequency multiplication that rely on techniques that are open-loop in nature such as combining multiple edges using an XOR, instead of a closed-loop feedback system like a PLL. We can see that by using reference multiplication, the noise shaping frequency for the DSM has increased from f_{REF} to Mf_{REF}, where M is the reference multiplication ratio. While reference multiplication allows us to suppress the quantization noise, it suffers from the extra noise and complexity from the reference multiplier. The noise and power consumption tends to be larger when an integer-N PLL is used. This is because the PLL tends to be a much more complex system since it uses feedback while reference multipliers usually just create delayed versions of the reference and then interpolate the edges. Also, the noise from a reference multiplier tends to be easier to deal with since it is just white noise, whereas for the integer-N PLL, we must deal with noise from the VCO which is integrated white noise that can only be suppressed adequately with a wide loop bandwidth. On the other hand, reference multipliers often require calibration schemes for duty cycle correction, and they also become impractical to use with high multiplication ratios. In prior arts, we often see reference doublers [48] and sometimes reference quadruplers [49, 50], but seldom see reference multipliers with a higher multiplication ratio than that.

Another clever way to increase the noise shaping frequency is to split the frequency divider into two parts, and clock the DSM with the intermediate frequency [51]. Fig. 2.15 shows a conceptual block diagram of such an architecture. It is very important to note that if we simply split the frequency divider and use the intermediate point to clock the DSM, this will not work because the

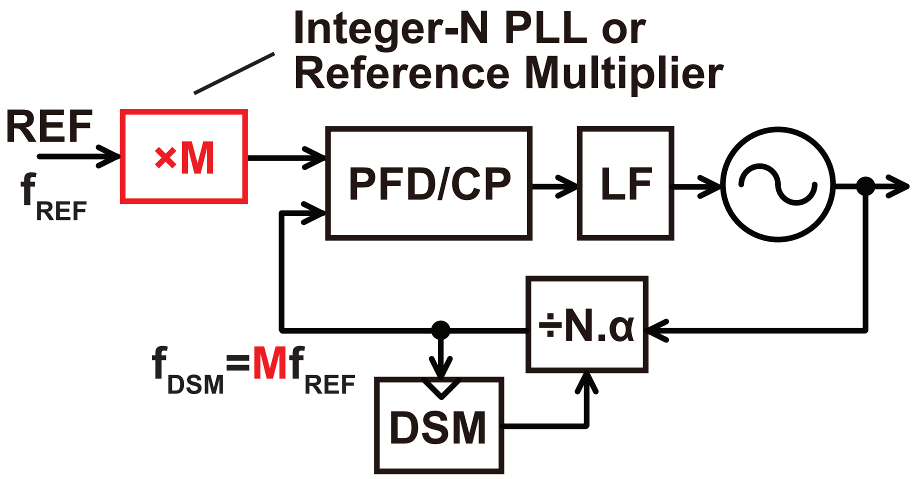


Figure 2.14: A block diagram of a fractional-N PLL with reference multiplication

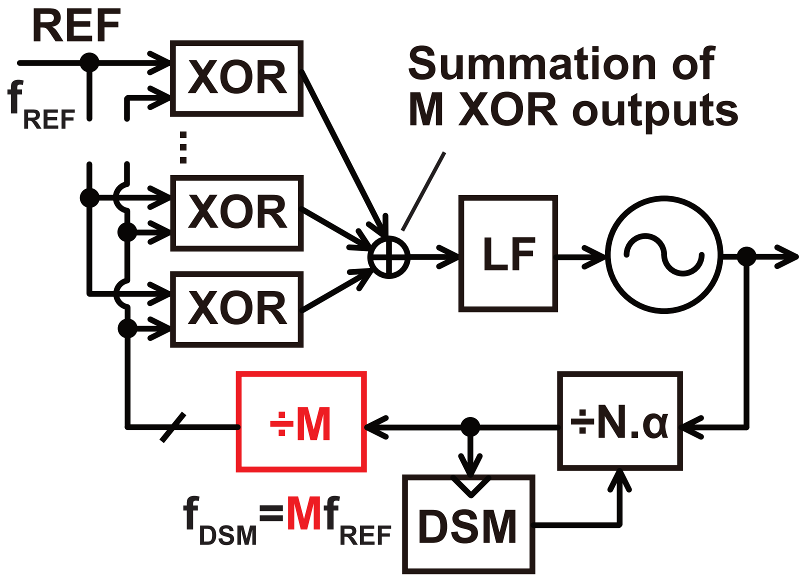


Figure 2.15: A block diagram of a fractional-N PLL with a split frequency divider [51]

second stage divider acts as an edge selector that samples the first stage divider output containing the quantization noise. This will cause aliasing, and the quantization noise shaped to a high frequency will get down-converted in-band, undermining the entire point of increasing the noise shaping frequency. To avoid this issue, the second stage divider in Fig. 2.15 generates M different phases at its output, meaning that all of the edges from the first stage divider are preserved. These M phases then enter M different XOR PDs, where they can all be compared with a single reference signal since XOR PDs can operate with an offset at the input. As a result, we have successfully increased the noise shaping frequency by a factor of M without the use of a reference multiplier. The drawback here is that firstly, the factor of M that can be used is limited because all of the XOR PDs must operate in their respective linear regions. Secondly, the XOR PDs will result in very large reference spurs, especially since they operate with large offsets, and this in turn can make it difficult to widen the bandwidth. It is worth noting that even though the noise shaping frequency has been increased, the reference spur still appears at the reference frequency.

There are some works that make use of Micro-Electro-Mechanical (MEMs) oscillators instead of conventional quartz XOs to directly generate a high frequency reference signal [52, 53]. MEMs oscillators are currently viewed as very promising alternatives to quartz XOs due to their great integration potential with silicon processes, wide range of attainable frequencies, and robustness to vibration and shock among others. However, due to the fact that it is not yet a main stream solution, and other issues such as temperature sensitivity that are not trivial to solve [54], the use of MEMs-oscillator-based references are not yet a practical solution to realize low noise fractional-N frequency synthesis. Also, a more fundamental limitation of increasing the noise shaping frequency is that it cannot be done indefinitely because the logic inside the DSM can fail to operate if the frequency is increased too much (around a few hundred MHz in 65nm CMOS process), and even if it operates correctly, the power consumption can become large depending on the bit size and order of the DSM. Nonetheless, increasing the noise shaping frequency is one of the most effective and frequently used ways to suppress the quantization noise without relying on calibration schemes.

Adding another means of filtering for the quantization noise is also an often-used approach. The key here is that the enhancement of the filtering characteristic must be done without changing the characteristic of the primary loop itself, otherwise it would be the same as just using a narrow bandwidth PLL which would not be good for suppressing the VCO PN. A way to realize this kind of filtering is to create multiple feedback signals that contain the same desired component from the PLL output, but quantization noise components with different delays. By adding these signals together with different weights, the quantization noise can be filtered by an equivalent Finite-Impulse-Response (FIR) characteristic, but the PLL output component is obtained unaltered, meaning that the characteristic of the primary loop is unaltered. Fig. 2.16 shows an example of an architecture employing such a concept, where we have multiple MMDs and corresponding PFD/CPs [55, 56]. Each of these MMDs are controlled by the output of a common DSM, except they are delayed by different multiples of the reference period. By doing so, the output of these MMDs contain quantization noise components that are also shifted by different delays, but the same PLL output PN component. By performing a weighted sum of these MMD outputs after the PFD/CPs (the weighting can be done by using different current values for the CPs), an FIR filter for the quantization noise can be realized. An issue with this architecture is the matching required between the gain and linearity of the different paths, and the simple complexity required to realize multiple MMDs and PDs. Another way to realize the FIR is shown in Fig. 2.17, where the MMD output goes through multiple delays equal to the reference period [57, 58]. The output of these delays are then compared to the reference signal with corresponding XOR PDs, and then summed at their outputs using resistive weights. Since the quantization noise components at the outputs of each delay are also delayed versions of each other, appropriate summation of these signals will again, lead to an equivalent FIR filter. Furthermore, since the delay lines are realized using flip-flops that are clocked by the PLL output, the outputs of each delay will contain the same PLL output component and therefore will be obtained unaltered. The use of XOR PDs and resistive summation offers better linearity than the PFD/CPs, although the mismatch between paths remain. The realization of multiple delay lines will increase the overall complexity and pose a power consumption overhead. The extent of the overhead from the delay lines are debatable. On one hand, they are scaling friendly since they depend on time-domain operation and are realized with flip-flops which are digital blocks. On the other hand, they must be clocked with the PLL output whose frequency can be high, and realizing a narrow filtering bandwidth will require many taps. Ultimately, the feasibility of this method will depend on many factors such the technology that the chip is being realized in, the desired filtering bandwidth, the frequency plan, and many others.

As we have seen, prior arts for suppressing the quantization noise in fractional-N PLLs take various approaches, such as noise cancellation, increasing the noise shaping frequency, and improving the filtering characteristic by realizing an equivalent FIR. All of these methods have their own pros and cons, with the improvement in quantization noise suppression and overhead from implementing these methods varying greatly depending on the conditions.

2.3 Chapter Summary

In this chapter, we looked at the basics of PLL design, established some of the important trade-offs and fundamental issues, and introduced some of the important prior works that attempt to overcome the issues in fractional-N frequency synthesis.

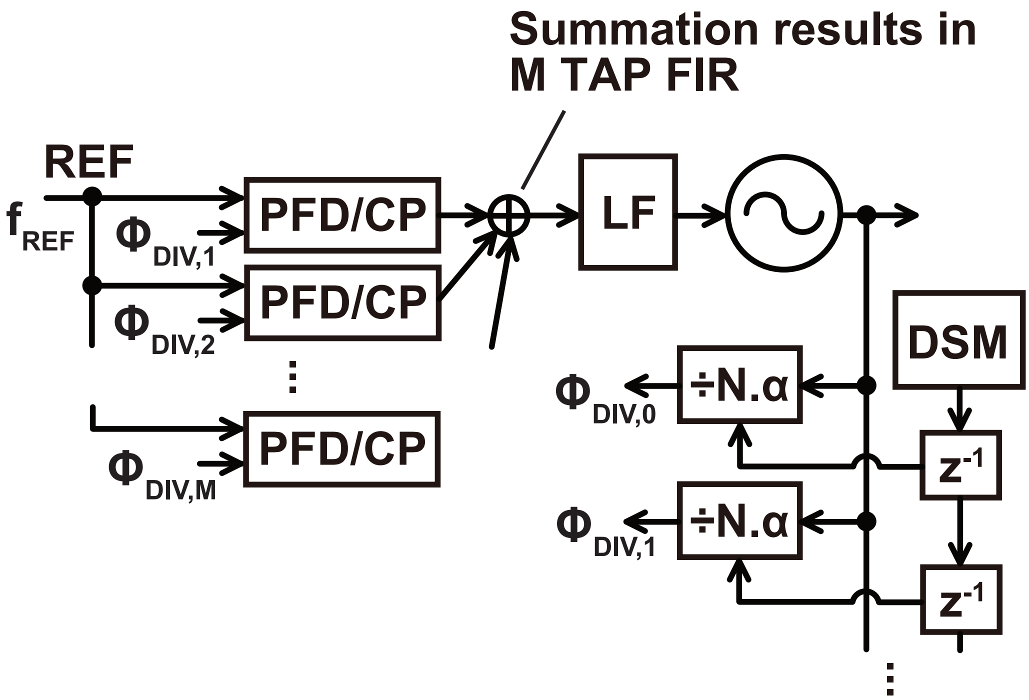


Figure 2.16: A block diagram of a fractional-N PLL with multi-path feedback achieving an equivalent FIR filter [55, 56]

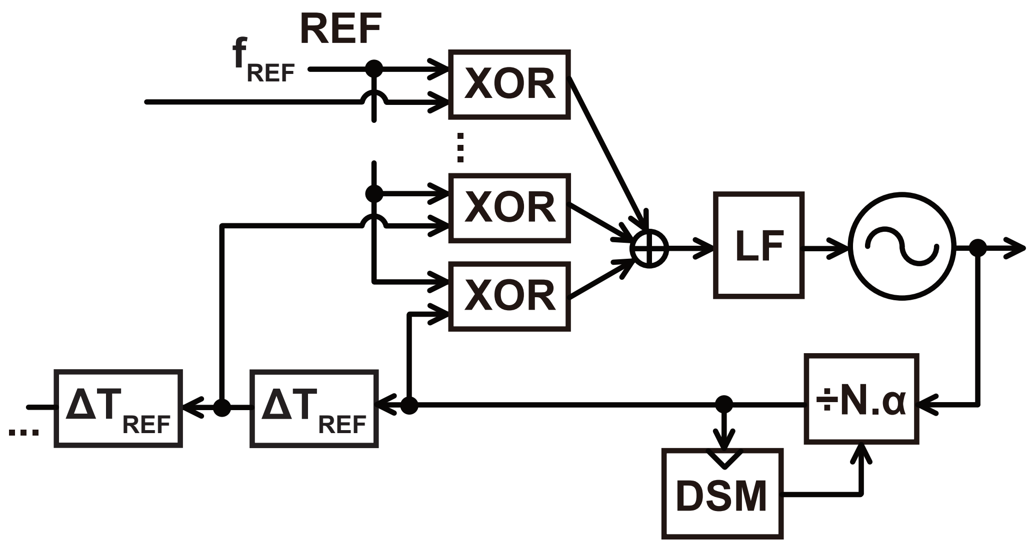


Figure 2.17: A block diagram of a fractional-N PLL with delay lines achieving an equivalent FIR filter in the time-domain [57, 58]

Chapter 3

The Dual-Feedback PLL

As was discussed in the previous chapter, fractional-N PLLs suffer from quantization noise generated by the modulated feedback divider, as well as fractional spurs induced by non-linearities in the loop. Harmonic-Mixer(HM)-based feedback is a highly effective method for combating these issues, and is a central concept of this dissertation. In this chapter, we will first look at how HM based feedback can be used inside of a fractional-N PLL to suppress quantization noise and fractional spurs by avoiding amplification within the loop. We will then introduce the triple-loop PLL which is a prior art and one of the first works to make use of HM-based feedback in a fractional-N PLL. Then, we will introduce the dual-feedback PLL which is one of the main contributions of this dissertation. Analysis, measurement results, and comparisons with other architectures will be given. This chapter is based off of our works published in [59, 60].

3.1 Use of Harmonic-Mixer-Based Feedback

The use of HM-based feedback is one of the central concepts of this dissertation and will be discussed in this section. We first explain the issue of noise amplification in PLLs, which is the fundamental issue that we wish to solve using HM-based feedback. Then we look at how realizing frequency subtraction with a HM can be used to avoid this amplification. Finally, we will look at how the HM operation is actually realized at the circuit level by employing Sample-and-Hold (S/H) operation.

3.1.1 Amplification of Input-referred Noise in Feedback Systems

A PLL is a feedback system that controls the output phase with the phase of the reference as the input. If the feed-forward gain from the input to the output is $A(s)$ and the feedback gain from the output to the input is β as depicted in Fig. 3.1(a), the open-loop gain will be $\beta A(s)$. Assuming that this is sufficiently bigger than 1, the closed loop gain from the input referred noise to the overall output will be the inverse of the feedback gain, $1/\beta$. Applying this general concept to a fractional-N PLL, we can see that since the feedback gain of a fractional-N PLL is the inverse of the division ratio as shown at the top of Fig. 3.1(b), the input-referred noise including the quantization noise will get amplified by the division ratio at frequencies lower than the unity gain bandwidth, where the open-loop gain is sufficiently large. We explained how the quantization noise from the MMD significantly degrades the PLL output PN in chapter 2. The fundamental concept of noise amplification is one of the reasons this degradation is so significant. Assuming that we are generating a 2.4GHz signal from a 100MHz reference like we did in the calculations for chapter 2, the quantization noise referred to the PLL input will get amplified by around 20log(24)≈27.6dB,

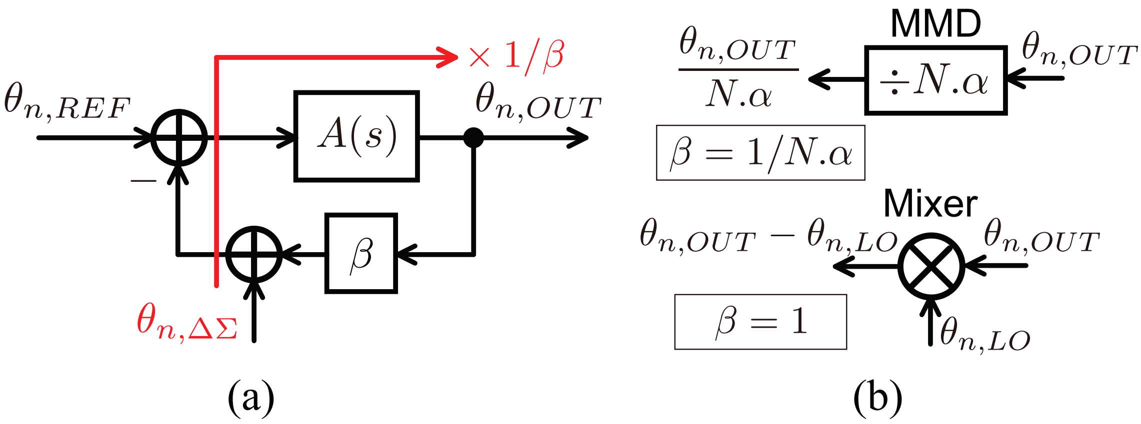


Figure 3.1: (a) The basic concept of noise amplification in a general feedback system and (b) the feedback gain of a divider and a mixer

which is a huge factor. On the other hand, if we can somehow avoid this amplification of noise or attenuate in some way, we will be able to suppress the quantization noise greatly and realize another approach for noise suppression in addition to the ones introduced in chapter 2. A feedback configuration based on frequency subtraction using a mixer is shown at the bottom of Fig. 3.1(b). We can see that in this case, the feedback gain in the phase domain is unity since the entirety of the feedback signal appears at the mixer output instead of it getting divided by the division ratio. As such, if we can somehow employ this concept in a fractional-N PLL, the amplification of quantization noise can be avoided and the performance can be greatly improved. A key contribution of this dissertation is the investigation and proposal of architectures that make use of this concept.

Although very basic, it is important to note the relationship between the phase noise, jitter, and operating frequency. If we take the output of a PLL and put it through a divide by N, the PN will be reduced by a factor of N, and its power spectrum by a factor of N^2, but the operating frequency will also decrease by a factor of N. If we try to get back the original operating frequency by using a PLL or even an ideal frequency multiplier to multiply the frequency by N, the PN spectrum will increase again by N^2, rendering the entire operation useless. From this we can see that just paying attention to the PN is insufficient. The important quantity here is the integrated jitter, which is the standard deviation of the time-domain error of the signal of interest when compared to an ideal one. After going through the divide by N, the PN of the signal will be reduced by a factor of N, but the period of the signal will be increased by a factor of N. Since the phase is a quantity that is defined relative to the length of one period i.e. one period contains 2π worth of phase. As such, the decrease in PN by a factor of N and the increase in the period length by a factor of N cancel out and the resulting integrated jitter is the same. The integrated jitter is proportional to the square root of the integrated phase noise divided by the frequency of operation. Our target in terms of lowering noise will be to obtain a better integrated jitter, which can get confusing when talking about noise amplification and avoiding it.

3.1.2 Use of Harmonic-Mixer-Based Feedback to Avoid Noise Amplification

Fig. 3.2 shows the most basic core of a PLL using mixer based feedback. Besides the usual blocks used in a PLL such as the PD, LF, and VCO, we have a mixer and a Local-Oscillator (LO) signal that is used to down-convert the output signal to the reference frequency. f_{REF}, f_{LO}, and f_{OUT} are

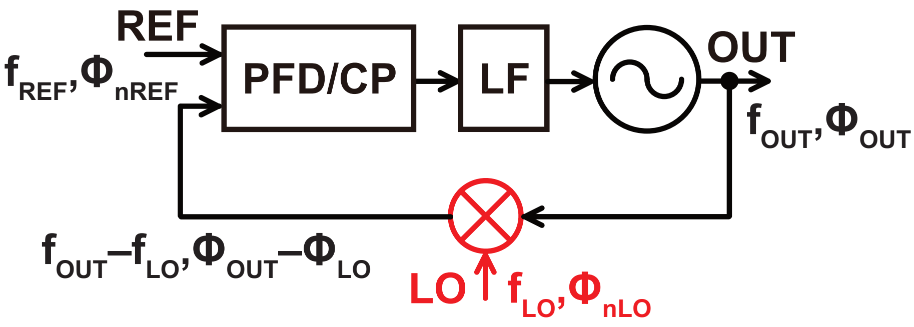


Figure 3.2: The core of a PLL using mixer based feedback

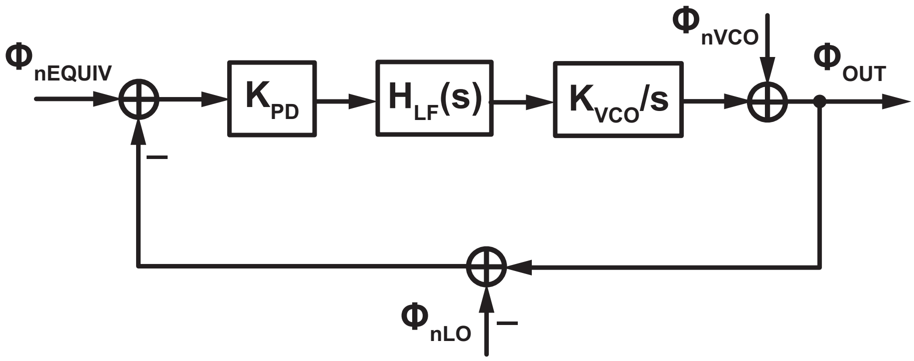


Figure 3.3: The noise model of a basic mixer based PLL

the frequencies of the reference, LO, and output respectively while their phase noise components are denoted as ϕ_{nREF}, ϕ_{nLO}, and ϕ_{OUT}. The output of the mixer is a signal with a frequency of $f_{\text{OUT}} - f_{\text{LO}}$ and a phase noise component of $\phi_{\text{OUT}} - \phi_{\text{LO}}$. Since the frequency of the feedback signal must be equal to that of the reference, $f_{\text{OUT}} = f_{\text{LO}} + f_{\text{REF}}$ i.e. the output frequency of the mixer-based PLL is equal to the sum of the LO and reference frequencies. Fig. 3.3 shows a noise model of the mixer-based PLL core. The mixer is modeled by a simple subtraction of the phase noise components, and ϕ_{nLO} is the phase noise contained in the LO signal. The definition for the rest of the symbols are the same as in the conventional architecture. The open-loop gain $H_{\text{OL}}(s)$ can be calculated as

$$H_{\text{OL}}(s) = \frac{K_{\text{PD}} K_{\text{VCO}} H_{\text{LF}}(s)}{s}. \tag{3.1}$$

This is very similar to the open-loop transfer function of the conventional PLL given in equation (2.1), except for the absence of the frequency division ratio N_1 since the feedback gain is unity. The other noise transfer functions can be calculated as follows.

$$H_{\text{VCO,OUT}}(s) \equiv \frac{\phi_{\text{OUT}}}{\phi_{\text{nVCO}}} = \frac{1}{1 + H_{\text{OL}}(s)} \tag{3.2}$$

$$H_{\text{EQUIV,OUT}}(s) \equiv \frac{\phi_{\text{OUT}}}{\phi_{\text{nEQUIV}}} = \frac{H_{\text{OL}}(s)}{1 + H_{\text{OL}}(s)} \tag{3.3}$$

$$H_{\text{LO,OUT}}(s) \equiv \frac{\phi_{\text{OUT}}}{\phi_{\text{nLO}}} = -\frac{H_{\text{OL}}(s)}{1 + H_{\text{OL}}(s)} \tag{3.4}$$

The transfer function from the VCO output to the PLL output is given by equation (3.2), and it is the same as that in the conventional architecture given by equation (2.6). This shows that in order to achieve the same amount of filtering for the VCO PN, the same open-loop gain as that in the conventional architecture is needed.

The transfer function from the equivalent reference-referred PN to the PLL output is given by equation (3.3). This expression is similar to that in equation (2.3), except for the fact that there is now no amplification by the division ratio N_1. This is the avoidance of noise amplification that was mentioned earlier, and is one of the key characteristics we will make use of to suppress the quantization noise. The frequency characteristic of $H_{\mathrm{EQUIV,OUT}}$ is the same as that in a conventional PLL: it is a low-pass characteristic with a cut-off frequency equal to the unity-gain bandwidth.

Finally, the transfer function from the LO output to the PLL output is given by equation (3.4). This is the same as the transfer function from the equivalent reference-referred noise to the PLL output, except that the polarity is reversed.

As we will explain in chapter 4, f_{REF} must be much smaller than f_{LO} due to circuit non-idealities. This in turn, means that f_{LO} will be close to f_{OUT}, giving rise to the issue of frequency pulling between oscillators. This issue is especially prominent when we use LC oscillators for the VCOs due to magnetic coupling, but it can also occur through power supply and substrate coupling. When two oscillators are operating at a similar frequency, they can affect each other and cause harmful effects such as PN degradation. The issue of frequency pulling can be greatly mitigated if we employ a Harmonic-Mixer (HM) where its output f_{HM} is equal to the difference between that of the input and some integer multiple of that of the LO i.e. $f_{\mathrm{HM}} = f_{\mathrm{OUT}} - kf_{\mathrm{LO}}$ where f_{OUT} and f_{LO} are the frequencies at the PLL output and the LO as depicted in Fig. 3.2, and k is an integer known as the harmonic index of the HM. By using a HM instead of a simple mixer, f_{OUT} and f_{LO} are no longer close to each other, and frequency pulling can be greatly mitigated. That being said, f_{OUT} is now close to kf_{LO}, meaning that there is still some risk of the oscillator at f_{OUT} getting pulled by the kth harmonic of f_{LO}. To further reduce the risk of frequency pulling, a frequency divider can be placed before the LO signal, whose frequency division ratio is co-prime with the harmonic index k. The concepts are summarized in Fig. 3.4. The simple configuration using a mixer as shown in Fig. 3.4(a) is prone to frequency pulling, so configurations shown in Figs. 3.4(b) and (c) that are based on HMs will be used often in the architecture proposed in this dissertation.

3.1.3 Architecture and Operation Principle of Harmonic-Mixer

Here we will explain how the HM can be realized. HM are commonly used circuits in wireless transceivers and measurement equipment for high speed signals among others. A common way to realize a HM is by generating and combining multiple phases from a single LO signal to effectively multiply the LO frequency, then use this as an LO for a standard mixer, such as one based on a Gilbert cell [61, 62, 63, 64]. This essentially is just a mixer + frequency multiplier. While it is a standard configuration used in many cases, the generation of multiple phases can be costly in terms of power consumption and complexity.

An efficient and commonly used method for realizing a HM in prior art is based on the Sample-and-Hold (S/H) operation [59, 60, 65, 66, 67]. The S/H operation can be broken down into two operations: sampling and holding. Fig. 3.5 shows the sampling operation in the time

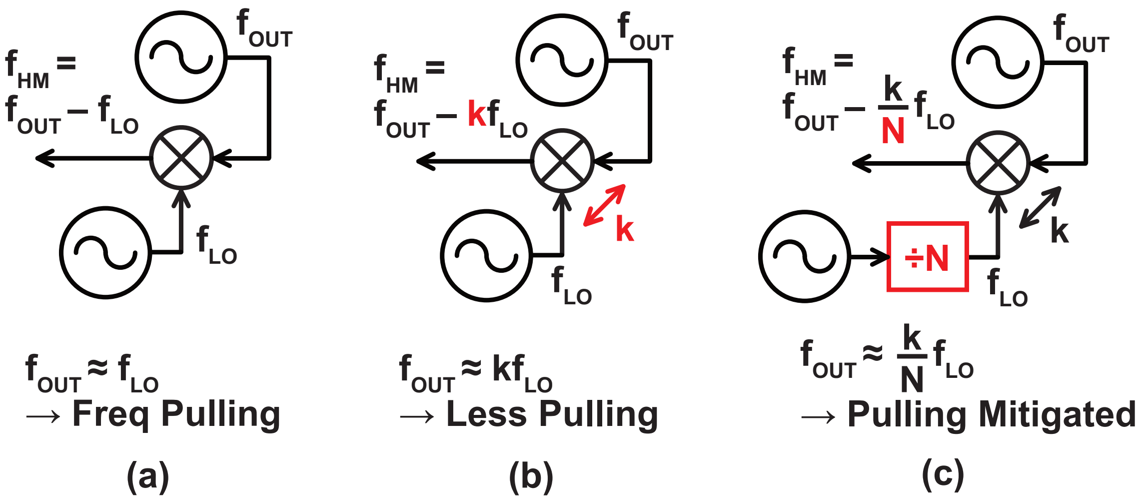


Figure 3.4: The effect of frequency pulling when (a) a fundamental mixer is used to down-convert the PLL output, (b) a HM is used, and (c) a HM and frequency divider are used

and frequency domains. Fig. 3.5(a) shows the sampling operation in the time domain, and we can see that it is equivalent to multiplying the input signal with a train of impulses and converting it to a sequence of discrete time values defined at the sampling instances. The impulse train has a frequency of f_{SH} which is equivalent to the frequency of the LO signal used to generate the sampling clock. On the other hand, Fig. 3.5(b) shows the same sampling operation but in the frequency domain. For the S/H input, we assume that the signal is close to a single tone as is the case when the S/H is used in the feedback path of a PLL. Its frequency is denoted as f_{PLL}. The impulse train in the time domain is also an impulse train in the frequency domain, with components at increments of f_{SH}. Since multiplying two signals with each other in the time domain is equivalent to performing a convolution between them in the frequency domain, the result of the sampling operation in the frequency domain is as shown in the bottom of Fig. 3.5(b). In this example, $f_{\mathrm{PLL}} \approx 3f_{\mathrm{SH}}$, and therefore the component at $\pm(f_{\mathrm{PLL}} - 3f_{\mathrm{SH}})$ appears at the lowest frequency. We also get components such as $\pm f_{\mathrm{SH}} \pm (f_{\mathrm{PLL}} - 3f_{\mathrm{SH}})$, $\pm 2f_{\mathrm{SH}} \pm (f_{\mathrm{PLL}} - 3f_{\mathrm{SH}})$, etc.

After the sampling operation, the signal experiences the hold operation, where the sampled signal is held for the duration of one period of the LO until the next sampling occurs. Fig. 3.6 illustrates the hold operation in the time and frequency domains. Fig. 3.6(a) shows the hold operation in the time domain, and we can see that holding the sampled signal for one period is equivalent to performing a convolution between the sampled signal and a rectangular function with a width of $T_{\mathrm{SH}} \equiv 1/f_{\mathrm{SH}}$. The result is the output we get from performing a S/H operation on the PLL output. On the other hand, Fig. 3.6(b) shows the same hold operation but in the frequency domain. The rectangular function is equivalent to a sinc function in the frequency domain (the figure shows the power spectrum which is its absolute value squared), and in this case the notches can be found at multiples of f_{SH}. Since the convolution of two signals in the time domain is equivalent to multiplying them in the frequency domain, the S/H output in the frequency domain can be obtained by multiplying the result of the sampling operation with the sinc characteristic. The result is shown at the bottom of Fig. 3.6(b). Assuming that $f_{\mathrm{PLL}} - 3f_{\mathrm{SH}} \ll f_{\mathrm{SH}}$, it can be seen that the lowest frequency components at $\pm(f_{\mathrm{PLL}} - 3f_{\mathrm{SH}})$ remain with relatively little attenuation, while the higher frequency components get attenuated by the sinc characteristic because they appear close to the notches. If we ignore these high frequency components, the resulting output of the S/H is a signal component at $f_{\mathrm{PLL}} - 3f_{\mathrm{SH}}$ which is the desired HM output signal if the

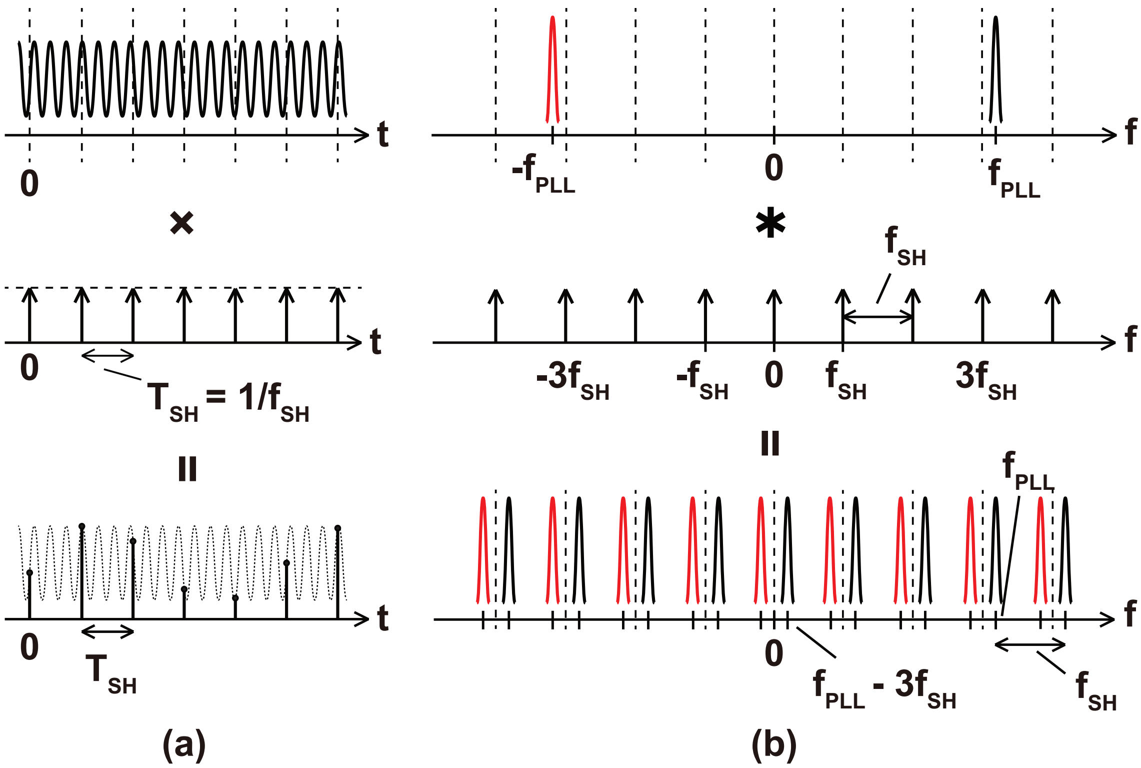


Figure 3.5: The sampling operation of a signal in the (a) time domain and (b) frequency domain

harmonic index is 3. HMs with different harmonic indexes can be realized depending on the relationship between f_{PLL} and f_{SH}. We will denote this frequency of $f_{\mathrm{PLL}} - kf_{\mathrm{SH}}$ as f_{HM} which is the frequency of the HM output. In summary, performing a S/H operation on the PLL output results in a signal component at $f_{\mathrm{PLL}} - kf_{\mathrm{SH}}$ which is what is desired by a HM. In reality, there are many non-idealities that degrade the output if we simply use a S/H as a HM and must be addressed for proper operation. This will be discussed in chapter 4.

3.2 Triple-Loop PLL and its Issues

So far, we have looked at the core of a mixer-based or HM-based PLL and their characteristic where the input phase noise does not get amplified in the loop. However, we have not yet explained how this structure can be used to actually realize a fractional-N PLL. In this section, we will look at the triple-loop PLL [65], which is a prior art that makes use of the HM-based feedback to suppress the quantization noise and fractional spurs in a fractional-N PLL. The qualitative characteristics of the triple-loop architecture will be explained, along with analysis based on noise transfer functions. We will also explain some issues with the triple-loop PLL.

3.2.1 Triple-Loop PLL Architecture

The basic architecture of the triple-loop PLL is shown in Fig. 3.7 [65]. It consists of three separate PLLs, which is why it is called the triple-loop architecture. The main PLL is based on the HM-based feedback that we discussed in section 3.1.2. The auxiliary integer-N PLL is used to generate the LO signal for the HM, while the auxiliary fractional-N PLL is used to generate the reference signal for the main PLL. Both auxiliary PLLs make use of a common reference signal. It should be noted that the actual architecture proposed in [65] was slightly different from this specific

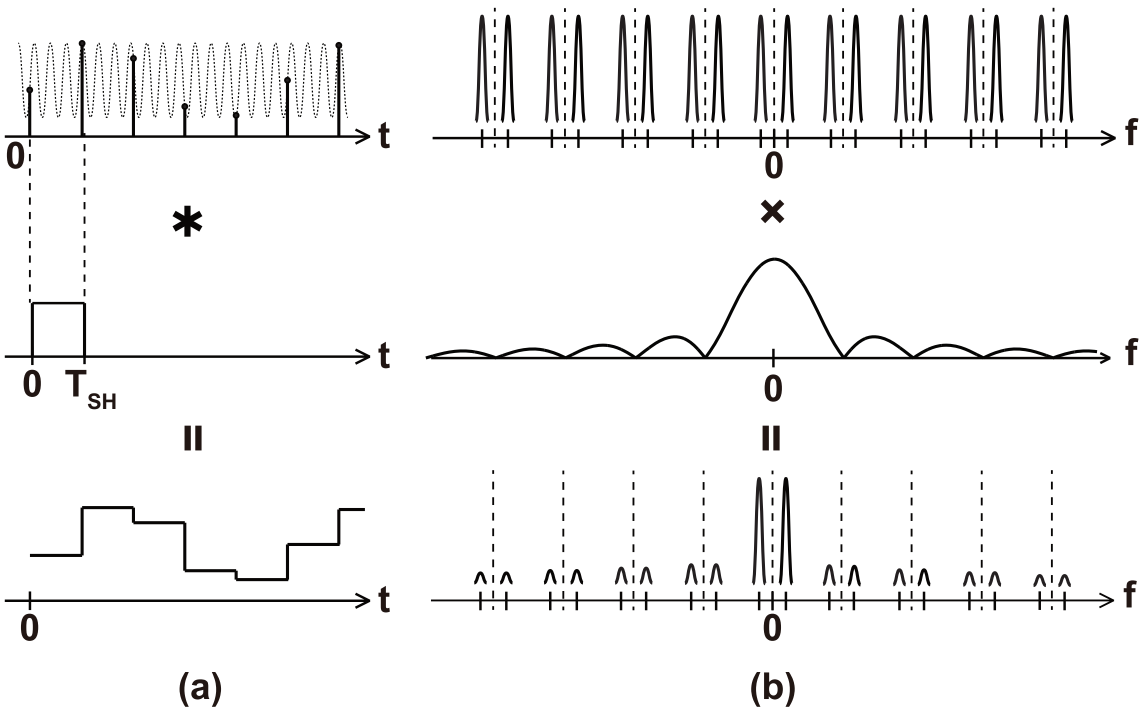


Figure 3.6: The hold operation of a signal in the (a) time domain and (b) frequency domain

architecture since a Bang-Bang-PD (BBPD) was used for the main PLL, while a Sampling-PD (SPD) was used for the auxiliary integer-N PLL. Nevertheless, the key concept regarding the triple-loop architecture remains the same.

The frequencies of the overall output, the auxiliary integer-N and fractional-N PLL outputs, and the reference are denoted as f_{OUT}, f_{INT}, f_{FRAC}, and f_{REF}, respectively. The frequency multiplication ratios of the integer-N and fractional-N PLLs are N_1 and $N_2 + \alpha$ respectively, where α is the fractional part of the Frequency-Control-Word (FCW), so $f_{\mathrm{INT}} = N_1 f_{\mathrm{REF}}$ and $f_{\mathrm{FRAC}} = (N_2 + \alpha) f_{\mathrm{REF}}$. M_1 and M_2 are the division ratios of the divider placed between the main PLL and the auxiliary integer-N and fractional-N PLLs respectively. k is the harmonic index of the HM. The output frequency can be expressed as follows

$$f_{\mathrm{OUT}} = \frac{k}{M_1} f_{\mathrm{INT}} + \frac{1}{M_2} f_{\mathrm{FRAC}} = \left(\frac{kN_1}{M_1} + \frac{N_2 + \alpha}{M_2} \right) f_{\mathrm{REF}}. \tag{3.5}$$

It can be seen from equation (3.5) that f_{OUT} is a linear combination of f_{INT} and f_{FRAC}. As such, the auxiliary integer-N PLL can be used for coarse frequency tuning while the auxiliary fractional-N PLL can be used for fine frequency tuning.

The principle of noise suppression in the triple-loop PLL is as follows. Since the auxiliary fractional-N PLL just uses a conventional architecture, its output contains quantization noise and fractional spurs that have been amplified by a factor of $N_2 + \alpha$ as we explained before due to the small feedback gain. The output of the auxiliary fractional-N PLL then enters the static frequency divider with a division ratio of M_2, meaning that the output phase noise including the quantization noise and fractional spurs are attenuated by a factor of M_2. This alone is not very meaningful, since the operation frequency also goes down by a factor of M_2 and the jitter remains constant. Amplifying this frequency by M_2 to obtain the original frequency would be completely meaningless since the phase noise would get re-amplified. However, the HM-based PLL generates

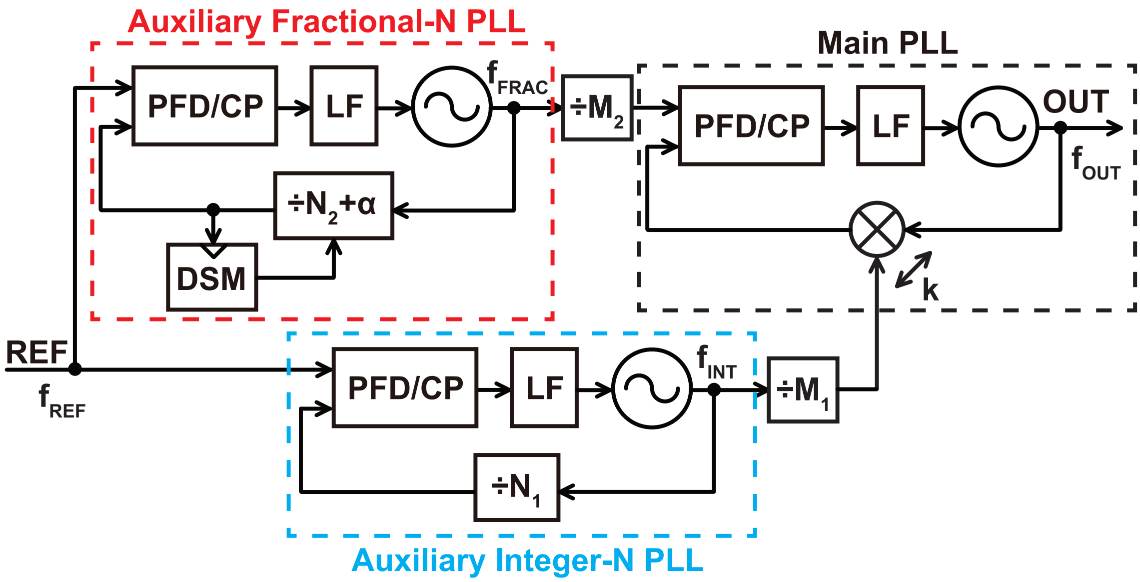


Figure 3.7: The triple-loop PLL architecture [65]

a high frequency signal by up-converting its reference frequency with the LO signal instead of multiplying it by some factor like in a conventional PLL, so the noise from the auxiliary fractional-N PLL does not get re-amplified. Equivalently, the noise from the auxiliary fractional-N PLL stays low after getting attenuated by M_2 because the main PLL input to output transfer function has a unity DC gain as we explained for equation (3.3). As a result, the triple-loop PLL can achieve a roughly M_2 times lower noise contribution from the quantization noise of the DSM and non-linearity induced fractional spurs. Of course, the main PLL output also contains the noise from the auxiliary integer-N PLL. However, this block does not suffer from quantization noise of fractional spurs, and therefore its noise is much easier to lower.

3.2.2 Noise Transfer Functions

Now that we have introduced the triple-loop architecture and gave a qualitative explanation of its noise suppression characteristics, we will look in to some more details by calculating the noise transfer functions. Fig. 3.8 shows a linear noise model of the triple-loop PLL. K_{PD1}, K_{PD2}, K_{PDM}, $H_{\mathrm{LF1}}(s)$, $H_{\mathrm{LF2}}(s)$, $H_{\mathrm{LFM}}(s)$, K_{VCO1}, K_{VCO2}, and K_{VCOM} are the PD gains, LF transfer functions, and VCO gains of the auxiliary integer-N PLL, auxiliary fractional-N PLL, and main PLL respectively. ϕ_{nEQUIV1}, ϕ_{nEQUIV2}, ϕ_{nEQUIVM}, ϕ_{nVCO1}, ϕ_{nVCO2}, and ϕ_{nVCOM} are the equivalent reference-referred PN and the VCO PN of the auxiliary integer-N PLL, auxiliary fractional-N PLL, and main PLL respectively. ϕ_{nDSM} and ϕ_{nREF} are the DSM quantization noise and fractional spurs, and reference PN respectively. Note that we have referred the fractional spur to the MMD output and included it inside ϕ_{nDSM} Finally, ϕ_{INT}, ϕ_{FRAC}, and ϕ_{OUT} are the resulting PN at the outputs of the auxiliary integer-N PLL, auxiliary fractional-N PLL, and main PLL respectively. Firstly the open-loop gains of the auxiliary integer-N PLL, auxiliary fractional-N PLL, and main PLL which are denoted as $H_{\mathrm{OL1}}(s)$, $H_{\mathrm{OL2}}(s)$, and $H_{\mathrm{OLM}}(s)$ respectively can be calculated as

$$H_{\mathrm{OL1}}(s) = \frac{K_{\mathrm{PD1}} K_{\mathrm{VCO1}} H_{\mathrm{LF1}}(s)}{N_1 s} \tag{3.6}$$

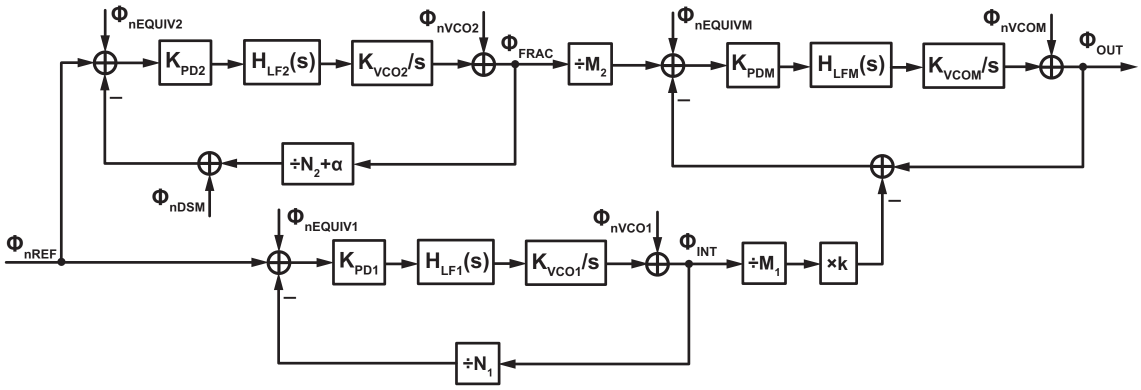


Figure 3.8: The noise model of the triple-loop PLL

$$H_{\mathrm{OL2}}(s) = \frac{K_{\mathrm{PD2}} K_{\mathrm{VCO2}} H_{\mathrm{LF2}}(s)}{(N_2 + \alpha)s} \tag{3.7}$$

$$H_{\mathrm{OLM}}(s) = \frac{K_{\mathrm{PDM}} K_{\mathrm{VCOM}} H_{\mathrm{LFM}}(s)}{s}. \tag{3.8}$$

Based on these expressions for the open-loop transfer function, the transfer functions from each noise source to the overall output can be calculated as shown below:

$$H_{\mathrm{VCOM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCOM}}} = \frac{1}{1 + H_{\mathrm{OLM}}(s)} \tag{3.9}$$

$$H_{\mathrm{EQUIVM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIVM}}} = \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{3.10}$$

$$H_{\mathrm{VCO1,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCO1}}} = \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{INT}}}\frac{\phi_{\mathrm{INT}}}{\phi_{\mathrm{nVCO1}}} = \frac{k}{M_1}\frac{1}{1 + H_{\mathrm{OL1}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{3.11}$$

$$H_{\mathrm{EQUIV1,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIV1}}} = \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{INT}}}\frac{\phi_{\mathrm{INT}}}{\phi_{\mathrm{nEQUIV1}}} = \frac{kN_1}{M_1}\frac{H_{\mathrm{OL1}}(s)}{1 + H_{\mathrm{OL1}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{3.12}$$

$$H_{\mathrm{VCO2,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCO2}}} = \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{FRAC}}}\frac{\phi_{\mathrm{FRAC}}}{\phi_{\mathrm{nVCO2}}} = \frac{1}{M_2}\frac{1}{1 + H_{\mathrm{OL2}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{3.13}$$

$$\begin{aligned} H_{\mathrm{EQUIV2,OUT}}(s) &\equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIV2}}} = \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{FRAC}}}\frac{\phi_{\mathrm{FRAC}}}{\phi_{\mathrm{nEQUIV2}}} \\ &= \frac{N_2 + \alpha}{M_2}\frac{H_{\mathrm{OL2}}(s)}{1 + H_{\mathrm{OL2}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \end{aligned} \tag{3.14}$$

$$H_{\mathrm{DSM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nDSM}}} = \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{FRAC}}}\frac{\phi_{\mathrm{FRAC}}}{\phi_{\mathrm{nDSM}}} = -\frac{N_2+\alpha}{M_2}\frac{H_{\mathrm{OL2}}(s)}{1+H_{\mathrm{OL2}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)} \tag{3.15}$$

$$\begin{aligned} H_{\mathrm{REF,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nREF}}} &= \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{INT}}}\frac{\phi_{\mathrm{INT}}}{\phi_{\mathrm{nREF}}} + \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{FRAC}}}\frac{\phi_{\mathrm{FRAC}}}{\phi_{\mathrm{nREF}}} \\ &= \frac{kN_1}{M_1}\frac{H_{\mathrm{OL1}}(s)}{1+H_{\mathrm{OL1}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)} + \frac{N_2+\alpha}{M_2}\frac{H_{\mathrm{OL2}}(s)}{1+H_{\mathrm{OL2}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)}. \end{aligned} \tag{3.16}$$

The transfer function from the main VCO to the overall output is given by $H_{\mathrm{VCOM,OUT}}(s)$ as shown in equation (3.9), and it is a high-pass characteristic similar to that in a conventional PLL.

The transfer function from the equivalent reference-referred PN of the main PLL to the overall output is given by $H_{\mathrm{EQUIVM,OUT}}(s)$ as shown in equation (3.10). As we observed for equation (3.3), this is a low-pass characteristic with unity DC gain, meaning that the noise from this node such as the main PLL PD noise or the noise from the HM referred to its output will not get amplified when it reaches the output. It is also important to note that this absence of noise amplification is meaningful because the operation frequency of this node is much lower than the overall output frequency.

The transfer function from the VCO of the auxiliary integer-N PLL to the overall output is given by $H_{\mathrm{VCO1,OUT}}(s)$ as shown in equation (3.11). The frequency independent gain of $H_{\mathrm{VCO1,OUT}}(s)$ is k/M_1 which is similar to the ratio between the operation frequencies of the auxiliary integer-N PLL output and the overall output, though they are not exactly the same due to the fact that the frequency from the auxiliary fractional-N PLL must also be summed to obtain that of the overall output. For the frequency dependent part, a high-pass characteristic is contributed by the auxiliary integer-N PLL, while a low-pass characteristic is contributed by the main PLL, resulting in a band-pass characteristic in total. The noise suppression offered by this band-pass characteristic varies greatly depending on the loop bandwidths of the auxiliary integer-N PLL $f_{\mathrm{UG,INT}}$ and the main PLL $f_{\mathrm{UG,MAIN}}$. If $f_{\mathrm{UG,MAIN}}$ is larger than $f_{\mathrm{UG,INT}}$, the band-pass characteristic will have a unity pass band gain with a low side cut-off frequency of $f_{\mathrm{UG,INT}}$ and a high side cut-off frequency of $f_{\mathrm{UG,MAIN}}$. On the other hand, if $f_{\mathrm{UG,MAIN}}$ is smaller than $f_{\mathrm{UG,INT}}$, the low side and high side cut-off frequencies will be reversed, and the pass band gain will be much lower. This can be understood by the fact that if $f_{\mathrm{UG,MAIN}}$ is narrower than $f_{\mathrm{UG,INT}}$, the main PLL will provide low-pass filtering for some of the in-band PN from the auxiliary integer-N PLL and greatly reducing its noise contribution, whereas if $f_{\mathrm{UG,MAIN}}$ is larger than $f_{\mathrm{UG,INT}}$, the main PLL bandwidth is too wide to provide meaningful low-pass filtering for the auxiliary integer-N PLL and the VCO noise will appear at the overall output. As such, designing the auxiliary integer-N PLL to have a much wider bandwidth than the main PLL is one way to reduce the PN overhead.

The transfer function from the equivalent reference-referred PN of the auxiliary integer-N PLL to the overall output is given by $H_{\mathrm{EQUIV1,OUT}}(s)$ as shown in equation (3.12). This is a cascaded low-pass characteristic from the auxiliary integer-N PLL and the main PLL, and the DC gain of kN_1/M_1 is similar to the ratio between the frequencies of the reference signal and the overall output. The cascaded low-pass characteristic can be useful for suppressing high frequency noise such as the reference spurs.

The transfer function from the VCO of the auxiliary fractional-N PLL to the overall output is given by $H_{\mathrm{VCO2,OUT}}(s)$ as shown in equation (3.13). The frequency dependence is similar to that of $H_{\mathrm{VCO1,OUT}}(s)$, with it being a band-pass characteristic. However, the frequency independent gain is just $1/M_2$, meaning that the PN contribution from the VCO of the auxiliary fractional-N PLL is mitigated greatly.

The transfer function from the equivalent reference-referred PN of the auxiliary fractional-N PLL to the overall output is given by $H_{\mathrm{EQUIV2,OUT}}(s)$ as shown in equation (3.14). This is a cascaded low-pass characteristic from the auxiliary fractional-N PLL and the main PLL, and the DC gain of $(N_2 + \alpha)/M_2$ is equal to the ratio between the frequencies of the reference signal and the main PLL input. In addition to the cascaded low-pass characteristic, the small DC gain greatly mitigates this noise source, just like the VCO of the auxiliary fractional-N PLL.

The transfer function from the DSM quantization noise and fractional spurs to the overall output is given by $H_{\mathrm{DSM,OUT}}(s)$ as shown in equation (3.15). This equation is the same as equation (3.14), except for the fact that the polarity is reversed. Again, we can see that in addition to getting low-pass filtered twice, the quantization noise gets amplified by $N_2+\alpha$ but then attenuated by M_2, then reaches the overall output without getting re-amplified. This matches our explanation in section 3.2.1. From equations (3.14) and (3.15), we can also see that the fractional spur enjoys a similar attenuation by M_2 when compared to a conventional fractional-N PLL.

Finally, the transfer function from the reference to the overall output is given by $H_{\mathrm{REF,OUT}}(s)$ as shown in equation (3.16). The DC gain of $H_{\mathrm{REF,OUT}}$ is $\left(\frac{kN_1}{M_1} + \frac{N_2+\alpha}{M_2}\right)$ which is equal to the ratio between the frequencies of the reference signal and the overall output, meaning that the PN contribution from the reference signal is in some sense the same as in a conventional PLL.

By calculating the noise transfer functions for the triple-loop PLL, we have confirmed analytically that the triple-loop PLL suppresses the quantization noise and fractional spurs.

3.2.3 Issues with the Triple-Loop PLL

As we explained both qualitatively and analytically, the triple-loop architecture is very useful for suppressing the the quantization noise from the DSM and non-linearity-induced fractional spurs. However, an obvious issue with this architecture is that a total of three individual PLLs are required, which can lead to an increase in the power consumption and design complexity. The chip area and coupling with other blocks can also become a serious issue if we use three inductors for the VCOs, which is one of the reasons why the work in [65] uses a ring VCO for the auxiliary fractional-N PLL. This is the issue we attempt to alleviate in our proposed dual-feedback architecture.

3.3 Dual-Feedback PLL

In the previous section, we introduced the triple-loop architecture and explained how it makes use of the HM-based feedback to attenuate the quantization noise from the DSM and fractional spur. In this section, we introduce the dual-feedback architecture [59, 60], which achieves similar characteristics as the triple-loop architecture in terms of quantization noise and spur suppression, but without suffering from the overhead and complexity of having three separate PLLs. As we did for the triple-loop PLL, we will show the proposed architecture, explain its characteristics qualitatively, and then analyze the architecture in detail by calculating noise transfer functions.

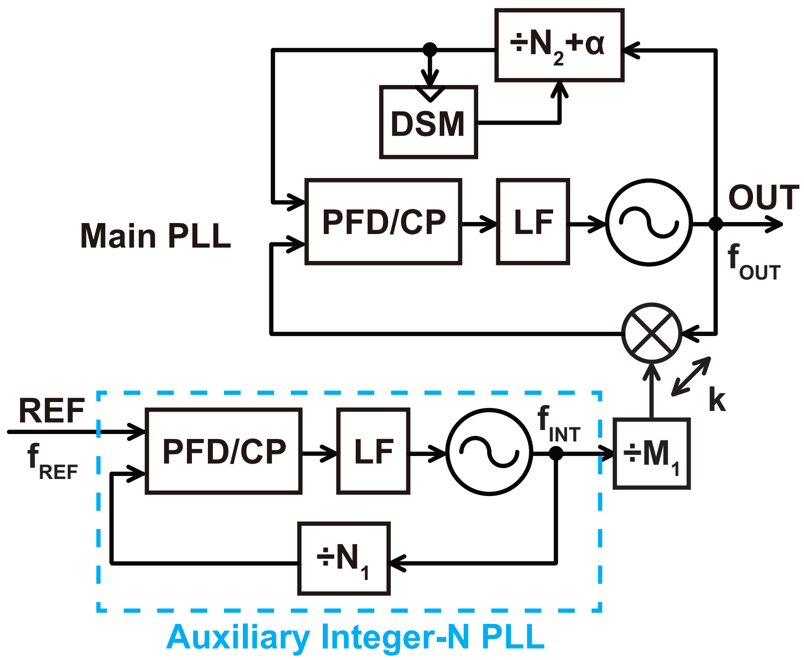


Figure 3.9: The dual-feedback PLL architecture

3.3.1 Proposed PLL Architecture

The basic architecture of the dual-feedback PLL is shown in Fig. 3.9. Unlike the triple-loop PLL, there is only one auxiliary PLL in addition to the main PLL. Instead however, the main PLL has two feedback paths. The auxiliary integer-N PLL is used to generate the LO signal for the HM, just like in the triple-loop PLL. This integer-N PLL can also be used for coarse frequency tuning. On the other hand, the reference for the main PLL must have fractional tunability. In the triple-loop PLL, this was achieved using an auxiliary fractional-N PLL. In the dual-feedback PLL, we instead generate this fractionally-tunable signal using the overall output and a fractional divider. Besides this additional feedback path, the rest of the HM-based PLL is the same as in the triple-loop PLL.

The frequencies of the overall output, the auxiliary integer-N PLL output, and the reference are denoted as f_{OUT}, f_{INT}, and f_{REF} respectively. N_1 and $N_2+\alpha$ are the frequency division ratios of the auxiliary integer-N PLL and the MMD in the main PLL respectively, so $f_{\mathrm{INT}} = N_1 f_{\mathrm{REF}}$. M_1 is the division ratio of the frequency divider placed between the auxiliary integer-N PLL and the main PLL, and k is the harmonic index of the HM. Once the PLL has locked, the output frequencies of the HM-based feedback $f_{\mathrm{OUT}} - \frac{k}{M_1} f_{\mathrm{INT}}$ and the MMD based feedback $\frac{f_{\mathrm{OUT}}}{N_2+\alpha}$ are the same. Therefore, the output frequency can be expressed as follows

$$f_{\mathrm{OUT}} = \frac{k}{M_1}\frac{N_2+\alpha}{N_2+\alpha-1} f_{\mathrm{INT}} = \frac{kN_1}{M_1}\frac{N_2+\alpha}{N_2+\alpha-1} f_{\mathrm{REF}}. \tag{3.17}$$

It is worth noting here the requirements posed here on the frequency plan in order to properly achieve seamless frequency tuning. The output frequency of the dual-feedback PLL can be coarsely tuned by changing the frequency multiplication ratio of the auxiliary integer-N PLL N_1 while it can be finely tuned by changing the division ratio of the MMD $N_2+\alpha$. Therefore, the range covered by fine frequency tuning has to cover the coarse tuning frequency step. For fixed values of k, N_1, M_1, and f_{REF}, the minimum and maximum achievable values of f_{OUT} are given by $f_{\mathrm{OUT,min,fine}}$ and $f_{\mathrm{OUT,max,fine}}$ given by the expressions below

$$f_{\mathrm{OUT,min,fine}} = \frac{kN_1}{M_1}\frac{(N_2+\alpha)_{\max}}{(N_2+\alpha)_{\max}-1}f_{\mathrm{REF}} \tag{3.18}$$

$$f_{\mathrm{OUT,max,fine}} = \frac{kN_1}{M_1}\frac{(N_2+\alpha)_{\min}}{(N_2+\alpha)_{\min}-1}f_{\mathrm{REF}} \tag{3.19}$$

where $(N_2+\alpha)_{min}$ and $(N_2+\alpha)_{max}$ are the minimum and maximum values of $N_2+\alpha$ respectively. On the other hand, if we fix $N_2+\alpha$ to its maximum value of $(N_2+\alpha)_{max}$ (which gives the minimum output frequency when we fix the other variables) and change N_1 to N_1+1, the resulting output frequency $f_{\mathrm{OUT,coarse}}$ is given by

$$f_{\mathrm{OUT,coarse}} = \frac{k(N_1+1)}{M_1}\frac{(N_2+\alpha)_{\max}}{(N_2+\alpha)_{\max}-1}f_{\mathrm{REF}}. \tag{3.20}$$

For seamless frequency tuning to be achieved, $f_{\mathrm{OUT,max,fine}}$ must be larger than or equal to $f_{\mathrm{OUT,coarse}}$. After comparing these two values, we end up with the inequality

$$N_1 \geq \frac{(N_2+\alpha)_{min}-1}{1-\frac{(N_2+\alpha)_{min}}{(N_2+\alpha)_{max}}}. \tag{3.21}$$

To satisfy this inequality, we require a large value of N_1 (which implies a low reference frequency and/or high output frequency for the auxiliary integer-N PLL) and/or a small $(N_2+\alpha)_{min}$ and large $(N_2+\alpha)_{max}$. The latter condition can sometimes be difficult to meet since the HM output frequency is limited as we will explain later, meaning that $(N_2+\alpha)_{min}$ cannot be too small. All of the frequency plans for the PLLs implemented in this thesis are designed to meet this requirement.

For the suppression of quantization noise in the dual-feedback PLL, it is useful to compare its characteristics to that of the triple-loop PLL. We explained in section 3.2 how in the triple-loop PLL, the quantization noise and fractional spurs first get amplified by a factor of $N_2+\alpha$ by the auxiliary fractional-N PLL, then gets lowered by a factor of M_2 by the static frequency divider, then reaches the overall output without getting re-amplified due to the unity DC gain transfer function of the HM-based main PLL. In the dual-feedback PLL, the quantization noise and fractional spurs are generated at the input of the main PLL, meaning that they reach the overall output without experiencing any amplification in the first place. As a result, the contribution from the quantization noise and fractional spurs in the dual-feedback PLL is lower than that in the conventional fractional-N PLL by approximately a factor of $N_2+\alpha$. It should be noted that since the HM-based main PLL now has an extra feedback path based on the MMD, its loop transfer function is slightly altered from that of the simple HM-based core with unity DC gain. This change is small however, since the gain of the MMD based feedback is only $1/(N_2+\alpha)$ while that of the HM-based feedback is unity.

3.3.2 Noise Transfer Functions

We will now analyze the dual-feedback architecture in more detail using noise transfer functions. Fig. 3.10 shows a linear noise model of the dual-feedback PLL. K_{PD1}, K_{PDM}, $H_{\mathrm{LF1}}(s)$, $H_{\mathrm{LFM}}(s)$, K_{VCO1}, and K_{VCOM} are the PD gains, LF transfer functions, and VCO gains of the auxiliary integer-N PLL and main PLL respectively. ϕ_{nEQUIV1}, ϕ_{nEQUIVM}, ϕ_{nVCO1}, and ϕ_{nVCOM} are the equivalent reference-referred PN and the VCO PN of the auxiliary integer-N PLL and the main PLL

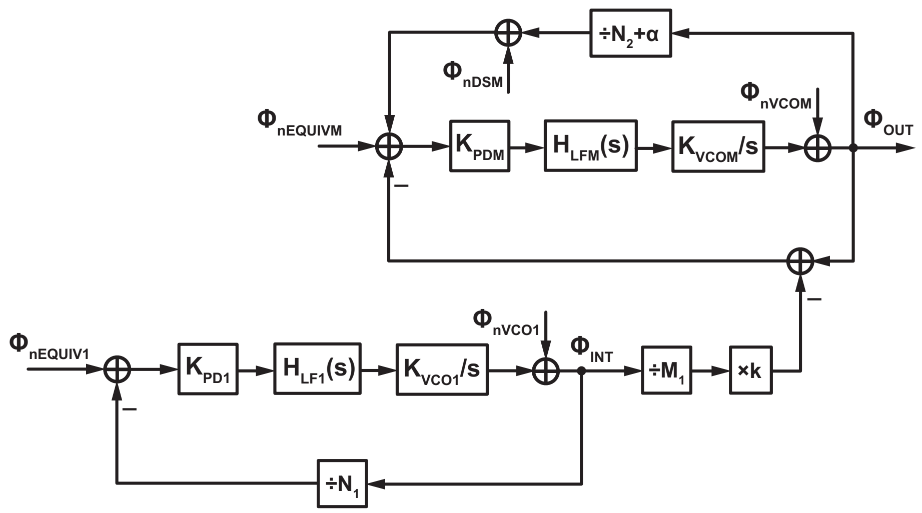


Figure 3.10: The noise model of the dual-feedback PLL

respectively. ϕ_{nDSM} is the DSM quantization noise and fractional spurs. Finally, ϕ_{INT} and ϕ_{OUT} are the resulting PN at the outputs of the auxiliary integer-N PLL and main PLL respectively. The open-loop gains of the auxiliary integer-N PLL and main PLL which are denoted as $H_{\mathrm{OL1}}(s)$ and $H_{\mathrm{OLM}}(s)$ respectively can be calculated as

$$H_{\mathrm{OL1}}(s) = \frac{K_{\mathrm{PD1}} K_{\mathrm{VCO1}} H_{\mathrm{LF1}}(s)}{N_1 s} \tag{3.22}$$

$$H_{\mathrm{OLM}}(s) = \frac{K_{\mathrm{PDM}} K_{\mathrm{VCOM}} H_{\mathrm{LFM}}(s)}{s} \left(1 - \frac{1}{N_2 + \alpha}\right). \tag{3.23}$$

An important thing to note here is that for the main PLL, the two feedback paths are merged into one and the main PLL is treated as one loop. The feedback gains of the HM-based path and MMD based path are unity and $1/(N_2 + \alpha)$ respectively, and since their polarities are opposite, the total feedback gain of the negative feedback system is $1 - 1/(N_2 + \alpha)$. Using these expressions for the open-loop gains, the transfer functions from each noise source to the overall output can be calculated as shown below:

$$H_{\mathrm{VCOM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCOM}}} = \frac{1}{1 + H_{\mathrm{OLM}}(s)} \tag{3.24}$$

$$H_{\mathrm{EQUIVM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIVM}}} = \frac{N_2 + \alpha}{N_2 + \alpha - 1} \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \approx \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{3.25}$$

$$H_{\mathrm{DSM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nDSM}}} = \frac{N_2 + \alpha}{N_2 + \alpha - 1} \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \approx \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{3.26}$$

$$\begin{aligned} H_{\mathrm{VCO1,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCO1}}} &= \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{INT}}} \frac{\phi_{\mathrm{INT}}}{\phi_{\mathrm{nVCO1}}} = \frac{k}{M_1} \frac{N_2 + \alpha}{N_2 + \alpha - 1} \frac{1}{1 + H_{\mathrm{OL1}}(s)} \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \\ &\approx \frac{k}{M_1} \frac{1}{1 + H_{\mathrm{OL1}}(s)} \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \end{aligned} \tag{3.27}$$

$$\begin{aligned} H_{\mathrm{EQUIV1,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIV1}}} &= \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{INT}}}\frac{\phi_{\mathrm{INT}}}{\phi_{\mathrm{nEQUIV1}}} = \frac{kN_1}{M_1}\frac{N_2+\alpha}{N_2+\alpha-1}\frac{H_{\mathrm{OL1}}(s)}{1+H_{\mathrm{OL1}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)} \\ &\approx \frac{kN_1}{M_1}\frac{H_{\mathrm{OL1}}(s)}{1+H_{\mathrm{OL1}}(s)}\frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)}. \end{aligned} \tag{3.28}$$

The transfer function from the main VCO to the overall output is given by $H_{\mathrm{VCOM,OUT}}(s)$ as shown in equation (3.24). This again, is a high-pass characteristic similar to that in a conventional PLL.

The transfer function from the equivalent reference-referred PN of the main PLL to the overall output is given by $H_{\mathrm{EQUIVM,OUT}}(s)$ as shown in equation (3.25). It is a low-pass characteristic similar to that in the triple-loop PLL as was explained for equation (3.10). However, the DC gain is slightly deviated from unity because of the MMD-based feedback. Since the feedback gain is $\left(1-\frac{1}{N_2+\alpha}\right)$, the DC gain of $H_{\mathrm{EQUIVM,OUT}}(s)$ is $1/\left(1-\frac{1}{N_2+\alpha}\right)=\frac{N_2+\alpha}{N_2+\alpha-1}$. However, for large values of $N_2+\alpha$, this can be approximated as unity. All the approximations in equations (3.25), (3.26), (3.27), and (3.28) are based on this assumption. As such, the equivalent reference-referred PN of the main PLL does not get amplified by the loop.

The transfer function from the DSM quantization noise and fractional spurs to the overall output is given by $H_{\mathrm{DSM,OUT}}(s)$ as shown in equation (3.26). The expression is the same as that for $H_{\mathrm{EQUIVM,OUT}}(s)$. Consequently, we can see that the quantization noise and fractional spur avoid amplification in the loop, and achieve a suppression by a factor of $N_2+\alpha-1$ when compared to a conventional fractional-N PLL. Again, this is in contrast to the triple-loop PLL where the quantization noise and fractional spurs first get amplified by a factor of $N_2+\alpha$ and then attenuated by a factor of M_2. In terms of the frequency characteristic, the dual-feedback PLL provides just one low-pass characteristic from the main PLL, while the triple-loop PLL provides two low-pass characteristics from the auxiliary fractional-N PLL and the main PLL. On the other hand, the noise shaping frequency in the triple-loop PLL must be equal to the reference frequency, while for the dual-feedback PLL, it can be configured to a higher frequency depending on the specific frequency plan being employed. All things considered, the dual-feedback PLL offers a similar potential for suppression of quantization noise and fractional spurs as the triple-loop PLL, while avoiding the complexity and overhead from the auxiliary fractional-N PLL.

The transfer functions from the VCO and equivalent reference-referred PN of the auxiliary integer-N PLL are given by $H_{\mathrm{VCO1,OUT}}(s)$ and $H_{\mathrm{EQUIV1,OUT}}(s)$ as shown in equations (3.27) and (3.28) respectively. These transfer functions have similar characteristics with that of from the auxiliary PLL in the triple-loop PLL, since we have low-pass and high-pass filtering from the auxiliary PLL, along with low-pass filtering from the main PLL. Again, one solution to suppress the PN from the VCO of the auxiliary integer-N PLL effectively is to design the bandwidth of the auxiliary integer-N PLL to be much wider than that of the main PLL.

Based on the analysis of the noise transfer functions, we can conclude that the dual-feedback PLL achieves similar noise suppression characteristics as the triple-loop PLL, while avoiding the complexity and overhead from the auxiliary fractional-N PLL.

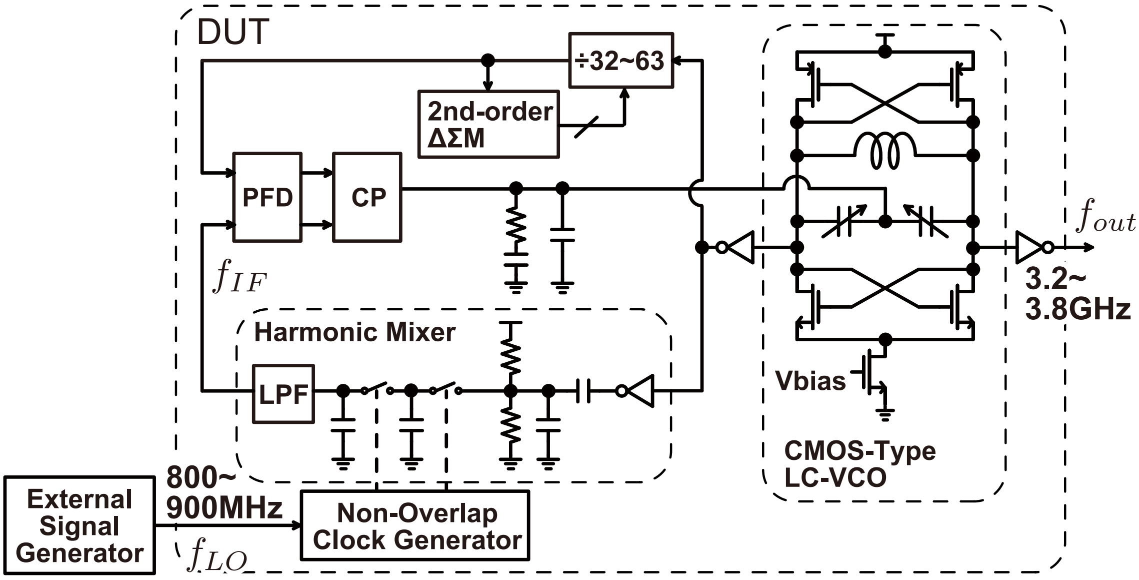


Figure 3.11: The implementation details of the dual-feedback PLL

3.4 Implementation and Measurement Results

In the previous section, we proposed the dual-feedback PLL architecture and explained how it retains the similar benefits of quantization noise and fractional spur suppression as the triple-loop PLL. We implemented a proof-of-concept fractional-N PLL based on the dual-feedback architecture. Here, we explain some of the implementation details, along with measurement results that demonstrate the effectiveness of the proposed architecture. We also provide some supplementary simulation results to help support our proposal.

3.4.1 Circuit Implementation Details

The prototype was implemented in the TSMC 65nm bulk CMOS process. Figs. 3.11 and 3.12 show the implementation details and the chip micrograph respectively. The minimum goal of this chip was to demonstrate that the dual-feedback architecture can successfully avoid the amplification of quantization noise. As such, we implemented just the main PLL, and used an external signal generator to generate the 800 to 900MHz LO signal instead of an auxiliary integer-N PLL. The LC-VCO was based on a conventional CMOS type cross-coupled architecture with an output frequency between 3.2 to 3.8GHz. The VCO gain is around 30MHz/V. The PLL is based on a type-II PFD/CP based architecture with a 1MHz bandwidth. The MMD is controlled by a 2nd-order DSM with 16 bits, with an operation frequency between 60 to 100MHz. A 2nd-order DSM was used to reduce the hardware complexity and overhead since the quantization noise can be reduced to a negligible level without increasing the noise shaping order thanks to the avoidance of noise amplification. In hindsight, a 3rd-order DSM may have been a better choice since it offers better randomization of the MMD control sequence and avoids unnecessary risk in terms of large fractional spurs. The 4th-order HM is realized based on a two stage S/H circuit with DC biasing and low-pass filtering before and after the S/H. The requirements for these filters are discussed in detail in chapter 4. The two-stage S/H is controlled by a non-overlap clock generator whose input is the LO signal. The active core area is around 0.24mm^2.

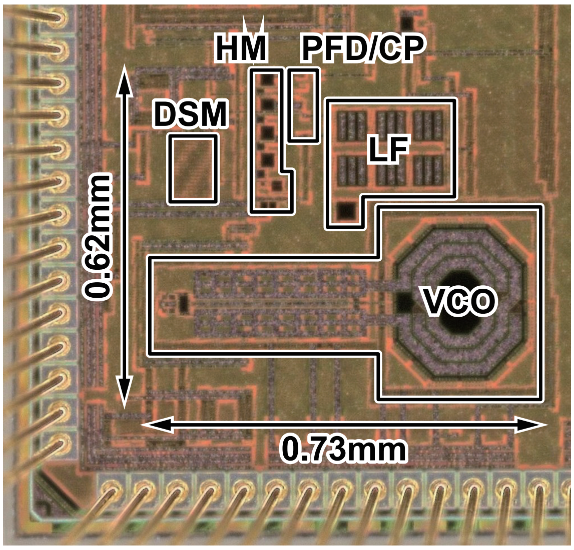


Figure 3.12: The chip micrograph of the dual-feedback PLL

3.4.2 Measurement Results and Performance Comparison

Here we will show some measurement results along with some supplementary simulation results to demonstrate the effectiveness of our work. Fig. 3.13 shows the simulated and calculated PN contribution from the DSM in a conventional fractional-N PLL (simulated results in red, calculated results in black dashed line) and in a dual-feedback PLL (simulated results in blue, calculated results in black dash-dotted line). Here, the output frequency in both cases is 3.588GHz, the noise shaping frequency for both cases is 77.97MHz, and the loop bandwidth in both cases is 800kHz. For the conventional case, we assumed a 77.97MHz signal being provided by the reference while the division ratio of the MMD was set to 46.01983 to generate the desired 3.588GHz output. For the dual-feedback PLL, we have one feedback path realized by an MMD also with a division ratio of 46.01983, and another feedback path realized by a 4th-order HM with an LO frequency of 877.5MHz. The VCO and PFD/CP gain are the same for both architectures, and the LF is adjusted to compensate for the difference in the feedback gain and realize the same open-loop transfer function. We can see from both the calculation and simulation results that compared to the conventional architecture, the dual-feedback architecture achieves an improvement in noise suppression by a factor equal to the division ratio of 46.01983 minus 1, or around 33dB. This is as expected from our qualitative assessment of the architecture and analysis. The slight discrepancy between the calculated and simulated noise at low frequencies is likely due to noise folding coming from non-linearities in the loop. Nevertheless, the effect on the overall jitter is minimal.

Fig. 3.14 shows the measured PN spectrum of the implemented PLL in integer (red) and fractional (blue) modes. We also show the expected noise contribution from the DSM quantization noise in the conventional and proposed PLLs. All of the settings such as the frequency plan and loop bandwidth are the same as for the simulation results in Fig. 3.13, except that for the measured phase noise, the division ratio of the MMD is 46 for integer mode and $46{+}2^{-6}$ for the fractional mode which are slightly different from that in the calculated results. The effect of this difference on the PN however, is negligible. The difference between the PN spectrum in integer and fractional modes corresponds to the PN contributed by the DSM quantization noise. In a conventional fractional-N PLL, this corresponds to the large PN profile shown in the black dashed line while for the dual-feedback PLL, it corresponds to the much smaller PN profile in the black dash-dotted

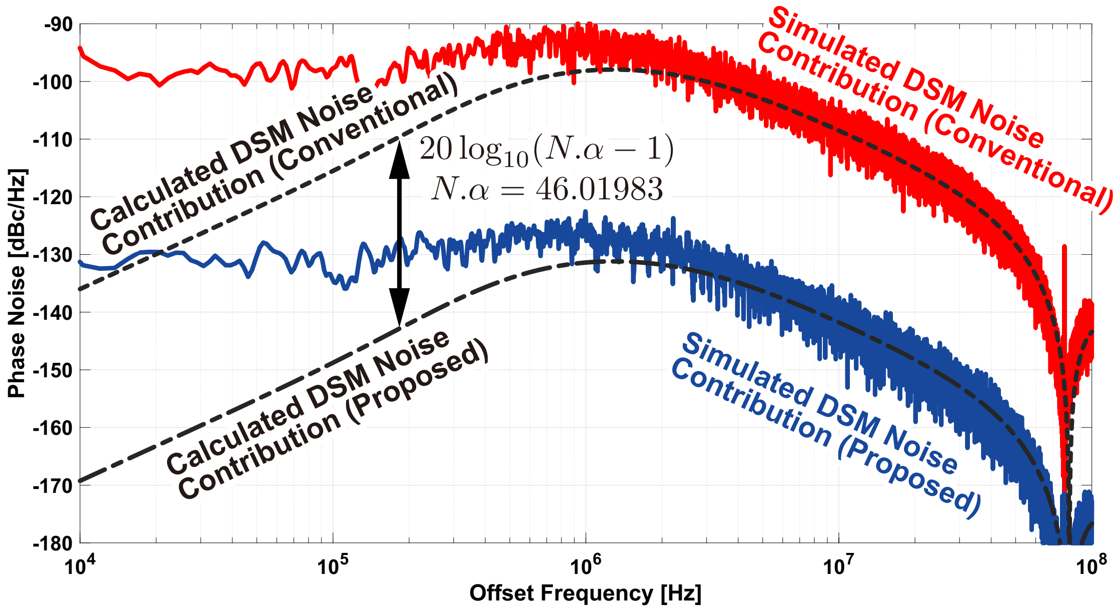


Figure 3.13: The simulated and calculated phase noise contribution from the DSM quantization noise in a conventional fractional-N PLL (red and black dashed lines) and dual-feedback PLL (blue and black dash-dotted lines) both with a loop bandwidth of 1MHz, a noise shaping frequency of 77.97MHz, an output frequency of 3.588GHz, and a loop bandwidth of around 800kHz

line. In the actual measurement results, we can see that the difference between the PN in integer and fractional modes is much smaller than the expected contribution in the conventional case, in fact it is almost negligible, and so we can conclude from this result that the quantization noise has been successfully suppressed by avoiding amplification.

Fig. 3.15 shows the simulated fractional spurs in the (a) conventional and (b) proposed architectures. The MMD division ratio has been set to 46.00386 since we wanted to make the fractional spur appear at a low offset frequency of 301kHz. All of the other settings such as the loop bandwidths and frequency plan are roughly the same as that used for Fig. 3.13. In addition, we have intentionally added a 5% CP current mismatch to create a non-linearity in the loop characteristic. We can see that for the conventional case, the fractional spur is around −33dBc while for the proposed case, it is around −66dBc. The actual value of the fractional spur is obviously dependent on the actual current mismatch and the specific non-linearity in the loop characteristic, but the difference in the fractional spur level between the two cases is 33dB, which matches the predicted factor of $N.\alpha - 1$ due to avoiding noise amplification.

Fig. 3.16 shows the measured output spectrum containing (a) fractional and (b) reference spurs. We unfortunately cannot make a comparison of the fractional spur with the conventional case since we did not implement a conventional fractional-N PLL with the same circuit blocks. However, the fractional spur value of −65dBc is very low with respect to other state-of-the-art works, and also matches well with the spur values achieved in the behavioral simulation. The reference spur also achieves a respectable value, thanks to the fact that the noise at the input of the main PLL does not get amplified. Fig. 3.17 shows a plot of the fractional spur level for different fractional parts of the FCW. We can see that the spur level is relatively constant for low FCWs, but it gets lower as the FCW gets large since the offset frequency of the fractional spur also increases and it experiences some low-pass filtering from the main PLL. The worst case fractional spur value is −65.3dBc.

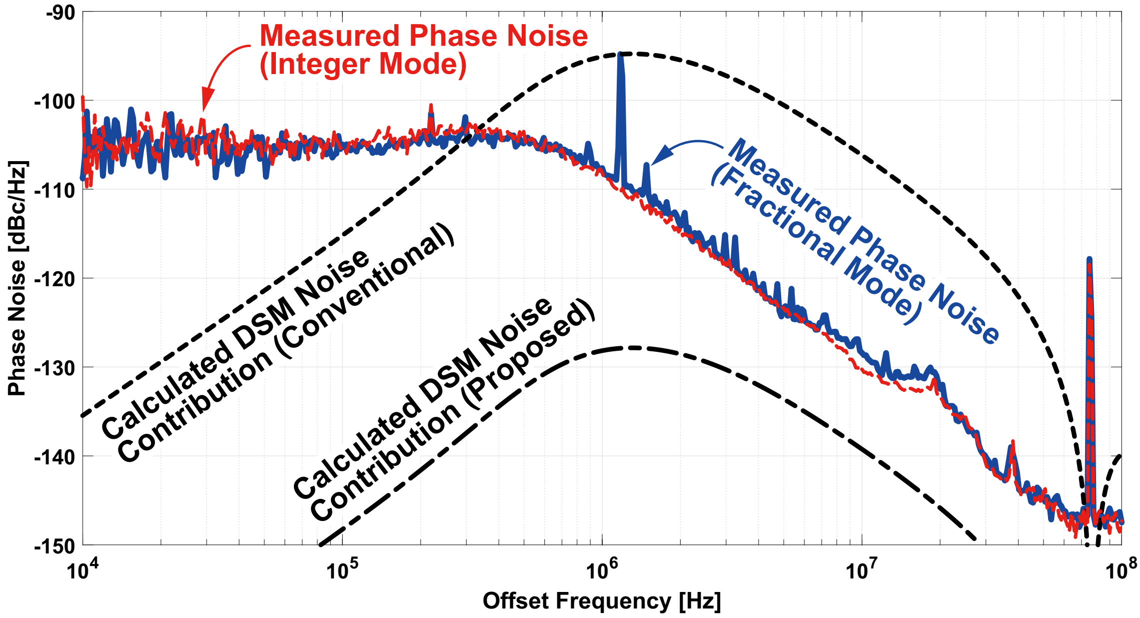


Figure 3.14: The measured PN spectrum of the implemented chip in integer mode (red) and fractional mode (blue), and the calculated PN contribution from the DSM quantization noise in the conventional and proposed architectures

Finally, we performed a measurement to observe the effectiveness of cascaded low-pass filtering to suppress the out-of-band spurs. Though we used an external signal generator for the LO signal in this implementation, in practice, an integer-N PLL should be used to generate this LO signal. This means that the LO signal will contain reference spurs which can potentially degrade the spurious performance of the overall PLL. However, as we saw in equation (3.28), the equivalent reference-referred PN from the auxiliary integer-N PLL which includes the reference spur experiences low-pass filtering from both the auxiliary integer-N PLL and the main PLL, and this cascaded filtering characteristic should suppress the reference spur and mitigate the degradation of the overall spurious performance. To demonstrate this, we intentionally injected spurious tones in the LO signal for the HM, and compared the spurs at the input and output of the PLL. Fig. 3.18 shows the result of this measurement, with the input LO signal being shown in (a), and the resulting PLL output being shown in (b). The input LO shown in red contains intentionally injected spurs, while they are greatly filtered in the output shown in blue. Looking at the spur at 16MHz offset, we can see that the attenuation of this spur due to the filtering characteristic is greater than 35dB, which is expected due to the 40dB/dec roll off of the type-II PLL characteristic. As a result, we can expect the spurious performance of the dual-feedback PLL to not degrade so much even if we use an integer-N PLL for the LO generation.

A performance comparison of the implemented dual-feedback PLL with other works is shown in table 3.1. The proposed circuit achieves a spurious performance similar to that of the other state-of-the-art works, but without relying on calibration and with just one auxiliary PLL.

3.5 Analytical Performance Comparison with Conventional Fractional-N PLL

In this chapter, we have introduced the dual-feedback architecture, and demonstrated how it makes use of the HM-based feedback by avoiding amplification of the quantization noise. This suppression

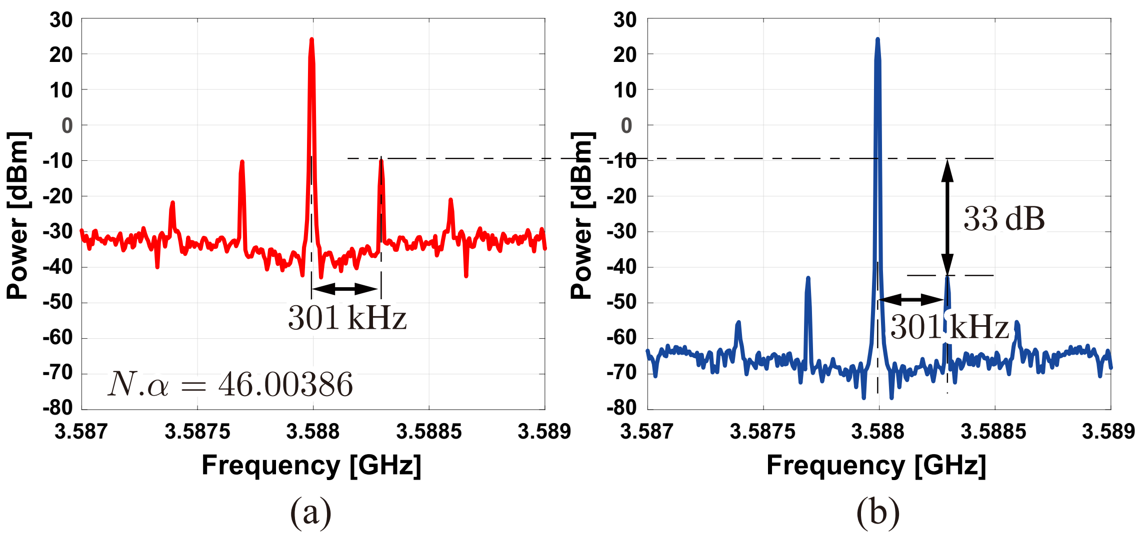


Figure 3.15: The simulated fractional spur when we deliberately added a 5% CP current mismatch for the (a) conventional architecture and (b) proposed architecture

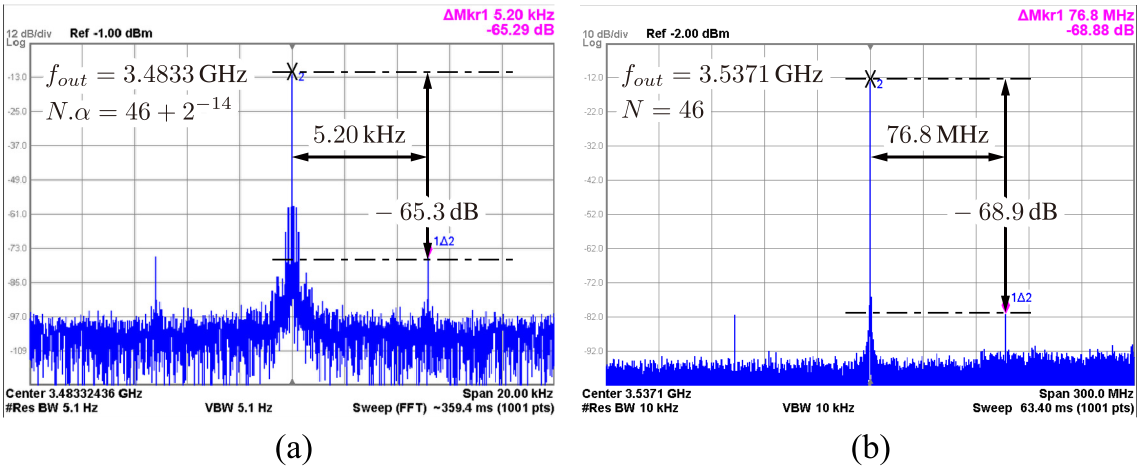


Figure 3.16: The measured output spectrum containing (a) fractional and (b) reference spurs

of quantization noise certainly provides an advantage over a conventional single-loop fractional-N PLL. In return however, an extra PLL is required to generate the LO signal for the HM. Is it really worth it to employ the dual-feedback architecture? And if so, under what circumstances? These are natural questions to ask, and ones that we attempt to answer in this section. Specifically, we will compare the dual-feedback architecture with the conventional single-loop fractional-N PLL when they are optimally designed for jitter and/or power, and make a comparison of the two. We will also make comparisons with a conventional integer-N PLL to serve as a sort of upper limit in terms of performance, and also a comparison with a DTC-based PLL.

3.5.1 Comparison Overview

We will first give an overview of the comparison that we are making. If we take a basic single-loop integer-N PLL as an example, we can see that if we have a given VCO PN profile and equivalent reference-referred PN profile and ignore all other noise sources, we can choose a certain PLL bandwidth that optimizes the overall integrated jitter. Furthermore, for a fixed circuit architecture, the PN profiles of the circuit blocks can be roughly derived based on its power consumption.

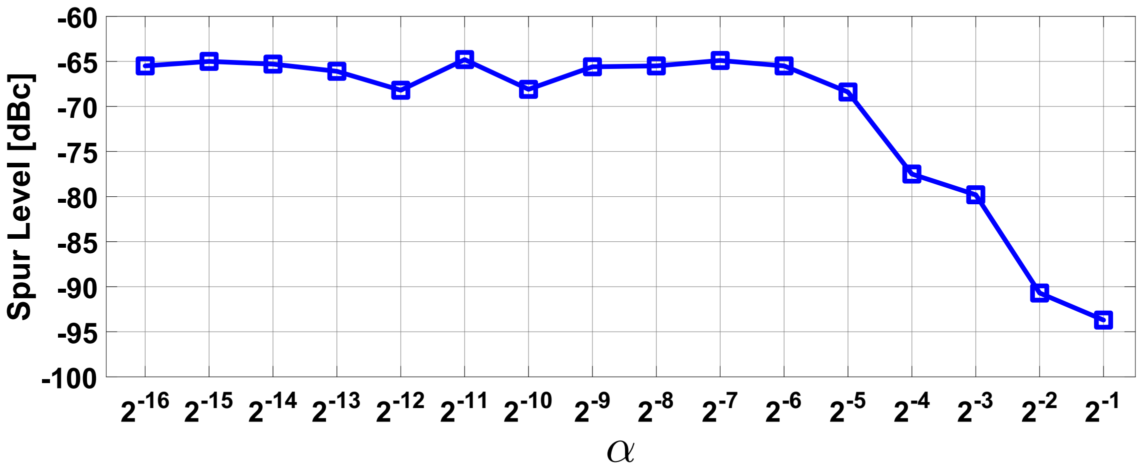


Figure 3.17: A plot of the fractional spur level for different fractional FCWs

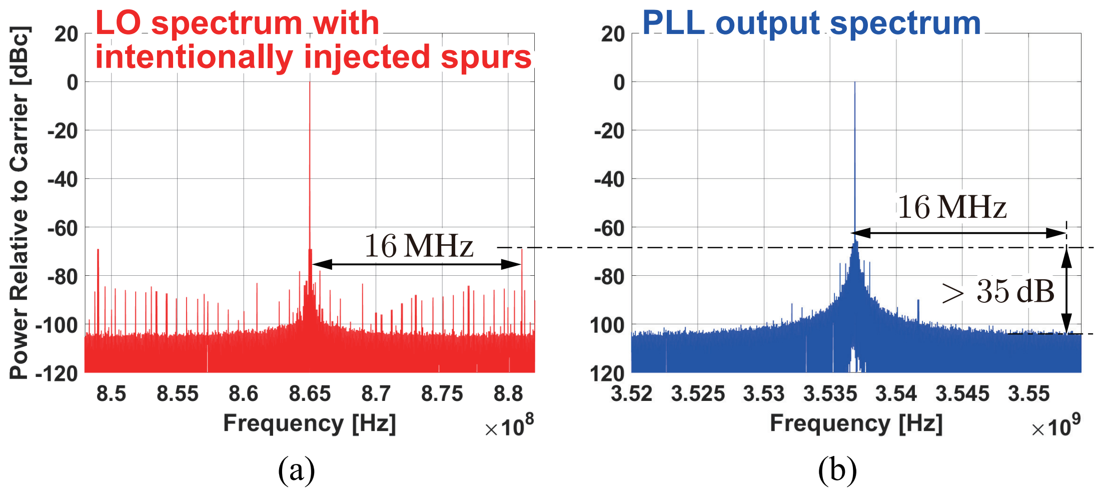


Figure 3.18: A plot of the fractional spur level for different fractional FCWs

As such, we can obtain a plot showing the relationship between the integrated jitter and power consumption under optimally designed conditions. Our goal is to make this kind of plot for all of the architectures that we want to compare, which include the conventional integer-N and fractional-N PLLs, the HM-based dual-feedback PLL, and the DTC-based fractional-N PLL.

3.5.2 Analysis Framework

Here, we show the framework that we base our comparison on. Fig. 3.19 shows a block diagram of the conventional fractional-N PLL and the dual-feedback PLL showing the conditions under which we calculate the integrated jitter. We use the same reference frequency of 50MHz, and both architectures generate the same output frequency of 3.103GHz. For the dual-feedback PLL, the external integer-N PLL generates a frequency of 4.5GHz, divides this by 3 to obtain the LO for the HM, and then down-converts the overall output to 103MHz with a second order HM. All of the PLLs besides the dual-feedback PLL are based on a type-II 3rd-order architecture, with the zero located at a frequency 5 times smaller than the unity gain bandwidth, and the pole located at a frequency 5 times larger than the unity gain bandwidth. This is because when we assume a practical design point with an LC-VCO, the optimal bandwidth ends up being sufficiently narrower than the reference frequency of 50MHz and so a type-I architecture is not necessary and a type-II

Table 3.1: A Performance Comparison of the Implemented PLL With Other Works

Reference	**This work**	Wu JSSC'19 [3]	Yang ISSCC'19 [8]	Ho ISSCC'18 [6]	Raczkowski JSSC'15 [4]
Architecture	Dual Feedback	DTC SPLL	Triple loop	Reference Dithering	DTC SSPLL
Freq. [GHz]	3.2-3.8	5.5-7.3	7.0-9.0	3.0-5.2	9.2-12.7
Calibration	**no**	yes	no	yes	yes
# of aux loops	**1**	N/A	2	N/A	N/A
Frac. Spur [dBc]	**<-65**	<-64	<-70	<-62.5	<-43
Ref. Spur [dBc]	**-69**	-70.2	<-66	-102	-60
RMS jitter [fs] (Integ. Range)	503 (10k~100MHz)	75 (10k~10MHz)	131 (10k~10MHz)	1780 (10k~100MHz)	280 (10kHz~60MHz)
IPN [dBc] (Integ. Range)	-42.2 (10k~100MHz)	-54 (10k~10MHz)	-46.9† (10k~10MHz)	-30.9† (10k~100MHz)	-39.8 ~ -38.1 (10kHz~60MHz)
Power [mW]	6.9*	18.9	13.4	18.1	13
Area [mm^2]	0.24	0.45	0.25	0.338	N/A
Process [nm]	65	28	16	65	28

*excluding the auxiliary PLL and I/O buffers †calculated from RMS jitter

architecture can be used for a high loop gain. For the dual-feedback PLL, we assume a type-I SPD-based architecture for the external PLL and a type-II 3rd order architecture for the main PLL. This is because while the main PLL ends up with a sufficiently narrow optimal bandwidth in a practical design, the external PLL may require a wide optimal bandwidth in order to take advantage of the noise filtering by the main PLL. Accordingly, we take into account the implicit sinc characteristic due to the S/H operation of the SPD. An XOR-PD or SPD can be used for the main PLL if the CP noise is too large, although it tends to be negligible since the CP noise does not get amplified due to the characteristic of the HM-based loop. The conventional integer-N PLL and the DTC-based fractional-N PLL use similar settings to the conventional fractional-N PLL shown in Fig. 3.19(a), except that the conventional integer-N PLL does not contain the quantization noise from the DSM (and has a slightly lower output frequency of 3.1GHz=62×50MHz since the multiplication ratio has to be an integer value), while the DTC-based fractional-N PLL also does not contain the quantization noise from the DSM (since we are assuming perfect noise cancellation by the DTC) but instead contains the thermal noise from the DTC.

The reference PN at 50MHz is −160dBc/Hz. For the VCO, we assume a simple LC-oscillator-based architecture with cross-coupled transistors. The PN spectrum of the VCO $S_{\mathrm{nVCO}}(f)$ can be calculated as

$$S_{\mathrm{nVCO}}(f) = \frac{4\pi kT(1+\gamma)}{P_{\mathrm{VCO}}}\left(\frac{f_{\mathrm{OUT}}}{2Qf}\right)^2 \tag{3.29}$$

where γ is the excess noise coefficient of the MOSFET that is assumed to be 1 in our calculations, P_{VCO} is the power consumption of the VCO, f_{OUT} is the output frequency of the VCO, and Q is the quality factor of the LC tank which we assume to be 10 in our calculations.

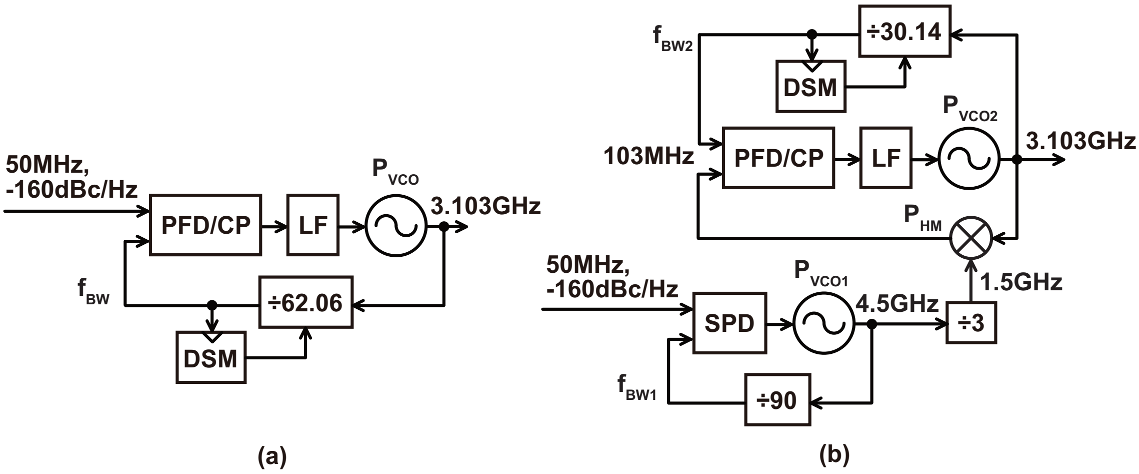


Figure 3.19: A block diagram of the (a) conventional fractional-N PLL and (b) the dual-feedback PLL used for the comparison

The white thermal noise from the PD referred to the reference is inversely proportional to the PD power consumption. Here, we will assume for all architectures that the PD adds white noise at the reference that is inversely proportional to the PD power consumption, and that the level of this white noise is −170dBc/Hz for a 1mW power consumption of the PD (The approximate values for all of the noise levels in this analysis are based on those in the prior arts such as [41, 68, 69]). It is important to note that even though the dual-feedback PLL contains two PDs, the total noise contribution from them is roughly the same as that from a single PD in the other architectures because the noise from the PD of the main PLL does not get amplified and is negligible compared to that from the external PLL. This noise $S_{\mathrm{nPD}}(f)$ can be calculated as

$$S_{\mathrm{nPD}}(f) = \frac{10^{-20}}{P_{\mathrm{PD}}}. \tag{3.30}$$

The white thermal noise generated by the DTC $S_{\mathrm{nDTC}}(f)$ is also inversely proportional to the DTC power consumption P_{DTC}. Assuming that a DTC with 1mW power consumption results in around a −160dBc/Hz PN profile referred to a 50MHz reference, $S_{\mathrm{nDTC}}(f)$ can be calculated as

$$S_{\mathrm{nDTC}}(f) = \frac{10^{-19}}{P_{\mathrm{DTC}}}. \tag{3.31}$$

It is important to note that the DTC output may also contain some residual quantization noise if we accept imperfect accuracy for the gain calibration in exchange for reduced settling time. In this analysis however, we will assume that the DTC calibration has perfect accuracy and will not consider anyresidual quantization noise. A typical target for the calibration accuracy can be around 1% [41].

Finally the HM also contributes white thermal noise, but at the input of the main PLL meaning it does not get amplified. We will assume here that this white noise is −150dBc/Hz when the HM consumes 1mW of power. The equation for the HM PN profile S_{nHM} can be calculated as

$$S_{\mathrm{nHM}}(f) = \frac{10^{-17}}{P_{\mathrm{HM}}}. \tag{3.32}$$

It is important to note that in a practical design, digital blocks such as the DSM can also consume a non-negligible amount of power, especially if it operates at high frequencies. That being said, we can expect the power consumed by digital logic blocks to scale with process since there is no trade-off between its power consumption and any sort of noise. As such, we will ignore the power consumed by digital logic in this analysis although it can be considered to gain a more accurate result depending on the process being used.

3.5.3 Derivation of Optimal Design Settings

The next task we face is the derivation of the optimal bandwidth conditions for a given VCO, DTC, and equivalent reference-referred PN profile. It would be ideal of course, if we could derive the optimal jitter for a given power consumption analytically, and create a plot for the relationship between the jitter and power based on calculation alone. This is may be doable, but we can see that this is not a simple task, especially for the more complex architecture like the dual-feedback PLL since there are multiple variables. Even the analysis in [70] that deals with just the jitter power trade-off for integer-N PLLs makes use of various approximations to obtain a useful and intuitive result with a reasonable amount of calculation. Therefore, here we will take a more hybrid approach where we first obtain the relation between the jitter and power through numerical methods, and then give explanations for the observed trends based on rough calculations with appropriate approximations.

The numerical method we will use is the genetic algorithm. It is often used in optimization problems and starts with a set of candidate solutions that gradually evolve toward better solutions. In our case, we start with a set of of conditions and plot the resulting jitter and power. Then we generate new points based on the genetic algorithm, but we keep the original points as long as there are no other points that achieve both a lower integrated jitter and a lower power consumption. Eventually, we obtain a set of points that make up a curve that represents the relationship between the jitter and power under optimal conditions. This is much better than simply optimizing for the jitter/power FoM and comparing different architectures based on just one design point. Often times in a practical design situation, the jitter or power is given as the design spec that must be met, and the other parameter is optimized within that requirement. If a system requires an integrated jitter of 100fs, a PLL with an integrated jitter of 200fs will not cut it, no matter how good the FoM is. By comparing the curve that represents the relationship between the jitter and power, we can see which architecture is better than the other, and also things such as which architectures are suited for low power designs, which are suited for low jitter designs, etc.

It is important to note that for a simple integer-N PLL, [70] has provided an analytical result based on pure calculation. The optimal bandwidth setting is proportional to the integrated jitter squared, and the VCO power consumption is inversely proportional to the jitter to the fourth power.

3.5.4 Jitter-Power Performance Comparison

Based on the settings and method we just explained, we will obtain the relationship between the optimal jitter and power for the different architectures. The genetic algorithm uses a population size of 600 that went through 300 generations. The bandwidths of the PLLs were varied from a minimum value of 10kHz to a maximum value of 40% of the reference frequency i.e. 20MHz for the single loop architectures since the reference frequency is 50MHz and 40MHz for the main PLL in the dual-feedback PLL since its reference frequency is around 100MHz. The power consumption of

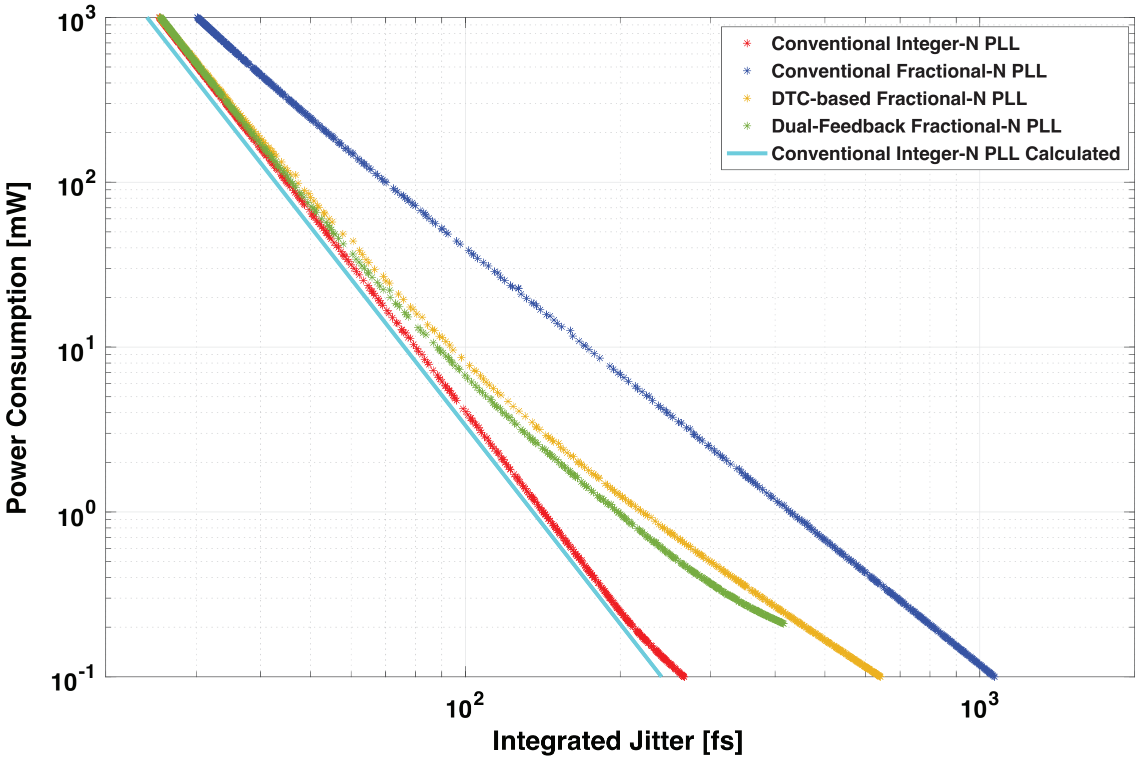


Figure 3.20: The relationship between the jitter and power under optimal design for the conventional integer-N PLL (red), the conventional fractional-N PLL (blue), the DTC-based fractional-N PLL (yellow), the dual-feedback PLL (green), and the calculated relationship between the optimal jitter and power for the conventional integer-N PLL based on [70] (cyan)

the VCOs are varied from 100uW to 1W, that of the DTC from 10uW to 100mW, that of the PD from 10uW to 100mW, and that of the HM from 10uW to 100mW. The power consumption range for the DTC, PD, and HM are set one order smaller than the VCOs since their noise contribution tends to be much smaller and varying the power consumption over a wide range will make the optimization take unnecessarily long. We first consider just the VCO and DTC for the total power consumption to get an idea of the overall trend, then also include the power consumption from the PD and HM to obtain more accurate results.

Fig. 3.20 shows the obtained results. The relationship between the jitter and power for the conventional integer-N PLL is shown in red, the conventional fractional-N PLL in blue, the DTC-based fractional-N PLL in yellow, and the dual-feedback PLL in green. We also have plotted the calculated relationship between the jitter and power based on [70] in cyan. Several interesting observations can be made from this figure.

Firstly, the calculation result for the integer-N PLL based on [70] matches the result of the numerical analysis well, with the power being proportional to $\frac{1}{\sigma^4}$ where σ is the integrated rms jitter. This serves as good evidence that both the calculation and numerical analysis results are valid. Secondly, both the dual-feedback PLL and the DTC-based fractional-N PLL offer clearly better achievable performances than the conventional fractional-N PLL over all jitter and power requirements within the analyzed region. This shows that the added cost of using an extra PLL in the dual-feedback PLL or a DTC in a DTC-based fractional-N PLL is worth the benefit of improved quantization noise suppression in both cases. We can also see that the dual-feedback PLL seems to offer better achievable performance than that of the DTC-based fractional-N PLL although this difference is not as big and may change depending on the settings. Thirdly, both the dual-feedback

PLL and the DTC-based fractional-N PLL obviously perform worse than the conventional integer-N PLL, but this performance gap becomes smaller for high-power, low-jitter designs. For the dual-feedback PLL, this is because as the total power consumption gets large, the power consumption for the main VCO can also be increased, meaning that the optimal bandwidth setting for the main PLL becomes narrow. When this happens, not only can the DSM quantization noise be suppressed to a negligible level, but the noise from the external PLL can also be suppressed to a negligible level because it experiences low-pass filtering from the main PLL. As for the DTC-based fractional-N PLL, the DTC power consumption in the optimal design increases as the total power consumption increases, until the DTC noise becomes smaller than the fixed reference-referred equivalent PN of -160dBc/Hz. Beyond this point, both the power consumption and PN contributed by the DTC become negligible compared to that of the overall PLL, and the performance becomes similar to that of an integer-N PLL. Actually, even for the conventional fractional-N PLL, we can expect the jitter power performance to eventually match that of the integer-N PLL since the optimal bandwidth becomes so narrow that the quantization noise becomes negligible. However, this does not happen within our analysis range which covers up to a power consumption of 1W, because the quantization noise is so large. Finally, the dual-feedback PLL performs slightly better than the DTC-based PLL for medium jitter and power designs, but becomes worse for very low power and high jitter designs. This is because for low to medium jitter, the overhead from the extra VCO in the dual-feedback PLL can be suppressed by employing a narrow main PLL bandwidth, but eventually, the VCO power consumption cannot be made any lower, and the optimal bandwidth for the main PLL becomes wider resulting in less filtering for the external PLL noise.

We can also give some explanations for some of the finer details in the plotted trends. For the integer-N PLL, the power is proportional to $\frac{1}{\sigma^4}$ based on the analysis performed in [70]. This may seem strange at first, because for most of the circuit blocks, its power consumption is proportional to $\frac{1}{\sigma_{\text{cont}}^2}$ where σ_{cont} is the contribution of the block to the overall output jitter. Therefore, the power consumption of the PLL should be proportional to $\frac{1}{\sigma^2}$ (This is actually the basis for the definition of the jitter power FoM, which is based on the quantity equal to the power consumption times the jitter squared). The difference arises from the fact that the equivalent reference-referred PN is kept constant instead of scaling with the power consumption. This is a reasonable assumption because once the reference-referred PN from the circuit blocks are suppressed to a negligible level, the PN from the reference signal itself becomes dominant, which is usually dependent on the device generating the reference signal (a crystal, MEMS device, etc) and cannot simply be lowered by increasing the power consumption.

The relationship for the fractional-N PLL also seems to be close to a somewhat straight line meaning that it can be approximated by the power being inversely proportional to the jitter being raised to some power. It seems that the trend lies somewhere between being proportional to $\frac{1}{\sigma^3}$ and $\frac{1}{\sigma^2}$. For relatively low jitter designs where the bandwidth is sufficiently narrower than the noise-shaping frequency, we can give a very crude explanation by assuming that the quantization noise contributes flat noise extending from the unity gain bandwidth of the PLL f_{UBW} to half of the noise shaping frequency as shown in Fig. 3.21. Here, we assumed that the quantization noise on its own is roughly proportional to $f^{2(l-1)}$ where l is the order of the DSM. Assuming that the DSM is of the 3rd order, the quantization noise is proportional to f^4. Then, this goes through the low-pass characteristic of the PLL. Since the PLL is a type-II third order PLL, its characteristic can roughly be approximated as being flat until f_{UBW}, and then being proportional to $1/f^4$ at higher frequencies. As a result, we obtain a PN contribution that has a flat portion extending from f_{UBW} to half of the reference frequency. Furthermore, the magnitude of this flat portion is proportional

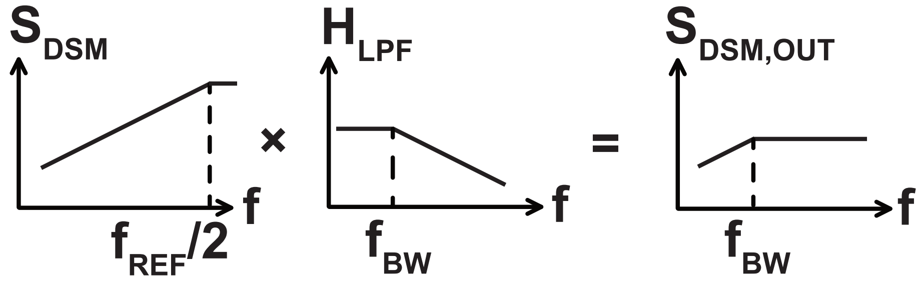


Figure 3.21: A rough approximation of the DSM PN contribution

to f_{UBW}^2 because the quantization noise rises proportionally to f^2 up until it reaches f_{UBW} at which point it becomes flat. Finally, assuming that f_{UBW} is sufficiently smaller than the half of the reference frequency, we can say that the frequency range covered by this flat noise spectrum is relatively independent of f_{UBW}, and therefore the PN contribution from the quantization noise at the overall PLL output is proportional to f_{UBW}^2. On the other hand, the PN contribution from the VCO is inversely proportional to $P_{\mathrm{VCO}} f_{\mathrm{UBW}}$ where P_{VCO} is the VCO power consumption [70]. Therefore, assuming that the PLL output PN is dominated by the VCO PN and the quantization noise i.e. we ignore the reference-referred equivalent white noise, we can say that

$$\sigma^2 \propto \alpha f_{\mathrm{UBW}}^2 + \frac{\beta}{P_{\mathrm{VCO}} f_{\mathrm{UBW}}} \tag{3.33}$$

where α and β are appropriately chosen constants. Based on the AM-GM inequality, we can transform this equation as

$$\sigma^2 \propto \alpha f_{\mathrm{UBW}}^2 + \frac{\beta}{P_{\mathrm{VCO}} f_{\mathrm{UBW}}} = \alpha f_{\mathrm{UBW}}^2 + \frac{\beta}{2P_{\mathrm{VCO}} f_{\mathrm{UBW}}} + \frac{\beta}{2P_{\mathrm{VCO}} f_{\mathrm{UBW}}} \geq 3\sqrt[3]{\frac{\alpha\beta^2}{4P_{\mathrm{VCO}}^2}} \propto P_{\mathrm{VCO}}^{-2/3}. \tag{3.34}$$

Therefore, if we just consider the power consumption from the VCO, the PLL power consumption is roughly proportional to $\frac{1}{\sigma^3}$ in the optimally designed case. This roughly matches the result in Fig. 3.20.

For the DTC-based fractional-N PLL, at low-jitter, high-power design points, the performance is similar to that of the integer-N PLL because the DTC incurs a negligible power and PN penalty. On the other hand, for the high-jitter, low-power design points, the DTC power consumption also becomes low and as a result, the DTC and VCO become the main contributors of PN. Since both the DTC PN and VCO PN are inversely proportional to their respective power consumptions, we can say that in this case, the PN of the entire DTC-based fractional-N PLL is inversely proportional to its power consumption. Therefore, the power consumption becomes proportional to $\frac{1}{\sigma^2}$ for high-jitter, low-power design points, and the curve shown in Fig. 3.20 is a combination of a characteristic proportional to $\frac{1}{\sigma^4}$ and $\frac{1}{\sigma^2}$.

For the dual-feedback PLL, at low jitter, high power design points, the performance is similar to that of the integer-N PLL just like the DTC-based fractional-N PLL. The deviation at high-jitter, low-power design points is the result of a combination of a few different things. If the bandwidth of the external PLL can be made indefinitely large, the noise overhead from the external VCO can always be made negligibly small by making the external PLL bandwidth much wider than that of the main PLL. In this case, trend for the dual-feedback PLL will follow the trend for the integer-N PLL until the main PLL bandwidth becomes sufficiently wide for the quantization noise to become one of the dominant sources of noise. From there, the relationship for the dual-feedback PLL should follow a similar trend as that for the conventional fractional-N PLL, except of course, that it will be shifted down because the quantization noise is not amplified. In reality however, the bandwidth of the external PLL is limited by the factors such as the reference frequency and cannot be increased beyond a certain point. As such, for very high-jitter and low-power designs, the external VCO will also contribute a significant amount of PN because it does not experience any filtering from the main PLL.

So far, we have only considered the power consumption from the VCO and the DTC. However, there is also some power consumed by the reference path (blocks such as the PD) and by the HM. To obtain a more accurate comparison, we will now include the noise contributed by the HM and PD. Fig. 3.22 shows the obtained results. The relationship for the conventional integer-N PLL is shown in red, the conventional fractional-N PLL in blue, the DTC-based PLL in yellow, and the dual-feedback PLL in green. We also made a contour plot for the jitter-power FoM in black and grey, and also a line demonstrating the jitter requirement for wireline (pink) and wireless (light green) applications. Specifically, the pink line specifies a jitter of 150fs which is required for the clock of a 28GHz ADC for it to achieve an Effective Number of Bits (ENOB) of 8. This kind of ADC is often used in 56Gb/s or 112Gb/s wireline receivers [71, 72, 73, 74]. The light green line specifies a jitter of 200fs which is required for the LO of a 256 QAM transceiver to achieve a 3.5% worst case EVM over the sub-7GHz band, a requirement specified in the 3GPP TS 5G standard [75]. At low-jitter, high-power design points, all of the trends are very similar with the one in Fig. 3.20 because the power overhead from the reference path and the HM are negligible. At high-jitter and low-power design points however, the power consumption saturates somewhat due to the minimum power required by the different blocks, as well as the fact that the noise from the PD becomes non-negligible compared to the fixed reference noise. Nonetheless, the performance of the DTC-based PLL and dual-feedback PLL are still significantly better than that of the conventional fractional-N PLL over a significantly wide range. Importantly, the range where a large advantage in performance is observed covers the important applications that we show i.e. the DTC-based PLL and dual-feedback PLL achieve significantly better performance than the conventional fractional-N PLL when the jitter is designed to be around 100 to 200fs. Also as a reference, the best achievable FoM for the dual-feedback PLL is roughly equal to that of the DTC-based fractional-N PLL, much better than that of the conventional fractional-N PLL, and worse than that of the conventional integer-N PLL.

Based on these results, we have demonstrated analytically with a hybrid approach using numerical methods that the dual-feedback architecture can achieve a significantly better performance than the conventional fractional-N PLL and a similar performance to a DTC-based fractional-N PLL depending on the settings and the target jitter/power performance. This, in addition to the fact that the dual-feedback architecture does not require calibration like the DTC-based architecture and therefore has a shorter settling time and simpler design means that the dual-feedback architecture and HM-based architectures in general can be very useful in various high performance applications.

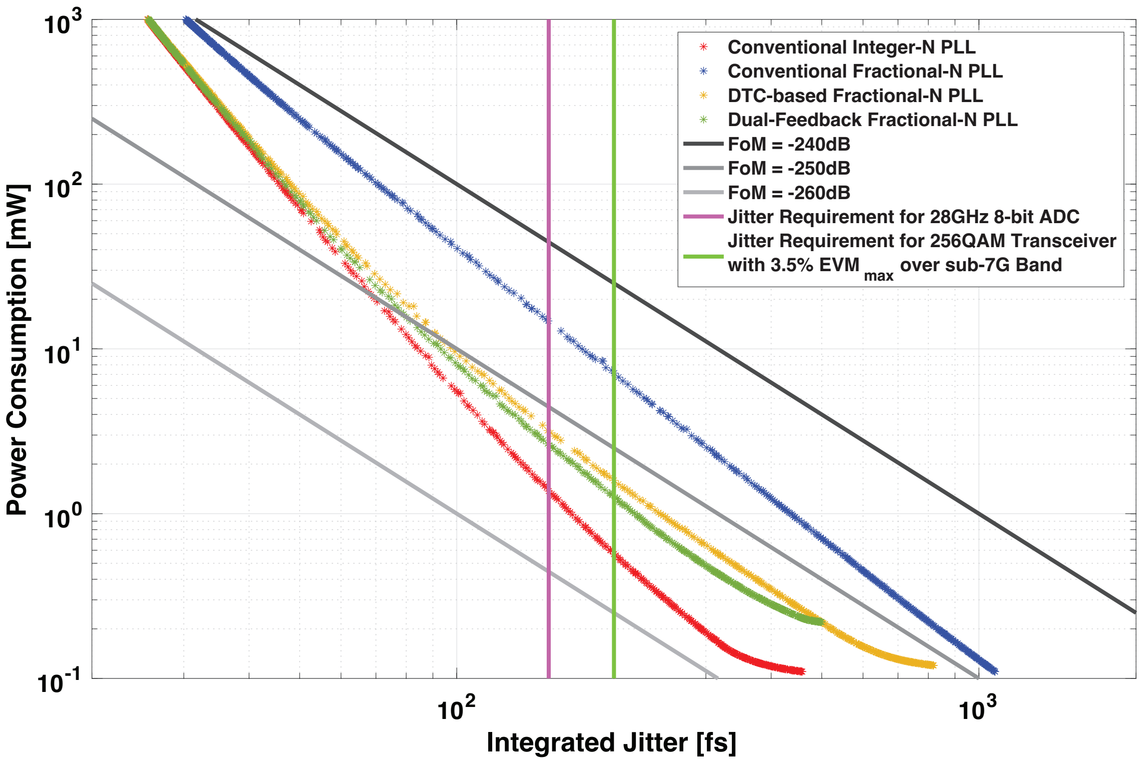


Figure 3.22: The relationship between the jitter and power under optimal design for the conventional integer-N PLL (red), the conventional fractional-N PLL (blue), the DTC-based fractional-N PLL (yellow), and the dual-feedback PLL (purple), a contour plot for when the FoM is −240dB (black), −250dB (grey), and −260dB (light grey), and a line indicating the jitter requirement for a wireline (pink) and wireless (light green) application

3.6 Chapter Summary

In this chapter, we introduced the concept of HM-based feedback which allows us to avoid the amplification of certain noise sources and then looked at the triple-loop architecture which is a prior work that makes use of this concept to suppress the quantization noise and fractional spurs. We then proposed the dual-feedback architecture which improves on the triple-loop architecture by realizing the same benefit of avoiding spur and quantization noise amplification with one less auxiliary PLL. We demonstrated the effectiveness of our proposed architecture through simulation and measurement results, and also made a comparison with other architectures based on a hybrid numerical analysis.

Chapter 4

Analysis of Offset Spurs in Harmonic-Mixer-Based PLLs

As a consequence of employing the S/H operation, HM-based PLLs suffer from the issue of offset spurs. These offset spurs can be suppressed with properly designed filters placed before and after the S/H, but prior works do not provide a detailed explanation for the mechanisms of offset spur generation or a design guideline for the filters to suppress these spurs. In this chapter, we provide a detailed explanation of the generation mechanisms of these offset spurs, and analytically derive their amplitude and location. We also provide a guideline on how the low-pass filters for the HM should be designed based on the results of this analysis. This chapter is based off of our work published in [76].

4.1 Issue of Offset Spurs in Harmonic-Mixer-Based PLLs

Before delving into the details, we will give a rough overview of the issue we are dealing with. Fig. 4.1 shows the concept of offset spurs in a HM-based PLL. As we can see in the upper half of Fig. 4.1, using just a simple S/H circuit as the HM can result in two kinds of unwanted tones at the HM output. We have some unwanted tones at high frequencies which we refer to in our dissertation as the type-A spurs, and we also have unwanted tones near the carrier itself which we refer to as the type-B spurs. The type-B spurs are obviously problematic since they are close to the desired signal and cannot be filtered out easily, so they will appear as spurious tones at the PLL output. The type-A spurs are also problematic because even though they exist at a high frequency, they can get down-converted to lower frequencies during phase detection and also end up degrading the output spurious performance. To suppress the type-A and type-B spurs, a Low-Pass-Filter (LPF) must be placed after and before the S/H respectively. In the following analysis, we will explain why these tones appear in the first place, how large these tones are, what frequency they occur at, and how the LPFs can help to suppress them.

4.2 Background Theory

In order to understand the offset spurs smoothly, it is useful to first look at some relevant background theory. In this section, we will first explain how the PFD/CP and other PDs deviate from the ideal characteristic in the Linear-Time-Invariant (LTI) model due to its discrete time operation at just the rising edges of the input signals [26]. This is relevant in understanding how the high frequency tones at the HM output get down-converted to lower frequencies. Then we will explain

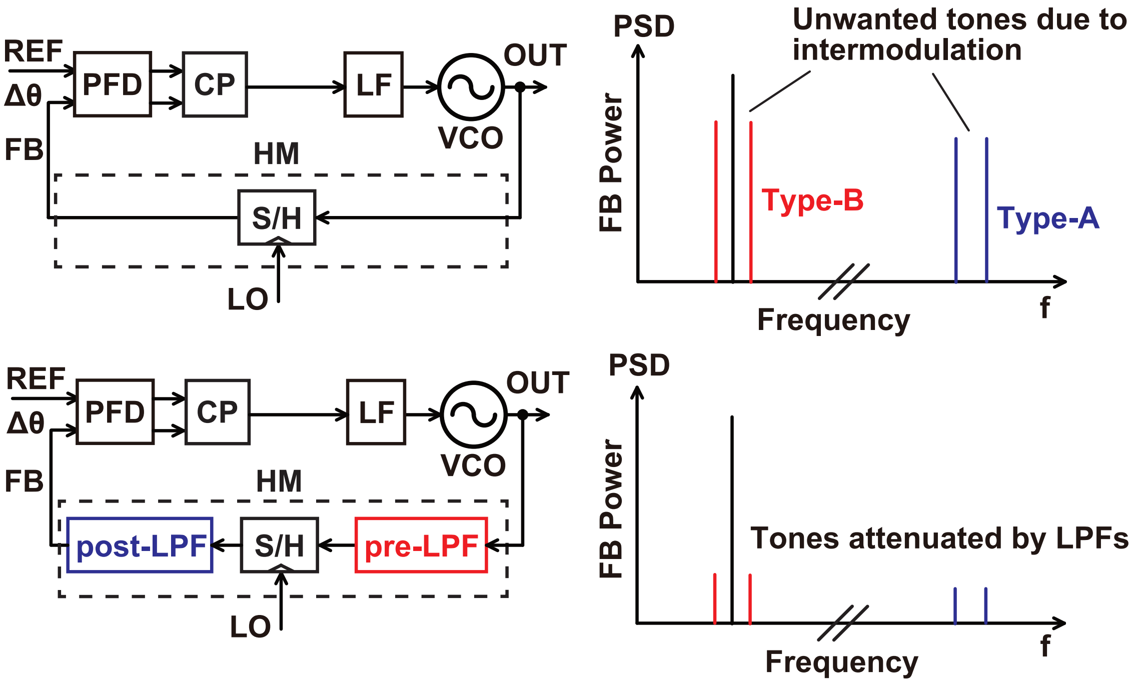


Figure 4.1: An overview of the issue of offset spurs in HM-based PLLs, and how they can be mitigated using properly designed LPFs

the relationship between the spectrum of a signal and the spectrum of its PN [77, 25]. This is relevant because we treat the PD as a phase-sampling block and the linear model of a PLL uses the phase as the variable of interest, while the output of the HM is treated in the voltage-domain. As such, we need to clarify how to convert the voltage-domain signal into the phase-domain.

4.2.1 Phase-Sampling Action in the PFD/CP

Often times, the PFD/CP and many other PDs in a PLL are treated as ideal blocks that simply take the phase difference between the two input signals and then multiply it by some fixed gain. This model is fine for predicting the basic behavior of PLLs in many practical cases. However, as we can see in Fig. 4.2(a), the PFD/CP does not continuously take the phase difference between the two inputs and produce a proportional output, but only does so at the rising edges of the two input signals. In other words, the PFD/CP operates in discrete time and as such, some sampling operation comes into play. In fact, considering the fact that the output current pulse of the CP is much narrower than one reference period, we can look at the PFD/CP as a block that takes the ”true” input phase difference which is a continuous time quantity, and multiplies it with an impulse train with a frequency equal to that of the reference. Fig. 4.2(b) shows the implications of this effect in the frequency domain. We can see that the spectrum of the input phase difference gets convoluted with an impulse train located at increments of the sampling frequency, resulting in copies of the input phase difference spectrum appearing at the sampling frequency and its harmonics. The signal then enters the LF and reaches the VCO. When the input phase difference does not contain any high frequency components and the loop bandwidth is sufficiently narrow, this does not pose much of an issue because there are no out-of-band components getting aliased in-band, and the copies of the input phase difference spectrum will get filtered out by the loop. However, this effect will be important in this analysis as we will see later since the HM output contains phase components near the HM LO frequency, which is much higher than the PFD/CP

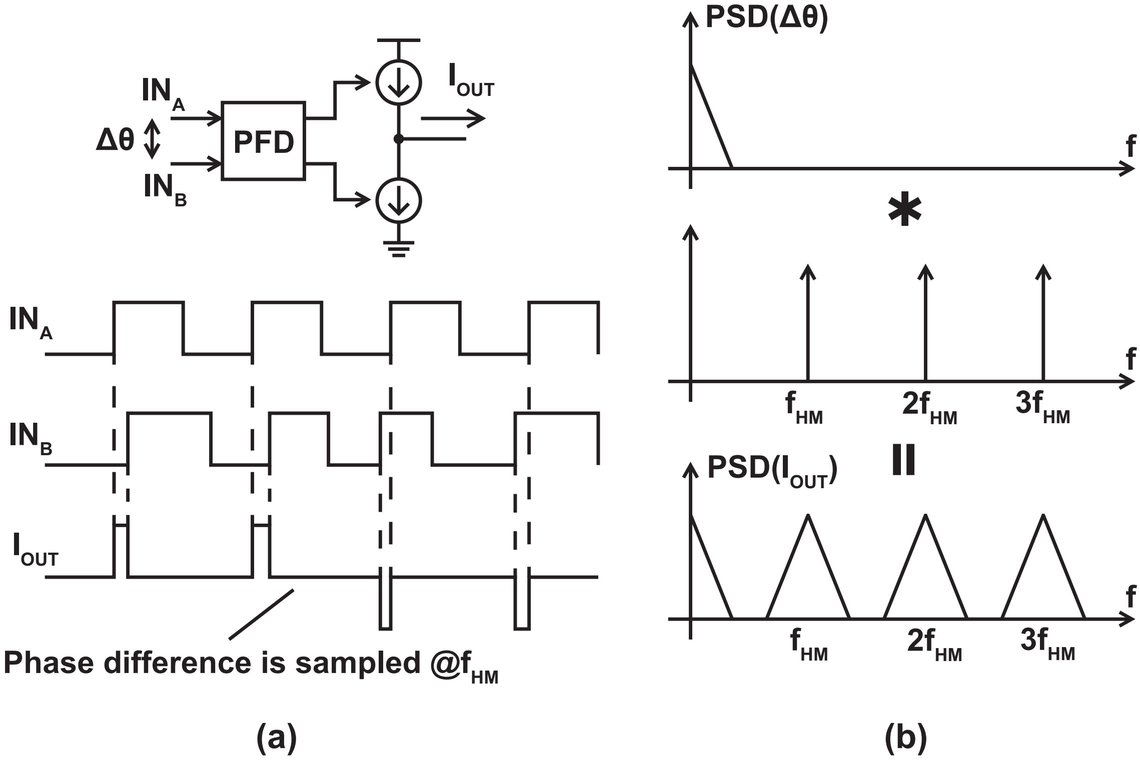


Figure 4.2: The effect of the discrete time operation of the PFD/CP

frequency which is equal to the HM output frequency. It should be noted that a similar thing can also be said for many other PD architectures such as XOR PDs and SPDs since they also operate in discrete time.

4.2.2 Relationship between Spectrum of a Signal and its Phase

Another important thing to understand is relationship between the spectrum of a signal and the spectrum of its excessive phase. We can obviously look at the frequency component of almost any signal by performing a Fourier Transform on it. However, for signals such as the outputs of PLLs that are sufficiently close to a perfectly periodic signal, we can deal with them by looking at just their deviation from an ideal sine-wave in the phase-domain. This quantity is called the excessive phase of a periodic signal, its Power-Spectral-Density (PSD) is often called the PN, and it is the quantity we usually use to treat the input and output signals of a PLL during analysis. The goal here is to derive the relationship between the original signal and the excessive phase, because the operation of the HM is analyzed in the voltage domain while the overall PLL treats the HM output using its excessive phase, and we need a way to convert between these two.

We will first assume that the voltage domain signal $V(t)$ can be expressed as follows

$$V(t) = A\sin(\omega_c t + \phi_n(t)). \tag{4.1}$$

We can see that $V(t)$ is a sine wave with an amplitude of A, an angular frequency of ω_c, and an excessive phase of $\phi_n(t)$. Assuming that $|\phi_n(t)| \ll 1$, this equation can be transformed as shown below

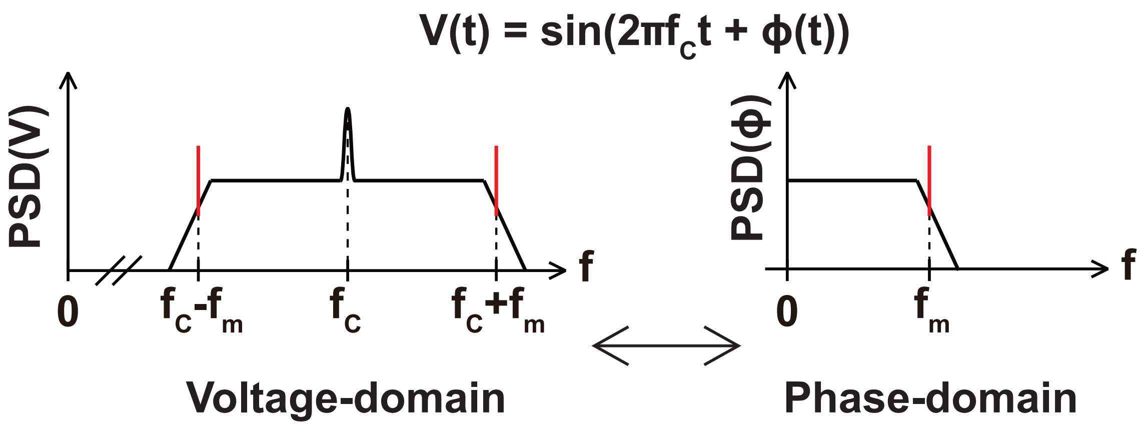


Figure 4.3: The relationship between the spectrum of a signal and that of its excessive phase

$$V(t) = A[\sin(\omega_{\mathrm{c}}t)\cos(\phi_{\mathrm{n}}(t)) + \cos(\omega_{\mathrm{c}}t)\sin(\phi_{\mathrm{n}}(t))] \approx A[\sin(\omega_{\mathrm{c}}t) + \phi_{\mathrm{n}}(t)\cos(\omega_{\mathrm{c}}t)]. \tag{4.2}$$

Finally, if we look at one specific frequency component in $\phi_{\mathrm{n}}(t)$ by expressing $\phi_{\mathrm{n}}(t) = A_{\mathrm{n}}\sin(\omega_{\mathrm{n}}t)$, $V(t)$ can be expressed as

$$V(t) \approx A[\sin(\omega_{\mathrm{c}}t) + A_{\mathrm{n}}\sin(\omega_{\mathrm{n}}t)\cos(\omega_{\mathrm{c}}t)] = A\sin(\omega_{\mathrm{c}}t) + \frac{AA_{\mathrm{n}}}{2}[\sin((\omega_{\mathrm{c}}+\omega_{\mathrm{n}})t) - \sin((\omega_{\mathrm{c}}-\omega_{\mathrm{n}})t)]. \tag{4.3}$$

In addition to the carrier at ω_{c}, we can see that $V(t)$ contains components at $\omega_{\mathrm{c}} \pm \omega_{\mathrm{n}}$ which resulted from the component at ω_{n} contained in the excessive phase. By generalizing this, we can see that the spectrum of a signal can be obtained by taking the PN spectrum and copying it to both sides of the carrier as is shown in Fig. 4.3.

4.3 Generation of Unwanted Tones

Based on our understanding of the S/H operation and also the relevant background theory such as the sampling action of the PFD/CP and the relationship between a voltage domain signal and its excessive phase, we can now explain in detail how the offset spurs are generated in a HM-based PLL. In this section, we will explain the generation mechanism of two types of offset spurs, which we refer to as type-A and type-B spurs.

4.3.1 High-Frequency Tones (Type-A Spurs)

In chapter 3, we looked at the principle of harmonic mixing realized by the S/H operation. Fig. 3.6(b) showed the effect of the hold operation in the frequency domain, and we saw that since the hold operation results in a sinc characteristic in the frequency domain, the high frequency components resulting from the sampling operation get attenuated and if we ignore these components, we obtain just the desired component at $f_{\mathrm{HM}} = f_{\mathrm{PLL}} - kf_{\mathrm{SH}}$. However, these high frequency tones are actually not filtered out completely because they exist at frequencies of $\pm f_{\mathrm{SH}} \pm f_{\mathrm{HM}}$, $\pm 2f_{\mathrm{SH}} \pm f_{\mathrm{HM}}$, $\pm 3f_{\mathrm{SH}} \pm f_{\mathrm{HM}}$ etc while the notch of the sinc characteristic exists at $\pm f_{\mathrm{SH}}$, $\pm 2f_{\mathrm{SH}}$, $\pm 3f_{\mathrm{SH}}$ etc. Basically, the tones are are located at offsets of $\pm f_{\mathrm{HM}}$ from the notches of the sinc,

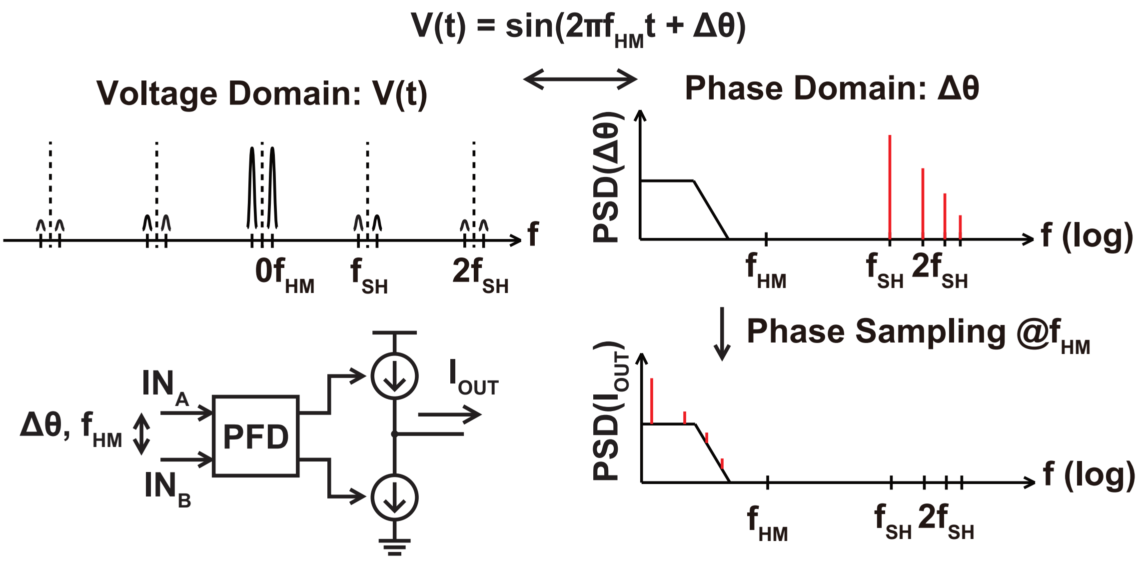


Figure 4.4: The mechanism of the generation of type-A spurs

and thus are not suppressed completely. As shown in the top half of Fig. 4.4, the high frequency tones in the spectrum of the HM output can be expressed as spurs at integer multiples of f_{SH} when converted to the phase-domain. Then, due to the sampling action of the PFD/CP, these tones get aliased to lower frequencies. If f_{SH} happens to be close to an integer multiple of f_{HM}, the spurs can appear at very low frequencies, where they cannot be suppressed by the low-pass characteristic of the PLL. As such, these spurs, which we will call type-A spurs, have to be suppressed before phase detection by using a LPF right after the S/H.

4.3.2 Low-Frequency Tones (Type-B Spurs)

We looked at the generation mechanism of type-A spurs, which come from high frequency tones at the HM output that then get down-converted during phase detection. Because the tones exist at high frequencies at the HM output, they can be suppressed by using an LPF at the HM output. However we can also see tones that are already close to the desired carrier signal at the HM output, and these tones cannot be suppressed with an LPF after the S/H. These are called the type-B spurs, and we will explain their generation mechanism here.

To understand the generation of type-B spurs, we have to look at the high order harmonics of the S/H input. Fig. 4.5 shows the sampling operation of the S/H in the frequency domain, but unlike in Fig. 3.5, we have also considered the high order harmonics of the S/H input i.e. the PLL output. The harmonic order k is 3 in this example. The resulting HM output contains various components. We obviously have the desired signal component at $f_{\mathrm{HM}} = f_{\mathrm{PLL}} - 3f_{\mathrm{SH}}$. We also have a component at $\pm(f_{\mathrm{SH}} - f_{\mathrm{HM}})$ (and $\pm(f_{\mathrm{SH}} + f_{\mathrm{HM}})$, $\pm(2f_{\mathrm{SH}} \pm f_{\mathrm{HM}})$, etc though they are not shown) which is the high frequency tone that leads to the type-A spur as we have already explained, and can be suppressed by the LPF right after the S/H. We also have components at integer multiples of f_{HM}, which are not problematic because they simply alter the shape of the waveform and will get down-converted to DC just like the fundamental signal. The issue here, comes from the other interactions between the harmonics of f_{PLL} and f_{SH}. When we look at the interaction between f_{PLL} and the harmonics of f_{SH}, we get $f_{\mathrm{PLL}} - 3f_{\mathrm{SH}}$ which is the desired signal and $4f_{\mathrm{SH}} - f_{\mathrm{PLL}} = f_{\mathrm{SH}} - f_{\mathrm{HM}}$ which is at a high frequency and can be filtered out. For $2f_{\mathrm{PLL}}$, we get $2f_{\mathrm{PLL}} - 6f_{\mathrm{SH}} = 2f_{\mathrm{HM}}$ which is a second order harmonic of the desired signal and also a

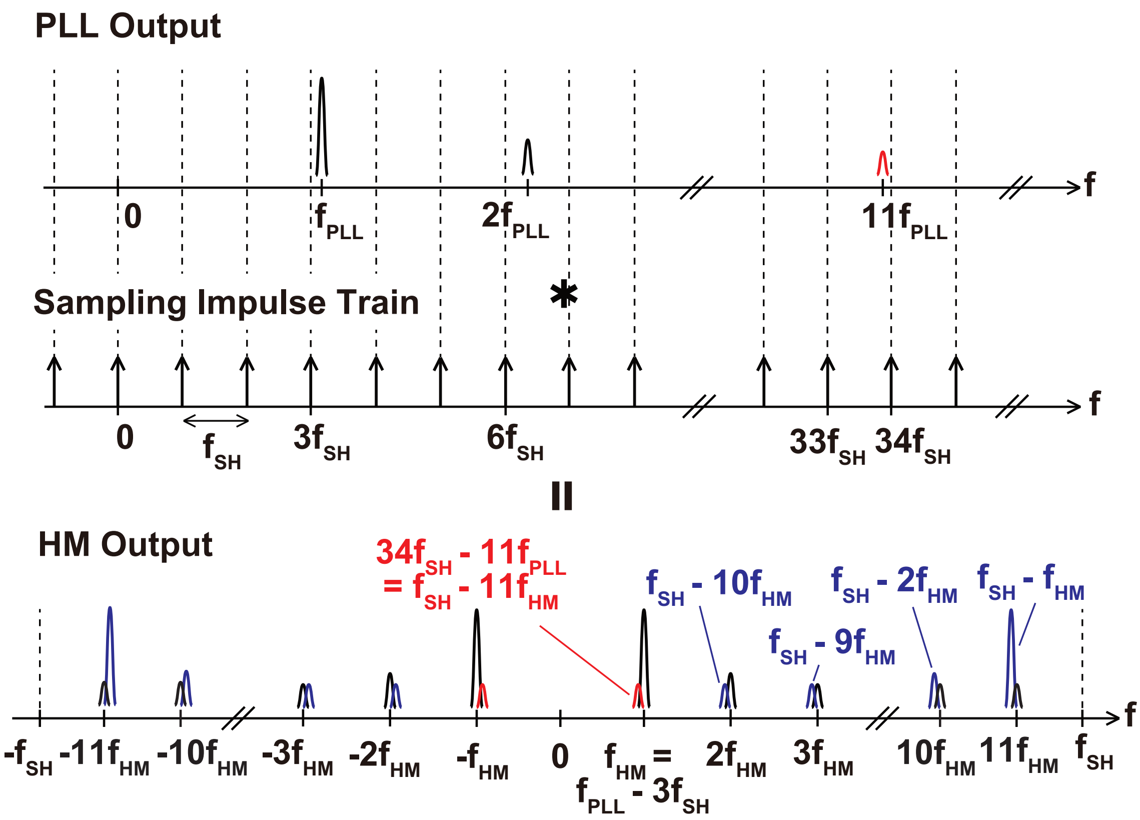


Figure 4.5: The frequency domain operation of a HM with high order harmonics

tone at $7f_{\mathrm{SH}} - 2f_{\mathrm{PLL}} = f_{\mathrm{SH}} - 2f_{\mathrm{HM}}$. Assuming that $f_{\mathrm{SH}} \gg f_{\mathrm{HM}}$, the latter tone is still at a high frequency and can be suppressed by the LPF after the S/H. However, it is f_{HM} smaller than the tone at $f_{\mathrm{SH}} - f_{\mathrm{HM}}$, and as we look at higher and higher harmonics, the resulting tones appear at lower and lower frequencies, and eventually it becomes so low that they become close to the desired signal and cannot be suppressed by an LPF. In the example in Fig. 4.5, f_{SH} happens to be close to $12f_{\mathrm{HM}}$. As a result, the tone at $34f_{\mathrm{SH}} - 11f_{\mathrm{PLL}} = f_{\mathrm{SH}} - 11f_{\mathrm{HM}}$, which results from the interaction between the 11th harmonic of f_{PLL} and the 34th harmonic of f_{SH} ends up being very close to the desired signal at f_{HM} and cannot be suppressed by the LPF after the S/H. This is the generation mechanism of the type-B spur. To suppress this, an LPF has to be placed before the S/H operation so that the high order harmonic of the S/H input is sufficiently suppressed before getting down-converted close to the desired signal.

4.4 Spur Location and Analysis of Spur Amplitude

In the previous section, we gave a qualitative explanation of the generation mechanism of the type-A and type-B spurs, and why LPFs placed before and after the S/H can be used to suppress them respectively. Here, we will delve into some more detail and derive expressions for the location and amplitude of these spurs.

4.4.1 Spur Location

The locations of the type-A and type-B spurs can be derived based on the spur generation mechanism that we explained in the previous section. For the type-A spur, we first can see that the spectrum of the HM output in the voltage domain contains tones at $\pm(f_{\mathrm{SH}} \pm f_{\mathrm{HM}})$ due to the

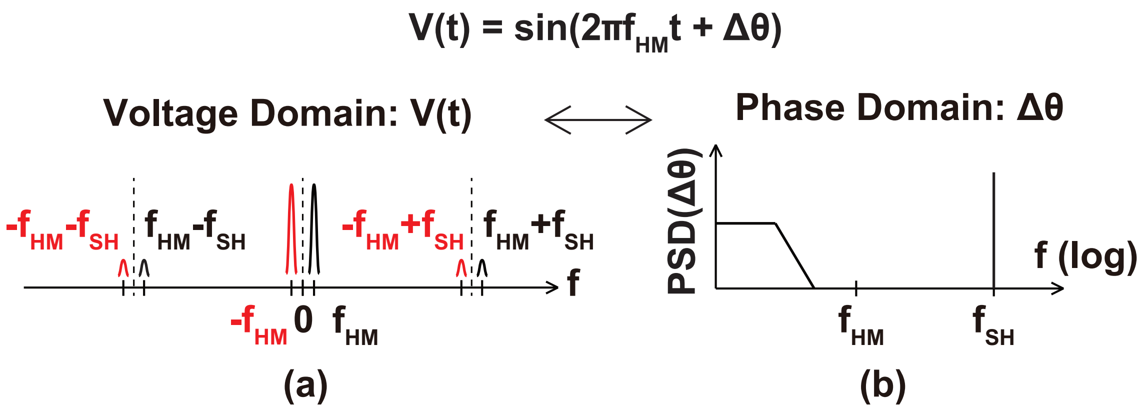


Figure 4.6: The conversion of the HM output from the voltage domain spectrum to the phase domain spectrum

interaction between f_{PLL} and the $k \pm 1$th harmonics of f_{SH}. As shown in Fig. 4.6, these tones in the voltage domain can be converted to a spur at f_{SH} in the phase-domain, because the tones $\pm f_{\mathrm{HM}} \pm f_{\mathrm{SH}}$ are at an equal frequency offset from the carrier frequency component at $\pm f_{\mathrm{HM}}$, and with roughly equal amplitudes. Similarly, we also have tones at $\pm(2f_{\mathrm{SH}} \pm f_{\mathrm{HM}})$ in the HM output spectrum which get converted to a spur at $2f_{\mathrm{SH}}$ in the phase-domain, though this spur is much smaller than the fundamental one due to stronger attenuation from the sinc characteristic. In fact, the spectrum of the HM output phase contains spurs at all integer multiples of f_{SH} that get smaller as the harmonic increases. During phase detection, this output from the HM experiences sampling at a frequency of f_{HM} when it is converted to an output current from the CP. Focusing on just the fundamental spur for now, we can see that the tone at f_{SH} gets down-converted to a frequency of Δf which can be calculated as

$$\Delta f = |f_{\mathrm{SH}} - l f_{\mathrm{HM}}| \tag{4.4}$$

where l is an integer that satisfies

$$\left(l - \frac{1}{2}\right) f_{\mathrm{HM}} < f_{\mathrm{SH}} < \left(l + \frac{1}{2}\right) f_{\mathrm{HM}} \tag{4.5}$$

because the sampling with f_{HM} creates copies of f_{SH} shifted by integer multiples of f_{HM}, and only the lowest frequency component remains since the higher frequency ones get filtered out by the PLL. Similarly, the phase components at $2f_{\mathrm{SH}}$, $3f_{\mathrm{SH}}$ etc in the HM output get down-converted to $2\Delta f$, $3\Delta f$ etc after phase detection, though they will be smaller than the fundamental spur at f_{SH}. Finally, these spurs simply experience the transfer function from the PLL and appear at the overall output. If f_{SH} happens to be close to an integer multiple of f_{HM}, the spur frequency Δf will be very small and it will not experience filtering from the PLL.

For the type-B spur, we can look at the example in Fig. 4.5. Here, the convolution between the harmonics of f_{PLL} and that of f_{SH} result in tones at f_{SH} minus integer multiples of f_{HM}. Since f_{SH} is close to $12f_{\mathrm{HM}}$ in this example, the tone at $f_{\mathrm{SH}} - 11f_{\mathrm{HM}}$ ends up being close to the desired signal at f_{HM}. Since this tone appears at the output without any further frequency translation, the type-B spur ends up appearing at a frequency offset of $|f_{\mathrm{SH}} - 12f_{\mathrm{HM}}|$. We can see that this is actually the same frequency as Δf, which is the location of the type-A spur. This of course, is assuming that all of the high order harmonics exist. If the HM input is a square wave for example, it only contains odd order harmonics and so the treatment of the spurs will be different depending on whether f_{SH} is close to an odd multiple or even multiple of f_{HM}. Nevertheless, the spur will still appear at Δf, and its amplitude can be calculated by paying attention to the HM operation principle. We will talk more about this when deriving analytical expressions for the amplitude of the type-B spurs.

In summary, both the type-A and type-B spurs appear at the difference between f_{SH} and the integer multiple of f_{HM} that is closest to it. When f_{SH} is very close to an integer multiple of f_{HM}, the resulting spur will appear at a low frequency and cannot be low-pass filtered by the PLL characteristic.

4.4.2 Amplitude of Type-A Spurs

Here, we will derive an expression for the amplitude of the type-A spurs. We will focus on just the fundamental spur at Δf since it is the largest. After the sampling operation, the desired signal at f_{HM} and the tones at $\pm(f_{\mathrm{SH}} \pm f_{\mathrm{HM}})$ have the same amplitude. They then experience a different amount of attenuation from the sinc characteristic. To simplify the notation, we first define the implicit sinc function as

$$F_{\mathrm{SINC}}(f, f_{\mathrm{SH}}) \equiv \frac{\sin(\pi f / f_{\mathrm{SH}})}{\pi f / f_{\mathrm{SH}}}. \tag{4.6}$$

Using this notation, the attenuation experienced by the desired signal is $F^2_{\mathrm{SINC}}(f_{\mathrm{HM}}, f_{\mathrm{SH}})$, the attenuation experienced by the signals at $\pm(f_{\mathrm{SH}} - f_{\mathrm{HM}})$ are $F^2_{\mathrm{SINC}}(f_{\mathrm{SH}} - f_{\mathrm{HM}}, f_{\mathrm{SH}})$, and the attenuation experienced by the signals at $\pm(f_{\mathrm{SH}} + f_{\mathrm{HM}})$ are $F^2_{\mathrm{SINC}}(f_{\mathrm{SH}} + f_{\mathrm{HM}}, f_{\mathrm{SH}})$. To convert the tones at $\pm(f_{\mathrm{SH}} \pm f_{\mathrm{HM}})$ into a single tone in the phase domain, the attenuation at $\pm(f_{\mathrm{SH}} - f_{\mathrm{HM}})$ and $\pm(f_{\mathrm{SH}} + f_{\mathrm{HM}})$ must be the same. From the closeup view of the notch of the sinc characteristic in Fig. 4.7, we can see that although they are not exactly the same, the amount of attenuation is roughly the same because the tones are at the same distance from the notch. As such, the amplitude of the spur at f_{SH} in the phase-domain can be approximated by the average of $F^2_{\mathrm{SINC}}(f_{\mathrm{SH}} - f_{\mathrm{HM}}, f_{\mathrm{SH}})$ and $F^2_{\mathrm{SINC}}(f_{\mathrm{SH}} + f_{\mathrm{HM}}, f_{\mathrm{SH}})$, relative to the attenuation for the carrier $F^2_{\mathrm{SINC}}(f_{\mathrm{HM}}, f_{\mathrm{SH}})$. Finally, this tone gets down-converted to Δf during phase detection and then experiences the low-pass characteristic of the PLL. Therefore, the amplitude U_{A}[dBc] of the type-A spur can be calculated as

$$U_{\mathrm{A}}[\mathrm{dBc}] = 10\log\left[\frac{|H_{\mathrm{PLL,cl}}(\Delta f)|^2 (F^2_{\mathrm{SINC}}(f_{\mathrm{HM}} - f_{\mathrm{SH}}, f_{\mathrm{SH}}) + F^2_{\mathrm{SINC}}(f_{\mathrm{HM}} + f_{\mathrm{SH}}, f_{\mathrm{SH}}))}{2F^2_{\mathrm{SINC}}(f_{\mathrm{HM}}, f_{\mathrm{SH}})}\right] \tag{4.7}$$

where $|H_{\mathrm{PLL,cl}}(\Delta f)|$ is the absolute value of the closed loop transfer function of the HM-based PLL at a frequency of Δf. We can also see from Fig. 4.7 that the tone remaining after the attenuation from the sinc characteristic gets larger as f_{HM} increases since the tone gets further away from the notch, and also gets larger as f_{SH} decreases since the slope of the sinc near the notch gets sharper. On the other hand, when f_{SH} decreases and f_{HM} increases, the desired signal gets smaller because the bandwidth of the sinc gets narrower while the signal appears at a higher frequency, increasing the amount of attenuation. Finally, the closed loop transfer function of the PLL $|H_{\mathrm{PLL,cl}}(\Delta f)|$ becomes largest when Δf is sufficiently small i.e. smaller than the bandwidth of the loop. Therefore, the type-A spur becomes largest when f_{SH} is smallest, and f_{HM} is largest,

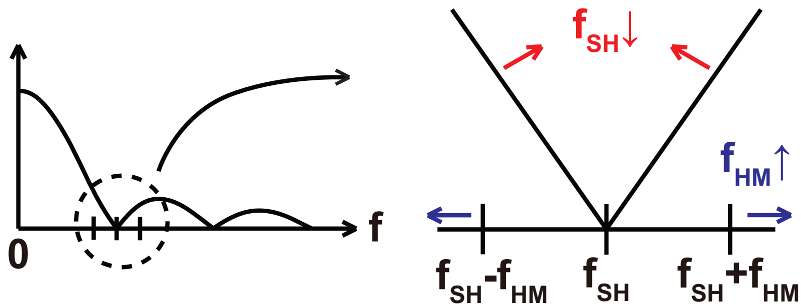


Figure 4.7: A closeup view of the notch in the sinc characteristic, and the dependence on f_{SH} and f_{HM}

under the condition that f_{SH} is close to an integer multiple of f_{HM} and Δf is small. The worst case type-A spur $U_{\mathrm{A,max}}$ can be calculated as shown below

$$U_{\mathrm{A,max}}[\mathrm{dBc}] = 10\log\left[\frac{F^2_{\mathrm{SINC}}(f_{\mathrm{HM,max}} - f_{\mathrm{SH,min}}, f_{\mathrm{SH,min}}) + F^2_{\mathrm{SINC}}(f_{\mathrm{HM,max}} + f_{\mathrm{SH,min}}, f_{\mathrm{SH,min}})}{2F^2_{\mathrm{SINC}}(f_{\mathrm{HM,max}}, f_{\mathrm{SH,min}})}\right] \tag{4.8}$$

where $f_{\mathrm{SH,min}}$ and $f_{\mathrm{HM,max}}$ are the minimum value of f_{SH} and the maximum value of f_{HM} respectively.

4.4.3 Amplitude of Type-B Spurs

For the type-B spurs, the calculation is simple since the sinc characteristic does not have much of an effect, but the situation is slightly complex since the spur amplitude is greatly dependent on the harmonic content of the S/H input and also somewhat on the characteristic of the LPF after the S/H. As such, instead of coming up with a general expression for the amplitude of the type-B spurs regardless of the situation, we will show how the type-B spur amplitude can be calculated based on the operation principle of the HM by looking at an example case where the input to the S/H is a square wave.

A square wave only has odd order harmonics. Therefore, the generation of the type-B spur is slightly different for when f_{SH} is close to an even multiple of f_{HM} and when it is close to an odd multiple. Fig. 4.8 shows what happens for the two cases. In Fig. 4.8(a), f_{SH} is close to $12f_{\mathrm{HM}}$. As we have explained, this means that $f_{\mathrm{SH}} - 11f_{\mathrm{HM}} \approx f_{\mathrm{HM}}$ and so the 11th harmonic of the S/H input gets down-converted near the desired signal. In addition, $f_{\mathrm{SH}} - 13f_{\mathrm{HM}} \approx -f_{\mathrm{HM}}$ and so the 13th harmonic gets down-converted close to the desired signal as well. In fact, we can see that the 11th and 13th harmonics appear at either side of the desired signal. Since the desired signal and tones are all close to each other, the attenuation from the post S/H LPF is the same for all of them and does not effect the amplitude of the resulting type-B spurs. To convert these tones into a single spur in the phase domain, they must have the same amplitude. For a square wave, the harmonics next to each other have roughly the same amplitude, so we can approximate them based on their average. By using the fourier series expansion, a square wave can be expressed in terms of a sine-wave and its odd order harmonics. The amplitude of the $2m+1$th harmonic is $\frac{1}{2m+1}$. Therefore, when $f_{\mathrm{SH}} \approx 2mf_{\mathrm{HM}}$, i.e. f_{SH} is close to an even multiple of f_{HM}, the amplitude of the type-B spur $U_{\mathrm{B}}[\mathrm{dBc}]$ is given by

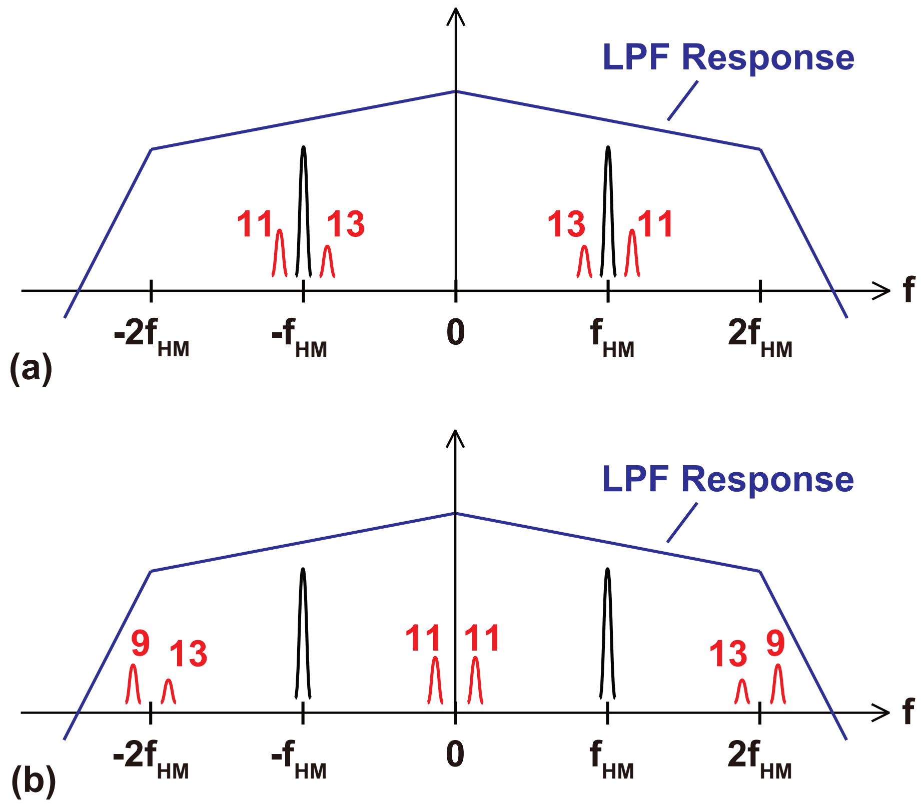


Figure 4.8: The conceptual spectrum of the HM output when (a) f_{SH} is close to an even (12th) multiple of f_{HM}, and (b) when it is close to an odd (11th) multiple of f_{HM}

$$U_{\mathrm{B}}[\mathrm{dBc}] = 10\log\left(\frac{(2m-1)^{-2} + (2m+1)^{-2}}{2}\right). \tag{4.9}$$

On the other hand, when f_{SH} is close to an odd multiple of f_{HM}, the conceptual spectrum of the HM output looks like Fig. 4.8(b) where $f_{\mathrm{SH}} \approx 11 f_{\mathrm{HM}}$. The 11th harmonic gets down-converted near DC, while the 9th and 13th harmonics get down-converted near $2f_{\mathrm{HM}}$. If the post S/H LPF has a sufficiently narrow BW, the tones near $2f_{\mathrm{HM}}$ will get suppressed, and only the desired signal and the tones near DC will remain. This is a problem because we technically cannot convert the tones into a single spur in the phase-domain unless we have two tones at an equal distance from the carrier and with equal amplitude, and in this case we have a tone on one side at a frequency offset of $f_{\mathrm{HM}} - \Delta f$ coming from the 11th harmonic, while the 9th and 13th harmonic on the other side is heavily attenuated. However, we will see later that a good approximation can be obtained if we ignore the 9th and 13th harmonics and assume that the 11th harmonic appears on both sides of the carrier except with just half the power. Using this approximation and also the fact that the desired signal at f_{HM} receives some attenuation from the post S/H LPF, the amplitude of the type-B spur $U_{\mathrm{B}}[\mathrm{dBc}]$ when $f_{\mathrm{SH}} \approx (2m+1) f_{\mathrm{HM}}$ i.e. f_{SH} is close to an odd multiple of f_{HM} can be expressed as

$$U_{\mathrm{B}}[\mathrm{dBc}] = 10\log\left(\frac{(2m+1)^{-2}}{2|H^2_{\mathrm{post,LPF}}(j2\pi f_{\mathrm{HM}})|}\right) \tag{4.10}$$

where $H_{\mathrm{post,LPF}}(j2\pi f)$ is the transfer function of the post S/H LPF.

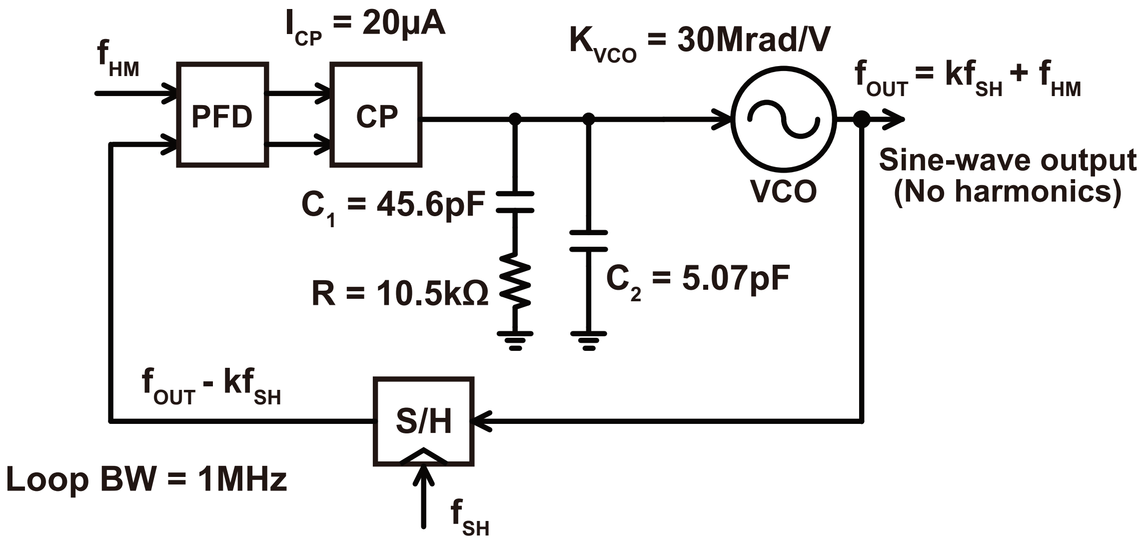


Figure 4.9: A block diagram of the PLL model used to simulate the type-A spurs

4.4.4 Behavioral Simulation Results

To demonstrate the accuracy of the expressions we derived for the spur amplitudes and also the correctness of the spur generation mechanism that we explained, some behavioral simulations were performed under different conditions.

We first performed simulations to verify the expressions for the type-A spurs shown in equation 4.7. Fig. 4.9 shows a block diagram of the model that we used for the simulation. The output of the VCO is a sine-wave and is directly connected to the S/H, meaning that we do not have to worry about the type-B spurs and can see the type-A spurs on their own. The PFD/CP, LF, and VCO are configured so that the loop bandwidth is 1MHz. f_{HM} and f_{SH} are varied in a way such that the resulting type-A spur appears at a sufficiently low offset frequency, and k is chosen so that the output frequency of the PLL remains relatively constant, even though the output frequency shouldn't have any major effect on the amplitude of the spurs at the output. Fig. 4.10 shows an example of the output spectrum containing the type-A spurs. Here, f_{SH} =800.1MHz and f_{HM} =100MHz meaning that $f_{\mathrm{SH}} \approx 8f_{\mathrm{HM}}$ and the type-A spur should appear at Δf =100kHz and its harmonics. We can see from Fig. 4.10 that the spurs indeed appear at 100kHz and its multiples. The amplitude of the spur is also very close to the value calculated by equation 4.7. We then looked at the dependency of the spur amplitude on f_{SH} and f_{HM}. Fig. 4.11(a) shows the calculated (red) and simulated (blue) amplitude of the type-A spur when we change f_{HM} while keeping f_{SH} constant. The values of f_{HM} are chosen so that Δf is always 100kHz. The actual values of f_{SH} and f_{HM} are also shown. We can see from the results that the simulated results match the calculation well. Furthermore, we can see that the spur amplitude gets larger as f_{HM} increases just like we predicted based on the fact that the tone gets further away from the notch of the sinc characteristic. Fig. 4.11(b) shows the calculated (red) and simulated (blue) amplitude of the type-A spur when we change f_{SH} while keeping f_{HM} constant. The values of f_{SH} are chosen so that Δf is always 100kHz. The actual values of f_{SH} and f_{HM} are also shown. We can see that here too, the simulated results match the calculation well. Furthermore, we can see that the spur amplitude gets larger as f_{SH} decreases just like we predicted based on the fact that the sinc characteristic gets sharper around the notch resulting in less attenuation for the same f_{HM}.

We also performed a simulation to see the effect of the post S/H LPF on suppressing the

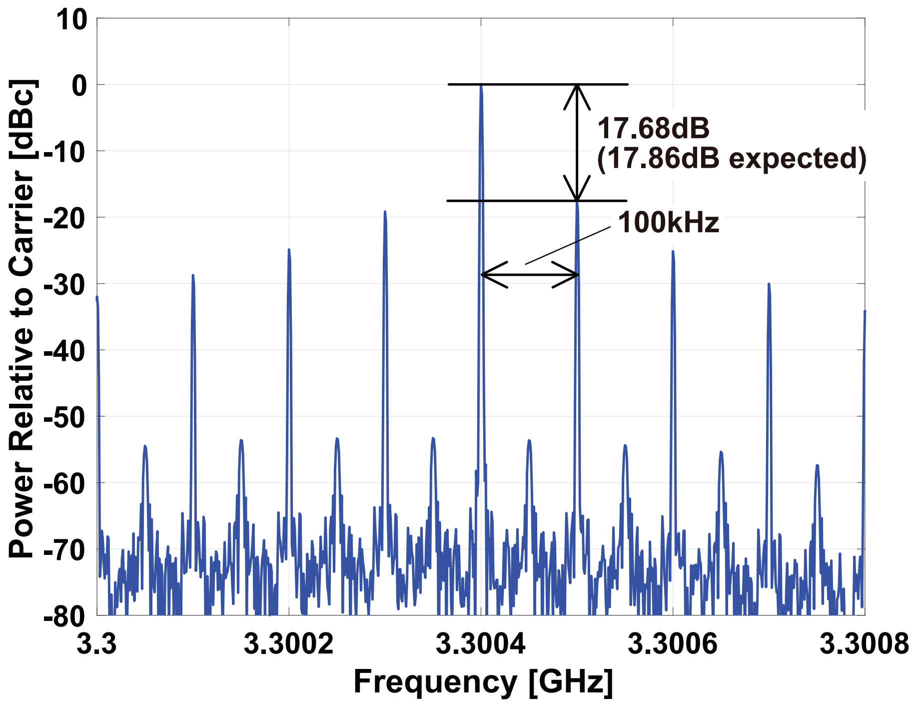
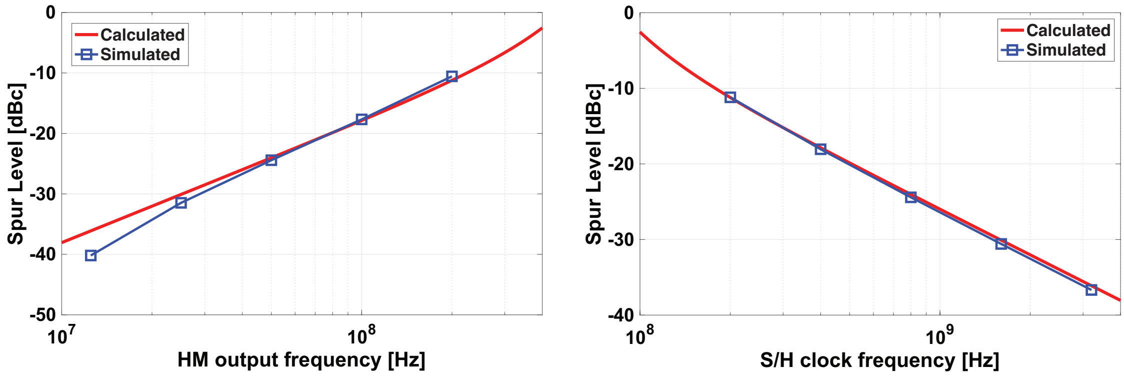


Figure 4.10: An example of the output spectrum of the HM-based PLL containing the type-A spurs (f_{SH} =800.1MHz, f_{HM} =100MHz)

type-A spurs. Fig. 4.12 shows a block diagram of the model being simulated. This is essentially the same as the model in Fig. 4.9 with f_{SH} =800.1MHz and f_{HM} =100MHz, except for the fact that there is a 1st order LPF placed after the S/H. Its bandwidth is 100MHz, meaning that it provides 3dB of attenuation for the desired signal at f_{HM} and 21dB of attenuation for the tones at around f_{SH}, resulting in 18dB of relative attenuation for the tones. Fig. 4.13 shows the resulting output spectrum with the type-A spurs. Since the frequency settings are the same as that of the spectrum shown in Fig. 4.10, the spurs are expected to have a magnitude of $-17.86-18= -35.86$dBc. The actual simulated result matches our prediction well, providing more evidence supporting that our explanation of the type-A spur generation mechanism is correct and a post S/H LPF can indeed be used to suppress them.

We also performed similar simulations to verify our analysis of the type-B spurs. Fig. 4.14 shows a block diagram of the model that we used for the simulation. The output of the VCO is a square-wave, and the goal here is to see how well the simulation results match our prediction based on equations 4.9 and 4.10. Since we want to see just the type-B spur for now, a 4th order LPF with a bandwidth of $2f_{\mathrm{HM}}$ was placed after the S/H to suppress the type-A spur. Just as it was for the type-A spur simulation, the PFD/CP, LF and VCO are configured so that the loop bandwidth is 1MHz and f_{HM} and f_{SH} are varied in a way such that the resulting type-B spur appears at a sufficiently low offset frequency to avoid the attenuation from the low-pass characteristic of the loop. k is chosen so that the output frequency of the PLL remains relatively constant although again, the output frequency shouldn't have any major effect on the amplitude of the spurs at the output. Fig. 4.15 shows an example of the output spectrum containing the type-B spurs. Here, f_{SH} =800.1MHz and f_{HM} =100MHz, meaning that the type-B spur should appear at Δf =100kHz and its harmonics. We can see from Fig. 4.15 that the spurs indeed appear at this frequency. Furthermore, since $f_{\mathrm{SH}} \approx 8f_{\mathrm{HM}}$ the amplitude of the type-B spurs can be predicted by equation 4.9, and the spectrum in Fig. 4.15 shows a good match with the predicted results. We



HM output frequency f_{HM} [MHz]	12.5	25	50	100	200
S/H clock frequency f_{SH} [MHz]	800.1	800.1	800.1	800.1	800.1
Harmonic Index k	4	4	4	4	4
PLL output frequency f_{OUT} [GHz]	3.2129	3.2254	3.2504	3.3004	3.4004
Simulated Spur Level [dBc]	-40.19	-31.48	-24.43	-17.68	-10.56

(a)

HM output frequency f_{HM} [MHz]	50	50	50	50	50
S/H clock frequency f_{SH} [MHz]	200.1	400.1	800.1	1600.1	3200.1
Harmonic Index k	16	8	4	2	1
PLL output frequency f_{OUT} [GHz]	3.2516	3.2508	3.2504	3.2502	3.2501
Simulated Spur Level [dBc]	-11.19	-18.06	-24.43	-30.59	-36.68

(b)

Figure 4.11: (a) The relationship between f_{HM} and the amplitude of the type-A spur, along with the details of the frequency settings that were used (b) The relationship between f_{SH} and the amplitude of the type-A spur, along with the details of the frequency settings that were used

then looked at how consistent the calculation results are with simulation when different harmonics of the PLL output get down-converted near the carrier. Fig. 4.16 shows the calculated (red) and simulated (blue) amplitude of the type-B spur when different harmonics of the PLL output get down-converted. f_{HM} is kept constant at 100MHz while f_{SH} is varied so that Δf is always equal to 100kHz. The details of the frequency settings used are also shown in the figure. We can see that the calculated results based on equations 4.9 and 4.10 match the simulated results well. It is worthy to note that the stair like characteristic of the calculation is due to the fact that two different equations are used for even and odd cases. Although both cases show good matching with the simulation, the even cases show a great match while the odd cases are about 1dB too small. This is likely due to two main causes. Firstly, we ignored the harmonics down-converted to around $2f_{\mathrm{HM}}$ when we derived the expressions, but in reality, the attenuation from the post S/H LPF is finite and there is some contribution to the type-B spur from those tones. If we take into account this effect, the spur level for the odd cases increase by around 0.26dB, making the calculated results slightly closer to the simulated ones. Secondly, the down-converted harmonics appear on just one side of the desired signal, but we assumed that it appears on both sides with half the power in order to convert the voltage-domain spectrum into the phase-domain one. This effect unfortunately is somewhat difficult to account for analytically. In a practical sense however, an error of 1dB is generally not an issue when using the analysis for a design and it can be said that the analysis matches the simulation results well.

Finally, we performed a simulation to see the effect of the pre S/H LPF on suppressing the type-B spurs. Fig. 4.17 shows a block diagram of the model being simulated. It is essentially the same as the model in Fig. 4.14 with f_{SH} =800.1MHz and f_{HM} =100MHz, except for the fact that there is a 1st order LPF placed before the S/H. Since $f_{\mathrm{SH}} \approx 8f_{\mathrm{HM}}$, we can see that the 7th and 9th harmonics of the PLL output will get down-converted near the carrier and therefore are the ones that need to be suppressed by the LPF. The pre S/H LPF used in this simulation has a bandwidth of 2.5GHz, and so the attenuation at the desired signal is 3dB while that for the 7th and 9th

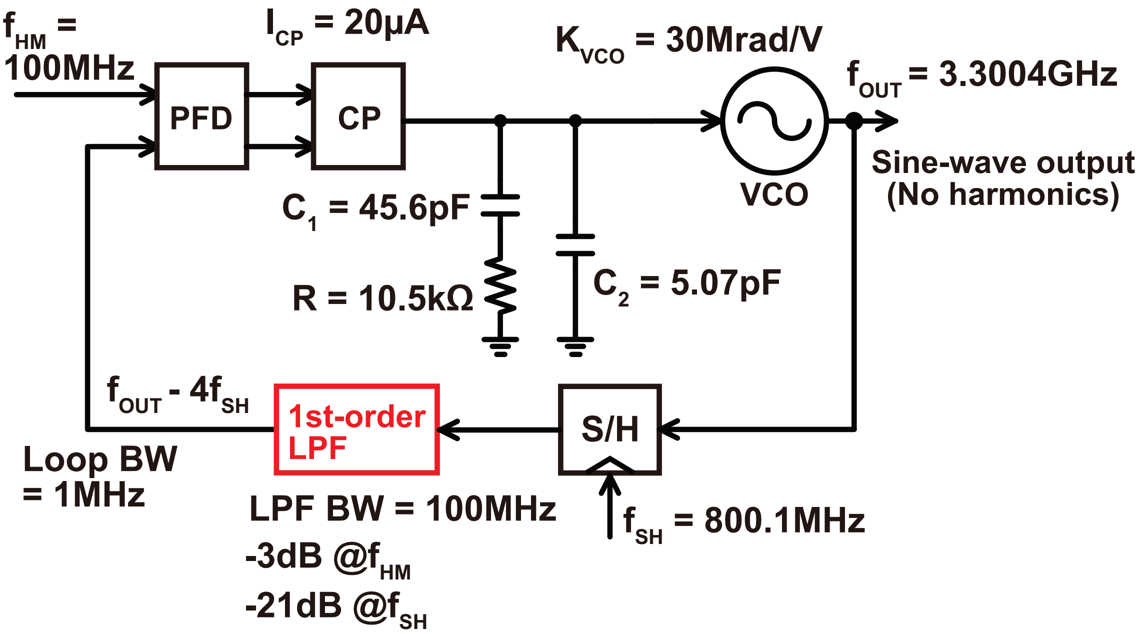


Figure 4.12: A block diagram of the PLL model used to simulate the type-A spurs with a 1st order LPF

harmonics are 17 and 19dB respectively, resulting in around 15dB of relative attenuation for the spurs. Fig. 4.18 shows the resulting output spectrum, and we can see that the results match the calculation relatively well.

By performing these behavioral simulations, we were able to show that the spurs appear at the expected offset frequencies, the expressions we derived for the spur amplitude are accurate, and that our understanding of the spur generation mechanism is correct.

4.5 Consideration of RC Constant in S/H

So far we have assumed that the S/H circuit is ideal, meaning that it has zero input impedance when the sampling switch is on, and infinite input impedance when it is off. In reality, the input impedance is not zero when the switch is on due to the on resistance of the switch and the output impedance of the isolation buffer, and the input impedance is not infinity when the switch is off because of switch leakage. The former non-ideality has an especially large effect on the characteristic of the S/H because the input impedance and the sampling capacitance form an RC constant that the input signal experiences. We will analyze this effect here by making use of the analysis performed in [66, 67], and come up with a more accurate model for predicting the offset spurs.

Fig. 4.19 shows a block diagram of the circuit model we are dealing with, the clock waveform, and the corresponding Signal Flow Graph (SFG). For the circuit model, we are dealing with a sampling capacitor and a switch, and there is a resistor in series with an ideal switch to represent the on-resistance of the switch and the output impedance of the isolation buffer. R represents this resistance, C the sampling capacitance, V_{in} and V_{out} are the input and output wave-forms, and v_{s} is the clock driving the switch. v_{S} is a clock with a period of T_{S} and a duty-cycle of $D = \tau/T_{\text{S}}$. The RC characteristic alters the input waveform in two ways. Firstly, it gets attenuated and phase shifted by the low-pass characteristic of the RC. Secondly, the sampled voltage from the previous period creates a transient response right after the switch turns on. The resulting effect

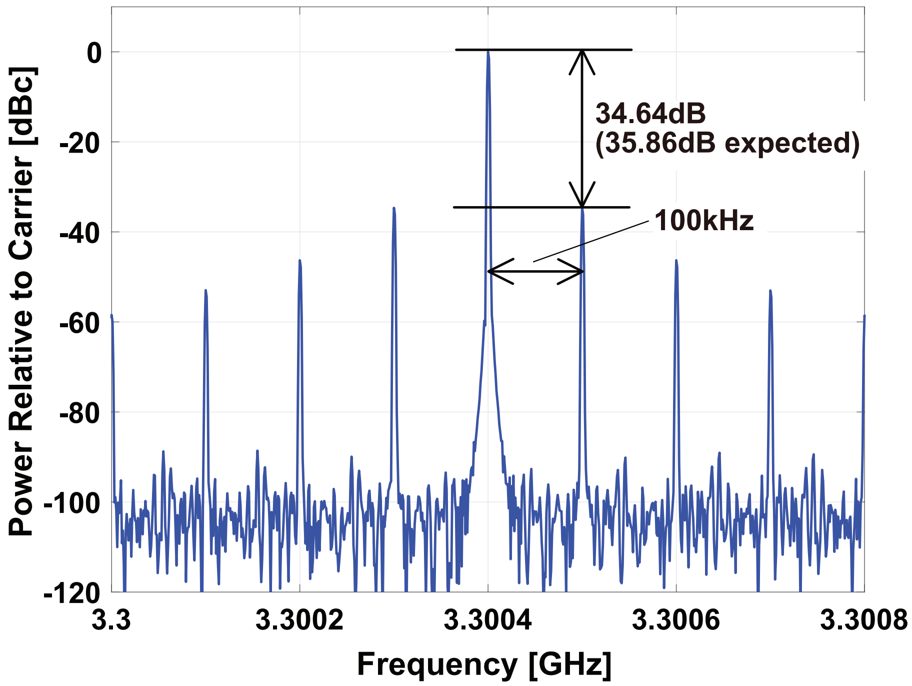


Figure 4.13: An example of the output spectrum of the HM-based PLL with a post S/H LPF containing the type-A spurs (f_{SH} =800.1MHz, f_{HM} =100MHz)

is summarized by the SFG. Furthermore, the SFG can be simplified for particular uses where certain approximations hold. Of particular use to us is the use of the S/H as a passive mixer that makes use of the frequency translation that occurs during impulse sampling. If the on time of the switch and the period of the input signal are sufficiently shorter than the R-C time constant, the on-time of the switch is sufficiently shorter than the input period to avoid attenuation from the sinc function, and the bandwidth of the post impulse sampling filter D/RC is wide enough so as not to attenuate the HM output too much, the SFG of the HM can be approximated as shown in Fig. 4.20. The Zero-Order-Hold (ZOH) corresponds to the second stage switch that performs the hold operation. We can see that in addition to the frequency translation and ZOH seen in an ideal S/H, we have an extra sinc filter before the frequency down-conversion, and an extra first order LPF after it. With proper design, these filters can actually contribute to the attenuation of the offset spurs. Otherwise, the spur generation mechanism and the way to suppress them is unchanged. Finally, using this SFG of the HM, the overall model of the HM-based PLL can be summarized as shown in Fig. 4.21.

4.6 Case Study for Low-Pass Filter Design and Circuit-Level Simulation Results

In this section, we use the analysis that we performed in the previous sections to form a guideline for designing a HM in an actual PLL. The flow of the HM design can be roughly outlined as follows:

1. Establish the worst-case frequency setting used during the design and circuit simulations
2. Create a HM without the pre/post LPFs. Then calculate the required amount of filtering based on the simulated bandwidth of the first stage sampler, the simulated harmonic level at the S/H input, and the clock duty-cycle

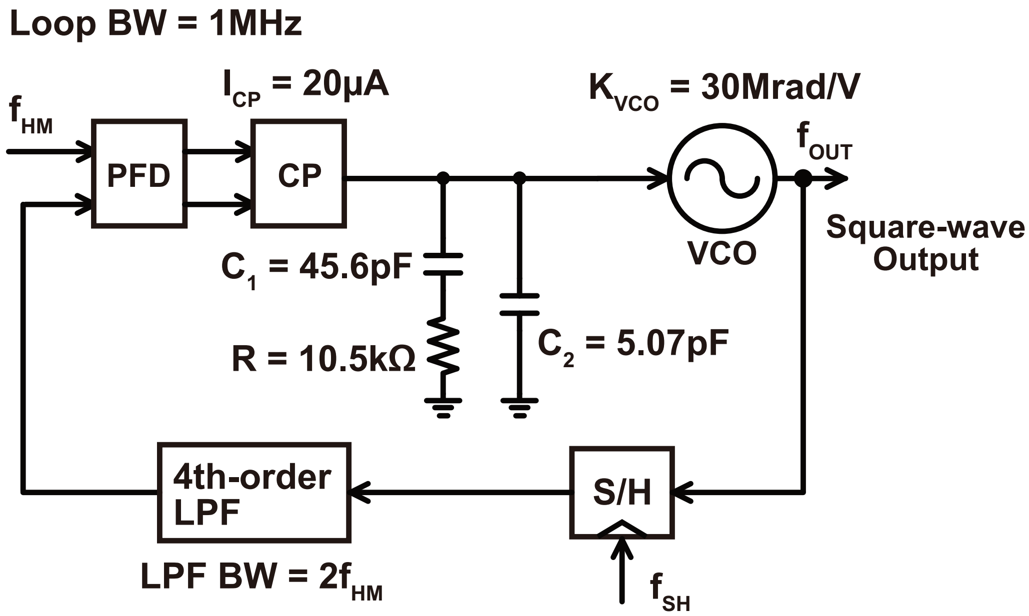


Figure 4.14: A block diagram of the PLL model used to simulate the type-B spurs

3. Design the pre/post LPFs based on the required specifications derived in step 2
4. Insert the LPFs and confirm the HM operation through simulation. Then close the PLL and confirm that the final offset spur meets the required specifications.

By following this procedure, we demonstrate how to make use of our analysis in detail by using a case study of a HM-based PLL at the transistor level and performing simulations accordingly.

To design the LPFs so that the offset spurs are suppressed below a certain spec for all frequency settings, we must first clarify which setting leads to the worst case offset spur. Both the type-A and type-B spurs appear at the same frequency, and experience attenuation from the low-pass characteristic of the PLL depending on its frequency. In the worst case, f_{SH} is close to an integer multiple of f_{HM} and receives no attenuation from the PLL characteristic. Furthermore, for the type-A spur, we saw that its amplitude increases as f_{SH} decreases and f_{HM} increases. Therefore, we should look at the frequency setting where f_{SH} is close to its minimum value, f_{HM} is close to its maximum value, and f_{SH} is close to an integer multiple of f_{HM}. For the type-B spur, the situation is slightly more complex because it is dependent on the specific harmonic profile of the S/H input. Nonetheless, the harmonics usually get smaller as their order increases. As such, we can assume that the worst case frequency setting for the type-B spur is also the same as that for the type-A spur. We can design the LPFs so that the offset spurs are suppressed below the required spec under this circumstance.

Fig. 4.22 shows the circuit model that was used to demonstrate the design flow. The HM including the S/H circuit and the LPFs before and after it are simulated in the schematic level, while the remaining blocks such as the PFD/CP, LF and VCO are simulated using behavioral models. The details of the post S/H LPF are shown in Fig. 4.23. The loop bandwidth is 2MHz, the PFD/CP input frequency is 60 to 80MHz, and the HM LO frequency is 800 to 900MHz with a duty cycle of D =0.125. The harmonic index of the HM is 3, and the R-C characteristic of the first stage of the S/H has a bandwidth of 380MHz. The PLL output whose frequency ranges from 2.46 to 2.78GHz is sine-wave shaped, and the harmonics arise from the HM input buffer. The 65nm

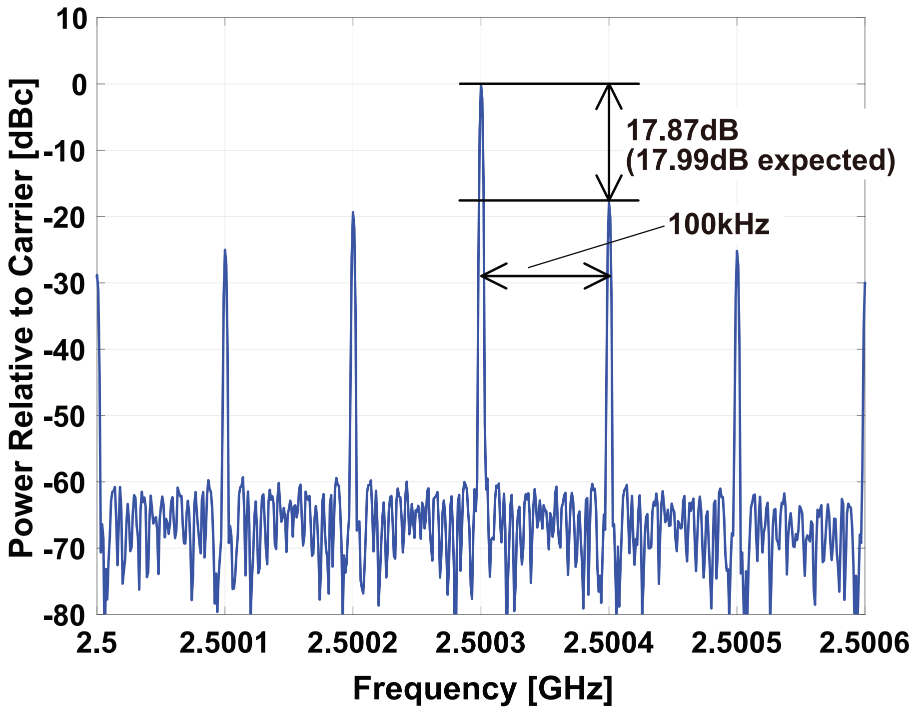
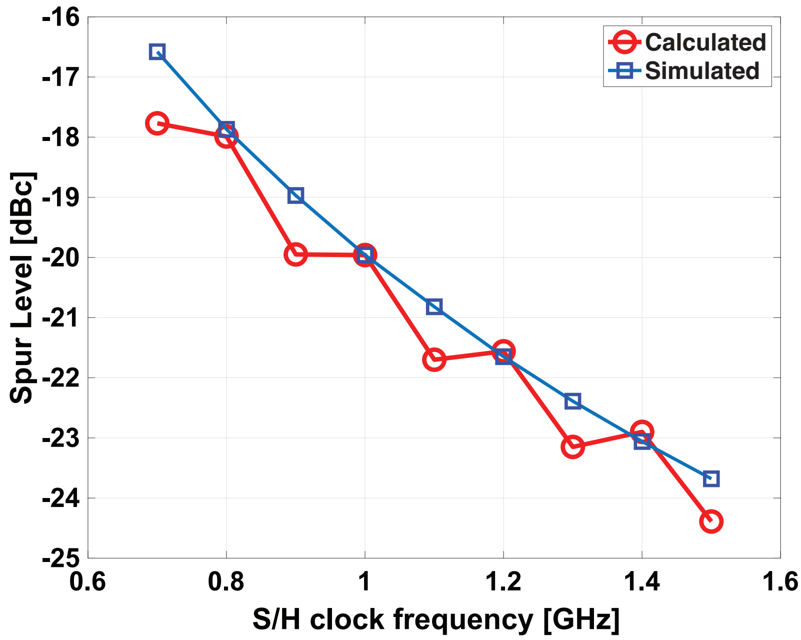


Figure 4.15: An example of the output spectrum of the HM-based PLL containing the type-B spurs (f_{SH} =800.1MHz, f_{HM} =100MHz)

CMOS process model is used. Two inverter-based buffers are placed between the VCO and the S/H as is done in an actual design to provide enough isolation, and the pre-LPF is placed between the two buffers to suppress the harmonics generated by the buffers. This configuration is the most practical in our example because placing the pre-LPF before the buffers means the LPF only suppresses the harmonics generated by the VCO, while placing the pre-LPF after both buffers will effect the output impedance seen by the S/H, complicating the design. Of course, the pre-LPF can be placed before the buffers if the VCO output already contains large harmonics (in a ring-VCO configuration for example), and the LPF can be placed after both buffers if the non-linearity of the second buffer is too large, though the latter will request us to carefully co-design the S/H circuit with the LPF. To simulate the offset spurs in the worst case condition, we will use a frequency setting where f_{SH} =800.5MHz and f_{HM} =80MHz. Here, f_{SH} and f_{HM} are close to their minimum and maximum values respectively and Δf is sufficiently lower than the PLL bandwidth.

As an example, if we have a target offset spur of −60dBc, since the type-A and type-B spur both appear at the same frequency, they individually must be suppressed to roughly −66dBc or lower. For the type-A spur, the attenuation without using a post S/H LPF comes from the sinc characteristic due to the hold operation of the S/H, and a low-pass characteristic with a bandwidth of $\frac{D}{2\pi RC}$ due to the R-C characteristic of the switch and sampling capacitor and the duty cycled operation of the switch. These add up to around −31dB, meaning that an extra 35dB of attenuation at f_{SH} should be provided by the post S/H LPF. For the type-B spur, since $f_{\mathrm{SH}} \approx 10 f_{\mathrm{HM}}$, we expect the 9th and 11th harmonics of the PLL output to get down-converted near f_{HM} at the HM output. Fig. 4.24 shows the spectrum of the HM input when we don't have a pre S/H LPF. We can see that the 9th harmonic has a relative power of around −41dBc. Furthermore, the equivalent sinc characteristic before the S/H as shown in Fig. 4.21 contributes around 19dB of attenuation for the 9th harmonic. This means that the type-B spur is around −60dBc without any extra filtering, and so the pre S/H LPF should provide atleast 6dB of extra suppression. Fig. 4.25 shows the



S/H clock frequency f_{SH} [MHz]	700.1	800.1	900.1	1000.1	1100.1	1200.1	1300.1	1400.1	1500.1
Dominant Harmonic(s)	7	7, 9	9	9, 11	11	11, 13	13	13, 15	15
Output frequency f_{OUT} [GHz]	2.2003	2.5003	2.8003	3.1003	3.4003	3.7003	4.0003	4.3003	4.6003

HM output frequency f_{HM} = 100MHz, Harmonic Index k = 3

Figure 4.16: The calculated (red) and simulated (blue) relationship between the f_{SH} and the resulting type-B spur amplitude

characteristics of the (a) pre and (b) post S/H LPFs respectively. We can see that both LPFs provide a sufficient amount of filtering to suppress the type-A and type-B spurs to well below the required spec. Finally, Fig. 4.26 shows the resulting output spectrum of the HM-based PLL. We can see that the offset spurs appear at 500kHz and its harmonics, and they are suppressed to well below the target we set of -60dBc.

4.7 Chapter Summary

In this chapter, we investigated the phenomenon of offset spurs in HM-based PLLs that arise due to the S/H operation in the HM. The two generation mechanisms of offset spurs were made clear, and the location and amplitude of the offset spurs were derived analytically. The effect of the RC constant in the S/H was also taken into account. Based on the analysis, we provided a guideline for designing LPFs to suppress the offset spurs, and demonstrated its feasibility through a case study. All of the results were backed up with simulation results that matched the derived expressions well.

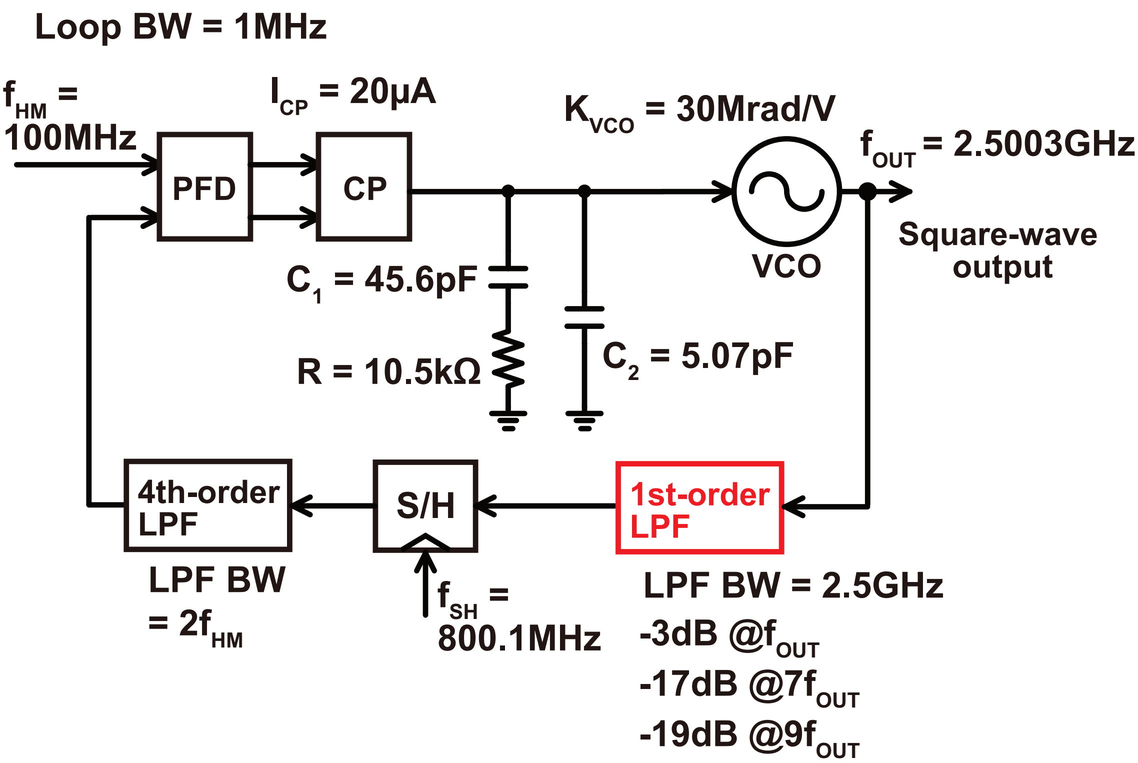


Figure 4.17: A block diagram of the PLL model used to simulate the type-B spurs with a 1st order LPF

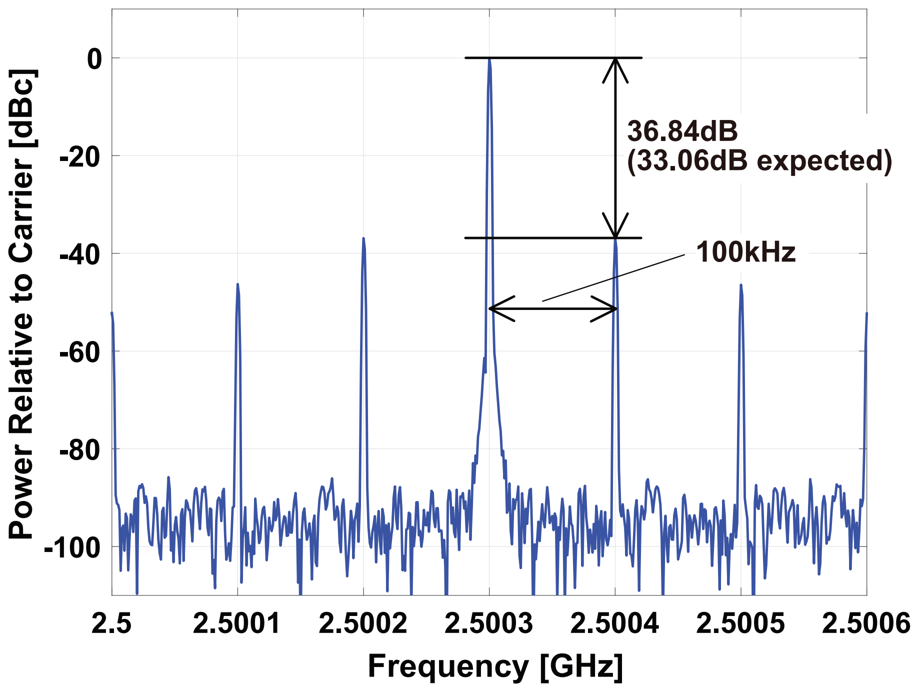


Figure 4.18: An example of the output spectrum of the HM-based PLL with a pre S/H LPF containing the type-B spurs (f_{SH} =800.1MHz, f_{HM} =100MHz)

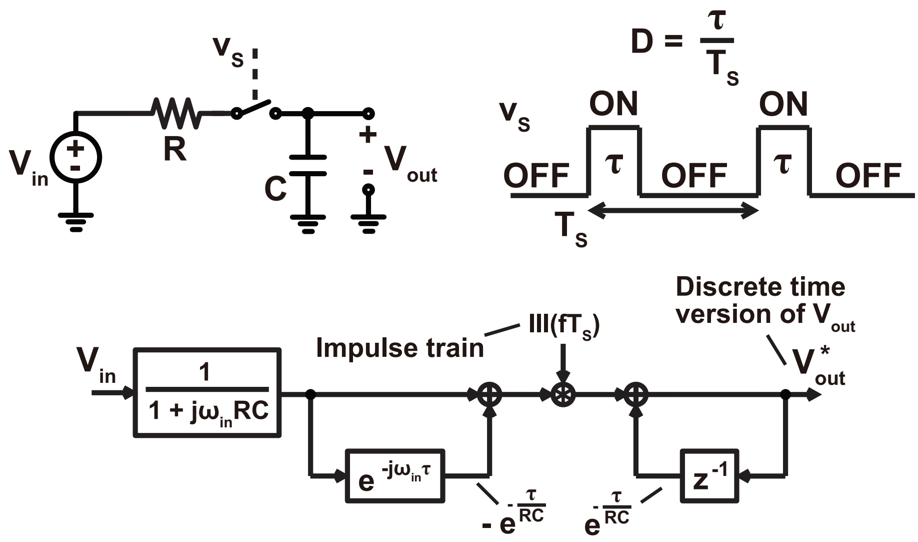


Figure 4.19: The circuit model we are dealing with, the clock waveform, and the resulting SFG of the switch + sampling capacitor

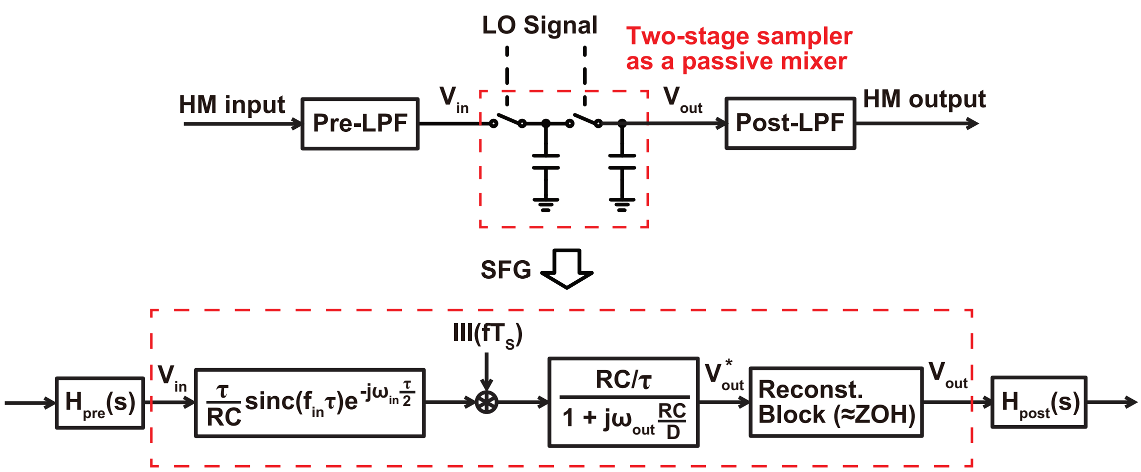


Figure 4.20: The HM and its SFG

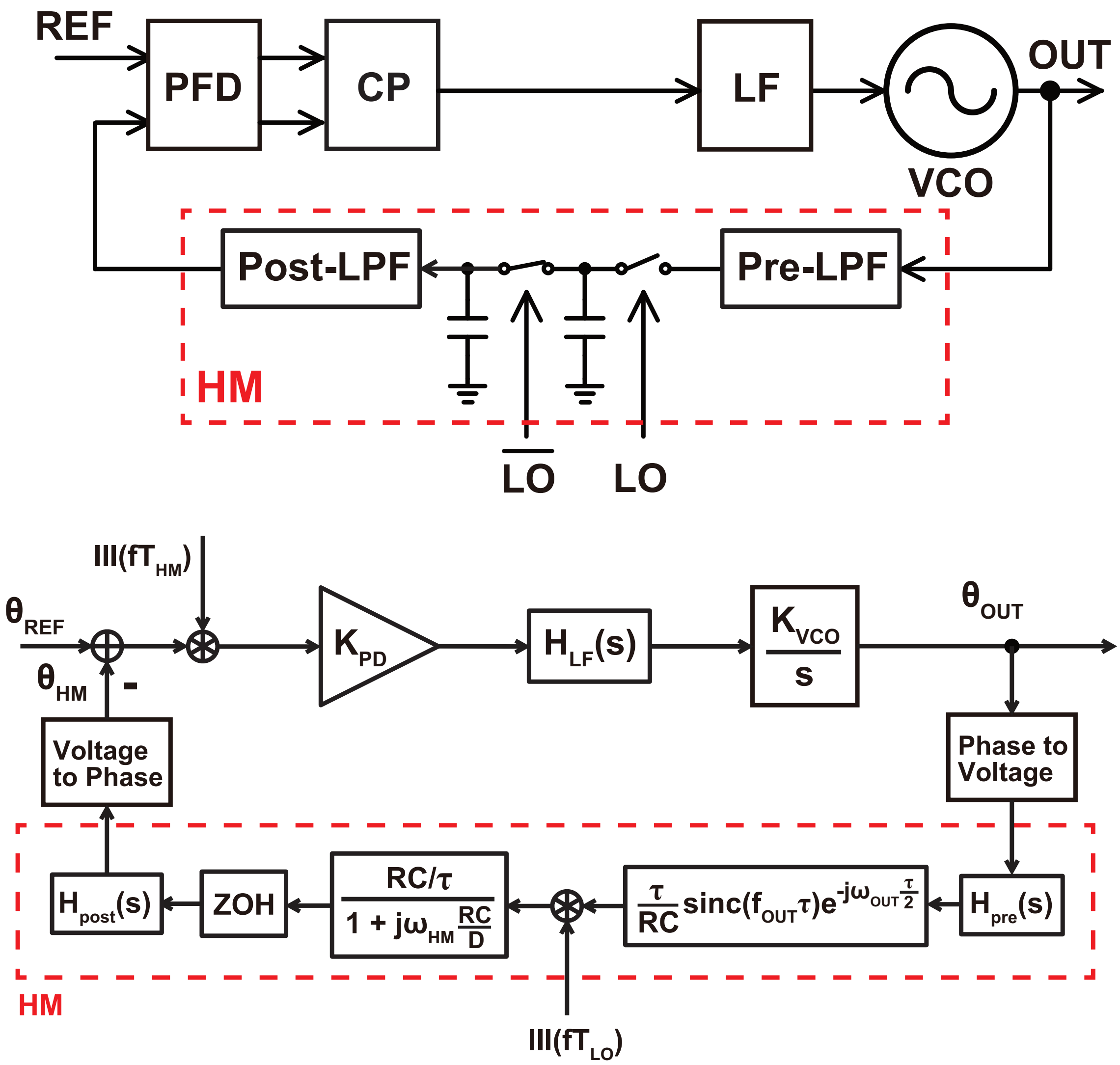


Figure 4.21: The overall model of the HM-based PLL

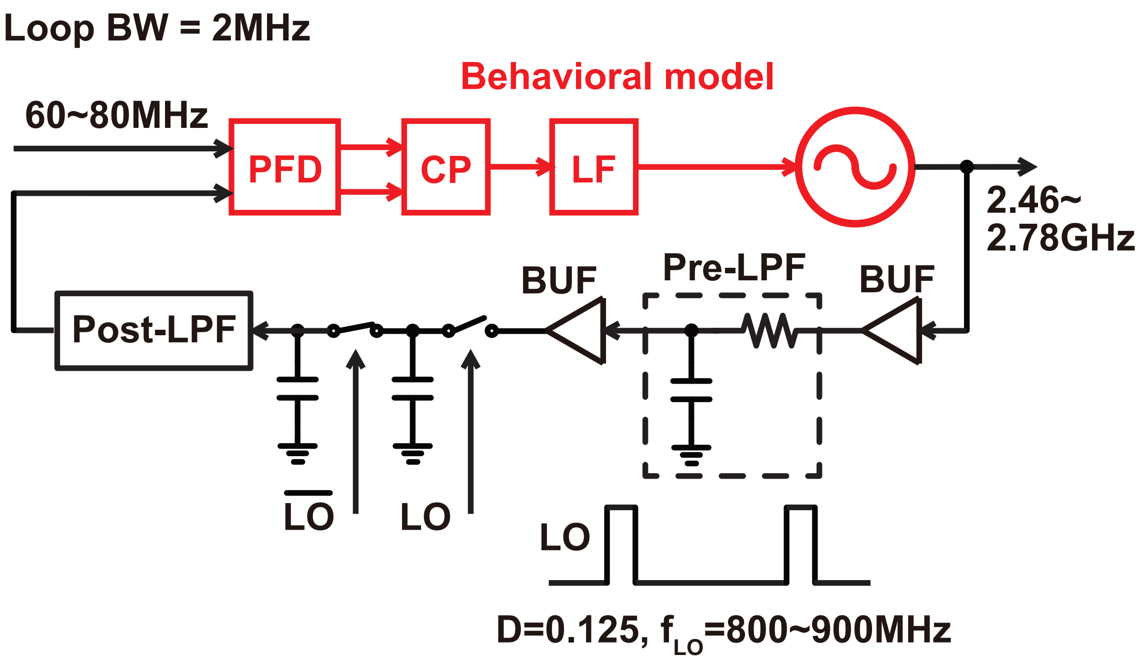


Figure 4.22: The model used to perform the semi circuit level simulation of the HM-based PLL

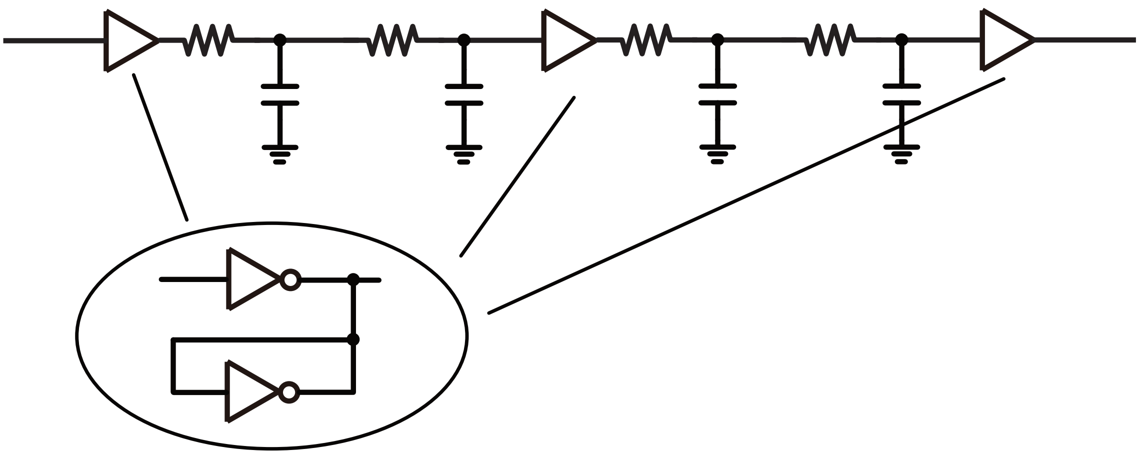


Figure 4.23: The circuit details of the post S/H LPF

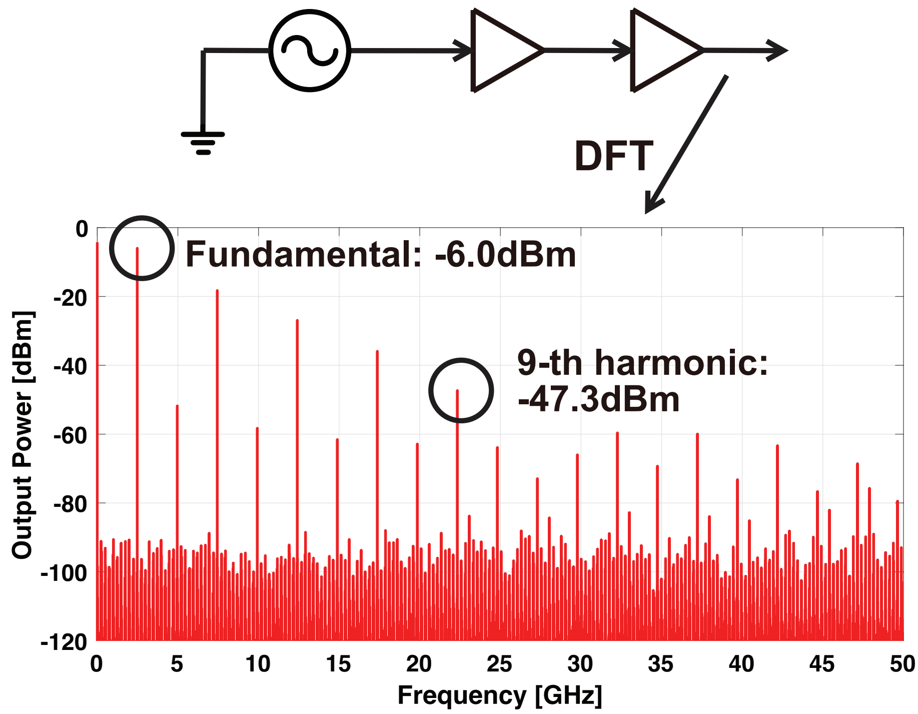


Figure 4.24: The spectrum of the HM input without using a pre S/H LPF

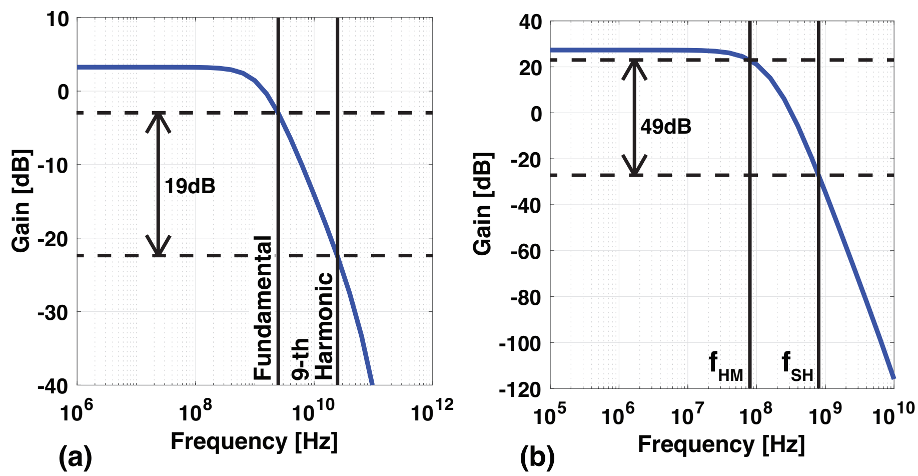


Figure 4.25: The characteristics of the (a) pre and (b) post S/H LPFs

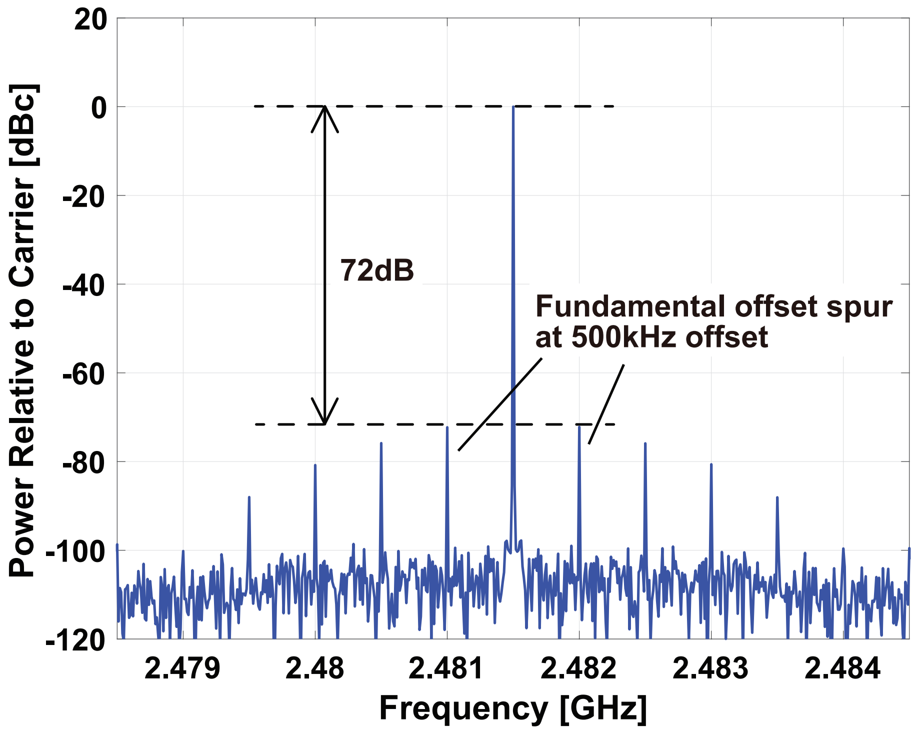


Figure 4.26: The spectrum of the output of the HM PLL

Chapter 5

Dual-Feedback PLL with Nested-PLL-Based Phase-Domain Filtering

In chapter 3, we introduced the dual-feedback architecture that achieves effective suppression of the quantization noise and fractional spurs for fractional-N synthesis. Along with the triple-loop PLL, it makes use of the HM based feedback to realize a unity DC gain transfer function and avoid the amplification of noise within the loop. It turns out however, that in some cases, the avoidance of noise amplification alone is insufficient to provide enough noise suppression. In this chapter, we propose a loop architecture that not only makes use of the HM-based feedback to avoid noise amplification, but also efficiently makes use of the other noise suppression approaches such as increasing the noise shaping frequency and adding noise filtering. The work in this chapter is based off of our work published in [78].

5.1 Use of Ring Oscillators in Frequency Synthesis

An important factor of motivation for this work is to realize a fractional-N PLL using ring VCOs, instead of relying on their LC based counterparts. Compared to LC VCOs, ring VCOs have many advantages, such as lower silicon area, avoidance of issues with magnetic coupling, a much wider achievable tuning range, and a convenient generation of multiple output phases among others. However, ring VCOs tend to suffer from much worse PN than LC VCOs assuming a similar power consumption. This makes the ring VCO difficult to use in frequency synthesizers for many applications, despite the various advantages that they offer.

As we explained in chapter 2, the suppression of the VCO PN by widening the bandwidth of the PLL is in a trade-off relationship with the suppression of the reference referred PN and quantization noise from the DSM that is achieved by narrowing the PLL bandwidth. This is why we want to suppress the quantization noise by some other means so that the bandwidth can be widened to suppress the VCO PN. When we want to realize a fractional-N PLL based on a ring VCO, this desire to suppress the quantization noise and widen the loop bandwidth is even more profound due to the inferior PN performance of the ring VCO.

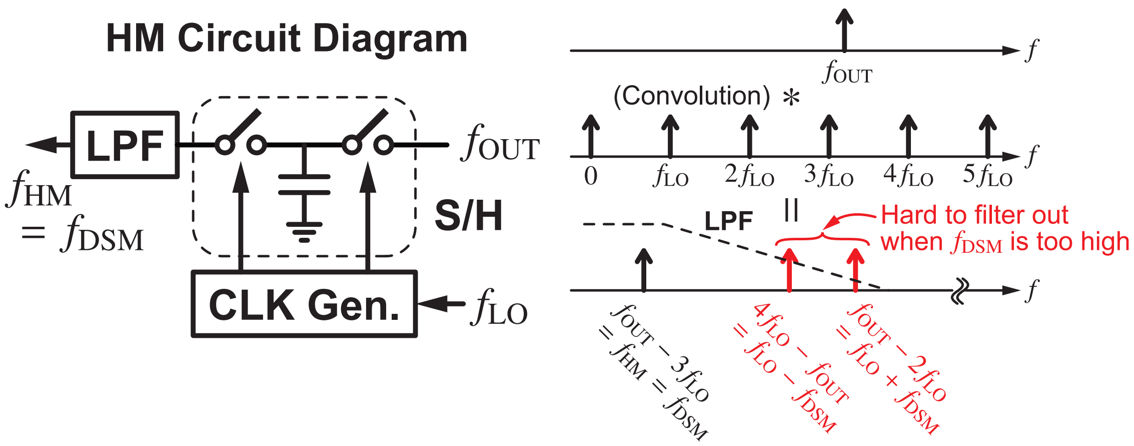


Figure 5.1: The limitation of the noise shaping frequency in the dual-feedback PLL

5.2 Prior Calibration-Free Approaches for Quantization Noise Suppression

As we discussed in chapter 2, there are many approaches to suppress the quantization noise without relying on calibration schemes. In this section, we look at two specific methods that are closely related to our proposal, and also discuss their limitations and disadvantages.

5.2.1 Avoiding Noise Amplification with HM-based Feedback

As we discussed in chapter 3, the dual-feedback PLL makes use of a feedback path based on a HM to help realize a closed-loop characteristic with unity DC gain and suppress the quantization noise effectively by avoiding its amplification. It is calibration-free, and certainly a candidate architecture when it comes to realizing a ring VCO based fractional-N PLL.

However, the down-side to the dual-feedback architecture especially when using it in a ring VCO based implementation is the limitation of the noise shaping frequency. As was explained in chapter 4 and shown in Fig. 5.1, the HM output contains not just the desired signal component at f_{HM}, but also some high frequency tones at $f_{\mathrm{LO}} \pm f_{\mathrm{HM}}$ which we called the type-A spurs, and these will appear at the overall output as spurious tones if they aren't filtered out before phase detection. We can see that if f_{HM} is not sufficiently smaller than f_{SH}, it will become very difficult to filter out the unwanted tones while not attenuating the desired signal component. Furthermore, f_{SH} is not easy to make large in an inductorless implementation because the ring VCO FoM degrades for higher frequencies due to the decreasing number of stages. Since f_{SH} cannot be made too high and f_{HM} must be sufficiently smaller than f_{SH}, f_{HM} cannot be made very high. And since the DSM noise shaping frequency is equal to the HM output frequency after lock, this in turn means that the noise shaping frequency is limited in the dual-feedback architecture due to the tones generated in the HM. In the design we implemented in chapter 3, this was not much of an issue because the avoidance of noise amplification was already enough to suppress the quantization noise to a negligible level. However, because the PN from a ring VCO is much worse than that from an LC VCO assuming a reasonable power consumption, the bandwidth of a ring VCO based PLL has to be very wide to achieve an acceptable integrated jitter, and the DSM quantization noise can heavily degrade the output PN, even if we avoid noise amplification.

5.2.2 Split-Feedback Division with Nested PLL based Filtering

Using a split-feedback divider with a nested PLL based filter is another interesting way to suppress the DSM quantization noise [79]. This method makes use of two approaches for quantization noise suppression: increasing the noise shaping frequency, and improving the noise filtering characteristic. Fig. 5.2 shows a conceptual block diagram of this architecture. Just like we saw for [51], the feedback divider is split into two parts to increase the noise shaping frequency. However, instead of using a multi-phase output for the second stage divider to avoid the aliasing of the quantization noise, this work uses a Phase-Domain-Low-Pass-Filter (PDLPF) which is realized by a nested PLL for use as an anti-aliasing filter. Since a PLL exhibits a low-pass characteristic from its input to its output, it can be used to suppress some of the high frequency quantization noise from the DSM before it enters the second stage divider, thus avoiding aliasing. This work actually predates [51], and was the first work to consider splitting the feedback divider into two parts to increase the noise shaping frequency. Not only that, but at first glance, it seems like the nested PLL based PDLPF can also provide extra filtering for the quantization noise. However, after more careful observation, we can see that the PDLPF cannot serve as a noise filter whose bandwidth can be tuned freely like the ones we discussed in chapter 2, and is in fact one of the things that make it difficult to design this PLL. A key difference between the nested-PLL based PDLPF in this work and the FIR filter in works such as [55, 56, 57, 58] is that the nested-PLL based PDLPF not only filters the quantization noise, but effects the open-loop transfer function of the PLL. This means that the nested PLL bandwidth cannot be designed freely to provide optimal attenuation for the quantization noise, but must be sufficiently wider than that of the overall PLL in order to maintain an appropriate phase margin. When we also consider the fact that the nested PLL bandwidth has to be narrow enough to serve as an anti-aliasing filter, we end up with the fact that the bandwidth of the overall PLL is quite limited after all, even though our original goal was to widen the loop bandwidth in order to suppress the ring VCO PN. Another issue posed by the nested PLL based PDLPF is that the noise contribution from the nested PLL itself can have a significant contribution to the overall PN since its noise gets amplified within the loop. Despite the fact that the nested PLL bandwidth is wider than that of the overall PLL, the PN contribution from the nested VCO and PD can still be significant, leading to a large power and/or noise overhead. So despite successfully increasing the noise shaping frequency and seemingly providing extra filtering for the quantization noise, this architecture is also not sufficient for achieving a satisfactory jitter performance using ring VCOs.

5.3 Dual-Feedback PLL with Nested-PLL based Phase-Domain Filtering

In this section, we will explain our proposed architecture. In the previous section, we discussed two approaches to suppress the DSM quantization noise. The HM-based dual-feedback architecture avoids the amplification of quantization noise in the loop, but suffers from a limitation of the noise shaping frequency. Split-feedback frequency division with nested PLL based PDLPF increases the noise shaping frequency and provides some extra filtering for the quantization noise, but suffers from stability issues and noise/power overhead due to the nested PLL itself. Our idea is actually very simple, we simply combine these two methods by using the split-feedback divider with PDLPF inside of the dual-feedback architecture. But by doing so, we are able to retain the benefits of both while overcoming their issues.

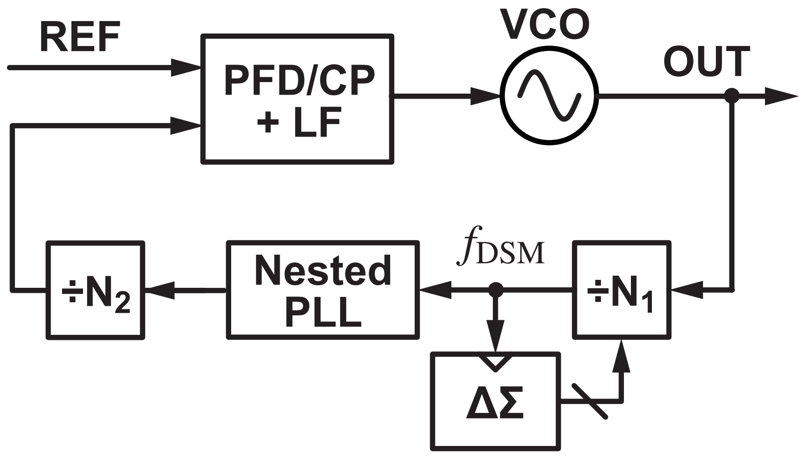


Figure 5.2: A conceptual block diagram of a fractional-N PLL employing a split-feedback divider and a nested PLL based anti-aliasing PDLPF

5.3.1 Proposed Architecture

The proposed architecture is shown in Fig. 5.3. As we explained, it is a combination of the dual-feedback architecture and the split-feedback frequency divider with PDLPF. It has several notable benefits. Firstly, this architecture retains the benefits of quantization noise suppression from both architectures. Noise amplification is avoided of course due to the characteristic of the dual-feedback architecture, the noise shaping frequency can be increased since it is no longer limited by the operation of the HM, and some filtering is provided by the PDLPF. In addition, the nested PLL does not effect the overall stability, and therefore its bandwidth can be designed freely. This is because the gain of the divider based path is $1/N_1N_2$ while that of the HM-based path is 1, meaning that the HM-based loop dominates the overall loop transfer function and the nested PLL characteristic has a negligible effect on the overall stability. As such, we can have a configuration for example where the main PLL has a very wide bandwidth to suppress the main VCO PN while the nested PLL can have a somewhat narrow bandwidth to provide filtering for the quantization noise. Finally, the noise contributed by the nested PLL itself is suppressed because of the avoidance of noise amplification due to the dual-feedback architecture, meaning that the nested PLL can be implemented with a low noise/power overhead. These advantages make this architecture very suitable potentially for realizing a ring VCO based fractional-N PLL, since it can effectively suppress the noise from both the ring VCO and the DSM quantization noise.

5.3.2 Noise and Stability Analysis

We will now look into the proposed architecture in a little more detail by calculating the transfer functions of the loop. Fig. 5.4 shows a linear noise model of the proposed PLL. We have represented the nested PLL in a black box fashion with just its closed loop transfer function and an equivalent noise source to simplify the analysis. The external PLL is also represented by just a single noise source. K_{PD}, $H_{\mathrm{LF}}(s)$, and K_{VCO} are the PD gain, LF transfer function, and VCO gain respectively. k, N_1, and N_2 are the harmonic index of the HM, and the division ratios of the first and second stage dividers respectively. $G_{CL,nest}(s)$ is the close loop transfer function of the nested PLL. It is a low-pass characteristic, and we will assume here that its DC gain is 1 i.e. it does not perform frequency multiplication although the result of the analysis will not change even if the nested PLL does perform frequency multiplication since it will be cancelled by an increase in N_2 assuming that the main PLL PD input frequency is constant. ϕ_{nVCO}, i_{nPD}, ϕ_{nHM}, $\phi_{\mathrm{nPLL,nest}}$, ϕ_{nDSM}, and

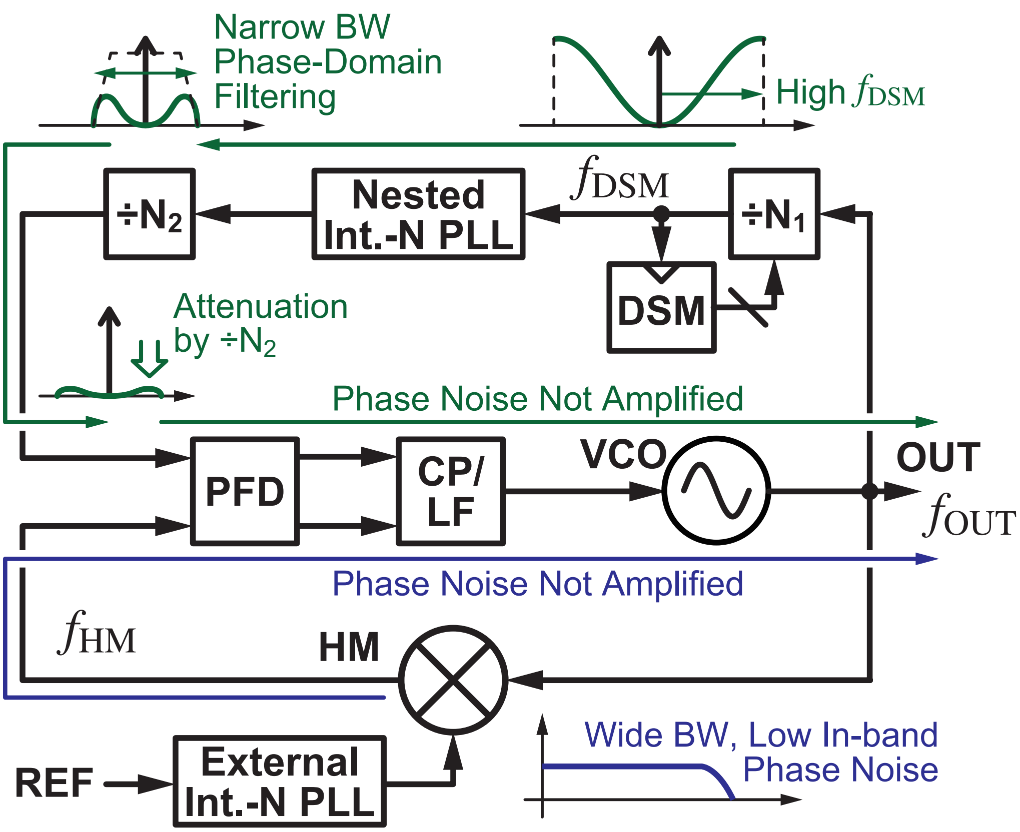


Figure 5.3: A block diagram of the proposed PLL that makes use of the split-feedback frequency divider with PDLPF inside of the dual-feedback architecture

$\phi_{\rm nPLL,ext}$ represent the noise contributed by the main VCO, the main PD, the HM, the nested PLL, the DSM, and the external PLL respectively. Finally, $f_{\rm OUT}$, $f_{\rm LO}$, $f_{\rm HM}$, and $f_{\rm DSM}$ are the frequencies of the overall output, the HM LO, the HM output, and the DSM respectively. The output frequency can be calculated as

$$f_{\rm OUT} = kf_{\rm LO} + f_{\rm HM} = \frac{kN_1N_2}{N_1N_2 - 1}f_{\rm LO}, \tag{5.1}$$

while the DSM noise shaping frequency can be calculated as

$$f_{\rm DSM} = N_2 f_{\rm HM}. \tag{5.2}$$

Equation 5.2 shows that the noise shaping frequency is N_2 times larger than the HM output frequency, meaning that it avoids the issue of limited noise shaping that the dual-feedback PLL suffered from.

We will start by looking at the stability of the loop. The open-loop gain of the main PLL $H_{\rm OL}(s)$ can be calculated as

$$H_{\rm OL}(s) = \frac{K_{\rm PD}K_{\rm VCO}H_{\rm LF}(s)}{s}\left(1 - \frac{G_{\rm CL,nest}(s)}{N_1N_2}\right). \tag{5.3}$$

The first part of the equation is the same as that of the HM-based PLL core in equation 3.1. As for the second part of the equation in parentheses, we can see that because the absolute value of

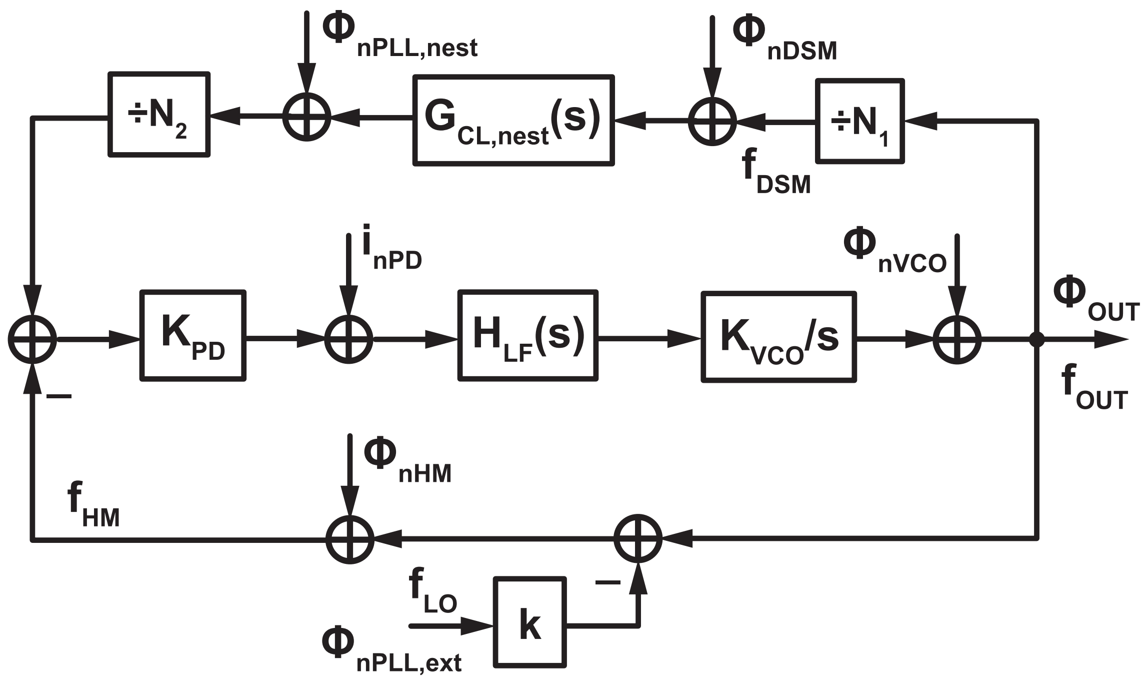


Figure 5.4: The noise model of the dual-feedback PLL with split-feedback frequency division and PDLPF

$G_{\mathrm{CL,nest}}(s)$ is less than or equal to the DC gain of 1 regardless of its bandwidth, $1 - \frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2} \approx 1$ for practical values of N_1 and N_2. Therefore, we can see that $H_{\mathrm{OL}}(s)$ can be approximated as

$$H_{\mathrm{OL}}(s) \approx \frac{K_{\mathrm{PD}} K_{\mathrm{VCO}} H_{\mathrm{LF}}(s)}{s} \tag{5.4}$$

and the overall stability has very little dependence on the bandwidth of the nested PLL. In contrast, the conventional architecture where we use the split-feedback divider and PDLPF inside of a single loop has an open-loop gain of $H'_{\mathrm{OL}}(s)$ which can be expressed as

$$H'_{\mathrm{OL}}(s) = \frac{K_{\mathrm{PD}} K_{\mathrm{VCO}} H_{\mathrm{LF}}(s)}{s} \frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2}. \tag{5.5}$$

Here, we can see that $G_{\mathrm{CL,nest}}$ is multiplied with the entire expression unlike in the proposed architecture where it made up just a small portion of it. Fig. 5.5 shows the relationship between the bandwidth of the nested PLL and the phase margin of the main PLL for the conventional architecture with the split-feedback divider and nested PLL in a single loop (blue) and in the proposed architecture (red) when the bandwidth of the main PLL on its own is fixed to 8MHz. We can see that for the conventional architecture, the phase margin of the main PLL degrades significantly as the bandwidth of the nested PLL gets narrower, and a bandwidth of around 12MHz is required just to achieve a 0 degree phase margin. In fact, according to [79], the bandwidth of the nested PLL has to be around 2.5 times wider than that of the main PLL to achieve a positive phase margin when considering a ±10% variation in the loop bandwidths. Of course, a design with 0 degree margin is unacceptable, and the bandwidth of the nested PLL in a practical design should be around 5 to 6 times wider than that of the main PLL to obtain reasonable stability. In contrast, the phase margin of the main PLL in the proposed PLL remains constant regardless of the bandwidth of the nested PLL, proving that it allows the bandwidths of the PLLs to be optimized much more freely. As we will explain later on, in our implementation of this architecture, we

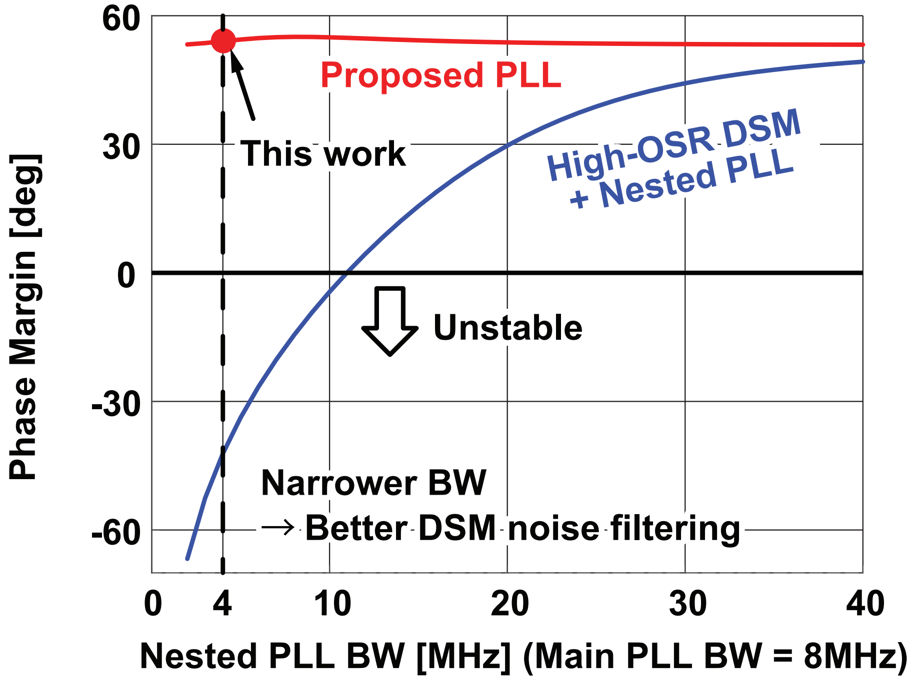


Figure 5.5: The relationship between the bandwidth of the nested PLL and the phase margin of the main PLL for the conventional architecture with the split-feedback divider and nested PLL in a single loop (blue) and in the proposed architecture (red) when the bandwidth of the main PLL on its own is fixed to 8MHz

employed a bandwidth of 4MHz for the nested PLL. For this case, we can also see from Fig. 5.5 that the proposed architecture achieves a good phase margin while the conventional architecture becomes completely unstable.

Next, we will look at the noise transfer functions. Based on the linear noise model shown in Fig. 5.4, the noise transfer functions can be calculated as shown below

$$H_{\mathrm{VCO,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCO}}} = \frac{1}{1+H_{\mathrm{OL}}(s)} \tag{5.6}$$

$$H_{\mathrm{PD,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{i_{\mathrm{nPD}}} = \frac{1}{K_{\mathrm{PD}}} \frac{1}{1-\frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2}} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \approx \frac{1}{K_{\mathrm{PD}}} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \tag{5.7}$$

$$H_{\mathrm{HM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nHM}}} = \frac{1}{1-\frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2}} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \approx \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \tag{5.8}$$

$$H_{\mathrm{NEST,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nPLL,nest}}} = \frac{1}{N_2} \frac{1}{1-\frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2}} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \approx \frac{1}{N_2} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \tag{5.9}$$

$$H_{\mathrm{DSM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nDSM}}} = \frac{G_{\mathrm{CL,nest}}(s)}{N_2} \frac{1}{1-\frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2}} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \approx \frac{G_{\mathrm{CL,nest}}(s)}{N_2} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \tag{5.10}$$

$$H_{\mathrm{EXT,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nPLL,ext}}} = -\frac{k}{1-\frac{G_{\mathrm{CL,nest}}(s)}{N_1 N_2}} \frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)} \approx -\frac{kH_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)}. \tag{5.11}$$

All of the approximations make use of the fact that the feedback gain is approximately 1.

The transfer function from the main VCO to the overall output is given by $H_{\mathrm{VCO,OUT}}(s)$ as shown in equation 5.6, and it is a high-pass characteristic similar to that in a conventional PLL.

The transfer function from the main PD to the overall output is given by $H_{\mathrm{PD,OUT}}(s)$ as shown in equation 5.7. We can see that it is a low-pass characteristic with a DC gain of around $1/K_{\mathrm{PD}}$. Similarly to the transfer function for the main PD in the dual-feedback PLL, the noise generated here does not get amplified by the feedback division ratio N.

The transfer function from the main HM to the overall output is given by $H_{\mathrm{HM,OUT}}(s)$ as shown in equation 5.8. We can see that it is a low-pass characteristic with a DC gain of around unity. This characteristic is also similar to the transfer function from the HM to the overall output in the dual-feedback PLL, meaning that the HM noise does not get amplified.

The transfer function from the nested PLL output to the overall output is given by $H_{\mathrm{NEST,OUT}}(s)$ as shown in equation 5.9. It is a low-pass characteristic with a DC gain of around $1/N_2$. This indicates that the noise from the nested PLL is suppressed to a negligible level just like we explained upon introducing this architecture. More specifically, assuming that the nested PLL does not perform any frequency amplification, the frequency of the nested PLL is N_1 times smaller than that of the overall output. Combining this with the fact that $H_{\mathrm{NEST,OUT}}(s)$ has a DC gain of $1/N_2$ and ignoring second order effects, we can say that the PN contribution from the nested PLL is suppressed roughly $N_1 N_2$ times more effectively than that of the main VCO. This is why the nested PLL in our proposed architecture can be implemented with a low noise/power overhead.

The transfer function from the DSM quantization noise to the overall output is given by $H_{\mathrm{DSM,OUT}}(s)$ as shown in equation 5.10. It is a cascaded low-pass characteristic consisting of the low-pass filtering from the nested PLL $G_{\mathrm{CL,nest}}(s)$ and that from the main PLL $\frac{H_{\mathrm{OL}}(s)}{1+H_{\mathrm{OL}}(s)}$, and a DC gain of $1/N_2$. We can see from this expression that the quantization noise is suppressed very effectively, from the cascaded low-pass characteristic and the attenuation by N_2 thanks to the HM-based feedback.

Finally, the transfer function from the external PLL output to the overall output is given by $H_{\mathrm{EXT,OUT}}(s)$ as shown in equation 5.11. It is a low-pass characteristic with a DC gain of around k. Similarly to in the dual-feedback PLL, the PN from the external PLL gets amplified by the ratio between f_{OUT} and f_{LO}, and then gets low-pass filtered by the main PLL.

Based on the loop transfer functions, we have analyzed the stability and noise characteristics of the proposed architecture. The proposed PLL achieves a better suppression of the DSM quantization noise with a low noise/power overhead, and otherwise it has similar characteristics to that of the dual-feedback PLL.

5.3.3 Comparison with Prior Arts

To make the benefits of our architecture clearer, it is useful to make comparisons with other architectures. Since our proposed architecture is a combination of the dual-feedback architecture and the split-feedback divider with PDLPF, we will first make comparisons with these standalone methods. Then we will also make comparisons with some other notable architectures such as the cascaded PLL and the DTC based PLL.

We first compare the proposed architecture with the dual-feedback PLL and the split-feedback divider with PDLPF in standalone operation. The comparison will be made in terms of quantization noise suppression. As we have already seen, the proposed architecture certainly achieves a much better suppression of the DSM quantization noise since it makes use of the noise suppression approaches from both methods. However, it is important to see how significant this improvement is in a practical design. Fig. 5.6 shows the circuits we are comparing, along with the calculated PN spectra. Fig. 5.6(a) shows block diagrams of the circuits being compared along with their frequency settings. The configuration used for the proposed architecture matches that of our actual implementation which will be explained in detail in the next section. A 940MHz clock for the HM is generated from the 1.88GHz external PLL output, which in-turn is generated from a 40MHz reference and a Sampling-Phase-Detector (SPD)-based PLL. The 940MHz clock signal is used for the 3rd order HM, which down-converts the 2.908GHz output signal to 88MHz. At the same time, the 2.908GHz is divided down by a factor of $N_1 = 11+\alpha$ with an MMD clocked by a 3rd order DSM to obtain a 264MHz signal containing quantization noise. The PDLPF multiplies this frequency by 4 to generate a 1.056GHz signal, and is then followed by a divide by $N_2 = 12$ to achieve an effective divide by 3. This configuration is used instead of a simple unity gain PLL and a divide by 3 to match our actual implementation[1], but this difference has no significant effect on the final result since the increase in N_2 and the increase in the nested PLL output frequency cancel each other out. The output frequency of the second stage divider is 88MHz which is the same as the HM output frequency, and any deviation between the two will be corrected by the feedback in the main PLL. The bandwidths of the external, nested, and main PLLs are 20, 4, and 8MHz respectively. The main and nested PLLs employ PFD/CP based type-II architectures. The dual-feedback PLL and the conventional PLL with split-feedback divider and nested PLL are configured to obtain a meaningful comparison with the proposed PLL. The dual-feedback PLL has the same configuration as the proposed PLL, except for the fact that the split-feedback divider and nested PLL are replaced by a single MMD and 3rd order DSM that divides down the 2.908GHz output signal to 88MHz directly. For the PLL with split-feedback divider and nested PLL, the HM-based feedback path is removed and an 88MHz signal is provided as a reference signal. The bandwidth of the main PLL is 8MHz, which is the same as that of the proposed PLL, but the nested PLL bandwidth is widened to 48MHz. This is necessary to maintain the stability of the overall loop.

Both conventional architectures have limitations that the proposed PLL solves. As was discussed in chapter 4, the output frequency of the HM cannot be made too large relative to the clock frequency for the S/H in the HM to avoid making the filtering requirements too high. At the same time, the S/H clock frequency itself cannot be made excessively large because for ring VCOs, the noise to power trade-off degrades substantially at higher frequencies due to the lowered number of stages that can be used. This means that for the dual-feedback PLL, the noise shaping frequency of the DSM is limited which in turn means that the loop bandwidth cannot be made very wide. The proposed PLL also offers significant advantages compared to the PLL with split-feedback divider and nested PLL filter. Firstly, thanks to the avoidance of noise amplification due to HM-based feedback, both the quantization noise from the DSM and the phase noise from the nested PLL are greatly suppressed. For the proposed architecture, the noise from the nested

[1] A ring VCO operating at around 1GHz was already available beforehand, so it was used instead of designing a new 264MHz ring VCO to save design effort.

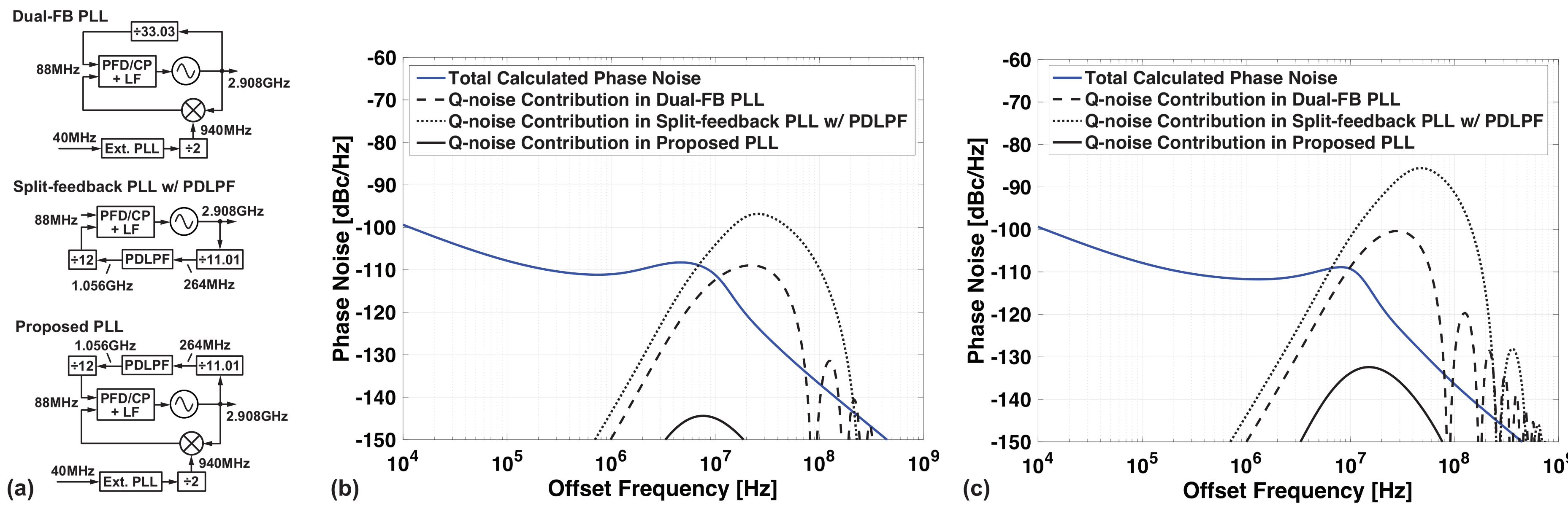


Figure 5.6: (a) A block diagram of the circuits being compared, (b) the calculated PN spectra of our design with the proposed architecture and the DSM PN contribution in the conventional architectures, and (c) the same calculated PN spectra expect with double the loop bandwidth for all architectures

PLL and the DSM are attenuated by a factor of N_2 as can be seen from equations 5.9 and 5.10. On the other hand, for the single-loop topology [79], both the DSM quantization noise and the nested PLL phase noise are amplified by a factor of N_1 because the feedback gain of the first stage divider is $\frac{1}{N_1}$. As a result, the proposed architecture achieves $N_1 N_2$ times better suppression of both the DSM quantization noise and the nested PLL phase noise. Another significant advantage of the proposed method is that the nested PLL has a negligible effect on the overall open-loop gain, and its bandwidth can be chosen independently of the main loop characteristics. Fig. 5.5 shows the relationship between the bandwidth of the nested PLL and the phase margin of the main PLL when the bandwidth of the main PLL is fixed to 8MHz. For the conventional method with just the split-feedback divider and nested PLL, the phase margin degrades substantially for lower bandwidths and around 12MHz bandwidth, which corresponds to 1.5 times that of the main loop, is required just to achieve a 0 degree phase margin. This factor can increase to 5 or 6 if we wish to obtain a more practical phase margin. Meanwhile, the proposed method suffers from no such degradation for the phase margin, and the nested PLL bandwidth can be chosen without worrying about the overall loop stability. A potential concern here is that a narrower nested PLL bandwidth would increase the nested VCO noise contribution and reduce the benefit of the proposed PLL. However, this will not be the case with an appropriate choice of the loop bandwidth. The nested PLL bandwidth does not have to be extremely narrow, and a bandwidth similar to or slightly narrower than that of the main PLL is enough to enhance the filtering of the quantization noise. If we combine this with the fact that the nested PLL noise contribution is attenuated by a factor of N_2 as opposed to the conventional architecture where it is amplified by N_1, we can conclude that with an appropriate loop bandwidth, the nested PLL can improve the quantization noise suppression while still contributing negligible noise to the overall output.

Fig. 5.6(b) shows the calculation results, where we show the calculated output phase noise of our PLL, along with the calculated phase noise contributions from the DSM quantization noise in the dual-feedback PLL, the PLL with the split-feedback divider and nested PLL filter, and the proposed PLL. For the two conventional PLLs, the noise degradation expected from the quantization noise greatly exceeds the total phase noise achieved in our design, while the proposed architecture suppresses the quantization noise to a negligible level. Furthermore, although the main PLL bandwidth is limited to around 1/10 of the HM output frequency because we are assuming a PFD/CP-based Type-II architecture, the loop bandwidth can be made even wider if we employ a Sampling-Phase-Detector (SPD)-based Type-I architecture, in which case the gap between the two cases will be even larger.

Fig. 5.6(c) shows a similar phase noise comparison where we extended the main PLL bandwidth to 16MHz (around 1/5 of the main PLL input frequency) and the nested PLL bandwidth in the proposed case to 8MHz. The nested PLL bandwidth in the conventional PLL was extended to 96MHz to maintain the bandwidth ratio of 6 required for stability. The noise degradation for all the architectures are even worse than before due to the reduced low-pass filtering of the quantization noise from the main and nested PLLs, but for the proposed PLL, it is still negligible compared to the overall phase noise.

Another architecture worth mentioning is the cascaded PLL, where an integer-N PLL is used as the first stage to provide a high frequency reference to increase the noise shaping frequency for the quantization noise in the second stage fractional-N PLL. In terms of quantization noise suppression, this method is similar to the split-feedback-divider-based PLL with PDLPF [79] because both architectures rely solely on increasing the noise shaping frequency to suppress the quantization noise although there are some other differences such as the absence of loop stability issues in the cascaded PLL and some extra noise filtering from the nested PLL in the split-feedback-divider-based PLL. This means that a similar noise degradation to that seen in Fig. 5.6(b) and (c) is expected in a cascaded PLL and the advantages of the proposed method still hold. The reader may also refer to a comparison between the cascaded PLL and the split-feedback-divider-based PLL given in [79].

Finally, it is important to further elaborate on the pros and cons of the proposed PLL when compared to the DTC-based PLL. The comparison will be made in terms of the phase noise and the settling time. The total settling time of the proposed PLL will be much less than that of the DTC-based PLL because the locking process only consists of the loop transient and time consuming processes such as gain and linearity calibration are not required. A potential concern here is that because the proposed architecture makes use of 3 PLLs, the overall locking time will increase because the individual PLLs must lock in order. This is not the case however, since there will be a large overlap between the 3 locking transients, and the overall locking time will be dominated by that of the slowest loop. To demonstrate this and also obtain an estimate of the worst case locking time, we performed behavioral simulations of the proposed PLL as well as the 3 individual PLLs, and compared the locking time of the proposed PLL with the sum of the locking times of the individual PLLs. The initial frequency error from the desired values were set to 100, 100, and 50MHz for the main, external, and nested PLLs respectively, which is the largest frequency error when assuming that the coarse tuning code for the VCOs are set to the correct values. Fig. 5.7 shows the locking transients along with block diagrams of the corresponding circuits. The locking time of the overall PLL within a $\pm$2MHz band error is 460ns, while the simple sum of the locking times of the individual PLLs within the same band error is 1.11μs. This shows that the degradation in locking time due to the use of 3 PLLs is small. Furthermore, the locking time of 460ns is significantly short compared to the reported locking times of the DTC based architectures, whose calibration can take tens of μs to even ms in the worst case to settle [40, 41, 42, 10, 11, 12].

As for the jitter power performance, both the proposed PLL and the DTC-based PLL suppress the quantization noise to a negligible level, and the proposed PLL suffers from the jitter and power contribution from the extra loops while the DTC-based PLL suffers from that of the DTC. The DTC contributes white noise at the PLL input whose noise is in a trade-off relationship with the power consumption. An optimization can be made based on the increase in in-band phase noise and the penalty in power consumption. For the proposed PLL, the overhead compared to a single loop PLL comes from having one extra VCO, since the nested PLL and the main PLL PD contribute negligible noise and power thanks to the HM-based feedback that avoids their noise being amplified. The actual effect of having an extra VCO depends on the bandwidths of the external and main PLLs. Assuming that the bandwidths are properly chosen to minimize the overall phase noise, if the bandwidth of one of the PLLs ends up much wider than that of the other, the overall noise and power will be dominated by the PLL with the narrower bandwidth and the penalty compared to the single loop case will be small. On the other hand, if the two PLLs end up having a similar bandwidth, the noise and power consumption will be degraded.

To summarize, both the DTC-based PLL and the proposed PLL suppress the quantization noise to a negligible level, but the proposed PLL achieves a faster locking time because it avoids calibration. The DTC-based PLL suffers from noise and/or power contribution from the DTC while the proposed PLL suffers from noise and/or power contribution from one extra VCO. The extent to which this extra VCO degrades the overall performance depends on the loop bandwidths.

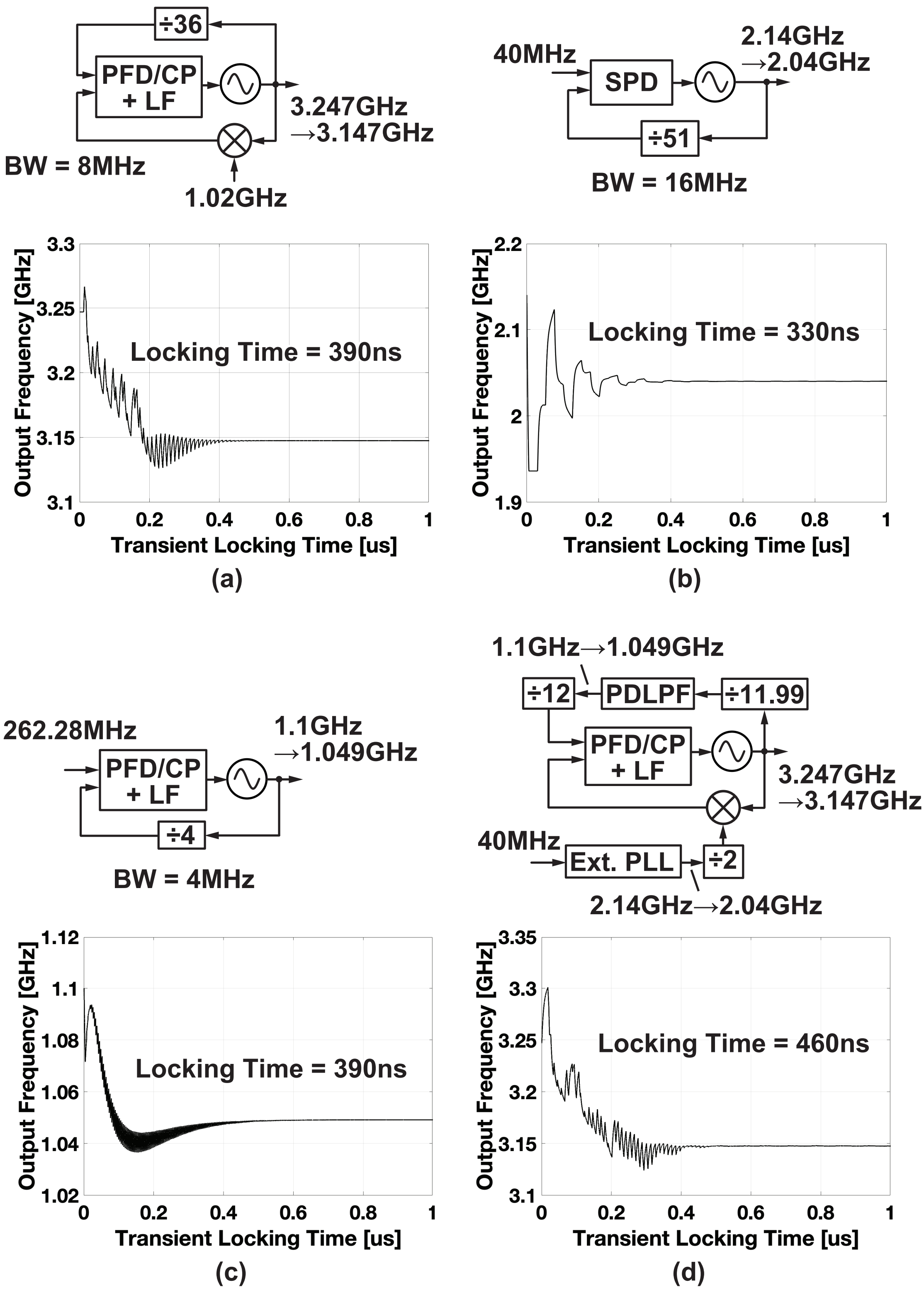


Figure 5.7: The simulated locking transients and corresponding block diagrams of the (a) standalone main PLL (b) standalone external PLL (c) standalone nested PLL and (d) overall PLL

In conclusion, our proposed architecture has distinct advantages over both the HM-based dual-feedback architecture and the split feedback divider with nested PLL based filter. This shows that by allowing the utilization of multiple approaches for quantization noise suppression, the proposed PLL is indeed effective for improving phase noise performance.

5.4 Implementation and Measurement Results

Based on the proposed architecture, we have implemented and measured a prototype chip to demonstrate the effectiveness of the quantization noise suppression scheme. In this section, we will give a detailed explanation of the implementation and the measurement results that were obtained.

5.4.1 Circuit Implementation Details

Fig. 5.8 shows the details of our design. The external auxiliary PLL employs a type-I SPD-based architecture to increase the PD gain and achieve a wide loop bandwidth. The nested and main PLLs employ a type-II PFD/CP based architecture. As was explained in the previous section, from a jitter/power performance stand point, it is preferable to use a SPD based architecture for the main PLL as well so that the loop bandwidth can be extended to its limit. However, this would require a coarse Frequency Locked Loop (FLL) since the frequency variation of the feedback signal from the HM-based path is much larger than that in a conventional PLL due to the direct down-conversion operation. As such, we settled for the somewhat limited bandwidth offered by the PFD/CP architecture, which still allows a wide enough bandwidth to demonstrate the effectiveness of the proposed method while being realizable with less design complexity. All 3 PLLs use VCOs based on differential ring oscillators. The DSM for the MMD is of the 3rd order, and its operating frequency is between 230 and 314MHz. The choice of this frequency is based on the fact that while a higher DSM frequency generally leads to improved noise shaping, it cannot be so high that the DSM fails to operate correctly or consumes too much power. Techniques such as TSPC based logic and the split-bus DSM architecture can be employed to increase the DSM frequency further[57].

The detailed diagram of the external auxiliary PLL is shown in Fig. 5.9. It is based on a simple type-I architecture using a two stage S/H circuit to achieve a wide bandwidth. An SPD is used instead of a Sub-Sampling PD (SSPD) to achieve robust locking, although in a similar situation to the main PLL, a SSPD with a FLL can be used if we wish to further reduce the noise and power consumption. The slope generator is based on an inverter with a tunable output resistance. It is worth noting that while it is not a focus of our work, the PVT variation of the SPD gain can have an impact on the overall performance of the proposed PLL. Techniques such as the one recently introduced in [80] can be used to mitigate this issue.

The ring VCOs in the main, external, and nested PLLs are differential ring VCOs with 4, 4, and 3 stages, respectively. All 3 VCOs are based on current-starved delay cells similar to the one used in [81] as shown in Fig. 5.10. The CPs in the main and nested PLLs employ cascode current sources with a complementary current path to suppress on/off transients. The detailed PFD/CP circuit diagram is shown in Fig. 5.11. The HM circuit is shown in Fig. 5.12. The S/H is realized using a simple two-stage sampler, based on an architecture similar to that in the SPD of the external PLL. This S/H is followed by a 4th-order LPF to suppress the unwanted high frequency tones at the S/H output [76].

Fig. 5.13 shows the chip micrograph along with the power breakdown. The design was fabricated in a 65nm bulk CMOS process, and the layout occupies a core area of 0.112mm^2. The

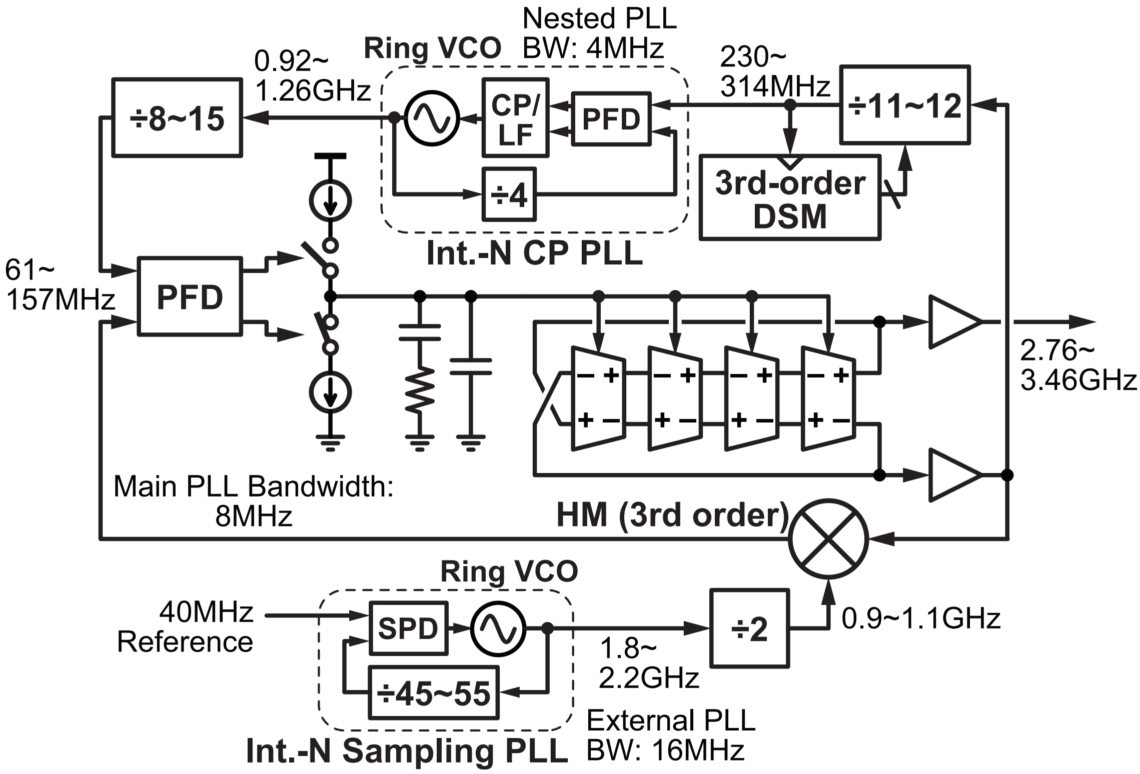


Figure 5.8: The details of the overall circuit implementation

total power consumption is 15.38mW (excluding the input and output buffers), of which just 0.87mW (5.7%) is consumed by the nested PLL. This shows that the split-feedback divider with PDLPF in our design can be implemented with a low power overhead.

The circuit was measured in a chip-on-board configuration. All the supply voltages have a nominal value of 1.2V, and are generated by on-board LDOs whose supplies come from a single external regulated voltage generator. The 40MHz reference signal is generated by an external signal generator (R&S SMA 100B).

5.4.2 Measurement Results and Performance Comparison

Figs. 5.14 and 5.15 show the measured phase noise spectra of the external and nested PLLs in standalone operation, their calculated phase noise, and the measured phase noise of their respective VCOs in free-running mode[2]. Reference spurs at 40MHz and its 2nd harmonic frequency are observed in Fig. 5.14 for the external PLL. For the standalone operation of the nested PLL, a 264.35MHz reference signal was given from an external signal generator to match the operation frequency during the overall operation. In Fig. 5.14, a slight deviation can be seen between the calculated and measured phase noise of the external PLL at around 20MHz. This likely is due to the process variation of the SPD gain, but the effect on the overall phase noise is minimal as we will see in Fig. 5.17. As for the nested PLL shown in Fig. 5.15, the measured phase noise matches the calculated phase noise well.

Fig. 5.16 shows the measured phase noise in integer and fractional modes in the red and blue solid lines respectively where the carrier frequency is around 2.908GHz. The fractional part of the

[2]The phase noise spectra of the free-running VCOs only extend to 40MHz due to the limitation of our signal source analyzer (Keysight E5052B) in wide acquisition mode.

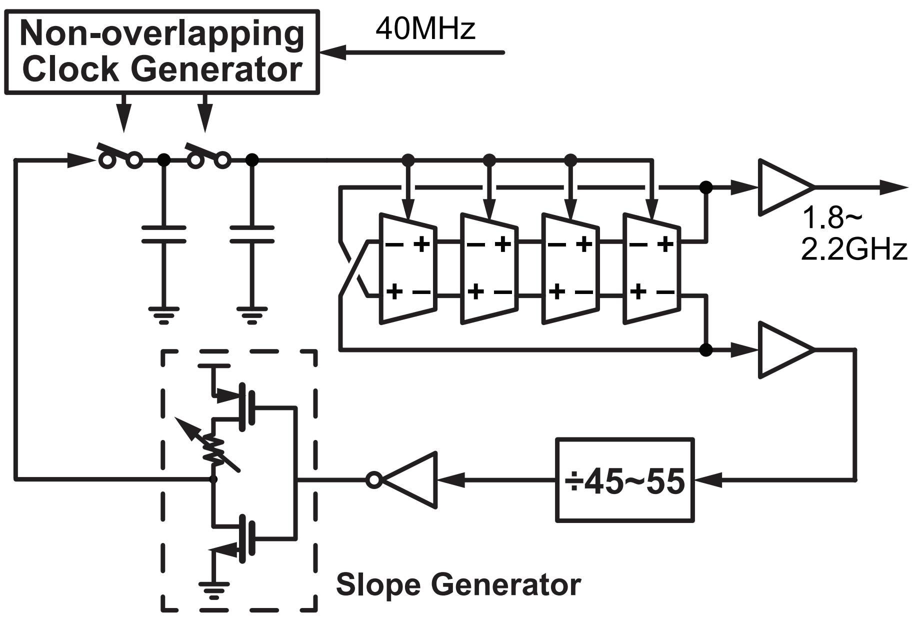


Figure 5.9: The details of the external PLL implementation

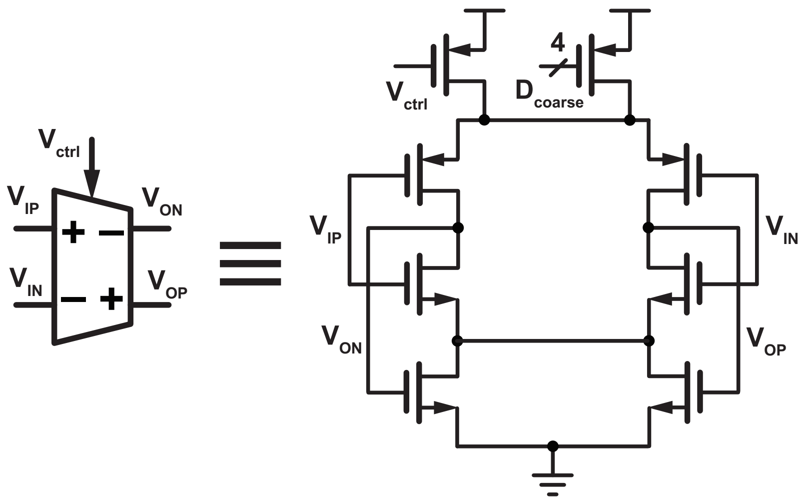


Figure 5.10: The architecture of the delay cell used in the VCOs

Frequency Control Word referred to as FCW_{FRAC} is equal to 2^{-16} for the fractional mode case. It also shows the expected noise contribution from the DSM in a dual feedback PLL, a PLL with split-feedback division and PDLPF, and a conventional fractional-N PLL in dashed, dotted, and dash-dotted black lines respectively assuming the same loop bandwidth and PFD/CP frequencies for these comparisons as that in our design. The expected noise contribution in the conventional architectures contain large peaks at high frequencies that are not observed in the measured phase noise. Since the difference in phase noise between the integer and fractional modes corresponds to the noise contribution from the DSM, this result shows that the proposed method successfully suppresses the quantization noise much more effectively than the prior arts. The expected DSM noise contribution in the proposed architecture is shown in the black solid line, and is well below the overall phase noise meaning that it leads to negligible noise degradation. The measured phase noise of the main VCO in free-running mode is also shown in grey[2]. The integrated jitter in integer mode is 838fs, while that in fractional mode is 869fs. The worst case jitter over the entire tuning range is 1.03ps, and the jitter-power FoM varies by around 2dB over this range.

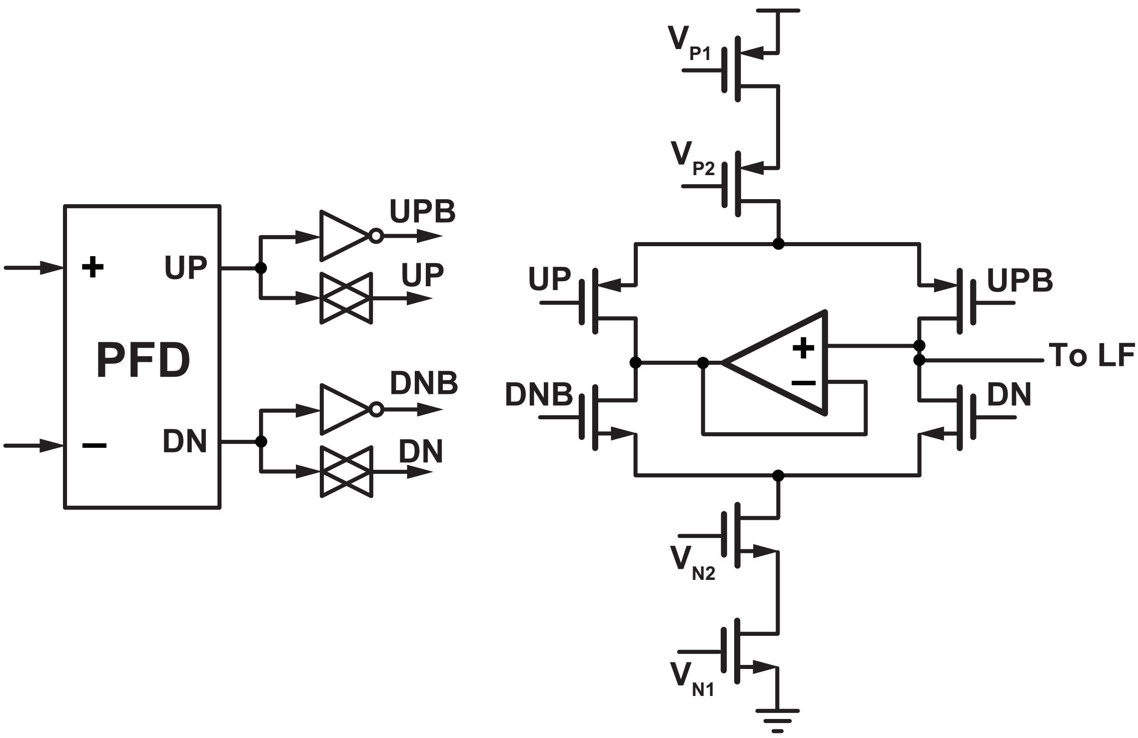


Figure 5.11: The circuit diagram of the CP employing cascode current sources and dummy current paths

Fig. 5.17 shows the measured phase noise in fractional mode along with the calculated overall phase noise and contributions from various blocks. The percentage contributed from each block to the overall Integrated Phase Noise (IPN) is also shown. We can see that the noise is mostly dominated by the main VCO and external PLL which contribute 42% and 54% of the IPN respectively, and the noise from other noise sources including the nested PLL and DSM are well suppressed with a total IPN contribution of less than 4%. It is worthy to note that the phase noise degradation due to peaking in the external PLL gets filtered by the main PLL and does not result in large performance degradation at the overall output. Also, the noise contribution from the main CP is suppressed to a negligible level thanks to the unity DC gain characteristic of the HM-based main loop. The bandwidth of the main PLL was slightly narrower than the design target of 8MHz, and the measurement and calculation results showed the best match when the main PLL bandwidth was set to around 5.5MHz. The calculation results in Figs. 5.16 and 5.17 are based on this adjusted bandwidth setting.

Fig. 5.18(a) shows the output spectrum of the PLL containing fractional spurs, and Fig. 5.18(b) shows a plot of the fractional spur values with respect to the corresponding FCW_{FRAC}. The worst case fractional spur is −49dBc. In Fig. 5.19(a), we can see an out-of-band spur at the external reference frequency of 40MHz. In addition, we can see in Fig. 5.19(b) an out-of-band spur at the main PLL input frequency of around 88MHz. All of the spurs achieve relatively low values thanks to a combination of cascaded low-pass filtering due to the use of multiple PLLs, and the avoidance of noise amplification due to the use of HM-based feedback. Specifically, the fractional spurs benefit from the avoidance of noise amplification, the reference spur at 40MHz benefits from the cascaded low-pass filtering from the external and main PLLs, and the spur at 88MHz benefits from the avoidance of noise amplification and the filtering from the main PLL. The spur values can be further improved based on the same techniques used to improve the spurs in single loop PLLs, such as adding dither in the DSM, linearizing the CP by adding an offset current, or employing

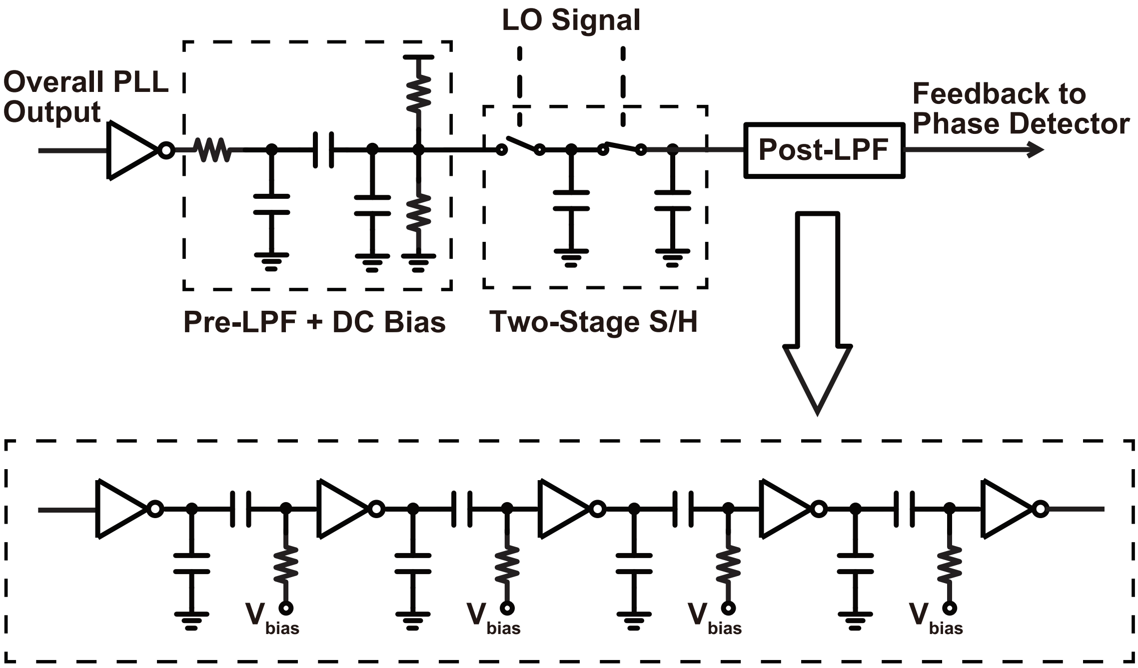


Figure 5.12: The circuit diagram of the HM based on a two-stage S/H

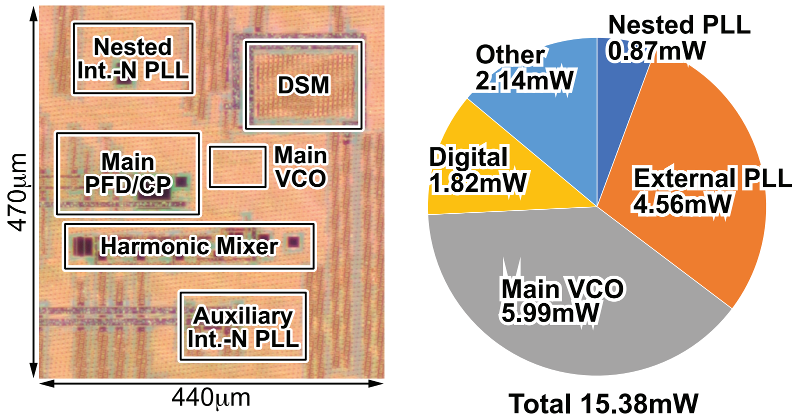


Figure 5.13: The chip micrograph along with the power consumption breakdown

a more linear PD in the first place such as an XOR PD. It should be noted that the unexpected tones at 4MHz and its harmonics seen in Fig. 5.19 are likely generated due to coupling between the DSM and the reference frequency blocks, since the frequency of the tones are always equal to the DSM frequency minus a multiple of the reference frequency, and the amplitude of the tones are sensitive to the reference and digital supplies. Therefore, they can be mitigated with a better substrate separation and possibly by separating the power supply and ground more carefully. The worst case coupling spurs are observed to be around −50dBc. We also observed some leakage from the external PLL output in the overall output spectrum causing around a −53dBc tone. This too can likely be suppressed by improving the substrate and power supply isolation.

It is also important to note that for the fractional spur, it benefits from the same avoidance of noise amplification as the quantization noise due to the HM-based feedback. This means that if we implemented a single loop fractional-N PLL with split-feedback division and PDLPF using the same circuit blocks, we would expect to obtain a fractional spur that is N times worse than the measured value. Since we did not implement a conventional single-loop fractional-N PLL with

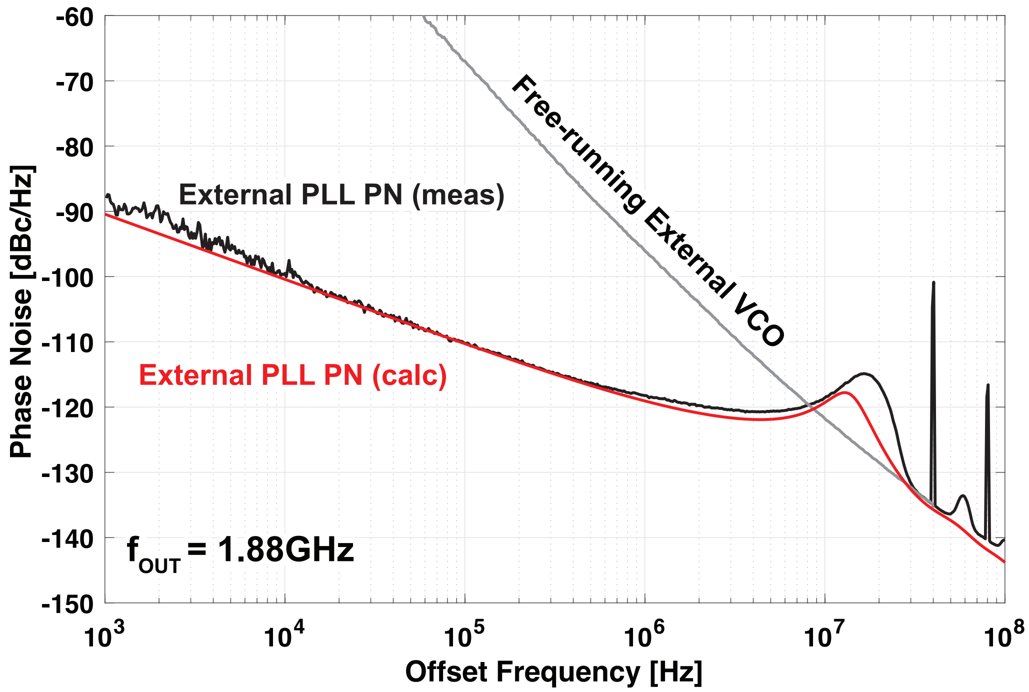


Figure 5.14: The measured output phase noise of the external PLL in standalone operation (black), the calculated phase noise of the external PLL (red), and the measured phase noise of the external VCO in free-running mode (grey)

split-feedback division and PDLPF, we made the comparison based on simulation. Fig. 5.20 shows the conceptual block diagram and the output spectrum containing fractional spurs for the (a) conventional single-loop architecture and (b) proposed architecture. Both PLLs have the same output frequency of around 3.2GHz, and the same division ratios for the split-feedback dividers. A reference frequency of around 100MHz is provided for the single-loop architecture, while for the proposed architecture, the output is down-converted by a 1.033GHz LO signal using a 3rd order HM. The bandwidths for the main and nested PLLs are set to the same values for both cases (1MHz and 10MHz respectively) to provide similar conditions and make the avoidance of spur amplification clearer. Non-linearity is realized by adding a 5% current mismatch to the CPs of both the main and nested PLLs. The resulting output spectrum shows a fractional spur value of -40dBc for the single-loop architecture and -70dBc for the proposed architecture. The improvement of 30dB corresponds to the avoidance of amplification since 20log(4*8.0018)$\approx$30.

Fig. 5.21 shows an example of the frequency transients of all the PLLs (black) measured at the same time, along with their smoothed out versions based on moving average filtering with an averaging factor of 50 (red). Here, the external PLL frequency changed from 1.88GHz to 2.04GHz, the nested PLL frequency changed from 1.057GHz to 1.049GHz, and the overall output frequency changed from 2.908GHz to 3.147GHz. The locking time within a $\pm$2MHz band of the desired output frequency is around 450ns, a value that is within the worst case locking time predicted based on the behavioral simulation. Once the coarse tuning codes of the VCOs are set to the correct values, this transient is the only thing that constitutes the locking process, and we can clearly see that this offers an advantage in locking time over calibration intensive architectures just like we demonstrated in the simulation.

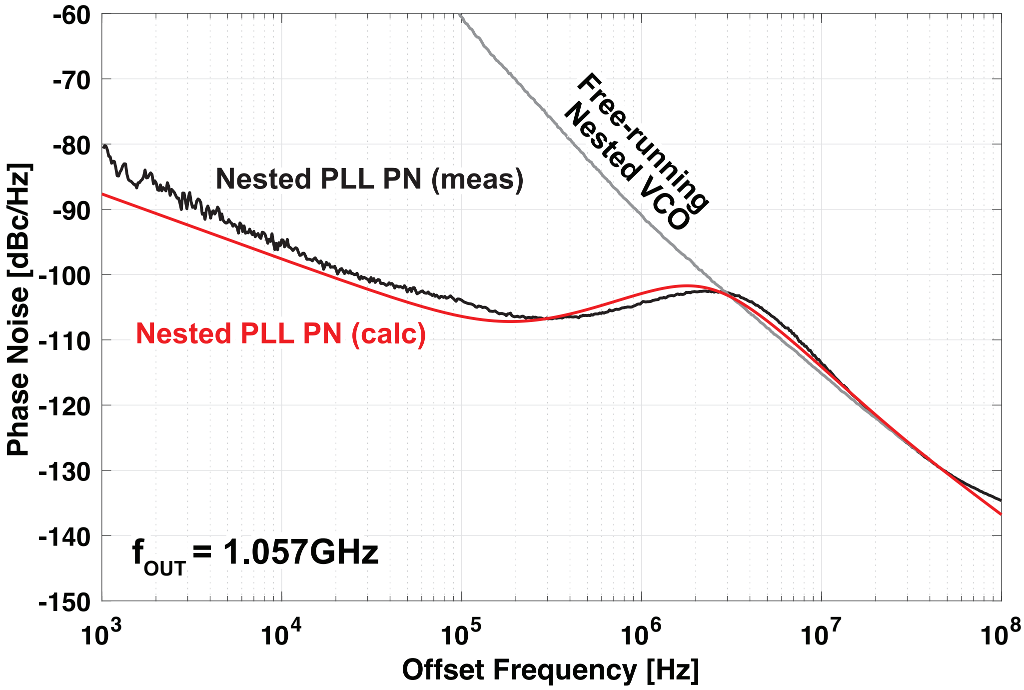


Figure 5.15: The measured phase noise of the nested PLL in standalone operation with a 264.35MHz external reference (black), the calculated phase noise of the nested PLL (red), and the measured phase noise of the nested VCO in free-running mode (grey)

Table I shows a comparison of our work with other state of the art ring-VCO-based fractional-N PLLs, fractional-N MDLLs, and also the works involving the HM-based dual-feedback architecture [59, 60] and the split-feedback + PDLPF inside a single-loop PLL [79]. Among the ring-VCO-based works that do not rely on background calibration [79, 57, 58, 56], our work achieves one of the best jitter-power FoMs thanks to its efficient quantization noise suppression mechanism. If we look at works that employ background calibration based on noise cancellation with DTCs [10, 43, 44, 11, 12], we can see that some of these works achieve a better FoM than that of our work. However, the advantage of calibration-free operation is significant in terms of reduced locking time and complexity. Furthermore, the performance gap can be closed further with an optimal architectural choice and further design optimization. One choice is to employ a type-I SPD based architecture instead of a type-II PFD/CP based architecture to allow a wider loop bandwidth and suppress the main VCO phase noise. In addition, there is more room for improvement in the performance of the VCOs used in the main and the external loops, as some recent state-of-the-art ring VCOs achieve around 8 dB better FoMs [57, 58].

The HM-based dual-feedback PLL [59, 60] achieves a better noise/power performance than our work, but is forced to use an LC-VCO due to the insufficient suppression of quantization noise. The work employing the split-feedback divider and PDLPF inside a single-loop PLL [79] has a much worse noise/power performance than our work, also due to the fact that it is forced to have a relatively narrow bandwidth because of the insufficient suppression of quantization noise. An estimation of the locking times based on the dominant factors such as the loop transient and calibration convergence is shown. We can see that our work achieves a far shorter locking time than the DTC-based architectures whose calibration takes hundreds of μs to even ms to converge [10, 11, 12]. Some of the works on DTC-based architectures did not report the convergence time [43, 44], but it is likely that the calibration convergence in those works also take in the order of

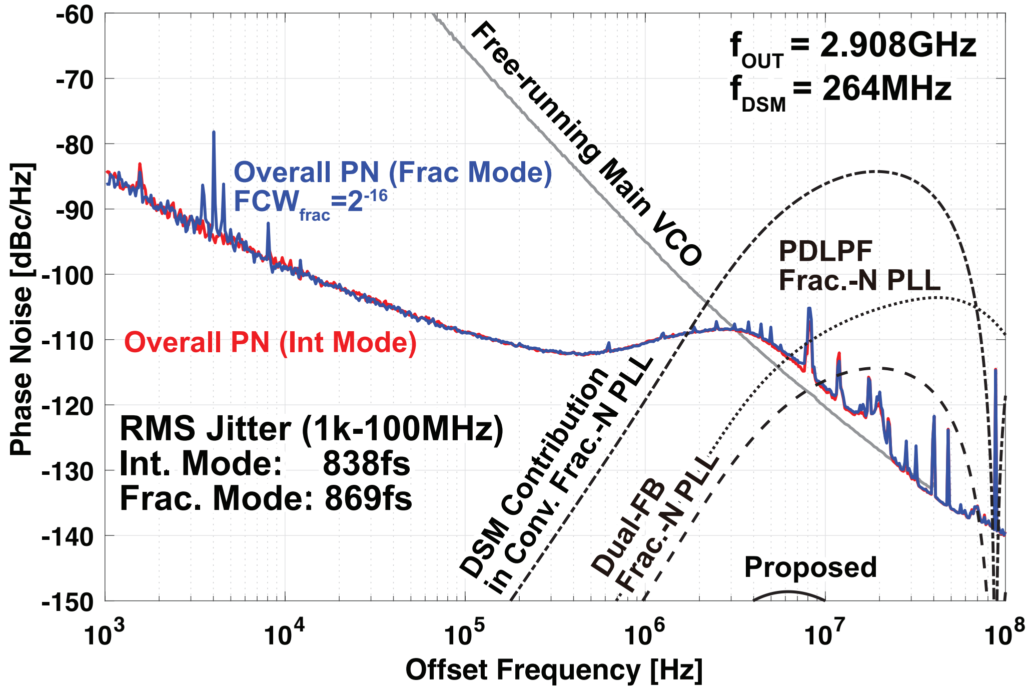


Figure 5.16: The measured output phase noise in integer (red) and fractional (blue) modes, the measured phase noise of the main VCO in free-running mode (grey), and the calculated contribution from the DSM quantization noise in the proposed PLL (solid black), dual-feedback fractional-N PLL (dashed black), fractional-N PLL using split feedback divider and PDLPF (dotted black), and single-loop fractional-N PLL (dash-dotted black)

tens of μs to ms since the general purpose and method of the calibration algorithms are the same. The works dealing with calibration-free architectures on the other hand [59, 79, 57, 58, 56] likely settle in the order of hundreds of ns to μs since their locking process mostly consists of only the loop transient like in our work. Finally, it is worth noting that the core area of our work is in line with those of the other ring-VCO-based PLLs, meaning that the use of multiple PLLs in our work does not ruin the advantage of low area for ring VCOs, because each VCO can be realized with sufficiently low area.

5.5 Analytical Performance Evaluation of Dual-Feedback PLL with PDLPF

Just like we did in section 3.5, we will analytically evaluate the dual-feedback PLL with split-feedback division and PDLPF. Similar settings will be used as the analysis in section 3.5, except we will assume a ring VCO instead of an LC-VCO, and we will include the case where the split-feedback divider and PDLPF is used inside the dual-feedback architecture. To account for the degraded PN of the ring VCO, we will make the tank quality factor equal to 1. For the split-feedback divider, the second stage divider has a division ratio of 6 meaning that the noise shaping frequency is increased to 6*103=618MHz. A conceptual diagram of the circuits being compared are shown in Fig. 5.22.

The obtained results are shown in Fig. 5.23. We can see that the DTC-based PLL and both dual-feedback PLLs perform considerably better than the conventional fractional-N PLL and worse than the conventional integer-N PLL. Furthermore, we can see that the performance of the

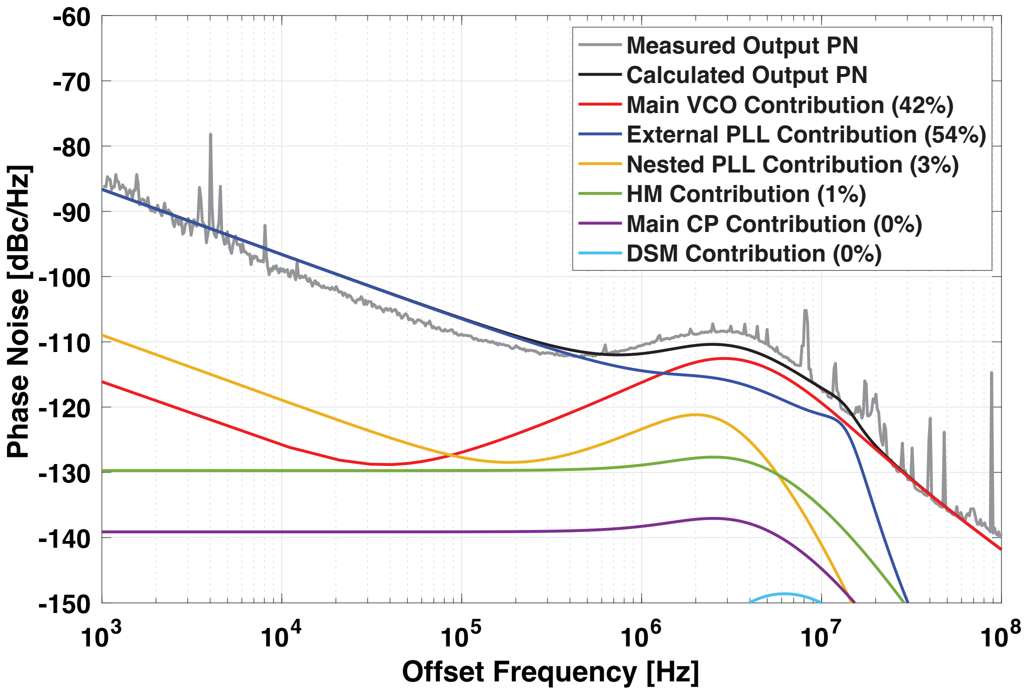


Figure 5.17: The measured output phase noise (grey) along with the calculated phase noise (black) and contributions from various blocks

dual-feedback PLL improves when the split-feedback division and PDLPF are employed, except at low power and high jitter design points where the minimum power consumption required by the 3 VCOs ends up degrading the performance. On the other hand, both of the dual-feedback PLLs perform worse than the DTC-based fractional-N PLL in this case. However, the gap is not so large especially if we employ the split-feedback division and PDLPF, and we of course have the added benefit of not requiring any calibration.

It is interesting to note the difference between this analysis where the HM-based PLL ends up performing worse than the DTC-based PLL, and the analysis performed where the use of an LC-VCO was assumed in chapter 3 where the dual-feedback PLL performed similar to, and even better than the DTC-based PLL in some conditions. This difference is due to the fact that since the ring VCO has a worse PN performance than the LC-VCO for the same power consumption, the optimal PLL bandwidth ends up being wider. This has a larger negative impact on the dual-feedback PLL than the DTC-based PLL because the extra noise source in the dual-feedback PLL is shaped quantization noise, while that in the DTC-based PLL is white noise with a flat profile.

5.6 Chapter Summary

In this chapter, we proposed a ring-VCO-based fractional-N PLL that employs a split-feedback divider with a nested-PLL-based PDLPF inside of the HM-based dual-feedback architecture. With the proposed architecture, we were able to achieve highly effective suppression of the quantization noise, and also solve various issues such as stability degradation and noise/power overhead that were present in the prior arts. Detailed analysis, comparisons with conventional architectures, and measurement results were given to demonstrate the feasibility of the proposed method and its effectiveness in improving the overall phase noise performance. Also, a comparison with other architectures was made using a similar hybrid numerical analysis as that in chapter 3.

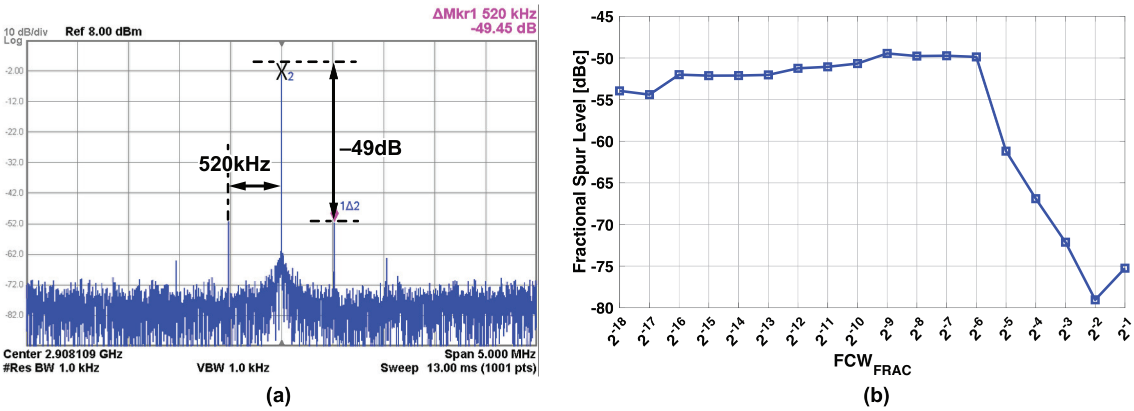


Figure 5.18: (a) The PLL output spectrum with low frequency fractional spurs ($FCW_{\mathrm{FRAC}} = 2^{-9}$) and (b) a plot of the relationship between FCW_{FRAC} and the fractional spur values

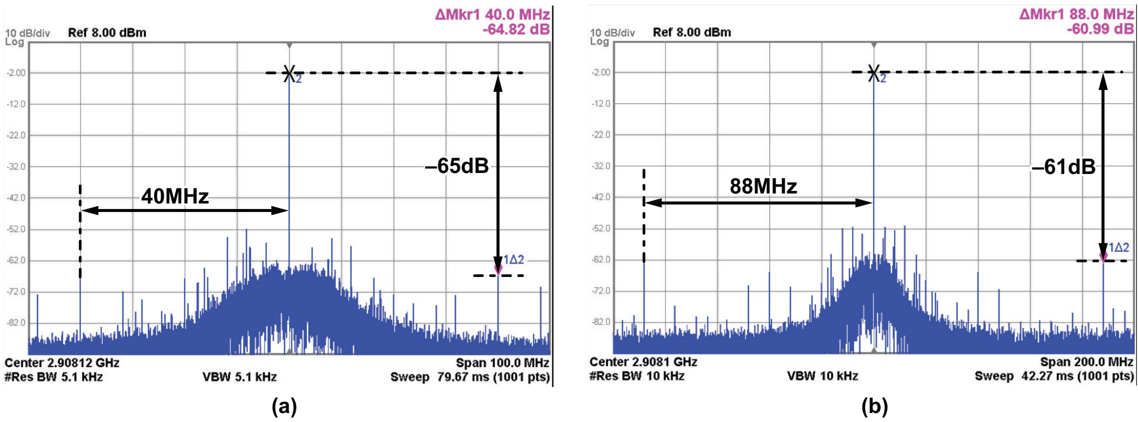


Figure 5.19: The PLL output spectrum containing the (a) overall reference spur at 40MHz and (b) out-of-band spur at the main PLL input frequency

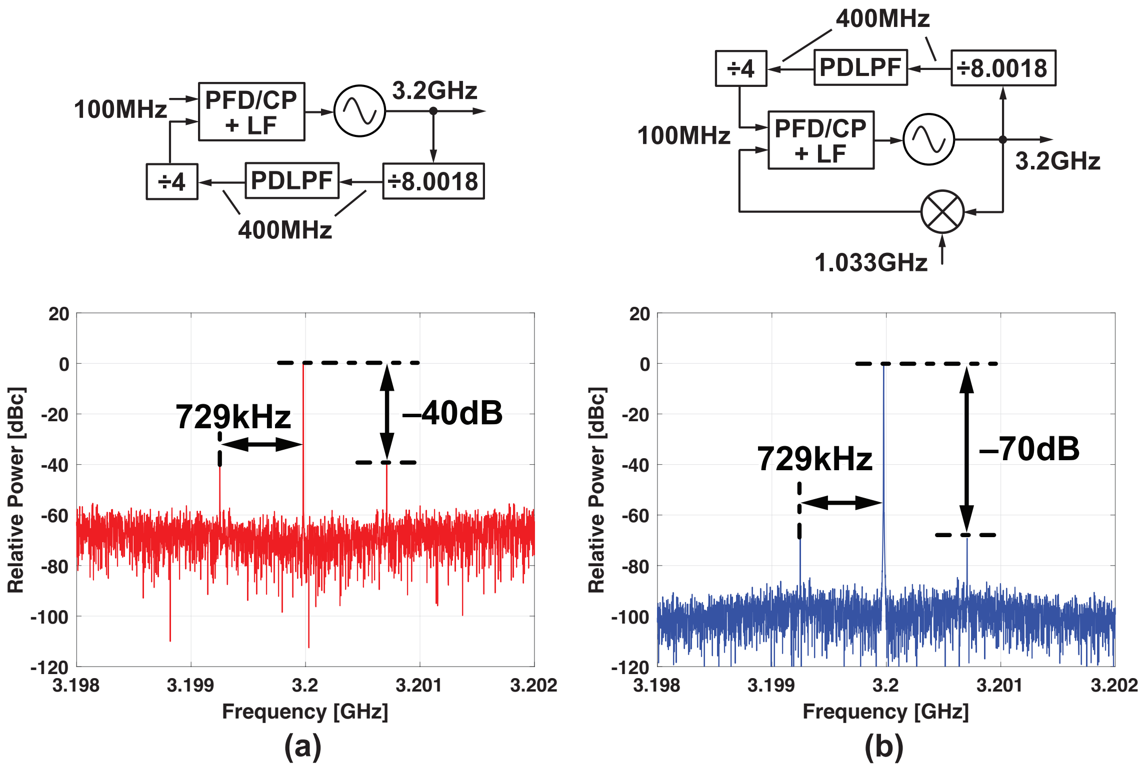


Figure 5.20: The conceptual block diagram and output spectrum containing fractional spurs for the (a) single-loop fractional-N PLL employing split-feedback division and PDLPF and (b) proposed architecture

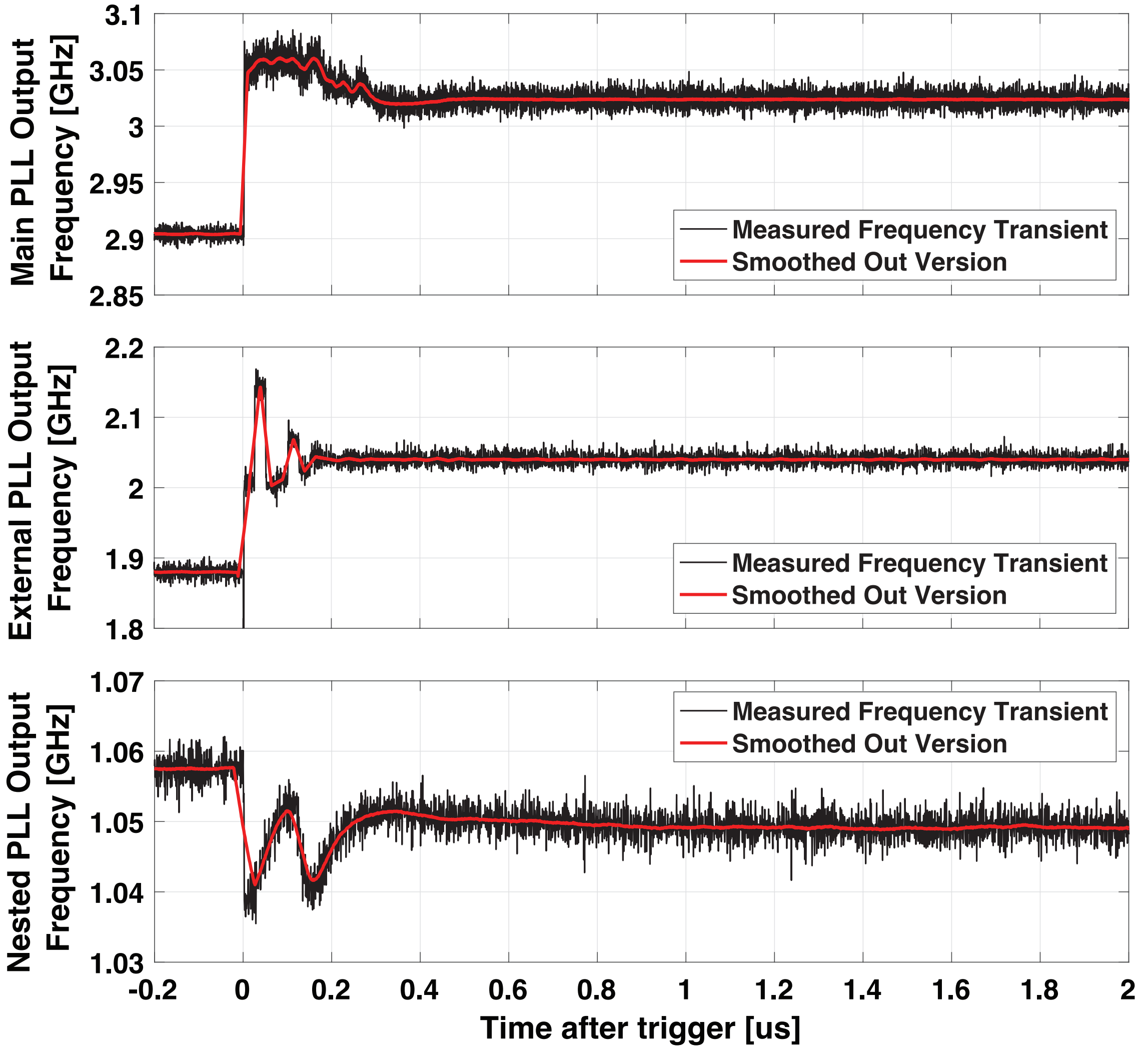


Figure 5.21: The measured frequency transients of the main, external, and nested PLLs (black) along with their smoothed out versions based on moving average filtering with an averaging factor of 50 (red)

Table 5.1: Performance Comparison

	This Work	[34] Park JSSC'22	[35] Zhang JSSC'22	[36] Elkholy JSSC'16	[37] Zhang JSSC'23	[38] Santiccioli JSSC'19		[53] Osada SSCL'20	[67] Park JSSC'12		[51] Kong JSSC'17	[52] Kong JSSC'18	[50] Zhang JSSC'20
Method	Dual-Feedback + Split Feedback + PDLPF	DTC w/ 1/8 range reduction	MDLL w/ DTC + TP Calibration	DTC + extended range MMD	MDLL w/ Injection Error Scrambling	MDLL w/ DTC range reduction		HM-based Dual-Feedback	Split-Feedback + PDLPF		Cascaded PLL w/ DSM + Time-Domain FIR	DSM + Time-Domain FIR	DSM + space-time averaging
VCO Topology	Ring	Ring	Ring	Ring	Ring	Ring		LC	Ring	LC	Ring	Ring	Ring
Calibration Required?	No	Yes	Yes	Yes	Yes	Yes		No	No	No	No	No	No
Ref. Freq. [MHz]	40	100	50	50	50	100		~800	32	60	22.6	22.6	50
Output Freq. [GHz]	2.9	5.2	1.5	5.2	1.5	1.65	3.05	3.5	2.4	2.4	2.4	2.4	2.4
BW [MHz]	5.5	70	10*	5	20*	30*		0.8	2	2	10	5.7	2.8
RMS Jitter [fs] (Integ. Range)	869 (1k-100M)	188 (1k-30M)	1690 (10k-10M)	1860 (10k-100M)	800/1000 (10k-10M/10k-30M)	397 (30k-30M)	376 (30k-30M)	503 (10k-100M)	4065**	3227**	1680 (10k-50M)	1500 (10k-50M)	2260 (1k-100M)
Ref. Spur [dBc]	–65/–61***	–64	–43	–44	–58	–56	–54.5	–69	–60	–54.7	–70	–60	–67
Frac. Spur [dBc]	–49	–59	–52	–42	–67	–51.5	54.8	–65	–37	–45	–52.5	–41	–47
Power [mW]	15.38	15.67	11.95	4	13.56	2.5	3.35	6.9	15.2	9.6	6.4	10	4.85
FoM† [dB]	–229.4	–242.6	–224.8	–228.5	–230.6/–228.7	–244	–243.2	–237.6	–216	–220	–227.4	–226.5	–226.1
Locking Time [us]	0.46††	500†††	N/A	N/A	2000†††	700†††		N/A	N/A		N/A	N/A	N/A
Core Area [mm^2]	0.112	0.139	0.18	0.084	0.23	0.0275		0.24	0.17	0.46	0.03	0.096	0.086
Process Node [nm]	65	65	65	65	65	65		65	130		45	45	40

*Estimated from PN spectrum **Calculated from Power Consumption and FoM ***Spur at Main PLL Input Frequency
†FoM = 10log10[(Jitter/1s)2 (Power/1mW)] ††Loop Transient Time †††Calibration Convergence Time

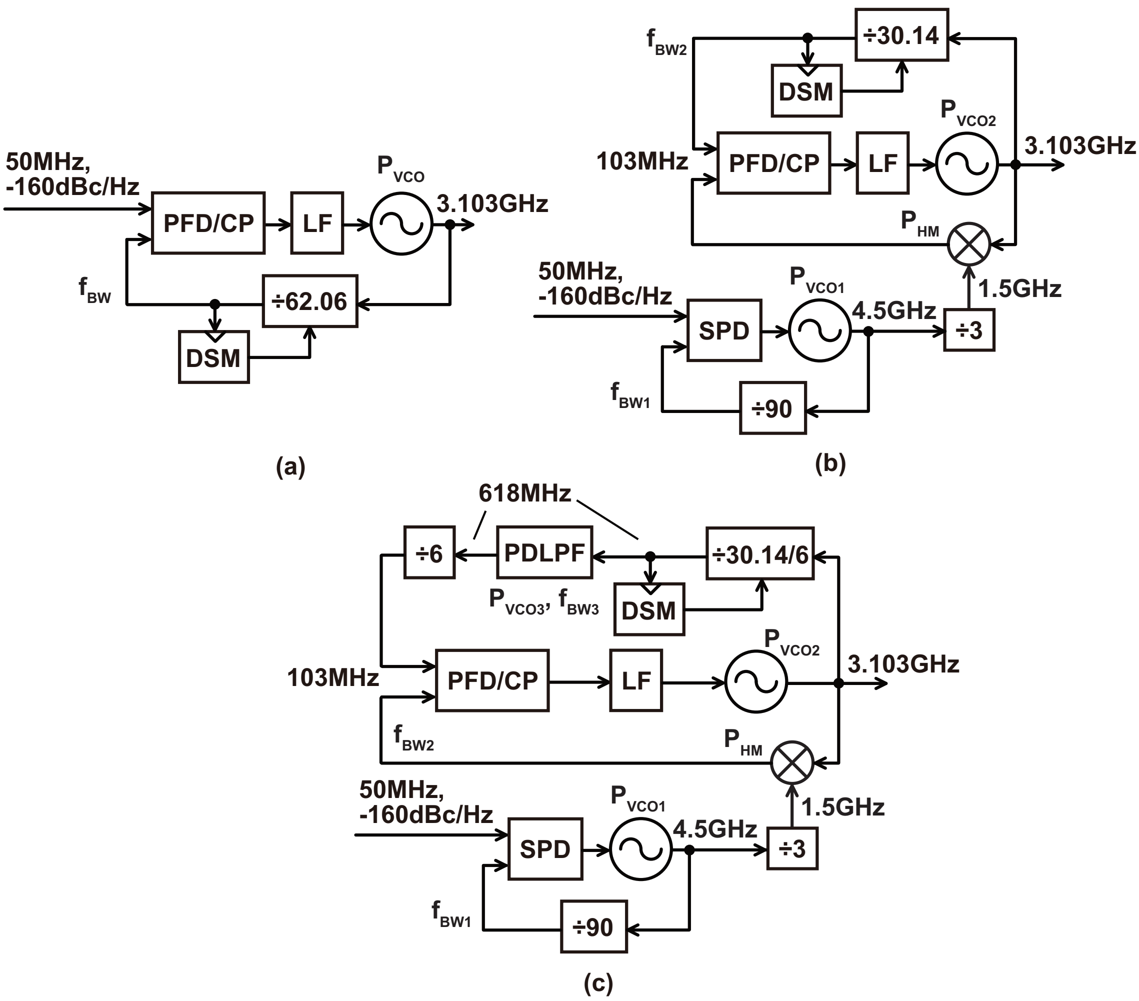


Figure 5.22: A block diagram of the (a) conventional fractional-N PLL (b) the dual-feedback PLL and (c) the dual-feedback PLL with PDLPF used for the comparison

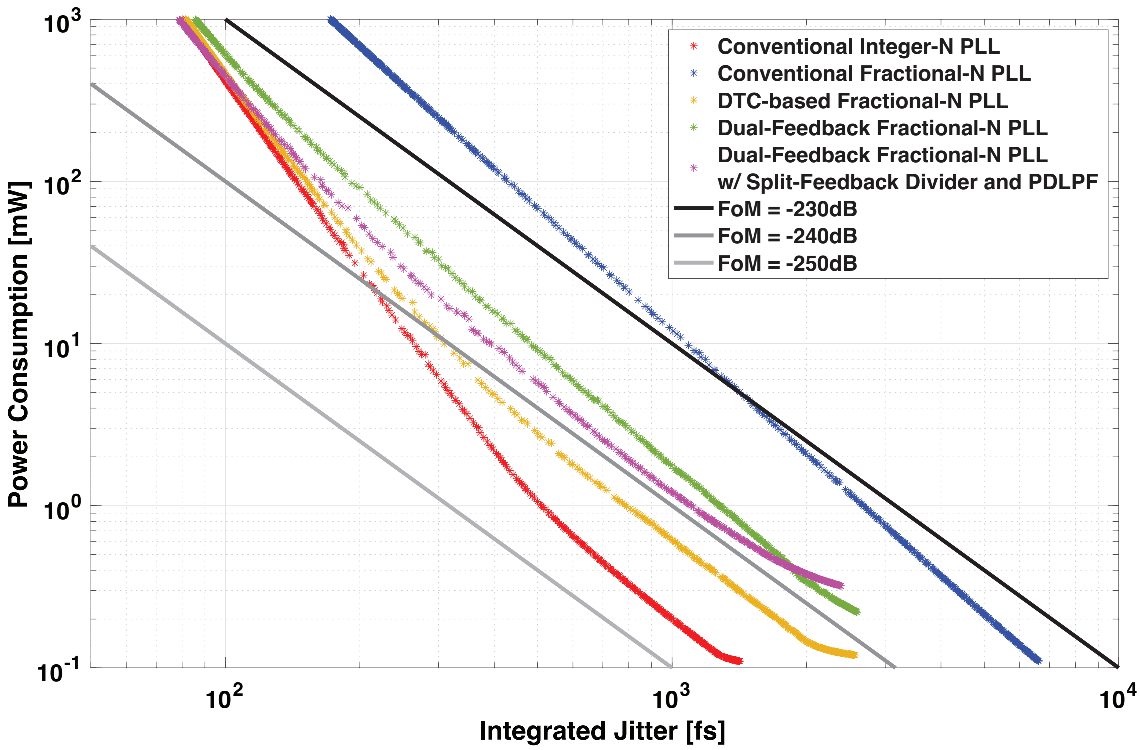


Figure 5.23: The relationship between the jitter and power under optimal design for the conventional integer-N PLL (red), conventional fractional-N PLL (blue), DTC-based fractional-N PLL (yellow), dual-feedback PLL (green), and the dual-feedback PLL employing split-feedback division and PDLPF (pink) where ring VCOs are assumed for all the PLLs

Chapter 6

Dual-Feedback PLL with Feed-forward Noise Cancellation

We have seen throughout this dissertation that HM-based PLLs are an effective architecture for suppressing the quantization noise from the DSM and fractional spurs. However, an obvious issue that remains is the need to use multiple PLLs. The dual-feedback PLL is slightly better than the triple-loop PLL in this sense because it does not require the auxiliary fractional-N PLL, but it still requires two PLLs in total since we need one PLL to generate the output signal and one more to generate the LO for the HM. In this chapter, we propose a feed-forward noise cancellation technique to mitigate this overhead from using an extra PLL.

6.1 Fundamental Issues in Multi-Loop PLLs

The power and/or noise overhead in frequency synthesizers that use multiple PLLs can be significant. If we look at a conventional single-loop fractional-N PLL, the sources of noise that can potentially degrade the power and/or noise performance of the PLL are the equivalent reference referred PN, the VCO PN, and the DSM quantization noise. What about for the dual-feedback PLL? Assuming that the equivalent reference referred PN of the main PLL and the HM PN are negligible since they don't get amplified, the main sources of PN here are the equivalent reference referred PN of the external PLL, the PN from the two VCOs, and the DSM quantization noise. The quantization noise is reduced thanks to the avoidance of noise amplification, but in return we have the PN contribution from an extra VCO. The goal of this work is to realize a dual-feedback PLL with negligible PN overhead from having an extra VCO.

6.2 Noise Suppression Techniques

Here, we will discuss some conventional techniques that can be used to suppress the PN and power overhead from the external PLL, along with their pros and cons.

6.2.1 Wide Bandwidth PLLs

A straightforward way to suppress the VCO PN is to use a wide bandwidth. As shown in Fig. 6.1, the main VCO and PLL can be designed so that it meets the jitter requirement with an optimal bandwidth setting, while the external PLL can then be designed with a much lower power consumption and much wider bandwidth. Since the external PLL PN experiences low-pass filtering

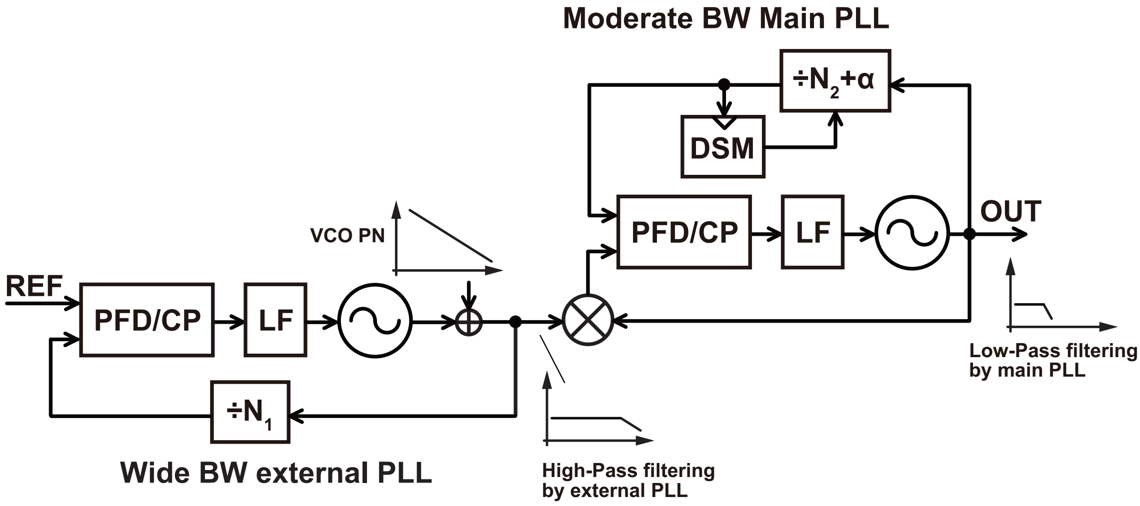


Figure 6.1: The use of a wide bandwidth external PLL and low-pass filtering from the main PLL to suppress the external VCO PN

from the main PLL, the high frequency PN from the external PLL will be suppressed, and so if we employ bandwidth for the external PLL that is sufficiently wider than that of the main PLL, the external VCO PN can be suppressed to a negligible level, and the overall PN will be dominated by that from the equivalent reference referred PN and the main VCO, meaning that the performance degradation compared to a single-loop integer-N PLL can actually be quite low. This simple method can be quite effective in some cases, allowing the dual-feedback PLL to achieve a jitter power performance close to that of an ideal single-loop solution. However, there are also some significant limitations. Making use of the low-pass filtering from the main PLL requires that the external PLL bandwidth is much wider than that of the main PLL. This is not always feasible, because it can be difficult to achieve such a wide bandwidth for the external PLL. High frequency references can be costly to realize, and it can also give rise to some difficulties in the frequency plan of a HM-based PLL. The output frequency of a HM-based PLL is given by $f_{\mathrm{OUT}} = kf_{\mathrm{SH}} + f_{\mathrm{HM}}$, and coarse frequency tuning can be done by changing f_{SH} while fine tuning can be done by changing f_{HM}. If the reference frequency is high, f_{HM} must cover a very wide range, making the frequency planning potentially difficult. If we cannot make the reference frequency large, a wide bandwidth will constitute a large portion of the reference frequency i.e. $f_{\mathrm{BW}}/f_{\mathrm{REF}}$ will be very large. Besides the hard limit of $f_{\mathrm{BW}} \leq 0.5 f_{\mathrm{REF}}$ unless we use techniques such as reference oversampling, achieving a large bandwidth to reference frequency ratio entails several difficulties. Firstly, if we design the bandwidth to be on the verge of the possible limit, the output PN can exhibit large peaking or even go unstable in the presence of loop gain variation over process. Secondly, a wide bandwidth PLL is often realized based on a type-I architecture in order to avoid issues such as gardner's limit. However, since a type-I architecture only has one pole at DC, it is not very efficient at suppressing the low frequency PN from the VCO. Another pole can be added to improve the low frequency PN suppression by employing an additional integral path, but its gain must be much smaller than that of the proportional path so its effectiveness is somewhat limited.

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6.2.2 Use VCDL Based Correction after PLL

Suppressing the external VCO PN using a wide bandwidth external PLL had two major issues. Firstly, a very wide bandwidth PLL is sensitive to PVT variation in terms of stability and secondly,

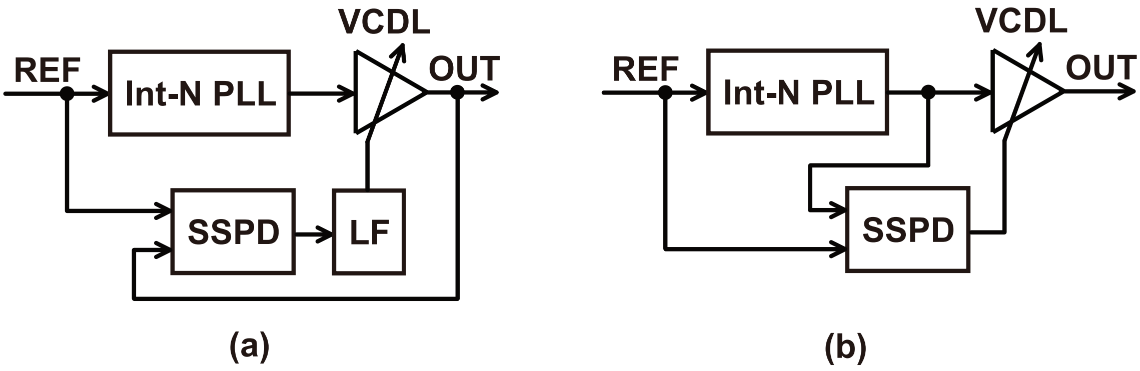

Figure 6.2: PLLs making use of VCDLs in a (a) feedback [82] and (b) feed-forward [83] configuration

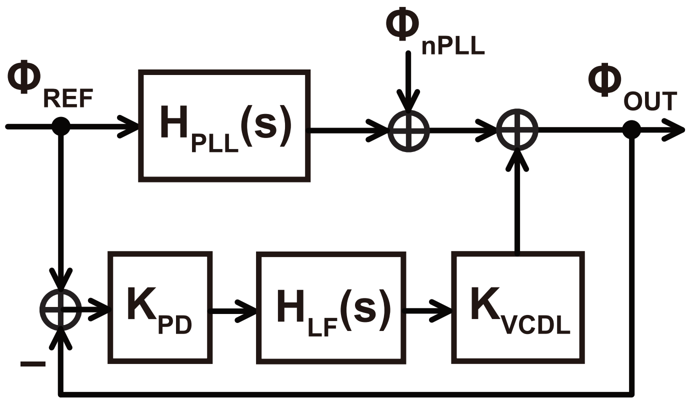

Figure 6.3: A linear noise model of the PLL employing a VCDL in a feedback configuration

a wide bandwidth PLL tends to be inefficient in terms of low frequency PN suppression because they employ type-I architectures meaning that the high-pass characteristic from the VCO output to the PLL output only has a 20dB slope. Therefore, what we want is a PVT robust way to perform wide bandwidth high-pass filtering on the external VCO PN. An interesting way to do this is to first realize a stable frequency signal by using a narrow to moderate bandwidth PLL, and then adding a Voltage-Controlled-Delay-Line (VCDL) at the PLL output that provides additional suppression for the VCO PN. The VCDL can be employed in two different ways: a feedback configuration [82] and a feed-forward configuration [83] as shown in Figs. 6.2(a) and (b) respectively. For the feedback configuration, the output phase of the VCDL is compared with the reference, resulting in an error signal that is used to control the VCDL. Fig. 6.3 shows a linear model of the PLL with a VCDL in feedback configuration. $H_{\text{PLL}}(s)$ is the transfer function of the PLL, K_{PD}, K_{VCDL}, and $H_{\text{LF}}(s)$ are the PD gain, VCDL gain, and LF transfer function all for the VCDL loop, and ϕ_{REF}, ϕ_{nPLL}, and ϕ_{OUT} are the phase noise contributed by the reference, PLL, and the resulting phase noise at the overall output respectively. If we calculate the transfer function from the PLL output to the overall output $H_{\text{PLL,OUT}}(s)$, we obtain the following expression

$$H_{\text{PLL,OUT}}(s) \equiv \frac{\phi_{\text{OUT}}}{\phi_{\text{nPLL}}} = \frac{1}{1 + K_{\text{PD}}K_{\text{VCDL}}H_{\text{LF}}(s)}. \tag{6.1}$$

Assuming that the quantity $K_{\text{PD}}K_{\text{VCDL}}H_{\text{LF}}(s)$ has a large gain at low frequencies (If $H_{\text{LF}}(s)$ contains an integrator for example), this will exhibit a high-pass characteristic. If we focus on the PN from the VCO, we can see that it first gets high-pass filtered by the PLL, and then high-pass filtered again by the feedback VCDL loop, improving the PN suppression at low frequencies as

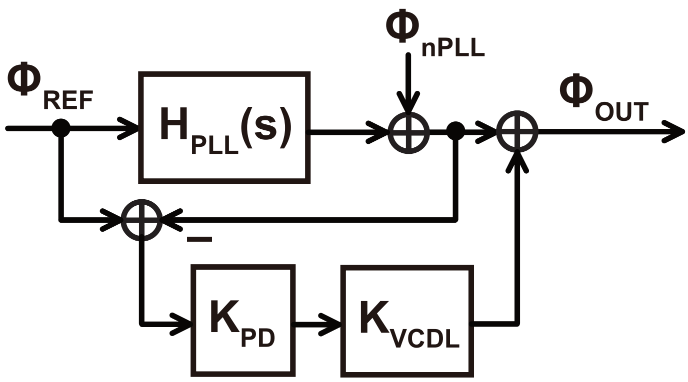

Figure 6.4: A linear noise model of the PLL employing a VCDL in a feed-forward configuration

we wanted. Increasing the order of the high-pass characteristic is an approach that is similar to adding an integral path in a single loop PLL. However, since the VCDL loop is separated from the PLL, its bandwidth is not limited by the stability of the PLL. As such, the VCDL loop can have a wide bandwidth to provide effective high-pass filtering for the VCO PN, while the addition of the integral path inside the PLL would be very limited in terms of the bandwidth. In terms of robustness towards PVT, the VCDL suffers from the same issues as a type-I wide bandwidth PLL. If its open loop gain varies and the original design at the edge of stability, the VCDL characteristic can exhibit some peaking and even lose stability.

For the feed-forward configuration, we take the output of the PLL instead of the VCDL, and generate the error signal by comparing it to the reference signal. Instead of trying to align the overall output like we did in the feedback configuration, here we anticipate the phase error that needs to be cancelled by looking at the PLL output, and then use this signal to control the VCDL. A linear model of the PLL with a VCDL in feed-forward configuration is shown in Fig. 6.4. The same notations are used as the ones in Fig. 6.3. If we calculate the transfer function from the PLL output to the overall output $H_{\mathrm{PLL,OUT}}(s)$, we obtain the following expression

$$H_{\mathrm{PLL,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nPLL}}} = 1 - K_{\mathrm{PD}} K_{\mathrm{VCDL}}. \tag{6.2}$$

This is an interesting result, because we can see that with perfect gain matching i.e. $K_{\mathrm{PD}} K_{\mathrm{VCDL}} = 1$, $H_{\mathrm{PLL,OUT}}(s)$ becomes zero! Unfortunately, this is too good to be true, and if we consider the fact that the phase detector only detects the input phase difference at the rising edge of the input signals and therefore performs sampling of the input phase difference, the transfer function of the PD will not just be a simple gain K_{PD} but will also contain a sinc characteristic that reduces the gain of the cancellation path at higher frequencies. In other words, the bandwidth of the feed-forward noise cancellation characteristic is limited by the sampling at the reference frequency just like a wide bandwidth PLL and the feedback configuration. Nonetheless, employing a VCDL in a feed-forward configuration in a PLL will provide additional noise suppression for the PLL output PN. We will analyze the details of this again later.

The use of a VCDL in a feed-forward configuration is also sensitive to loop gain variations due to PVT. However, because of the fact that there is no feedback loop, we do not have to worry about loss of stability and phase margin. Instead, the noise suppression will just become slightly worse if there are any gain errors. This means that the feed-forward noise cancellation scheme can be designed to have a wide bandwidth, without having to worry about maintaining stability over PVT and failure of operation. We will also analyze the detailed effects of gain error later on.

In short, there are two ways to use a VCDL in a PLL. Despite the fact that they have some challenges, both methods succeed in improving the suppression of the VCO PN compared to simply using a wide bandwidth PLL. However, one issue that we have yet to mention is the difficulty of designing the VCDLs in the first place. Since the VCDLs are attached to the output of the PLL, they must operate at a very high frequency. Combine this with the fact that the delay range of the VCDL must be wide enough to cover the phase error of the PLL output and systematic skews means that it can be very difficult to realize the VCDLs themselves with a low PN. Furthermore, we saw that even with the use of VCDLs, loop gain variation due to PVT is still an issue. The gain of the PD can be regulated quite well over PVT, based on prior works such as [80]. However, the gain of a high speed delay line can be very hard to regulate, and may require additional calibration to keep within a tolerable range. Though the use of a VCDL is very good for improved VCO PN suppression, the use of the high speed VCDL itself is something we would like to avoid.

6.3 Proposed Voltage-Domain Feed-forward Phase Noise Cancellation Technique

So far, we have looked at two major directions for suppressing the external VCO PN. The wide bandwidth PLL is straight forward, but the suppression of the VCO PN has some tough limitations. The use of a VCDL can help to greatly improve the PN suppression, but the realization of the VCDL itself creates additional challenges. In this section, we show our proposed method for improving the VCO PN suppression.

6.3.1 Proposed Concept

A block diagram showing the basic concept of our proposed technique is shown in Fig. 6.5. Similar to the PLL employing a VCDL in a feed-forward configuration, we take the PD output since it contains the information for the PLL output PN. Instead of using the error signal to control a VCDL however, we instead add it to the ouput of the main PLL PD. By doing so, we can cancel the PN component contributed by the external VCO. We also do not need any high speed VCDLs to perform this cancellation of PN. This is because the proposed technique makes use the fact that in a cascaded PLLs, the output PN component of the external PLL appears at the output of the second stage PD, where the signal is down-converted to DC and is low frequency. In contrast, the output of the VCO in a single-loop PLL is the final signal of the system, meaning that there are no additional opportunities to cancel the PN and it must be addressed directly, which necessitates the use of a VCDL since the signal is in the time-domain. Although we used it in the dual-feedback PLL, the proposed feed-forward noise cancellation technique can be used in other types of noise cancellation architectures as well.

6.3.2 Noise Cancellation Analysis

Next, we will give a more detailed analysis of the dual-feedback PLL with feed-forward noise cancellation. Fig. 6.6 shows a linear model of the architecture. K_{PD1}, K_{PDM}, $H_{\mathrm{LF1}}(s)$, $H_{\mathrm{LFM}}(s)$, K_{VCO1}, and K_{VCOM} are the PD gains, LF transfer functions, and VCO gains of the auxiliary integer-N PLL and main PLL respectively. K_{FF} is the gain of the feed-forward noise cancellation

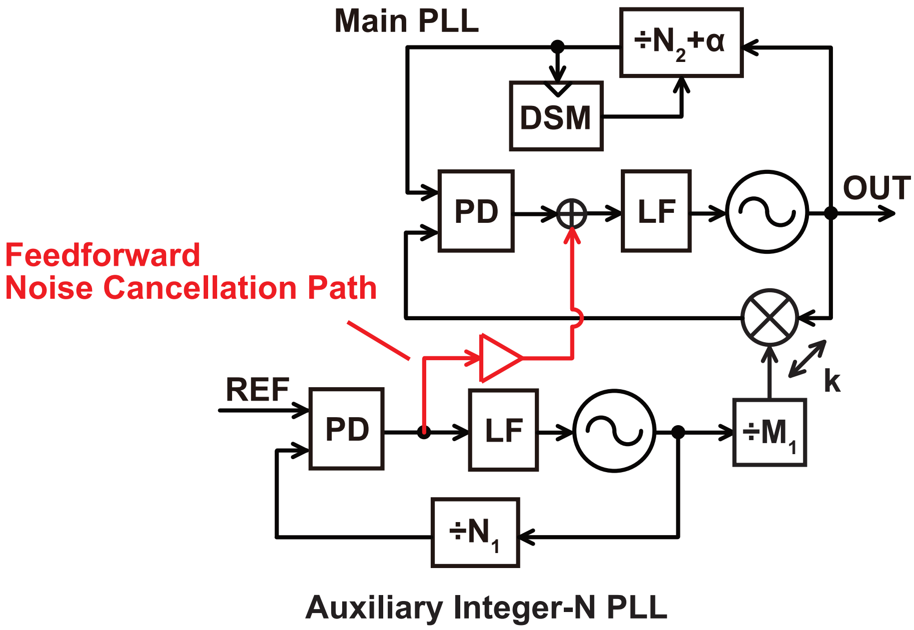

Figure 6.5: The basic concept of a dual-feedback PLL with voltage domain feed-forward noise cancellation

path. ϕ_{nEQUIV1}, ϕ_{nEQUIVM}, ϕ_{nVCO1}, and ϕ_{nVCOM} are the equivalent reference referred PN and the VCO PN of the auxiliary integer-N PLL and the main PLL respectively. ϕ_{nDSM} is the DSM quantization noise and fractional spurs. Finally, ϕ_{INT} and ϕ_{OUT} are the resulting PN at the outputs of the auxiliary integer-N PLL and main PLL respectively. Just like the dual-feedback PLL without feed-forward noise cancellation, the open-loop gains of the auxiliary integer-N PLL and main PLL which are denoted as $H_{\mathrm{OL1}}(s)$ and $H_{\mathrm{OLM}}(s)$ respectively can be calculated as

$$H_{\mathrm{OL1}}(s) = \frac{K_{\mathrm{PD1}} K_{\mathrm{VCO1}} H_{\mathrm{LF1}}(s)}{N_1 s} \tag{6.3}$$

$$H_{\mathrm{OLM}}(s) = \frac{K_{\mathrm{PDM}} K_{\mathrm{VCOM}} H_{\mathrm{LFM}}(s)}{s} \left(1 - \frac{1}{N_2 + \alpha}\right). \tag{6.4}$$

Like we did for the dual-feedback PLL, the two feedback loops in the main PLL are merged into one. Using these expressions for the open-loop gains, the transfer functions from each noise source to the overall output can be calculated as shown below

$$H_{\mathrm{VCOM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCOM}}} = \frac{1}{1 + H_{\mathrm{OLM}}(s)} \tag{6.5}$$

$$H_{\mathrm{EQUIVM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIVM}}} = \frac{N_2 + \alpha}{N_2 + \alpha - 1} \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \approx \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{6.6}$$

$$H_{\mathrm{DSM,OUT}}(s) \equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nDSM}}} = \frac{N_2 + \alpha}{N_2 + \alpha - 1} \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \approx \frac{H_{\mathrm{OLM}}(s)}{1 + H_{\mathrm{OLM}}(s)} \tag{6.7}$$

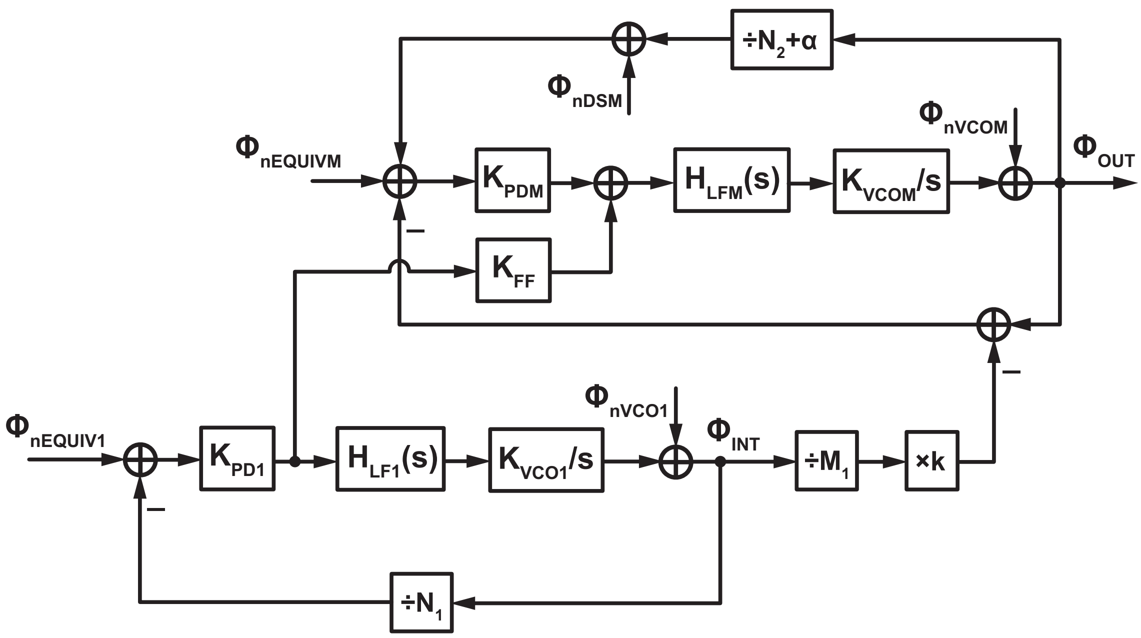

Figure 6.6: A linear noise model of the dual-feedback PLL with feed-forward noise cancellation

$$\begin{aligned} H_{\mathrm{VCO1,OUT}}(s) &\equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nVCO1}}} \\ &= \left(\frac{kK_{\mathrm{PDM}}}{M_1} - \frac{K_{\mathrm{FF}}K_{\mathrm{PD1}}}{N_1}\right) \frac{1}{1+H_{\mathrm{OL1}}(s)} \frac{1}{K_{\mathrm{PDM}}} \frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)} \end{aligned} \tag{6.8}$$

$$\begin{aligned} H_{\mathrm{EQUIV1,OUT}}(s) &\equiv \frac{\phi_{\mathrm{OUT}}}{\phi_{\mathrm{nEQUIV1}}} \\ &= \frac{N_1}{K_{\mathrm{PDM}}} \frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)} \left(\frac{kK_{\mathrm{PDM}}}{M_1} \frac{H_{\mathrm{OL1}}(s)}{1+H_{\mathrm{OL1}}(s)} + \frac{K_{\mathrm{FF}}K_{\mathrm{PD1}}}{N_1} \frac{1}{1+H_{\mathrm{OL1}}(s)}\right). \end{aligned} \tag{6.9}$$

The transfer functions from the main VCO PN, the equivalent reference referred PN of the main PLL, and the DSM noise to the output are given by $H_{\mathrm{VCOM,OUT}}(s)$, $H_{\mathrm{EQUIVM,OUT}}(s)$, and $H_{\mathrm{DSM,OUT}}(s)$ respectively, and can be expressed as shown in equations 6.5, 6.6, and 6.7 respectively. The transfer function from these noise sources in the main PLL to the overall output are the same as that for the dual-feedback PLL without noise cancellation, with the DSM noise and main PLL input referred PN enjoying the same avoidance of noise amplification.

The transfer function from the external VCO PN to the overall output is given by $H_{\mathrm{VCO1,OUT}}(s)$ as shown in equation 6.8. The second, third, and fourth parts of the right hand side are the same as that of a dual-feedback PLL without noise cancellation, with the second part corresponding to the high-pass filtering from the external PLL and the third and fourth parts corresponding to the low-pass filtering from the main PLL. The first part however, is a gain that is a difference between two quantities kK_{PDM}/M_1 and $K_{\mathrm{FF}}K_{\mathrm{PD1}}/N_1$. If the gain of the feed-forward path satisfies $K_{\mathrm{FF}} = \frac{kN_1K_{\mathrm{PDM}}}{M_1K_{\mathrm{PD1}}}$, the two quantities will become equal and $H_{\mathrm{VCO1,OUT}}(s)$ becomes equivalent to zero. This result shows that with proper gain matching, the proposed technique effectively suppresses the external VCO PN. Of course, just like we explained for the PLL employing the VCDL in a feed-forward configuration, the transfer function will not actually become zero because of the sampling action in the PD. A more accurate expression can be obtained multiplying the PD gain with a sinc characteristic that has a notch at the reference frequency [68]. This makes a difference at high frequencies, since the sinc characteristic provides some attenuation at frequencies close to the reference frequency. The equivalent filtering characteristic will become like a high-pass filter as a result, because the cancellation signal becomes more attenuated than the main signal at higher frequencies, while at lower frequencies, proper noise cancellation occurs because the sinc characteristic has a magnitude close to unity. Plots of the detailed calculation results will be shown later on.

Finally, the transfer function from the equivalent reference referred PN of the external PLL to the output is given by $H_{\mathrm{EQUIV1,OUT}}(s)$ as shown in equation 6.9. It consists of the low-pass characteristic of the main PLL, multiplied by the sum of the low-pass characteristic and the high-pass characteristic of the external PLL, but with different gains. If the gain matching condition for noise cancellation is met, the gain for the low-pass and high-pass characteristic becomes the same, and the term inside the parenthesis just simplifies to kK_{PDM}/M_1. Then, $H_{\mathrm{EQUIV1,OUT}}(s)$ gets simplified to

$$H_{\mathrm{EQUIV1,OUT}}(s) = \frac{kN_1}{M_1}\frac{H_{\mathrm{OLM}}(s)}{1+H_{\mathrm{OLM}}(s)} \tag{6.10}$$

which is the same as the transfer function from the equivalent reference referred PN of the external PLL to the output in the dual-feedback PLL without noise cancellation, except for the fact that there is no low-pass filtering from the external PLL. This result shows that the noise cancellation does not apply to the equivalent reference referred PN of the external PLL. More precisely, the noise cancellation only applies to the blocks such as the external VCO and external LF that come after the point where the signal is used for the feed-forward path at the PD output, and does not apply for the blocks that come before it, such as the PD itself, the reference signal, or the divider.

6.3.3 Effect of Gain Error

We explained both qualitatively and analytically how perfect gain matching will lead to perfect noise cancellation (at sufficiently low frequencies at least) and a gain error will lead to some residual noise. Here, we look at what the actual effect of this gain error is.

We will present some calculation results based on the settings used for our actual design that we will explain in detail later. Firstly, Fig. 6.7 shows the characteristic of the equivalent high-pass filtering that is provided for the external VCO by the feed-forward noise cancellation scheme, when we consider different gain errors. When there is perfect gain matching, the characteristic is like an ideal high-pass filter, with perfect noise suppression at DC, and a cut-off frequency at around 40MHz which is half of the reference frequency which has been set to 80MHz here. When some gain error exists, the noise suppression at low frequencies gets degraded. When the gain error is 10% for instance, the gain at low frequencies gets saturated at 20log(0.1)=−20dB. Therefore, gain matching is certainly important. An important thing to note here however, is that unlike a feedback configuration such as a wide bandwidth PLL, no additional peaking is observed, even with a very large gain error. This is because there is no issue of stability due to the feed-forward nature of the technique. This allows us to realize wide bandwidth noise suppression without any risk of operation failure.

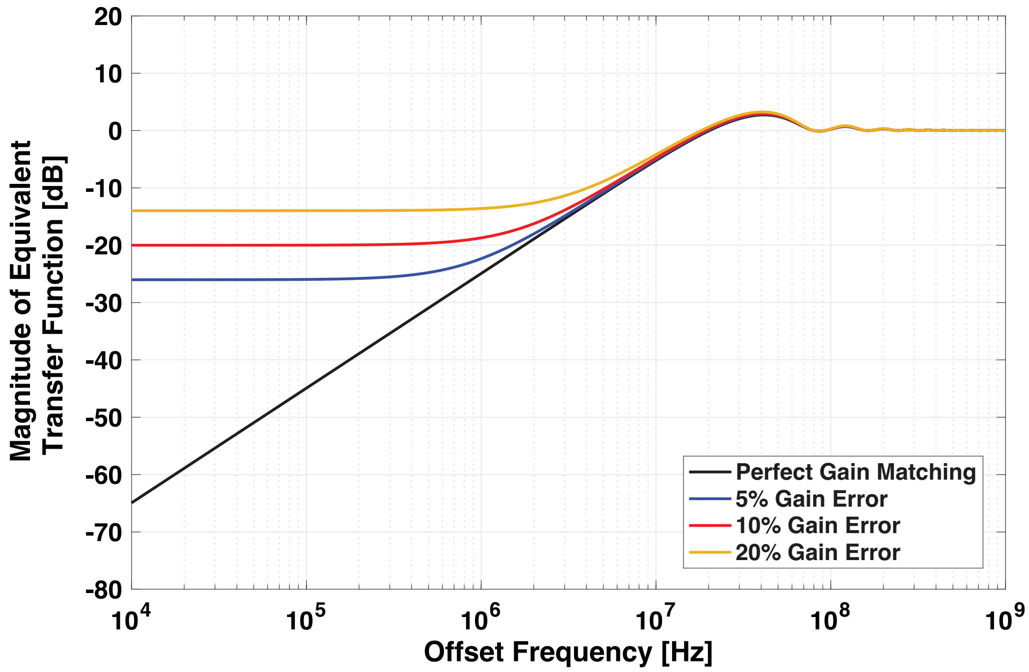

Figure 6.7: The characteristic of the equivalent high-pass filtering provided for the external VCO by the feed-forward noise cancellation scheme with different gain errors

What are the implications of this gain error on the output of the overall PLL? Fig. 6.8 shows the calculated output PN of the overall PLL with different gain errors. We can see that when the gain error is below 20%, the effect on the output PN spectrum is very small. On the other hand, if there is no noise cancellation, the PN from the external VCO significantly degrades the output PN spectrum. Here, the bandwidth of the main PLL is 3MHz, while the bandwidth of the external PLL is 8MHz (Further details will be explained later). From these results, we can see that the proposed noise cancellation scheme is very effective for suppressing the PN contribution from the external VCO, but as long as we make proper use of the low-pass filtering characteristic of the main PLL, the method is quite robust towards gain error. Also as reference, Fig. 6.9 shows the calculated external PLL PN spectrum multiplied by the filtering characteristic of the noise cancellation scheme shown in Fig. 6.7. Here too, we can see that the external VCO PN is greatly suppressed thanks to the noise cancellation scheme.

Fig. 6.10 shows the relationship between the gain error and the resulting output integrated jitter. We can see that although the noise cancellation helps to improve the jitter greatly compared to the case with no noise cancellation, even with a relatively large gain error, the jitter does not degrade too much. In our design, we shoot for a gain error within ±20%. In this range, the jitter degradation is only around 3%.

6.4 Implementation and Measurement Results

To demonstrate the effect of the proposed idea, we implemented a prototype dual-feedback PLL with feed-forward noise cancellation. In this section, we will explain the details of the implementation, along with measurement results.

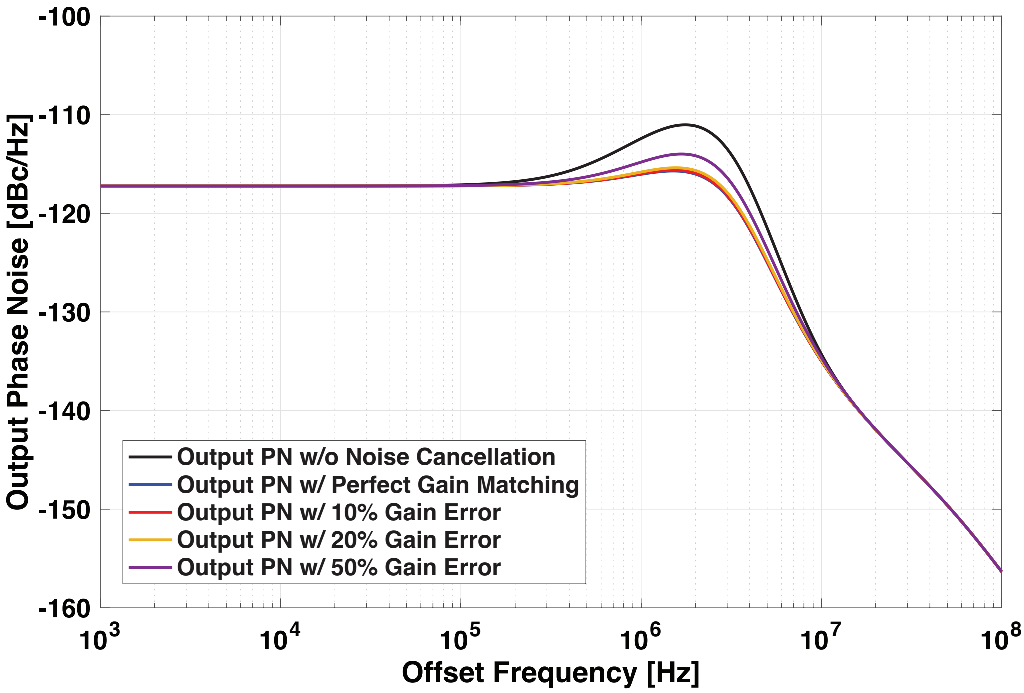

Figure 6.8: The calculated output PN of the overall PLL with different gain errors

6.4.1 Circuit Implementation and Details

The prototype was implemented in TSMC 65nm bulk CMOS process. Fig. 6.11 shows the details of the implementation. The external PLL is based on a type-II SPD based architecture, with a slight novelty for the LF which we will explain later. The LC VCOs for the main and external PLLs employed 8-shaped inductors to further reduce the risk of magnetic coupling. A ring VCO was also implemented for the external PLL to help demonstrate the effectiveness of the feed-forward noise cancellation even when the phase noise contribution from the first stage is very large. The output of the SPD is then fed to a gain stage, which is simply a two stage Operational-Transconductance-Amplifier (OTA) with resistive feedback to realize an accurate gain. The output of the gain stage is then summed with the main PLL PD with a resistive network. The main PLL uses a 3rd order HM, and a type-I architecture based on an XOR-PD and a simple LF consisting of two RC stages, similar to the architecture used in [53]. This fits well with the resistive summation with the output of the feed-forward path. The divide by 2s placed before the XOR-PD are used to convert the rising edges of the MMD and HM output into rising and falling edges at the XOR-PD input, since the XOR-PD will react to both the rising and falling edges of its input by creating a pulse width modulated signal. The bandwidth of the external PLL is designed to be around 8MHz, which is sufficiently narrower than the reference frequency of 80MHz so that there is no risk of PN degradation due to loop gain variation. The BW of the main PLL is designed to be around 3MHz to optimize the trade-off between the main VCO PN, the quantization noise, and the equivalent reference referred PN.

The external PLL architecture is chosen like this for a few reasons. Firstly, since we are using the SPD output for the feed-forward noise cancellation, we do not want to connect it directly to the VCO because it may suffer from kick-back, which may have some negative unexpected effects on the noise cancellation scheme. Secondly, we want to avoid offset at the input of the gain stage of the feed-forward path, since there is a risk that the output can saturate and prevent the main PLL

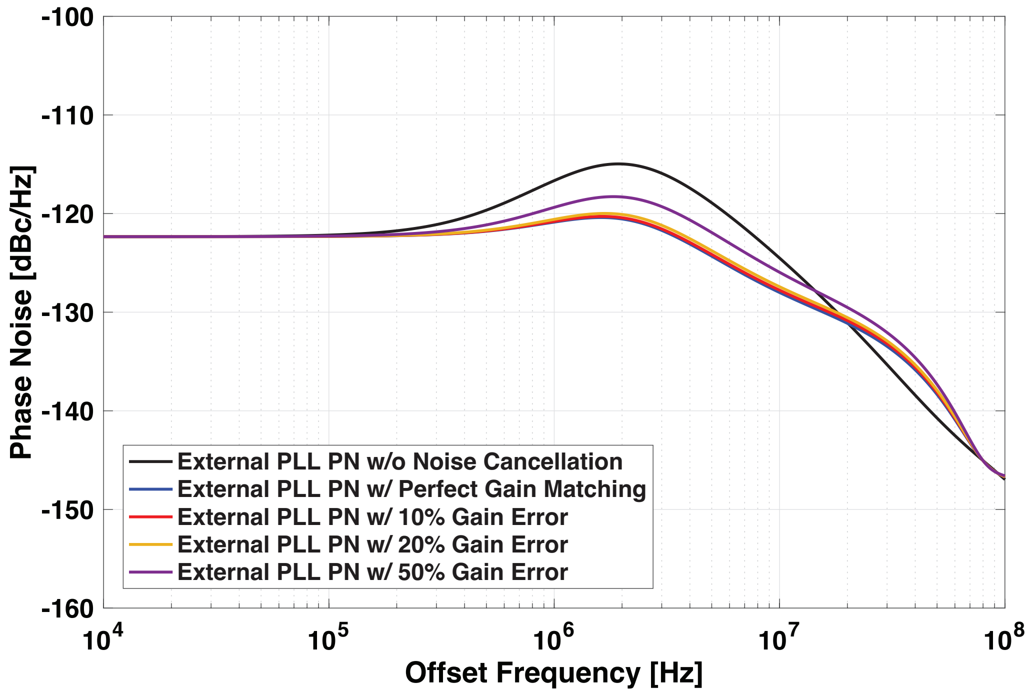

Figure 6.9: The calculated equivalent PN of the external PLL with different gain errors

from working properly. To do this, we want the external LF to have a large DC gain. This way, we can set the DC voltage of the SPD output after lock based on an external reference voltage, and then use the same reference voltage as the AC ground for the gain stage of the feed-forward path. By doing so, the input of the gain stage becomes small as long as the amplifier gain is sufficiently large. Therefore, we want the LF to contain an integrator. A typical architecture that can be used in this case is shown in Fig. 6.12, where we use a gm cell to convert the SPD output into a current, and then put the current through a second order LF. This architecture however, is not very robust to PVT because its loop characteristic relies heavily on the open-loop transconductance of the gm cell. Instead, the external PLL in our design implements a LF based on a closed-loop amplifier with a much more robust loop characteristic. Fig. 6.13(a) shows the proposed LF alone, with V_{IN} and V_{OUT} being its input and output, A the open-loop voltage gain of the OTA, and R and C being the values of the resistor and capacitor. The transfer function of the LF $H_{\mathrm{LF,prop}}(s)$ can be calculated as

$$H_{\mathrm{LF,prop}}(s) \equiv \frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}} = \frac{A}{1+A}\frac{1+sRC}{\frac{1}{1+A}+sRC} \approx \frac{1+sRC}{sRC} \tag{6.11}$$

where the approximation is based on $A \gg 1$. We can see that as long as the A is sufficiently large, we obtain the desired LF characteristic with a zero and a pole at DC, and furthermore, the characteristic is only dependent on the values of the passive components. In contrast, for the conventional LF shown in Fig. 6.13(b) has a transfer function $H_{\mathrm{LF,conv}}(s)$ that can be expressed as

$$H_{\mathrm{LF,conv}}(s) \equiv \frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}} = \frac{1+sRC}{sC/G_{\mathrm{m}}}. \tag{6.12}$$

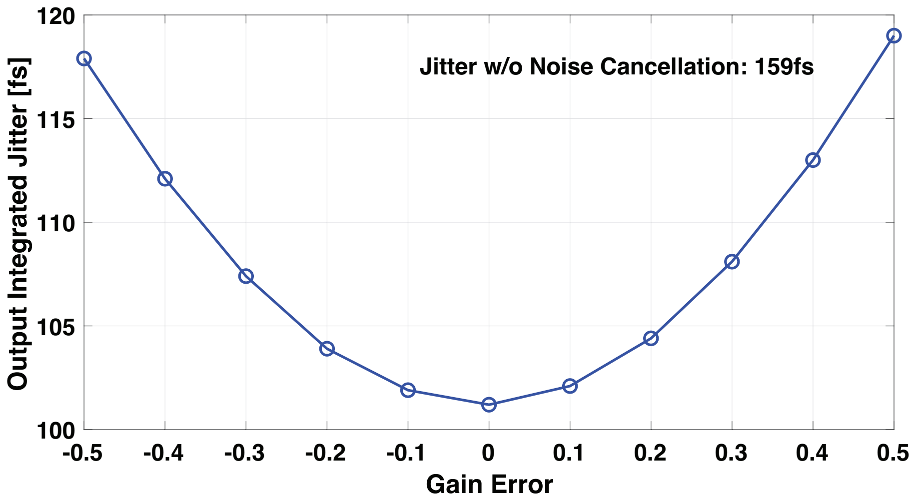

Figure 6.10: The relationship between the gain error and the output integrated jitter

We can see here that the DC gain of the LF characteristic is completely dependent on the transconductance G_{m}. Although not seen in the equation, the LF based on the gm cell also has a requirement similar to the large open-loop gain required by the OTA in the proposed LF. Specifically, if the output impedance r_{o} of the gm cell is too small, it will effect the characteristic of the LF. The DC gain of the proposed LF is limited by A in the proposed architecture, while it is limited by $g_{\mathrm{m}}r_{\mathrm{o}}$ in the conventional architecture. Actually, the requirement for the output impedance in the proposed LF is more scaling friendly than that in the conventional LF. Since the OTA in the proposed LF generates a voltage domain output, a low output impedance leads to a more ideal characteristic. In contrast, the gm cell in a conventional LF must have a high output impedance to achieve an ideal characteristic, which can be very difficult in advanced processes since the supply voltage will be low.

It is also very important to explain how we dealt with the gain variation. As we discussed during our explanation of equation 6.8, for gain matching to be satisfied, the relation

$$\frac{kK_{\mathrm{PDM}}}{M_1} = \frac{K_{\mathrm{FF}}K_{\mathrm{PD1}}}{N_1} \tag{6.13}$$

must be satisfied. What are the challenges present here? k, M_1, and N_1 will not suffer from any variation unless the circuit is completely malfunctioning, because they are well defined discrete values that are fixed by the circuit operation and settings. K_{PDM} is the gain of the XOR-PD which is given by $K_{\mathrm{PDM}} = V_{\mathrm{DD}}/2\pi$, and this is also quite robust to variations as long as its supply can be well regulated with an Low-Drop-Out (LDO) regulator for example. This leaves K_{FF} and K_{PD1} which are the gain of the gain stage in the feed-forward noise cancellation path and the gain of the SPD in the external PLL respectively. K_{FF} is relatively robust to variations with proper design because it is based on a feedback configuration whose gain is dependent on the ratio of two resistors, which is quite robust to PVT. Corner simulations showed that K_{FF} varies by just $\pm 0.7\%$. The SPD gain K_{PD1} is the most challenging of all these parameters, and can vary by more than 50% over PVT if we just use a very simple configuration such as an inverter with a capacitive load. To solve this issue, we used a current starved inverter with a Switched-Capacitor (SC)-based bias generator proposed in [80] as the slope generator for the SPD. The circuit details of the SC-based bias generator can be seen in Fig. 6.14. REF and FB are the reference and feedback signals of the

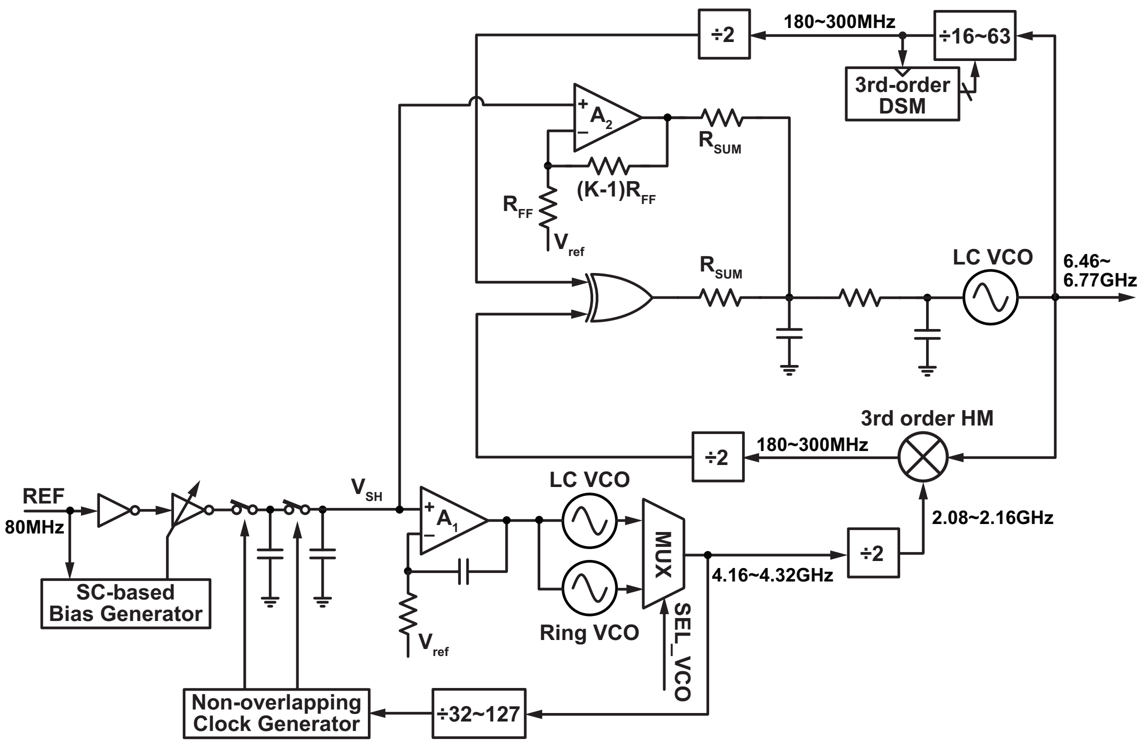

Figure 6.11: The implementation details of the dual-feedback PLL with the proposed feed-forward noise cancellation technique

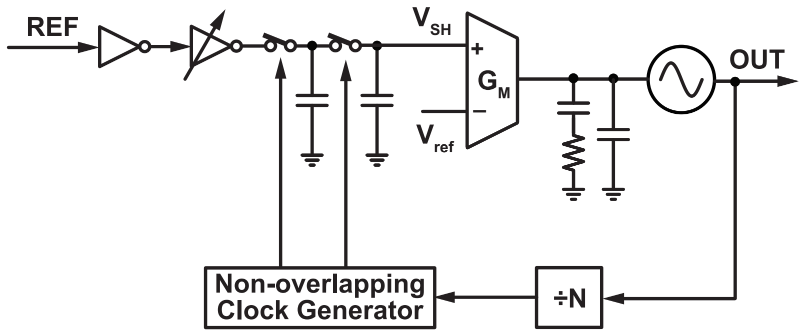

Figure 6.12: The gm cell based PLL architecture typically used to realize a type-II PLL with an SPD

external PLL respectively, C_1 and C_2 are the capacitors in the two-stage sampler of the SPD, C_{SCB} is the sampling capacitor used in the SC-based bias generator, C_{BIG} is the capacitor connected to the output of the SC to reduce the voltage ripple, V_{BandGap} is a stable voltage reference, A is the open-loop gain of the amplifier used in the feedback loop of the SC-based bias generator, and I_{tail} is the tail current of the slope generator. In a conventional SPD with a slope generator and a sampling capacitor, the slope can vary largely in part due to the variation of the pull down current of the inverter and the process variation of the sampling capacitor. In the SC-based bias generator however, the tail current of the current starved inverter I_{tail} is proportional to C_{SCB} if we assume that the reference voltage V_{BandGap} is generated by a Band Gap Reference (BGR) for example and is robust to any variations. On the other-hand, the slope of sampled voltage is proportional to the pull-down current I_{tail} divided by the sampling capacitance C_1. Therefore, the slope is no longer dependent on the absolute value of a capacitor but rather the ratio of two capacitances C_1 and C_{SCB}, which is much more robust to PVT. The design in [80] achieved a slope variation within $\pm 2\%$ over all corners. Our design was only able to achieve a variation within

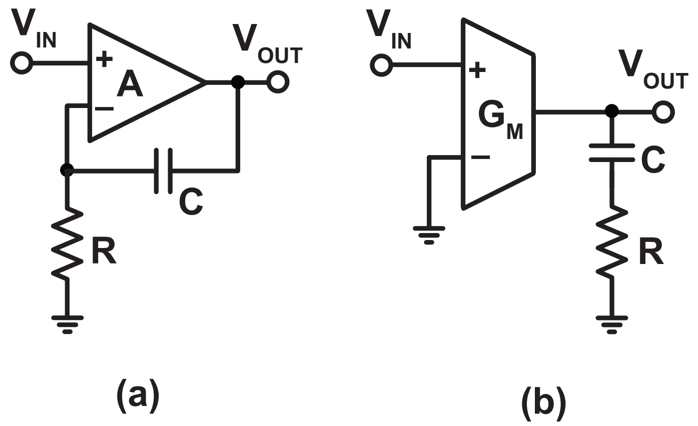

Figure 6.13: A linear model of the (a) proposed and (b) conventional LF

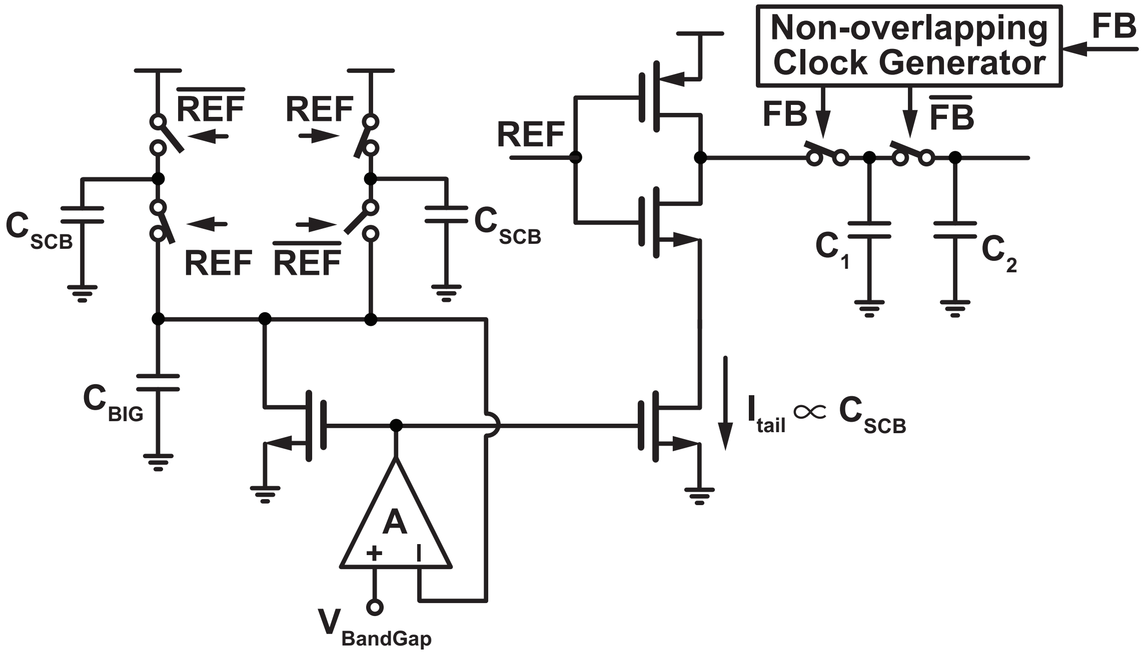

Figure 6.14: The circuit details of the SC-based bias generator and the SPD

$\pm 16\%$ due to non-ideal design (the transistor used for the inverter was too small compared to that for the tail current source). This however, is still much better than if we just used a simple inverter for the slope generator, and allows us to achieve a total gain error within $\pm 17\%$ considering the variation of both K_{FF} and K_{PD1}, which is within our original target of $\pm 20\%$. Fig. 6.15 shows the waveforms resulting from corner simulations of the slope generator using (a) the conventional tail current biasing scheme based on a current source and (b) the SC-based biasing scheme. It can be seen that the SC-based biasing scheme results in variation.

It is worthy to note the possibility of employing the same dual-feedback architecture with feed-forward noise cancellation, but with a more digitally intensive implementation i.e. using TDCs/BBPDs and digital loop control for the external and main PLLs. This is potentially a very useful idea not just because of the obvious benefits of digital PLLs but also due to the improved robustness of the noise cancellation scheme towards PVT variation. We no longer have to worry about any variation of the feed-forward gain, signal summation is very easy, and TDCs can be realized with a relatively well defined and robust gain. That being said, digital PLLs have

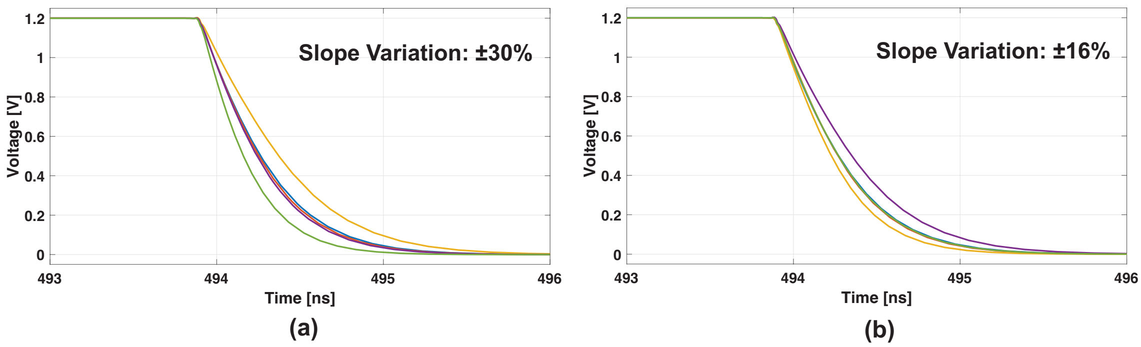

Figure 6.15: The waveforms resulting from corner simulations of the slope generator using (a) the conventional tail current biasing scheme using a current source and (b) the SC-based biasing scheme

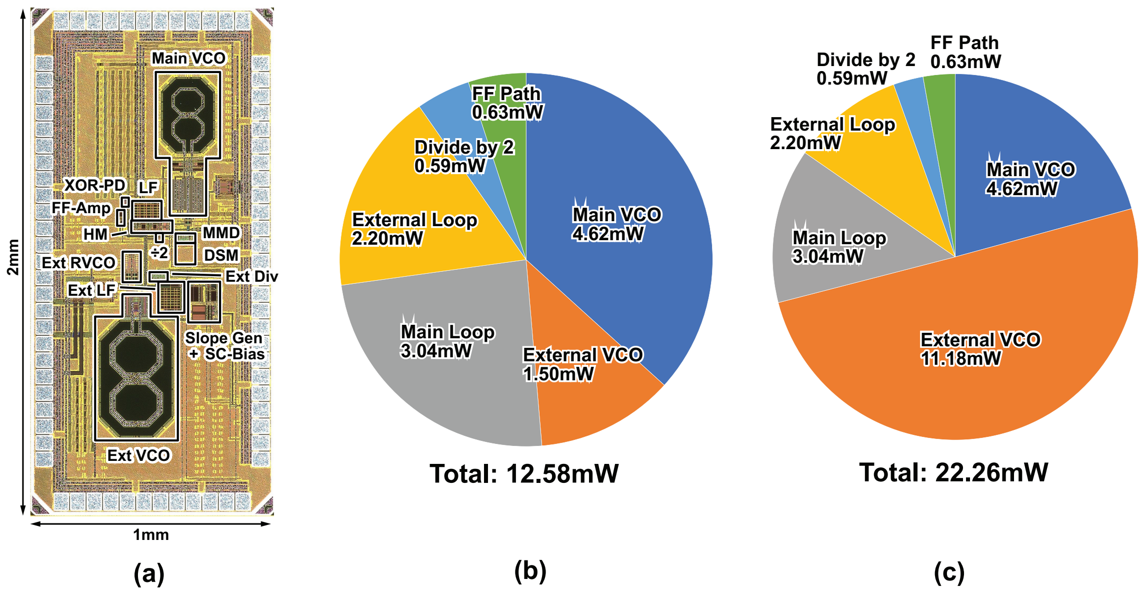

Figure 6.16: (a) The chip micrograph of the implemented chip along with the power consumption breakdown for when the (b) LC-VCO is used and (c) ring VCO is used for the external PLL

additional challenges of their own, such as quantization noise in the PD and limit cycles among others. As such, we took a safer approach with a more analog implementation in this particular design since our main goal was to demonstrate the concept of the feed-forward noise cancellation scheme.

6.4.2 Measurement Results and Performance Comparison

Fig. 6.16(a) shows the chip micrograph and which regions were allocated to the different circuit blocks. Figs. 6.16(b) and (c) show the power consumption breakdown of the implemented chip when we use the LC VCO and ring VCO for the external PLL respectively. The power consumption of the VCOs include that from the core and the output buffer, while the external loop and main loop refers to the blocks from the external and main PLL respectively other than the VCO (Divider, LF, PD). The divide by 2 refers to the divider placed between the external PLL and the HM. For the LC-VCO case total consumption is 12.58mW, of which just 1.50mW (12%) is consumed by the external VCO.

Fig. 6.17 shows the output spectrum of the external PLL (black) when the LC-VCO is being used. The free-running phase noise spectrum of the LC-VCO is also shown in grey. The integrated

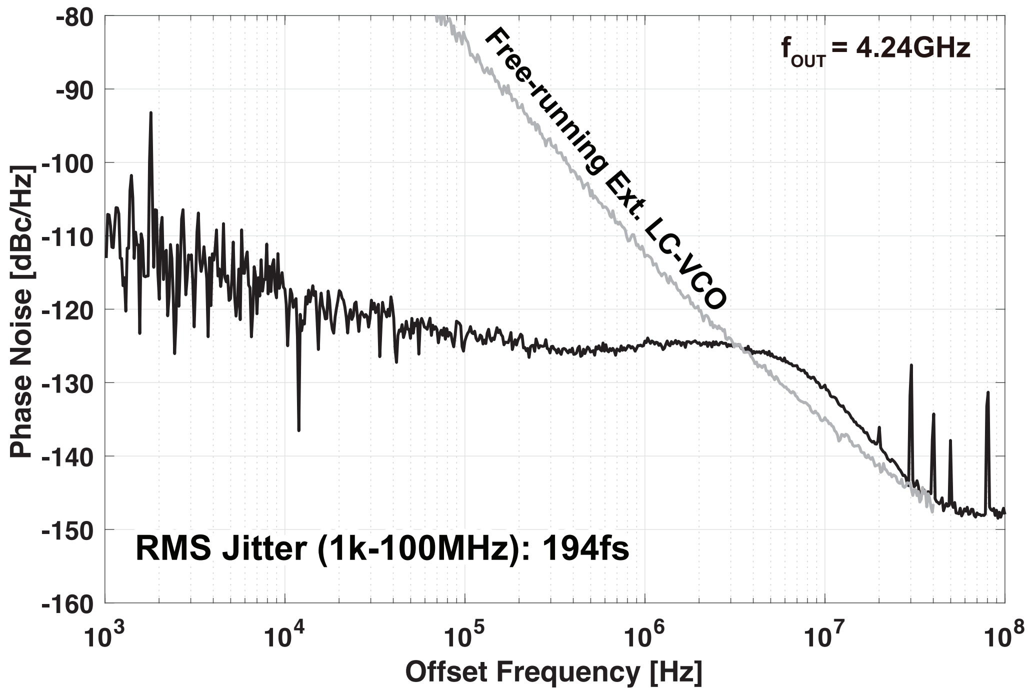

Figure 6.17: The phase noise spectrum of the external PLL when an LC-VCO is being used (black) along with the free-running phase noise spectrum of the LC-VCO

jitter is 194fs. The external LC-VCO has a phase noise of −106dBc/Hz at a 4.24GHz output frequency and 1MHz offset, which corresponds to an FoM of −177dB.

Fig. 6.18 shows the output spectrum of the external PLL (black) when the ring-VCO is used. The free-running phase noise spectrum of the ring VCO is also shown in grey. The integrated jitter here is 783fs, which is much worse than when we used the LC-VCO due to the inferior phase noise performance of the ring VCO.

Fig. 6.19 shows the phase noise spectrum of the implemented PLL when the noise cancellation scheme is turned off (red) and when it is on (blue). The LC VCO is used for the external PLL. The free-running phase noise spectrum of the main VCO is shown in grey. It should be noted that this is the spectrum of the divide by 4 output, and so the entire spectrum should be shifted up by 20log(4)=12dB to obtain the spectrum at the actual PLL output. We can see a clear improvement when the noise cancellation is turned on. The integrated jitter improves from 125fs to 106fs. No calibration or trimming was used to adjust the gain of the feed-forward noise cancellation path. The main VCO has a phase noise of −116dBc/Hz at a 6.6GHz output frequency and 1MHz offset, which corresponds to an FoM of −186dB.

Fig. 6.20 shows the phase noise spectrum with the noise cancellation scheme turned off (red) and on (blue) when we use the ring VCO for the external PLL. The free-running spectrum of the main VCO is shown in grey. An improvement in the suppression of noise contribution from the external PLL, similar to that in Fig. 6.19 but more significant in terms of magnitude can be observed. The integrated jitter is 410fs without noise cancellation and 127fs with noise cancellation. It is worthy to note that even with the huge phase noise contributed by the ring VCO of the external PLL, the integrated jitter is almost on par with the LC-VCO case thanks to the highly effective noise suppression of the noise cancellation technique.

Next we move onto the spurious tone measurements. All of these measurements were done

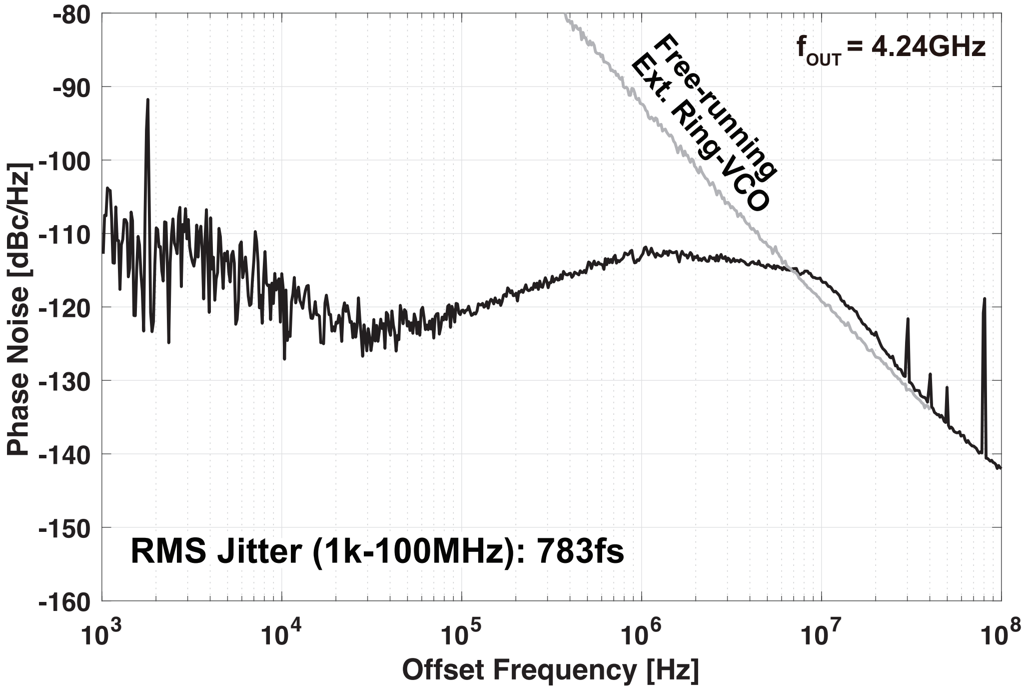

Figure 6.18: The phase noise spectrum of the external PLL when a ring VCO is being used (black) along with the free-running phase noise spectrum of the ring VCO

with the external PLL using the LC-VCO. Fig. 6.21 shows the (a) spectrum of the output containing the fractional spurs and (b) a plot of the fractional spur values for different fractional values of the FCW. Again, this is after the divide by 4, so the actual spur values when referred to the PLL output are 12dB higher than the values shown. The worst case fractional spur is around −63dBc when referred to the PLL output.

Fig. 6.22 shows the spectrum of the output containing the (a) reference spurs at 80MHz and (b) coupling spurs. The reference spur has a magnitude of around −89dBc, which is around -77dBc when referred to the PLL output. As for the coupling spur, based on the frequency it appears at, it is likely due to the coupling between the DSM and the reference circuitry. The DSM frequency here was 244.56MHz, the reference frequency is 80MHz, and the coupling spur appears at 4.56MHz. The disturbance of the substrate or power supply by the DSM is likely getting down-converted by the third harmonic of the reference frequency. The amplitude of the coupling spurs are around −66dBc in the worst case (−54dBc when referred to the PLL output), and they can be suppressed further with more careful layout design.

Table 6.1 shows a performance comparison of this work with other state-of-the-art fractional-N PLLs [84, 85, 86, 41, 69, 65, 87] in terms of jitter-power performance and spurious performance. We can see that both the jitter-power performance and the spurious performance are in-line with that of the other state-of-the art works. Our work is also calibration-free, unlike some of the other works that employ digital cancellation schemes such as DTCs.

The HM-PLL in [69] achieves a better jitter-power performance than our work, even though it does not make use of feed-forward noise cancellation. However, this is not an issue with our proposed technique or circuit architecture, but rather due to the fact that the individual circuit blocks such as the VCO, HM, and MMD/DSM have a better performance, which in turn is due to more optimal design and access to the very advanced 7nm process (The advanced process is

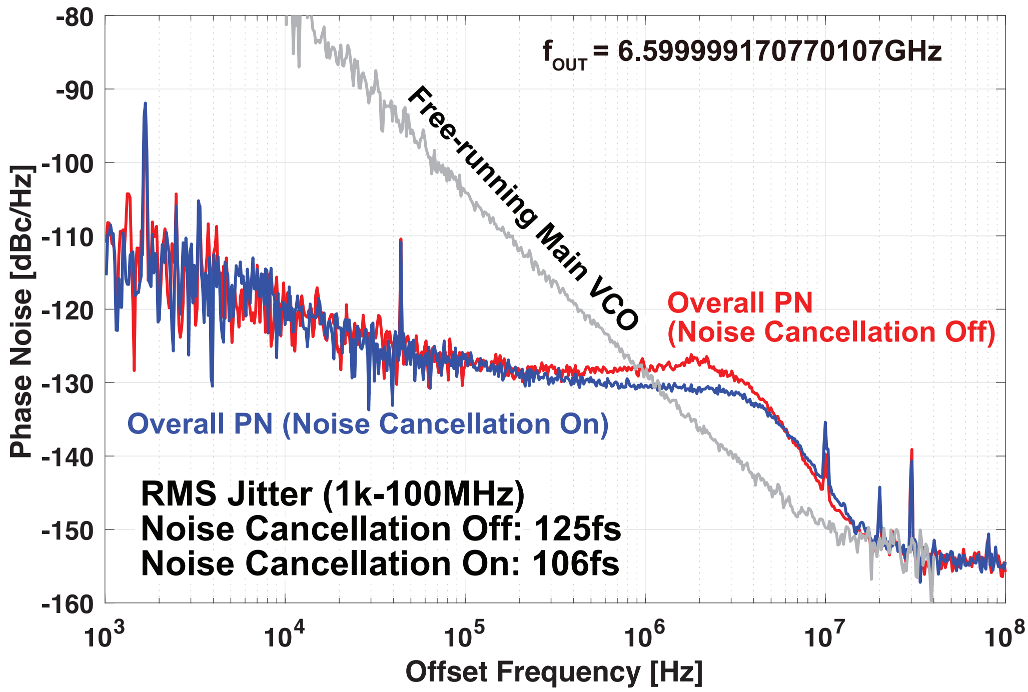

Figure 6.19: The output phase noise spectrum when the noise cancellation scheme is off (red), when it is on (blue), and the free-running phase noise of the main VCO (grey)

especially helpful to reduce the power consumption of the MMD and DSM). Based on the fact that the external PLL consumes around half the power of the entire PLL and has a significant contribution on the overall PN while the external PLL in our work only consumes around 15% of the power consumption of the entire PLL and its noise contribution is suppressed to a negligible level, we can expect our proposed PLL to have a better performance under the same conditions as [69].

Also for reference, our work when the ring VCO is used for the external PLL exhibits a very respectable jitter-power FoM of around −244dB.

6.5 Chapter Summary

In this chapter, we proposed a novel feed-forward noise cancellation technique for cascaded PLLs that does not require any high-speed delay lines. The effect of feed-forward noise cancellation was analyzed, and a HM-based PLL employing the technique was implemented. By achieving a state-of-the-art jitter-power and spurious performance with a low power consumption from the external VCO, we demonstrated how the proposed PLL can greatly mitigate the overhead from the extra VCO which is one of the big bottlenecks of improving the performance of HM-based PLLs and cascaded PLLs in general. The issue of gain error between the signal path and noise cancellation path was also addressed. The SPD of the external PLL was realized based on a current starved slope generator that is biased with a switched capacitor-based resistor, and an XOR-PD was used for the main PLL resulting in a PVT robust gain for all the signal paths. Consequently, the calibration-free characteristic of the HM-based PLL is maintained.

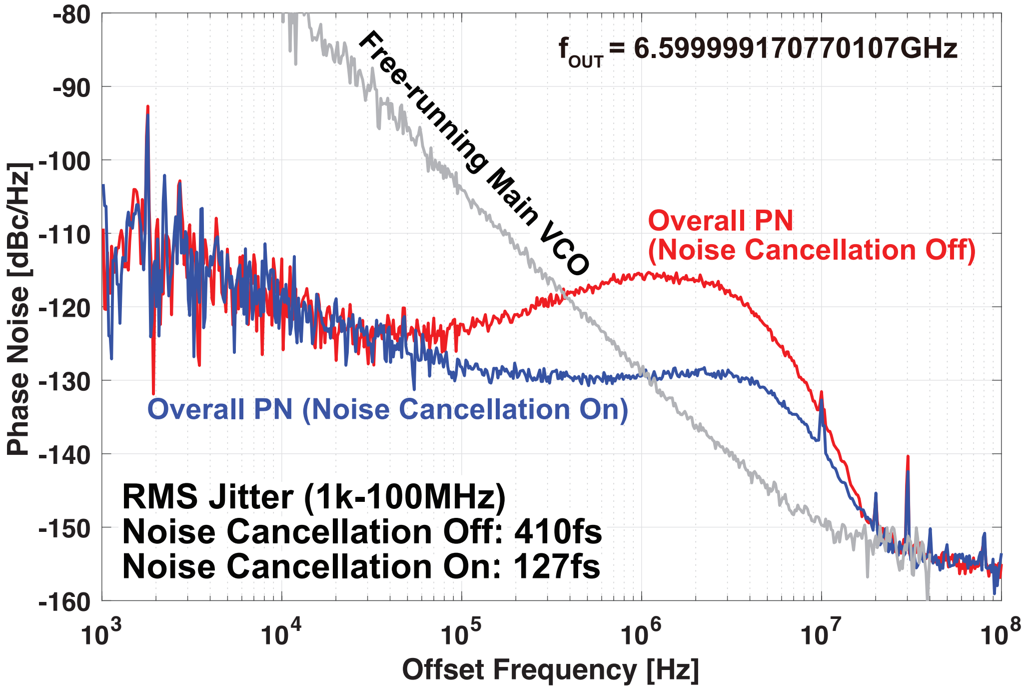

Figure 6.20: The output phase noise spectrum when the noise cancellation scheme is off (red) and when it is on (blue) while a ring VCO is being used for the external PLL

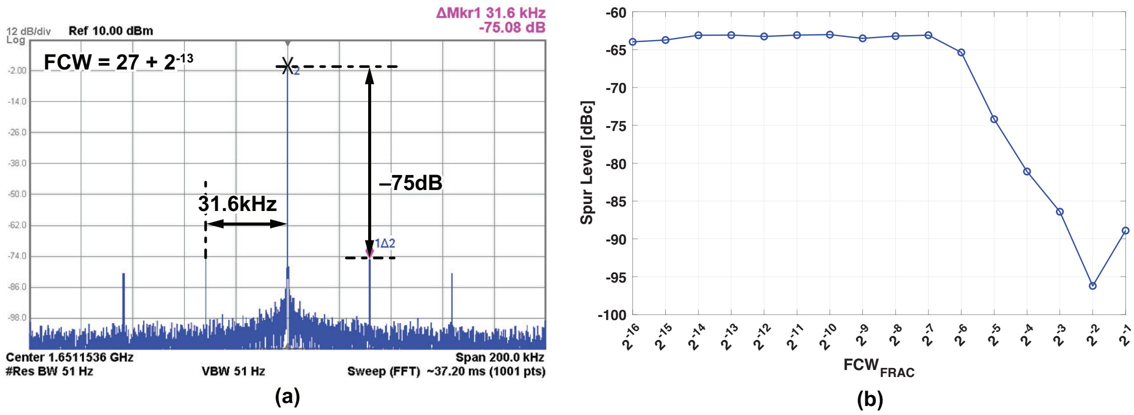

Figure 6.21: (a) The output spectrum of the PLL containing the fractional spurs and (b) a plot of the fractional spur level for different fractional values of the FCW

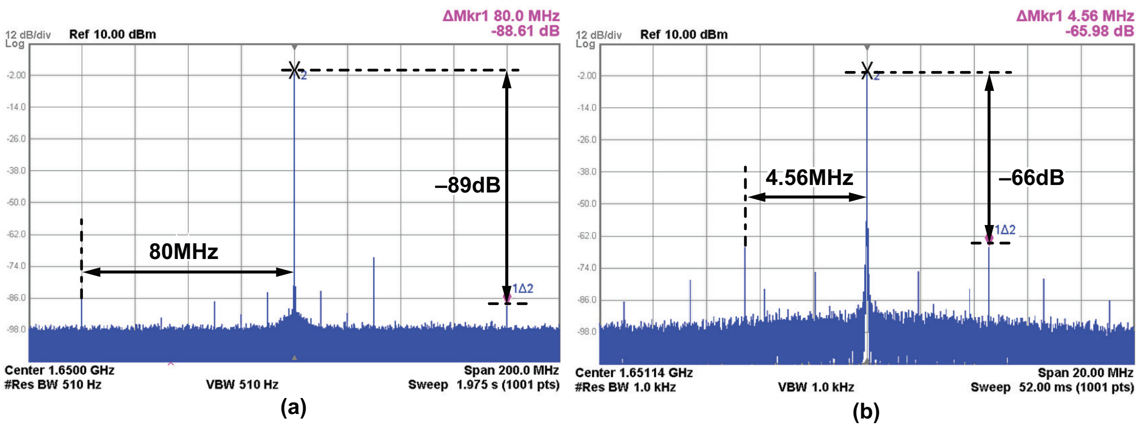

Figure 6.22: The output spectrum of the PLL containing (a) reference spurs and (b) coupling spurs

Table 6.1: A Performance Comparison of the Implemented PLL With Other Works

	This Work	[73] Castoro ISSCC23'	[74] Kim JSSC23'	[75] Xu VLSI21'	[32] Wu JSSC19'	[62] Yang JSSC22'	[59] Yang ISSCC19'	[76] Gao JSSC23'
PLL Architecture	HM-based w/ Noise Cancellation	Mutli-DTC	DAC w/ SCF DPD	Hybrid w/ DTC + SPD	DTC + SPD	HM-based	HM-based Triple-Loop	TAU + TDC
Calibration Free?	Yes	No	No	No	No	Yes	Yes	No
Reference Frequency [MHz]	80	250	150	50	52×2	74	60	40
Output Frequency [GHz]	6.5-6.8	9.25-10.75	12.8-15.0	3.3	5.5-7.3	25-28	7.0-9.0	2.56-4.1
Integrated Jitter [fs]	106	77	104	236	75	88	131	182
Power Consumption [mW]	12.58	17.9	7.3	4.6	18.9	12.9	13.4	3.48
FoM [dB]	−248.5	−249.7	−251.0	−246	−249.7	−250	−246.4	−249.4
Fractional Spur [dBc]	−63	−60.3	−58	−53	−64	−61	−70	−59
Reference Spur [dBc]	−76.6	−71.1	N/A	−80	−70.2	−83	−66	−73.5
Process [nm]	65	28	65	65	28	7	16	40
Active Area [mm^2]	0.36	0.36	0.21	N/A	0.45	0.24	0.25	0.31

Chapter 7

Conclusions and Future Perspectives

In this dissertation, we have investigated the possibilities offered by the use of HM-based feedback in fractional-N PLLs by proposing new architectures that make use of this concept and analyzing the expected performance of these architectures, as well as the phenomenon of offset spurs that can arise. In this chapter, we summarize and conclude our dissertation, and also look at future perspectives and possible directions that may be worth looking at for future research.

7.1 Conclusion

In this dissertation, we thoroughly investigated the possibilities of using HM-based feedback inside of a fractional-N PLL to realize high performance frequency synthesizers. We first began in chapter 2 with the basics of PLL design and the trade-offs involved with obtaining optimal PN performance. We then explained how a fractional-N PLL will suffer additionally from the quantization noise in the MMD, and since this quantization noise significantly degrades the output PN of the PLL, it must be suppressed in some way. Upon discussing some of the prior arts for quantization noise suppression along with their pros and cons, the HM-based architecture was introduced as a new approach. We explained how, although the HM-based architecture certainly has drawbacks of its own, it also has many distinct advantages over the prior arts, and also mentioned how it can be combined with other techniques since it makes use of a new "knob" by reducing the DC gain of the noise transfer function to the PLL output. Then in chapter 3 we introduced the dual-feedback architecture, which makes use of the HM-based feedback in a very simple and elegant way, and demonstrated its effectiveness through calculation, simulation, and measurement. After that, in chapter 4 we analyzed the phenomenon of offset spurs that arise in HM-based PLLs due to the S/H operation inside the HM. We gave an explanation for the generation mechanism of the offset spurs, predicted their positions and amplitude based on closed expressions, and validated their accuracy based on calculation and simulation. In chapter 5, we looked at the possibility of combining HM-based feedback with other approaches for quantization noise suppression by introducing the dual-feedback PLL with split-feedback division and PDLPF, and how this architecture makes use of multiple different approaches that work well with each other to suppress the quantization noise in a very effective way. Finally in chapter 6, we introduced a technique to realize feed-forward noise cancellation without relying on a high speed delay line, and used it in the dual-feedback PLL to reduce the PN overhead from having an extra VCO.

7.2 Future Directions

There are still many potential advantages of HM-based PLLs that were not investigated in this dissertation. We'd like to provide some future directions for HM-based PLL research here.

The first potential direction for future research on HM-based PLLs is its use in mm-wave frequency synthesis. Throughout the dissertation, we looked at the dual-feedback architecture as one of the core structures for making use of HM-based feedback. There is actually another architecture that achieves a very similar suppression of quantization noise as the dual-feedback architecture which is shown in Fig. 7.1 [69, 88]. We will refer to this architecture as the dual-feed-forward architecture here. Both the dual-feedback and dual-feed-forward architectures offer similar advantages in terms of quantization noise suppression: if the main PLL output has a higher frequency than the external PLL, the dual-feedback PLL offers lower quantization noise, and if the external PLL has a higher frequency, the dual-feed-forward PLL offers lower quantization noise. One significant advantage of the dual-feed-forward architecture however, is that there is no divider required in the feedback path of the main PLL. This characteristic is very useful for mm-wave frequency synthesis, where the design of high frequency dividers can be very challenging. At very high frequencies, it becomes very difficult for MMDs based on digital logic to operate. Therefore, the Injection-Locked-Frequency-Divider (ILFD) is often employed which is essentially an LC-tank whose second harmonic is locked to the input frequency, resulting in oscillation at half the frequency of the input signal. The difficult part about this architecture is not only the large area overhead from using LC tanks, but the limited locking range due to the fact that the ILFD has to oscillate at the divider output frequency. If the desired divider output frequency is too far away from the natural oscillation frequency of the LC tank, it will not oscillate. Furthermore, the ILFD has very low programmability in terms of the division ratio. These issues can be especially problematic if we need to cascade multiple ILFDs as prescalers. The HM on the other hand, does not suffer from this kind of issue because the output signal of the HM is already down-converted to the reference frequency which is much lower than the mm-wave output signal, and therefore does not require an LC tank to oscillate which can hinder the locking range and tunability. The use of HM-based PLLs in mm-wave frequency synthesis seems very promising, and is a good direction for future research efforts.

The second potential direction is the investigation of more scaling friendly HM architectures. Like we said in chapter 1, the goal of using HM-based feedback in a PLL is certainly not to go against the trend of process scaling and digital-friendly architectures. For example, the main PLL in a HM-based PLL can make use of a digital PD and LF, and this will actually have a low penalty in terms of noise because the quantization noise from the digital PD (usually a TDC or BBPD) will not get amplified in the loop. However, the HM itself is currently not a very scaling-friendly block, mainly because many passive components are required in the LPFs accompanying the S/H, especially the one used after the S/H operation. As we explained in chapter 4, the S/H in the HM must be accompanied by LPFs before and after it in order to avoid offset spurs. The LPFs before the S/H are usually not too problematic because they only have to filter out high order harmonics of the overall PLL output which means that they can have low complexity and a high bandwidth, which in turn means that they can have a low area. The LPF after the S/H on the other hand, must deal with tones at around the LO frequency, requiring much higher complexity and a relatively narrow bandwidth, which in turn can lead to a large area. A more scaling-friendly method for suppressing the offset spurs would greatly increase the usefulness of the HM-based PLL. One potential solution maybe to employ N-path sampling [67, 89, 90, 91, 92], which is a technique often employed in wireless receivers to provide harmonic rejection. The use of N-path sampling with multi-phase clocks and a greatly-simplified LPF would be a much more scaling-friendly way to suppress the offset spurs than employing a single path sampler with a very bulky LPF. The challenge here would be the efficient generation of multi-phase clocks, and the recombination of the outputs of the N sampling paths. One potential way to realize this concept is shown in Fig. 7.2. The N phases for the S/H clock can be generated efficiently by using an N stage ring oscillator for the external PLL. Furthermore, by also employing an N stage ring oscillator for the main PLL, we can shift the phases of the input to the N path S/H to obtain the same effect as shifting the phase after the S/H, which will be much more difficult. As a result of using the N-path S/H, the offset spurs can be cancelled and a much lower complexity LPF can be used after the S/H, leading to a much more scaling friendly implementation. It is also worthy to note that the HM-based feedback in many ways is similar to the signal path of a wireless receiver. As such, it may be useful to look towards innovations made in the area of wireless communication circuits to gain inspiration for new architectures in HM-based PLLs.

The third potential direction is the use of HM-based PLLs in applications. The key here is that we not only use the HM-based PLL as a black box, high performance frequency synthesizer, but take advantage of the uniqueness of the HM-based architecture over a conventional single-loop PLL. One potential usage of HM-based PLLs that can be thought of here is in multi-channel applications. A unique part about HM-based PLLs compared to conventional single-loop PLLs is the fact that they are cascaded PLLs. Furthermore, unlike some prior works where the first stage is fractional-N and the second stage is integer-N [93, 75, 94, 95], the HM-based PLL has an integer-N first stage and a fractional-N second stage. In a multi-channel wireline transceiver [96, 97] or a wireless phased array transceiver [98, 99, 100] for example, we have multiple units that each require separate clocks, often times with individually tunable frequencies. We can use HM-based PLLs to generate these clocks, except they can all use a common external PLL as shown in Fig. 7.3. That way, we can improve the jitter/power performance of the synthesizer by expending a large amount of power for the external PLL because the power consumption per channel is lowered by the number of channels. Also, since the second stage of the HM-based PLL is fractional-N, the frequencies of the individual clocks can be individually tuned with fine resolution.

Finally, HM-based PLLs can be combined with other methods of noise suppression to further improve the performance. The dual-feedback PLL with PDLPF that we discussed in chapter 5 is one such example. Another possible idea is to use a DTC for quantization noise cancellation inside of the dual-feedback architecture. This may seem excessive at first, but it can actually lead to jitter power performance that is superior to that of both the dual-feedback PLL and the DTC-based PLL. Fig. 7.5 shows the result of an analysis similar to that performed in section 3.5. The jitter power performance of the DTC-based PLL, the dual-feedback PLL, the dual-feedback PLL with a DTC, and a simple integer-N PLL are compared. The power consumption from the VCOs, the DTC, and the HM are considered. We can see that the combination of the dual-feedback architecture with the DTC offers potentially a significantly better jitter power performance than the standalone dual-feedback PLL or DTC-based PLL. This is because not only is the quantization noise eliminated thanks to the DTC, but the thermal and quantization noise from the DTC does not get amplified thanks to the dual-feedback architecture. The remaining overhead comes from the external VCO, but as we saw earlier, the overhead from the external VCO can be suppressed effectively by employing a much wider bandwidth for the external PLL than for the main PLL, and/or making use of the feed-forward noise cancellation technique that we explained in chapter 6. The downside of this idea is that we will lose the benefit of the calibration-free characteristic of the

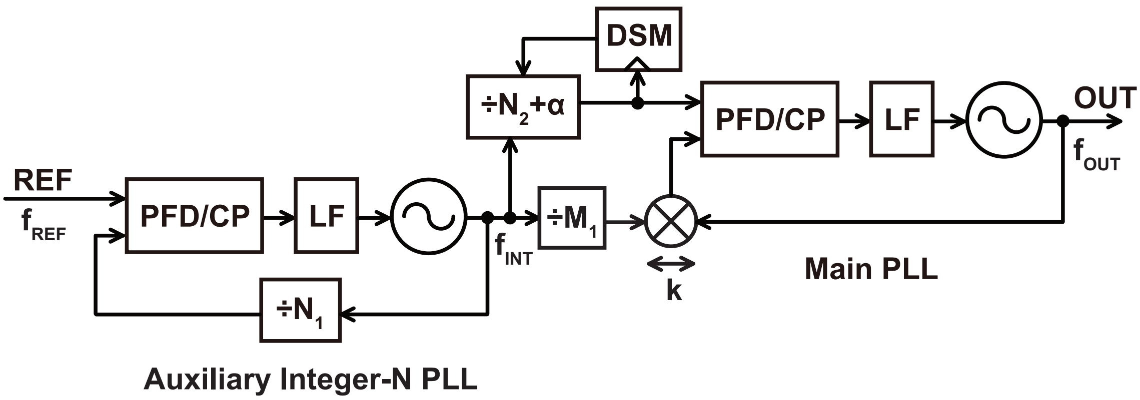

Figure 7.1: The dual-feed-forward PLL architecture [69, 88]

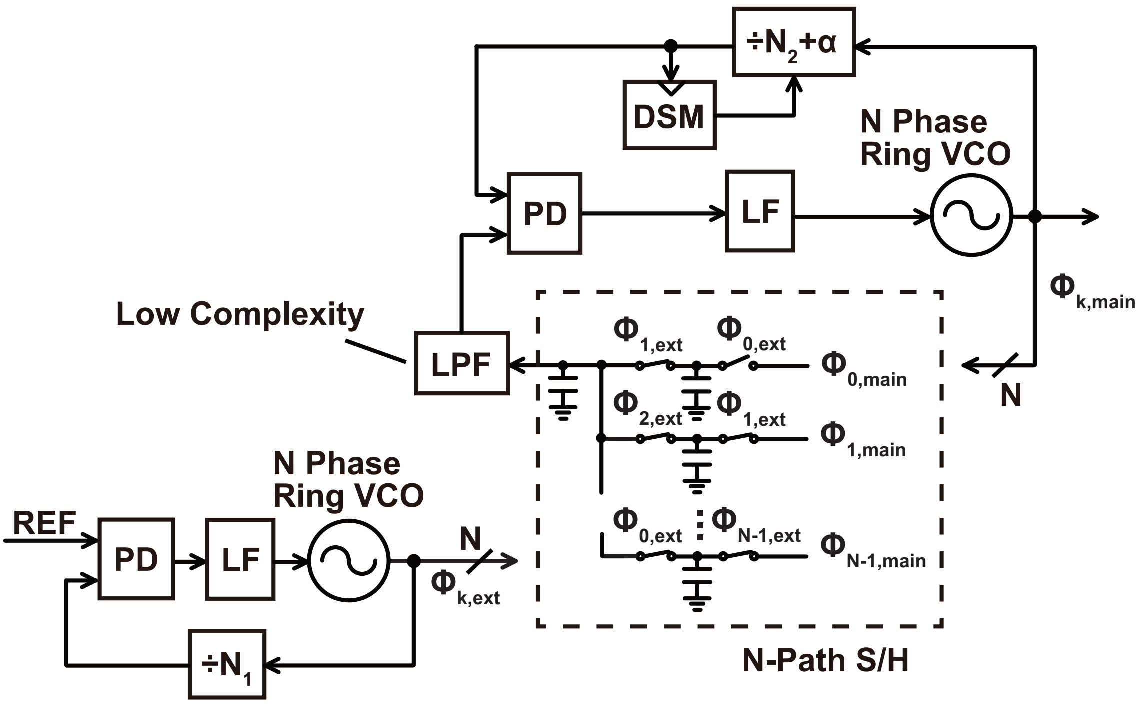

Figure 7.2: The concept of using an N-path S/H to realize a scaling friendly HM

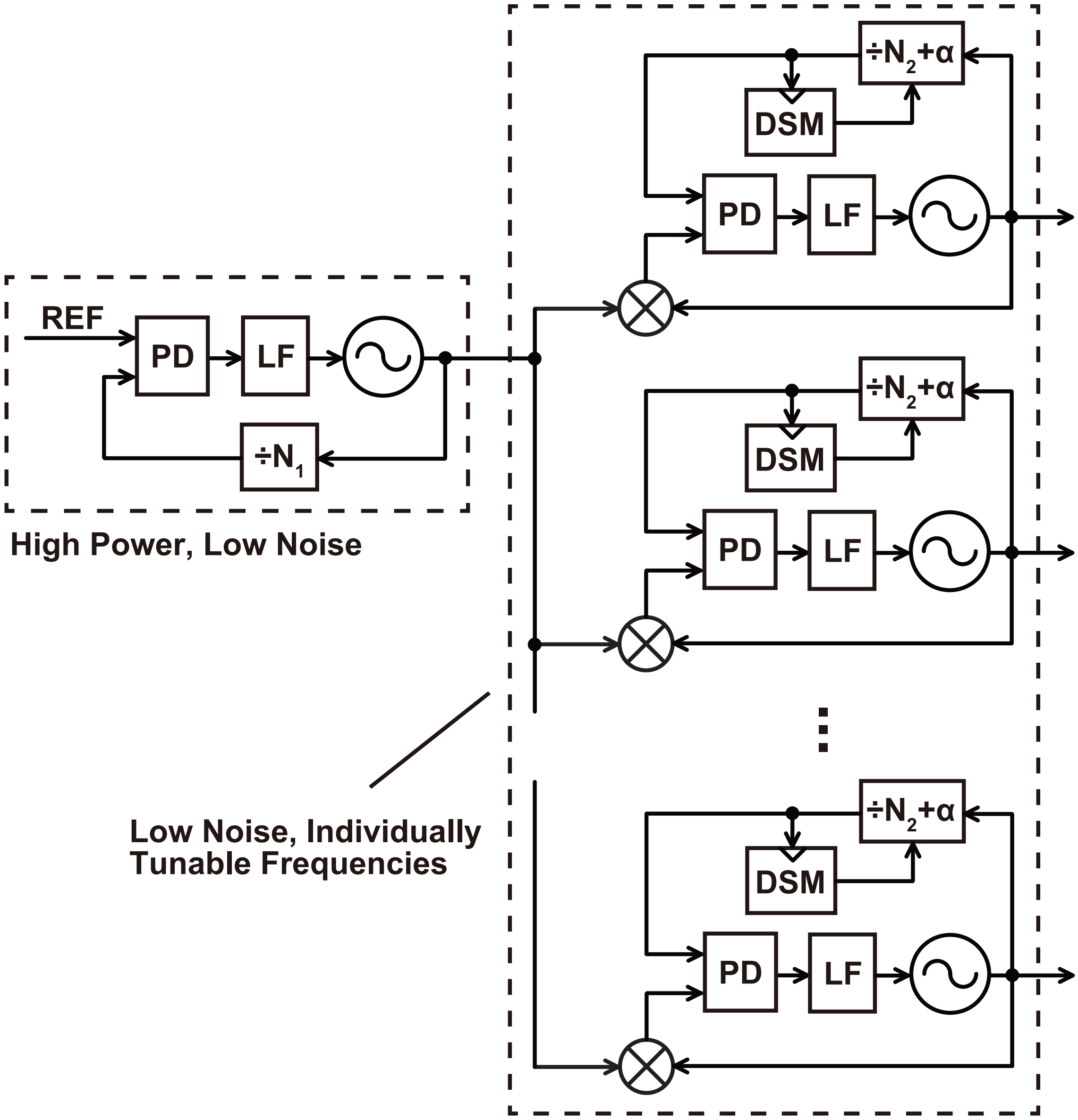

Figure 7.3: The use of a HM-based PLL to generate clocks/LOs for multi-channel applications

dual-feedback PLL. Even this is somewhat arguable however, because the linearity calibration of the DTC in this case may be avoidable since the fractional spurs resulting from its non-linearity will not get amplified. From this example, we can see that investigating the combination of HM-based PLLs with other methods can be a very promising direction.

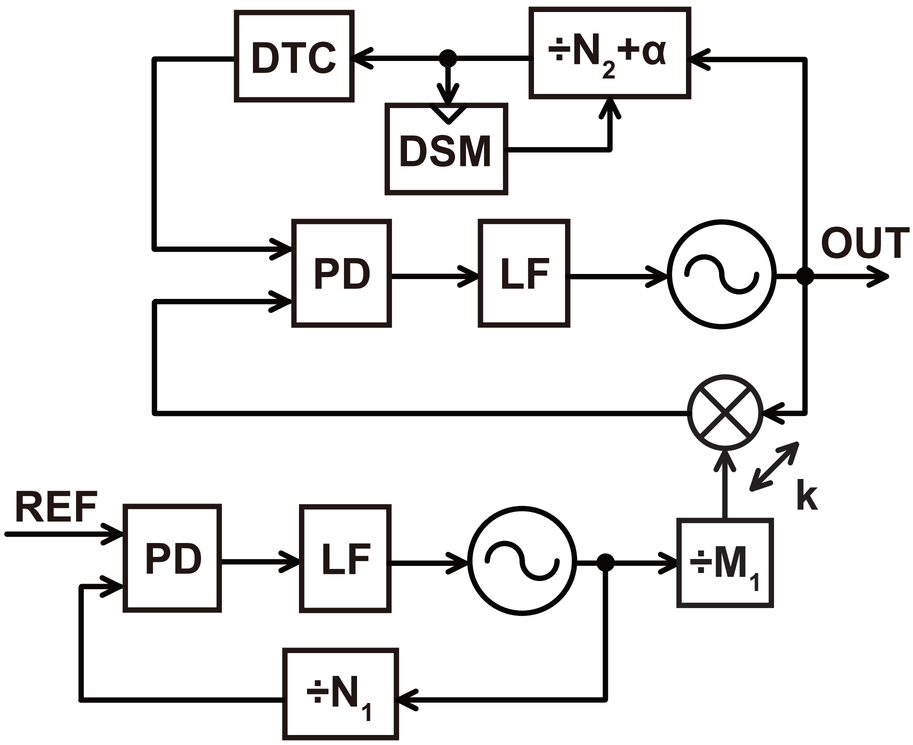

Figure 7.4: A block diagram of a dual-feedback PLL using a DTC

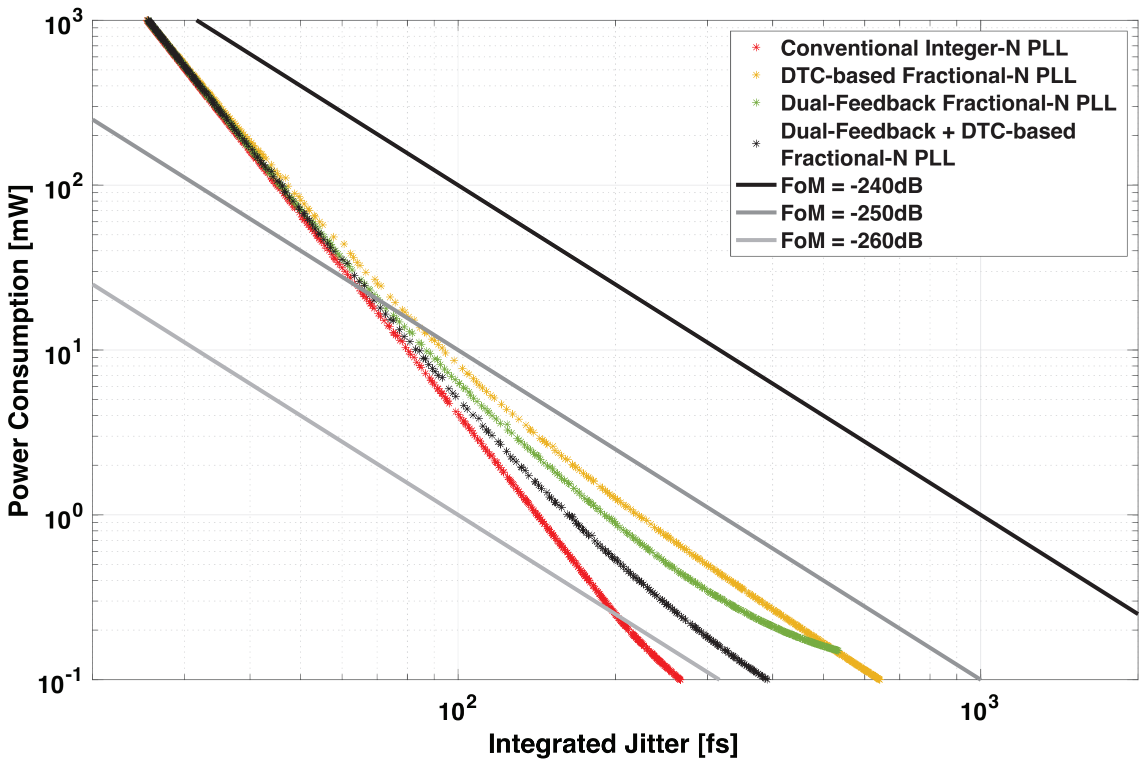

Figure 7.5: The relationship between the optimal jitter and power for the conventional integer-N PLL (red), DTC-based fractional-N PLL (yellow), dual-feedback fractional-N PLL (green), and the dual-feedback fractional-N PLL with DTC (black)